Tax Heterogeneity, Duration Risk, and the Term Structure: An Application to the Municipal Bond Puzzle*

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October 28, 2026

Abstract

I develop a term-structure model in which investors with heterogeneous taxes and balance sheets jointly choose taxable and tax-exempt bonds across maturities under short-sale constraints. These differences endogenously segment ownership and shape both yield curves. The model explains the municipal bond puzzle: short tax-exempt municipal yields closely match taxable yields after the highest household tax rates, while thirty-year yields exceed this benchmark by 118 basis points, implying lower marginal tax rates. Lower-tax investors whose liabilities hedge interest-rate risk bear duration risk more cheaply than households and hold long municipal bonds despite smaller exemption benefits. Long yields therefore reflect their lower taxes. Without targeting municipal yields, the calibrated model explains 83 percent of this puzzle. Empirical tests support the model's prediction that household tax effects weaken with maturity while those of lower-tax investors strengthen. Counterfactuals show how tax reforms affect financing costs in both markets through portfolio substitution.

Keywords: Term Structure, Investor Heterogeneity, Taxes, Endogenous Segmentation, Asset Ownership, Municipal Bonds
JEL Codes: G11, G12, G22, H24, H71, H74

*I am grateful to Moritz Lenel, Wei Xiong, and Motohiro Yogo for their guidance. I also thank Yacine Aït-Sahalia, Caio Almeida, Markus Brunnermeier, Fousseni Chabi-Yo, Aditya Chaudhry, Daniel Chen, Zhiyu (Julie) Fu, Nicolae Gârleanu, Sergei Glebkin, Zhouzhou Gu, Nicolas Hommel, Paymon Khorrami, Jane Li, Shiqi Li, Ziang Li, Jonathan Payne, Yinan Qiu, Julia Selgrad, Amy Huber Wang, Lingxuan (Sean) Wu, Yinjie Yu, Yao Zeng, William (Yinhong) Zhao, Haonan Zhou, and Haoxing Zhu, and participants at the MPSS Conference and Princeton University seminars for constructive feedback.

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1 Introduction

Whose taxes determine bond yields when investors pay different tax rates on the same bonds? U.S. municipal bonds provide a natural setting to study this question because the same tax exemption offers different benefits to investors facing different tax rates. These bonds finance state and local governments through a market of about $4 trillion. Municipal bond interest is exempt from federal income tax, and the exemption is worth most to top-bracket households, who face the highest tax rates. The standard benchmark in the literature therefore uses these households' tax rate to determine municipal yields: at every maturity, the municipal yield should equal their after-tax yield on a comparable taxable bond, such as a treasury or corporate bond (Schwert, 2017; Fleckenstein and Longstaff, 2025). This household benchmark fits short maturities, but thirty-year municipal yields exceed it by 118 basis points. The tax rate implied by these municipal yields and comparable taxable yields falls from about 38% at two years, close to the top-bracket household rate, to just 21% at thirty years. This municipal bond puzzle has been documented in the literature for over thirty years (Feenberg and Poterba, 1991). Under the U.S. tax code, however, investors face very different tax rates, and the data suggest that top-bracket households hold only about half the municipal market. Lower-tax households, banks and insurers hold large amounts even though the exemption is worth less to them. Understanding relative yields therefore calls for a model that accounts for this heterogeneity.

Motivated by this heterogeneity in ownership, I explain the puzzle with an equilibrium term-structure model in which multiple investor groups differ in their taxes and the duration of their liabilities. They choose portfolios jointly across taxable and tax-exempt markets and maturities, subject to short-sale constraints. Their portfolio choices endogenously segment ownership and determine both yield curves. Tax differences draw high-tax investors toward tax-exempt municipal bonds and lower-tax investors toward taxable bonds. However, long-term liabilities allow some lower-tax investors to bear interest-rate risk more cheaply than high-tax households. When rates rise, the value of these liabilities falls, offsetting losses on their bonds. This advantage grows with maturity and draws them into long municipal bonds despite their smaller exemption benefits. Because the exemption saves these investors less tax, they accept a smaller reduction in yield relative to taxable bonds than high-tax households do. Long municipal yields therefore reflect their smaller tax benefits, explaining why they exceed the household benchmark. Calibrated to U.S. taxes, balance sheets and holdings, without targeting municipal yields, the model explains 83% of the thirty-year puzzle. Untargeted empirical tests support its prediction that household tax effects weaken with maturity while those of lower-tax investors, like banks, strengthen. The framework applies beyond municipals because it jointly determines relative yields across taxable and tax-exempt markets. In the counterfactuals, investor substitution transmits tax reforms to the financing costs of both municipal and taxable issuers.

Household mortgage debt, banks' assets and deposits, and insurers' long-term obligations create different net exposures to rate movements. A two-investor illustration with common risk aversion shows how these balance-sheet differences interact with exemption benefits to determine ownership

across maturities. The high-tax household gains more from the exemption, while the lower-tax investor's stronger liability hedge lowers its cost of bearing rate risk. The resulting duration benefit grows with a bond's rate exposure and hence with maturity, whereas the difference in exemption benefits stays constant. Beyond a maturity threshold, the duration benefit outweighs the exemption disadvantage. In this illustration, the household therefore holds shorter municipal bonds, while the lower-tax investor holds longer municipal bonds and risky taxable bonds.

This sorting raises municipal yields above the household benchmark through three channels. First, lower-tax investors value the exemption less and therefore accept a smaller yield reduction, keeping the long municipal yields they price closer to taxable yields than the benchmark predicts. Second, the household alone holds shorter risky municipal bonds and bears rate risk at a higher cost than the lower-tax investor holding comparable taxable bonds. The household's higher required compensation raises municipal yields above the benchmark even at these maturities. Third, because taxes scale returns, they also scale risk. Under the household benchmark, the municipal yield is the taxable yield multiplied by the fraction of taxable interest the household keeps, so its sensitivity to interest rates is scaled by the same fraction. Lower-tax investors keep a larger fraction, so the long municipal yields they price move more with interest rates than the benchmark predicts. Holders require compensation for that additional rate risk, and household portfolio substitution carries the effect to shorter municipal bonds. All three channels vanish at the riskless shortest maturity, where the household's tax advantage determines relative yields.

Other motives can also lead lower-tax investors to hold long municipal bonds and generate the same qualitative pricing pattern. Lower trading costs or better default diversification can do so when the advantage grows with maturity and applies equally to taxable and tax-exempt bonds (Section 4.7). I focus on liability hedging because investors' widely differing liability durations provide an economically important and measurable source of differences in the cost of bearing interest-rate risk. Observed balance sheets and maturity holdings discipline these differences in the quantitative model.

The quantitative application asks how much of the puzzle observed taxes and portfolios can explain under my model. It includes high- and lower-tax households in each state, banks, property-and-casualty insurers, life insurers and tax-neutral investors trading in state municipal markets and a common taxable market. I measure tax treatment, bond supplies and investor balance-sheet sizes directly from data and estimate rate persistence and volatility from historical short rates. Observed investor ownership and allocation mainly discipline their unobserved exposure to interest rates, while taxable yields help discipline risk aversion. Municipal yields and empirical test coefficients are not targeted in calibration. The model accounts for 97 of the 118 basis points at thirty years, and measured default, liquidity and capital-gains differentials explain most of the remainder.

I evaluate the model against the household benchmark using three empirical tests, none targeted in the calibration. The model predicts that variation in top-bracket households' exemption benefits is fully reflected in implied taxes at the shortest maturity and less strongly at longer ma-

turities. Changes in lower-tax investors' exemption benefits have no effect at the shortest maturity and matter more at longer maturities. First, most states exempt residents' interest on their own municipal bonds from state income tax. Cross-state variation in state tax rates is reflected almost one-for-one in implied state taxes from observed municipal bond yields at short maturities and progressively less at longer maturities. Second, bank-qualified designation gives banks an additional tax benefit on a subset of municipal bonds issued by small borrowers. This bank-only benefit is barely reflected in short municipal yields and increasingly reflected at longer maturities. Third, I compare municipal yield changes with changes in the treasury reference yield at the same maturity. This relative response rises with maturity, consistent with lower-tax holders making long municipal yields more sensitive to interest rates.

Two counterfactual reforms show how removing different tax benefits changes financing costs and tax receipts. The first removes a state's exemption for residents' interest on its own municipal bonds, as Illinois, Iowa and Wisconsin generally do. Residents retain the federal benefit of holding municipal bonds but, as in states without an income tax, no longer have a tax incentive to favor home-state over other municipal bonds. High-tax households therefore mainly switch between municipal markets, with relatively modest changes in the maturities they hold. The state taxes the interest on these replacement investments, so selling local bonds does not substantially erode its tax base. Meanwhile, long municipal yields also reflect the valuations of institutions and nonresidents whose tax treatment does not change. Long yields therefore respond only partly to the lost household exemption, consistent with the partial pass-through of state taxes at long maturities documented in the first empirical test. Across seven state experiments averaged over 2000–2017, added state receipts exceed local issuers' added financing costs: own-state issuers incur $0.68 to $0.88 in added annual financing costs per dollar of added state receipts. Portfolio substitution also raises borrowing costs for other states' municipal issuers and taxable issuers.

The second reform also removes the federal exemption, taxing municipal interest like taxable interest for every investor. Taxable municipal issuance has a precedent in the 2009–2010 Build America Bonds program, which subsidized issuers' interest payments. Unlike under state repeal, high-tax investors can no longer keep an exemption by switching to other states' municipal bonds. With no tax reason left to bear the rate risk of long bonds, they shift toward shorter bonds and cash, which pay less interest and so generate less tax revenue. Lower-tax and tax-neutral investors take over more of the long bonds but pay little or no additional tax on the interest. Bearing more rate risk, they require higher yields in both markets, so financing costs rise by more than tax receipts. Averaged over 2000–2017, the reform raises annual financing costs by $46.3 billion and annual tax receipts by $38.7 billion. Municipal and taxable issuers together incur $1.20 in added costs per dollar of added federal and state receipts, including $0.13 borne by taxable issuers. The excess becomes higher after-tax interest income, distributed unevenly across investors. Lower-tax households, life insurers and tax-neutral investors gain on average, while high-tax households, banks and property-and-casualty insurers, which used to benefit more from the exemption, lose.

The framework is not only about municipal bonds. It explains relative bond pricing across taxable and tax-exempt markets, with investor portfolio substitution transmitting tax changes in one market to yields in the other. I apply it to municipal and taxable bonds because municipal tax rules provide clean variation to test the mechanism and address a longstanding pricing puzzle. Its applications extend to other settings where tax treatment differs across investors and markets, including state-tax differences between treasury and corporate bonds, tax-favored accounts and cross-border withholding. More broadly, heterogeneous taxes, together with balance-sheet differences and limits to arbitrage, shape endogenous market segmentation and are important for understanding bond yields and the effects of tax policy.

Section 2 reviews related work, and Section 3 presents the tax, ownership and yield evidence. Section 4 develops the model. Section 5 calibrates it and accounts for the municipal puzzle, and Section 6 tests its empirical predictions. Section 7 studies tax policy, and Section 8 concludes. Formal results, proofs and supporting evidence are in the appendix.

2 Related Literature

The paper contributes to research on how taxes affect portfolio choice and asset prices. Theoretical work studies how tax differences shape portfolios and equilibrium prices (Miller, 1977; Auerbach and King, 1983; Dybvig and Ross, 1986; Dammon and Green, 1987), and Sialm (2009) provides evidence that taxes are reflected in equity valuations. I study how these differences shape the bond yield curve. Under short-sale constraints, investors choose jointly across markets and maturities. Their taxes and costs of bearing rate risk determine who holds each bond and whose tax treatment its yield reflects.

Research on market segmentation links the yield curve to bond supply, investors' maturity preferences and their capacity to bear risk (Vayanos and Vila, 2021; Greenwood and Vayanos, 2014; Guibaud et al., 2013). Portfolio adjustment also transmits supply shocks across asset classes and currencies (Greenwood et al., 2018, 2023).[1] I explain how taxes and investor balance sheets generate partial segmentation across markets and maturities. Participation responds to taxes and balance-sheet exposures, so tax changes alter both yield curves through joint portfolio choice.

The model's explanation of participation connects to research on institutional balance sheets and asset pricing. Intermediaries' capital and exposure to interest-rate movements affect the risk premia they require (He and Krishnamurthy, 2013; Haddad and Sraer, 2020). Kekre et al. (2024) show how losses on arbitrageurs' bond portfolios can amplify the effect of monetary tightening on long yields. Insurers' long-term liabilities also shape their demand for long bonds and their capacity to bear credit risk (Domanski et al., 2017; Li, 2026).[2] I show how a balance-sheet advantage in

[1] Constraints on arbitrage can limit this adjustment (Gromb and Vayanos, 2018; Siriwardane et al., 2025). Related empirical work measures investors' asset demand and household rebalancing (Koijen and Yogo, 2019; Gabaix et al., 2026).

[2] Municipal offering spreads reflect banks' rate exposure and funds' liquidity needs (Pati, 2023). Liability structures also help explain investors' responses to credit-market downturns (Coppola, 2025).

bearing rate risk can offset a smaller tax benefit, explaining why some lower-tax investors hold long tax-exempt bonds and how their holdings affect equilibrium yields.

The municipal bond puzzle has motivated a longstanding literature on relative bond yields. Early explanations relied on a tax arbitrage available only to commercial banks, which before 1986 could deduct most of the interest on funds used to carry municipal bonds. In these accounts, bank demand set short municipal yields and household demand set long ones (Mussa and Kormendi, 1979; Hendershott and Koch, 1980; Kidwell and Koch, 1983; Skelton, 1983). The Tax Reform Act of 1986 removed this deduction for most new purchases and reduced banks' municipal holdings (Hines and Horton, 2024), yet long municipal yields remained high relative to after-tax taxable yields (Green, 1993). Later work studied default risk, liquidity, callability and the taxation of discount bonds (Ang et al., 2010; Schwert, 2017; Brown, 2011).[3] The maturity pattern survives careful adjustment for these features: in default-free pre-refunded bonds (Chalmers, 1998), in swap rates free of bond-specific credit and liquidity risk (Longstaff, 2011) and in option-free bonds adjusted for credit and liquidity (Fleckenstein and Longstaff, 2025). On a common household after-tax scale, my bond-side adjustments leave a large long-maturity gap. My model explains most of this remaining gap through endogenous participation by investors with different taxes and costs of bearing rate risk.

Empirical research on municipal tax policy links state tax benefits to local ownership, financing conditions and risk sharing across states (Hendershott and Kidwell, 1978; Pirinsky and Wang, 2011; Babina et al., 2021; Moussawi, 2022, 2024). Changes in banks' tax benefits likewise affect municipal investment and access to financing (Hines and Horton, 2024; Dagostino, 2025; Benson and Marks, 2024).[4] I test how household and institutional tax effects vary with maturity. Household state-tax pass-through declines, while the yield reduction associated with bank qualification rises. This opposite pattern tests the participation mechanism against a single household tax rate applied across the curve.

Fiscal analyses of the municipal tax exemption examine its revenue cost and benefits to borrowers. Poterba and Ramírez Verdugo (2011) show that estimates of forgone revenue depend on how households would reinvest without the exemption. Garrett et al. (2023) show how auction competition affects the borrowing-cost savings from tax benefits. I evaluate reforms by running counterfactuals that incorporate the equilibrium impact of investor reallocation, capturing changes in tax receipts and financing costs.

3 Taxes, Ownership and the Municipal Puzzle

In this section, I describe the data and the tax benefits available to different investors, then show that municipal ownership extends well beyond high-tax households. I measure the municipal puzzle

[3] Related pricing, tax-policy and maturity-clientele analyses include Fama (1977), Livingston (1982), Poterba (1986) and Brown and Sirmans (2011). Trading-cost evidence informs my liquidity adjustment (Harris and Piwowar, 2006; Green et al., 2007; Bagley et al., 2023). For surveys, Cestau et al. (2019) and Spreen and Gerrish (2022) summarize the long stream of literature on the municipal bond puzzle. Appendix C.4 compares the model with Green's benchmark.

[4] Municipal fund flows and fragility also affect yields and financing (Li et al., 2024; Adelino et al., 2026).

and examine whether bond characteristics or the choice of taxable benchmark explain its maturity pattern.

3.1 Data

I combine data on bond issuance, investor holdings and taxes. Mergent provides issuance records, ratings, bond characteristics and call schedules for municipal bonds and corporate bonds. Insurers' Schedule D filings with the National Association of Insurance Commissioners (NAIC) and a pension holdings panel collected through Freedom of Information Act requests allow me to examine institutional portfolios at the level of individual bond holdings. I use the Financial Accounts to measure aggregate holdings, the Survey of Consumer Finances to measure household balance sheets, and IRS Statistics of Income to document household tax-exempt interest income across income brackets.

The main yield comparison covers 2000–2020. Its taxable benchmark is the A-or-above High Quality Market (HQM) corporate zero-coupon curve from FRED. Federal Reserve treasury curves (Gürkaynak et al., 2007) provide the reference for the interest-rate response test. I measure trading and liquidity using municipal transactions from the Municipal Securities Rulemaking Board (MSRB) and corporate transactions from the Trade Reporting and Compliance Engine (TRACE) Enhanced database. Markit provides municipal and corporate credit default swap quotes, and the Tax Foundation provides state tax schedules. Appendix D.6 details the sources and their coverage.

3.2 Municipal issuance

Table 1: Municipal issuance: descriptive statistics (sample means).

Variable	Full sample	Tax-exempt, non-AMT	
		AA and above (score ≤ 3)	A or above (score ≤ 7)
Offering yield (%)	3.11	2.82	2.96
Corporate benchmark yield (%)	4.47	4.33	4.41
Maturity at issuance (years)	10.15	10.76	10.86
Offering price (per \$100 par)	102.88	104.68	103.85
Issue rating (1 = AAA, 3 = AA, 7 = A−)	3.80	2.18	3.55
Implied tax rate $1 - y^M/y^C$	0.29	0.35	0.33
Share callable	0.46	0.51	0.51
Share tax-exempt, non-AMT	0.92	1.00	1.00
Sample size (CUSIPs)	3,014,420	984,210	1,990,791

Notes. The table reports bond-level means for issues of 2000–2020 before the call and issuance-premium adjustments of Section 3.6, with one observation per municipal issue (CUSIP). The full sample contains all issues. The AA-and-above and A-or-above columns restrict to federally tax-exempt, non-AMT issues (not subject to the alternative minimum tax) with average agency score ≤ 3 and ≤ 7, the latter matching the A-or-above corporate credit band. The implied-tax row averages the bond-level ratio $1 - y^M/y^C$, winsorized at the first and ninety-ninth percentiles, so it need not equal one minus the ratio of the displayed yield means. Each row uses non-missing observations, so the full-sample rating mean uses the 2,292,092 rated issues. The corporate benchmark is the corporate zero yield for the issue month and maturity.

Table 1 describes municipal bonds issued during 2000–2020, with one observation per bond identifier (CUSIP). The full sample contains more than three million issues, with an average maturity of about ten years. Most pay federally exempt interest: taxable municipal bonds account for 5.8% of issues, bonds subject to the alternative minimum tax (AMT) account for 1.9%, and the remaining issues are tax-exempt, non-AMT bonds. Call provisions allow issuers to redeem bonds before maturity and are common in the data. Callable bonds account for 46% of all issues and 51% of the high-grade, tax-exempt samples, making callability important for comparing yields.

The table also reports two high-grade samples of tax-exempt, non-AMT bonds. The broader sample includes A-or-above issues (average agency score ≤ 7 on the twenty-two-notch scale) and matches the credit band of the corporate benchmark used for the main yield curves. The state-tax and interest-rate tests of Section 6 use the AA-and-above sample (≤ 3) to minimize the impact of default risk.

3.3 Bond taxation and the representative-tax benchmark

For a taxable household, interest on taxable bonds is ordinary income, while interest on tax-exempt municipal bonds is federally exempt. Interest includes both coupon payments and annual original-issue-discount accrual. Institutions generally do not receive the full exemption benefit from holding municipal bonds. For example, property-and-casualty (P&C) insurers pay an effective federal tax of 5.25% on municipal interest, both before and after the 2017 Tax Cuts and Jobs Act (TCJA). Let τ_i^C and τ_i^M denote investor i's effective tax rates on taxable and municipal interest. I define the relative after-tax share as the fraction of taxable interest retained divided by the fraction of municipal interest retained:

$$\xi_i \equiv \frac{1-\tau_i^C}{1-\tau_i^M}. \tag{1}$$

Investor i's exemption benefit is $1-\xi_i = (\tau_i^C - \tau_i^M)/(1-\tau_i^M)$. A smaller ξ_i means a larger benefit. When municipal interest is fully exempt, $\xi_i = 1-\tau_i^C$, so the benefit equals the tax rate on taxable interest. Appendix D explains the underlying tax rules.

The representative-tax benchmark uses the top-bracket household's tax treatment at every maturity. Write y_T^M and y_T^C for municipal and taxable yields at maturity T, and H for this household. Under the benchmark, the two bonds offer the household the same after-tax yield, $y_T^M = \xi_H y_T^C$. I call $\xi_H y_T^C$ the household benchmark and $1-\xi_H$ the household wedge. The benchmark's yield-implied tax is therefore $1 - y_T^M/y_T^C = 1-\xi_H$. With fully exempt municipal interest, it equals τ_H^C. Investors with different ξ_i cannot all be indifferent at this yield ratio. Their participation in the municipal bond market is the central question of the model.

3.4 Investor taxes and ownership

The tax benefit of municipal interest varies substantially across investors in the roughly $4 trillion municipal market. Table 2 summarizes these differences before the 2017 TCJA. For households and insurers, the net municipal bond exemption benefit is $1-\xi_i$ in (1). Households in higher tax

Table 2: Investor heterogeneity in the municipal market before the 2017 Tax Cuts and Jobs Act

Investor type	Federal treatment	State treatment	Net exemption benefit
High-tax household	Interest exempt	Exempt on in-state bonds only	0.35–0.43
Middle-tax household	Interest exempt	Exempt on in-state bonds only	0.25–0.33
Low-tax household	Interest exempt	Exempt on in-state bonds only	0.10–0.15
P&C insurer	Exempt, less proration	Premium taxes, no distinction	0.31
Life insurer	Exempt, less proration	Premium taxes, no distinction	0.21
Bank	Exempt, financing cost disallowed	Varies by state	0.15–0.33
Other (pensions, etc.)	Varies	Varies	< 0.05

Notes. For households and insurers, the net exemption benefit is $1 - \xi_i$, from statutory rates and proration over 2000–2017. For banks it also deducts the financing-tax cost described below. Appendix Table 37 gives the values after the 2017 Act. Household values are federal marginal rates in three groups (35% and above, 25% to 33%, below 25%), and the top value includes the 3.8% net investment income tax from 2013. The quantitative model pools middle- and low-tax households into one lower-tax group, whose benefit with home-state taxes is 0.21 to 0.27 across states in 2015–2017, against 0.43 to 0.50 for the high-tax group (Table 3). Proration cuts a P&C insurer's losses-incurred deduction by 15% of its exempt interest and a life insurer's reserve deduction by the policyholders' share. Section 265(b) disallows the share of a bank's interest expense allocable to exempt holdings, of which 80% stays deductible for bank-qualified bonds. The bank value, net of that disallowance, is the break-even implied tax, equal to the bank's tax rate minus its annual financing-tax cost per dollar of municipal holdings divided by the taxable yield. That cost is 1.02% in 2000–2002 and 0.11% in 2015–2017, and the bank range covers the six three-year periods of 2000–2017 and maturities of two to thirty years. States generally tax insurers through premium taxes, which do not distinguish municipal from taxable interest, and banks through franchise or income taxes.

brackets receive a larger benefit because they pay more tax on taxable interest. Exemption from home-state income tax adds to the benefit on in-state municipal bonds. Appendix D gives the detailed tax treatment of each investor group and the changes after the 2017 Act.

Insurers receive smaller benefits than high-tax households because exempt interest reduces their deductions. The reduction is less severe for P&C insurers than for life insurers, giving P&C insurers the larger benefit. Banks generally lose the interest deduction allocated to municipal holdings, with partial relief for bank-qualified bonds, so the table reports their benefit net of this financing-tax cost.[5] Tax-exempt institutions such as qualified pension funds and endowment funds already receive taxable interest free of current income tax and gain no additional benefit from the municipal exemption. Public pension funds in the sample hold almost no municipal bonds (Appendix C.8).

Figure 1 shows that municipal ownership spans these tax groups. Panel (a) reports aggregate holdings: households and municipal funds hold the largest amounts, while banks and P&C insurers also hold substantial portfolios despite smaller exemption benefits. P&C insurers allocate about 40% of their bond portfolios to tax-exempt municipals and maintain a larger municipal portfolio share than life insurers throughout the sample (Appendix C.8). Panel (b) documents household tax-exempt interest income across income brackets. Assigning municipal-fund holdings to households and allocating household holdings using tax-exempt interest shares puts roughly half the

[5]The quantitative model charges the bank financing-tax cost separately. Table 3 therefore reports the bank's gross exemption benefit $1 - \xi_i$ before that cost.

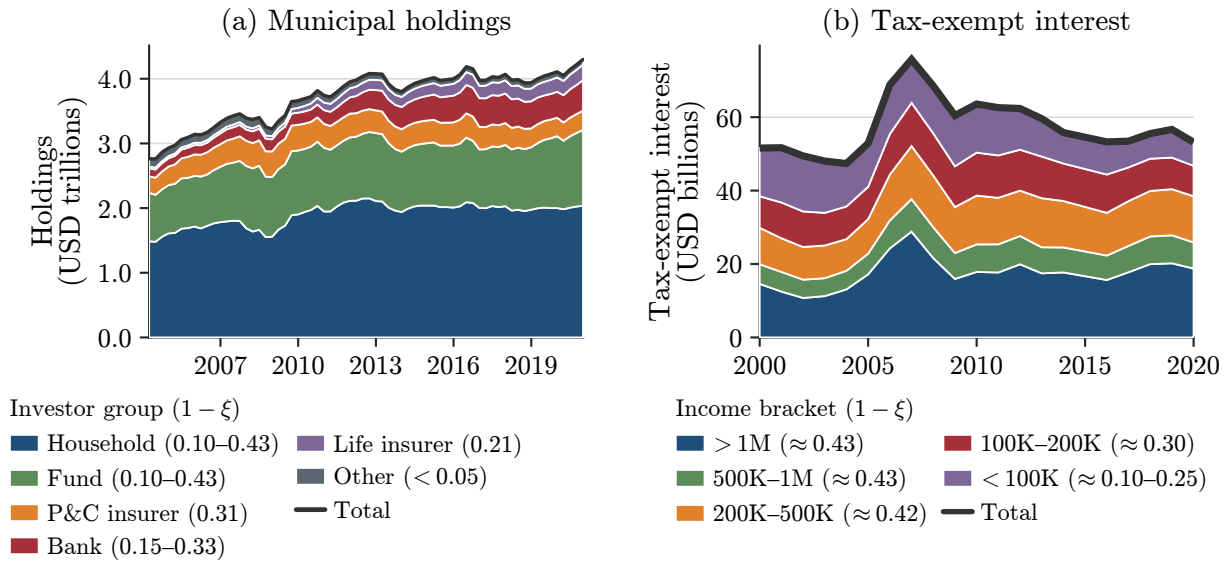


Figure 1: Municipal holdings and tax-exempt interest by investor group. Panel (a): direct holdings of taxable and tax-exempt municipal securities, 2005–2021 (USD trillions, Financial Accounts, with households including nonprofits). Panel (b): tax-exempt interest, an income flow, on taxable individual returns by adjusted-gross-income bracket, 2000–2020 (USD billions, IRS Statistics of Income). 2015 lacks comparable bracket detail and is omitted. Legend values in parentheses are net exemption benefits: in panel (a), the values before the 2017 Act from Table 2, with funds assigned their household shareholders' range, and in panel (b), approximate values by bracket.

market with high-tax households.[6] The remaining holdings by investors with smaller exemption benefits motivate a model in which investors with different tax treatment choose municipal and taxable bonds jointly. The quantitative model includes household groups by state of residence, insurers, banks and a residual group and uses this observed ownership to discipline the calibration (Section 5.1).

3.5 Limits to arbitrage

Tax rules and trading frictions limit the trades that could exploit yield differences or equalize investors' costs of bearing rate risk. Interest and short-sale payments incurred to carry exempt bonds generally lose their deduction. Consider a household that shorts a long-maturity corporate bond to hedge its municipal holdings. With the deduction denied, an equal-principal position earns coupon carry equal to the pretax municipal-minus-corporate yield difference. This carry is negative along the sample-average par curves and about -95 basis points a year at thirty years. Municipal shorts are scarce and costly because lenders owe tax on the payments that replace exempt coupons. Off-exchange swaps face eligibility restrictions, while futures losses can face deduction limits and deferral. The model summarizes these tax asymmetries and trading frictions through nonnegative bond positions and cash balances, ruling out short sales and borrowing. Removing the swap

[6] I use IRS Statistics of Income shares of tax-exempt interest on taxable returns, including fund distributions, to approximate household holdings by income group in 2010–2017. Adjusted gross income of at least $200,000 proxies for the high-tax group.

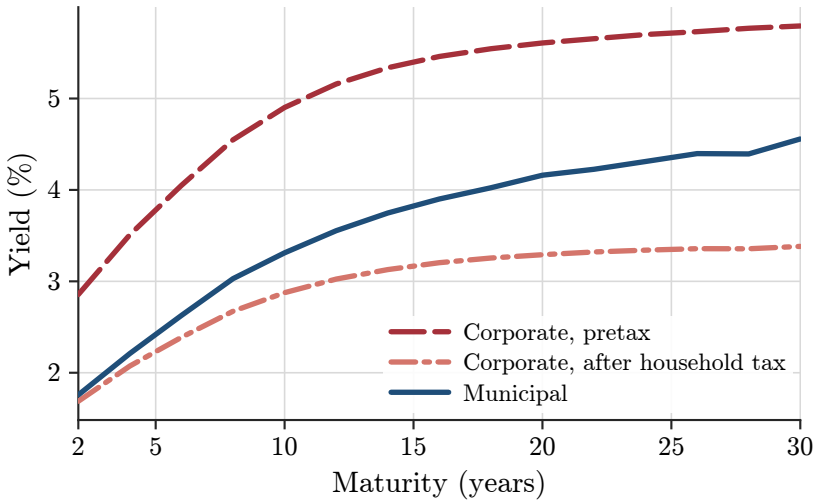


Figure 2: The municipal puzzle: A-or-above zero-coupon yields, 2000–2020. Municipal yields remove callability and the issuance premium. Averaging follows Appendix C.1.1. Appendix C.10.1 gives the AA-and-above counterpart.

frictions for households leaves most of the gap: in the quantitative model for 2015–2017, giving every household fee-free swap access lowers the thirty-year gap by only 9.1% (Appendix D.5.5). Appendix D.5 examines the tax treatment, carry and residual risk of these trades.

3.6 The municipal puzzle

The municipal puzzle is a maturity pattern: short municipal yields are close to the household benchmark, while long municipal yields substantially exceed it. To measure this pattern, I use the seasoned, option-free A-or-above corporate zero-coupon curve as the taxable comparison. High-grade corporate bonds are closer to municipals in credit and liquidity than treasuries. Treasury yields also contain a premium for safety and liquidity (Krishnamurthy and Vissing-Jorgensen, 2012), which would enter the measured gap if treasuries supplied the benchmark. Studies using treasury yields (Chalmers, 1998) or swap rates (Longstaff, 2011) as the taxable benchmark also find the municipal puzzle.

I construct municipal zero-coupon yields in the same credit band from offering prices and contractual cash flows. Issuance data provide broad coverage and a common measurement date. Offering yields also avoid the direct tax effects of capital gains and market discounts arising from prior secondary-market trades. I remove callability and the issuance premium to make municipal yields comparable with the seasoned, option-free corporate curve. The issuance premium is the extra yield at issuance relative to comparable secondary-market trades.[7] Let $y_t^C(T)$ denote the taxable yield and $y_{s,t}^M(T)$ the municipal yield after these two adjustments, where T is maturity, s is the issuer's state and t is the issue date. The municipal puzzle is the gap between $y_{s,t}^M(T)$ and the household benchmark $\xi_{H,s,t}\, y_t^C(T)$, where $\xi_{H,s,t}$ is the relative after-tax share of a top-

[7]The market-discount tax applies to later secondary purchasers. Its expected effect on resale values can also affect offering prices. The capital-gains adjustment includes this effect as well as taxes on realized gains.

bracket household resident in state s at date t, combining federal and home-state income taxes. The yield-implied tax expresses the same comparison in tax-rate units:

$$ImpliedTax_{s,t}(T) \equiv 1 - \frac{y^M_{s,t}(T)}{y^C_t(T)}. \tag{2}$$

The two measures satisfy

$$\underbrace{y^M_{s,t}(T) - \xi_{H,s,t}\, y^C_t(T)}_{\text{municipal yield gap}} = \underbrace{\left[(1-\xi_{H,s,t}) - ImpliedTax_{s,t}(T)\right]}_{\text{shortfall from the household wedge}}\, y^C_t(T). \tag{3}$$

For positive taxable yields, a positive gap means an implied tax below the household wedge $1-\xi_{H,s,t}$. In Figure 2, the municipal yield is near the household benchmark at the short end but about 118 basis points above it at thirty years. The implied tax falls from about 38% to 21% while the average household wedge remains near 41% (Figure 7(d)).

The declining implied-tax pattern also appears within issuer-matched tax-exempt and taxable municipals (Section 3.8) and in the AA-and-above and unlimited-tax general-obligation samples (Appendices C.10.1 and C.3).

3.7 Comparability and bond-side adjustments

Studies such as Fleckenstein and Longstaff (2025) use secondary-market prices of option-free bonds to examine the municipal puzzle. Consistent with that comparison, the puzzle already reflects two comparability adjustments, which remove callability and the issuance premium. I then assess three bond-side adjustments, for capital-gains taxation, liquidity and default risk, as explanations of the puzzle. Each bond-side adjustment measures the difference in compensation between municipal and corporate bonds, expressed in the household's after-tax yield units. Compensation common to both markets therefore cancels. The household wedge, one minus the ratio of averaged after-tax to pretax corporate yields, averages 41.3% and is nearly flat across maturities. Appendix C.1 explains the adjustments, and Appendix Table 14 reports their contributions by maturity. Appendix C.4 discusses the tax timing of taxable bonds in Green (1993).

At thirty years, the call and issuance-premium adjustments reduce the 160-basis-point spread over the household benchmark to the 118-basis-point municipal puzzle. The default, liquidity and capital-gains differentials explain 40% of the puzzle at ten years but only 12% at thirty. Subtracting them leaves a residual before the model of 103 basis points at thirty years for comparison with the model in Section 5.3. At two years, their sum exceeds the small puzzle, leaving −7 basis points, consistent with the convenience premium discussed in Fleckenstein and Longstaff (2025). The adjustments leave the central pattern intact: the residual before the model widens with maturity.

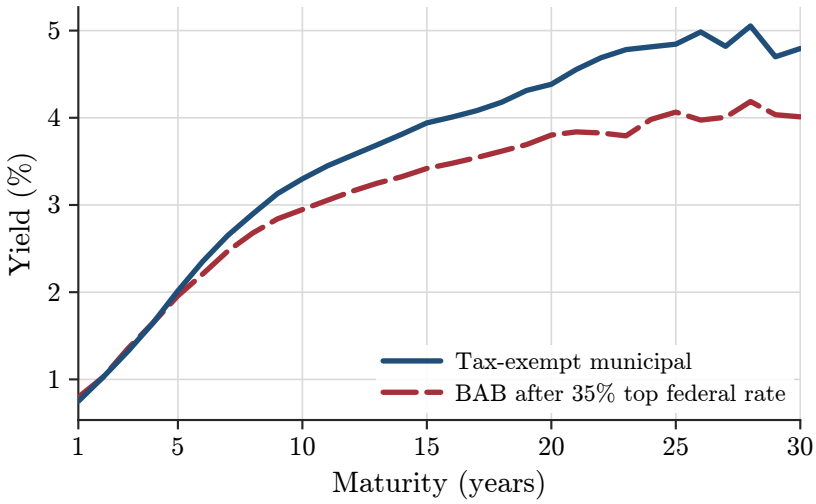


Figure 3: Tax-exempt and Build America Bond (BAB) offering yields by maturity, AA-and-above issues, April 2009–December 2010. BAB yields are multiplied by 0.65, one minus the 35% top federal rate in 2009–2010. Appendix Table 21 reports regressions with and without call-status matching. Appendix C.2 describes the sample.

3.8 Issuer-matched evidence

Because the puzzle compares municipal with corporate bonds, its maturity pattern could come from differences between the two sets of borrowers, such as credit risk or liquidity, which the bond-side adjustments of Section 3.7 address. A cleaner test compares taxable and tax-exempt bonds sold by the same municipal issuer, which share its default risk and are matched on maturity, issue quarter and, in one specification, call status. Their main difference is tax treatment, so a yield gap that still widens with maturity shows that the pattern is not explained by issuer characteristics such as default risk. The Build America Bonds (BAB) program of 2009–2010 makes this comparison possible: it let municipal issuers sell taxable bonds, with a federal subsidy of 35% of their interest. Because the top federal household rate was also 35%, I multiply BAB yields by 0.65 to put them on the top-bracket household's federal after-tax scale and leave tax-exempt offering yields unchanged. Under the household benchmark, the two yields are then equal at every maturity. I estimate how their difference varies with maturity among bonds from the same issuer:

$$y_b \;=\; \alpha_1\, EXMP_b \;+\; \beta\,(EXMP_b \times n_b) \;+\; \varphi_C\, y^{C}_{n_b,t_b} \;+\; \lambda_{j(b)\times n_b\times q_b} \;+\; \varepsilon_b, \tag{4}$$

where y_b is the offering yield for a tax-exempt bond and 0.65 times the offering yield for a BAB. The indicator $EXMP_b$ equals one for a tax-exempt bond and zero for a BAB, and n_b is maturity in years. The fixed effect $\lambda_{j(b)\times n_b\times q_b}$ compares bonds from issuer $j(b)$ at the same maturity and in the same issue quarter q_b. The corporate yield $y^{C}_{n_b,t_b}$, measured at the bond's maturity and issue month t_b, controls for market yield movements within that quarter. The coefficient β measures how the tax-exempt yield minus the household after-tax BAB yield changes with maturity. The household benchmark implies $\beta = 0$, and a positive β is the municipal puzzle within a single issuer.

The estimated gradient β is 3.4 basis points per year of maturity in the AA-and-above sample and 3.3 in the A-or-above sample, and 3.1 and 3.0 with call matching. With five-year maturity bins, the coefficients rise monotonically: in the AA-and-above sample with call matching, the $[25, 30]$-year coefficient is 63.9 basis points above the $[1, 5)$-year bin (Appendix Table 21). Figure 3 shows the same widening gap in mean yields. Long tax-exempt yields therefore exceed the household benchmark even against taxable bonds of the same issuer, so the puzzle does not come from differences between municipal and corporate borrowers.

4 Model

Motivated by the ownership structure of municipal bonds, I develop a term-structure model in which investors differ in tax treatment and balance-sheet exposure to interest rates. They choose bonds jointly across markets and maturities, subject to short-sale constraints. The model answers two questions: why investors with smaller exemption benefits hold tax-exempt bonds, and how their participation changes yields relative to the household benchmark.

I first present the general framework. A two-investor, two-market affine benchmark then isolates the mechanism, with formal conditions and proofs in Appendix A. Section 5 extends the framework for calibration.

4.1 Bonds, investors and portfolio choice

Time is discrete. Each investor solves an after-tax mean–variance portfolio choice problem across all assets and maturities, subject to short-sale constraints.

Interest rates. A single rate state z_t drives the short rate r, the pretax return on cash, which is subject to a zero lower bound:

$$z_{t+1} = \rho z_t + \sigma_r \varepsilon_{t+1}, \qquad r(z_t) = \max\{\bar{r} + z_t,\, 0\}, \qquad 0 < \rho < 1. \tag{5}$$

Here ε_{t+1} is standard normal, ρ measures persistence, σ_r the size of rate shocks and $\bar{r}$ the mean rate before the floor. Taxable cash earns this known log gross return $r(z_t)$ over the next period.

Bonds and returns. Bonds are zero-coupon and grouped into markets a. Tax rates differ by investor and market. For a given investor, all maturities in a market receive the same tax treatment. A bond of market a with T periods to maturity has price $P_T^a(z)$, pays one unit at maturity and becomes a $(T-1)$-period bond next period. Cash is the one-period taxable asset. Bond dollar supplies Q_T^a are fixed.

An investor who buys a T-period bond today and sells it next period earns

$$\text{gross holding return} = 1 + R_{T,t+1}^a = \frac{P_{T-1}^a(z_{t+1})}{P_T^a(z_t)}, \qquad P_0^a = 1.$$

The price ratio is the resale value per dollar invested. It carries rate risk because next period's price depends on next period's rate state.

With one period to maturity, the payment is known, so the bond is fully riskless. A zero-coupon bond's yield is $y_T^a = -\log P_T^a/T$. Investors optimize over the one-period holding horizon.

As in the literature, I approximate the net holding return for portfolio choice as[8]

$$R_{T,t+1}^a \approx \mathcal{R}_{T,t+1}^a = \mu_T^a(z_t) + u_{T,t+1}^a. \tag{6}$$

Here $\mu_T^a(z_t) = \log \mathbb{E}_t[1 + R_{T,t+1}^a]$ is the mean of the approximated return used in the portfolio problem. The shock $u_{T,t+1}^a$ is the unexpected component of the log holding return, with conditional mean zero. Thus μ determines expected portfolio income, while u determines risk. The approximation becomes exact in the continuous-time limit.

Investors. Investor i has funding W_i that can be invested in any bonds or cash, risk aversion γ_i and a liability hedge ℓ_i from assets and liabilities beyond its bond portfolio. A positive ℓ_i raises net wealth when rates rise, as does a long-term liability whose value falls. A negative ℓ_i adds to bond losses when rates rise. Investor i keeps the after-tax share $\xi_i^a = 1 - \tau_i^a$ of returns in market a, where τ_i^a is its tax rate on bonds in market a. Cash is taxed like taxable bonds, which form market C, so its after-tax share is ξ_i^C.

The investor chooses dollar positions $x_{i,T}^a \geq 0$ across all markets a and all maturities T, keeps the rest of its funding in cash, $x_i^0 = W_i - \sum_{a,T} x_{i,T}^a \geq 0$, and maximizes expected next-period wealth less a penalty for its variance. With pretax returns measured by $\mathcal{R}$, next-period after-tax wealth is $\widehat{W}_i'$, and the problem is

$$\begin{aligned} \max_{x_i \geq 0,\, x_i^0 \geq 0} \quad & \mathbb{E}_t[\widehat{W}_i'] - \frac{\gamma_i}{2} \operatorname{Var}_t(\widehat{W}_i'), \\ \text{subject to} \quad & \widehat{W}_i' = W_i + \xi_i^C r(z_t)\, x_i^0 + \sum_{a,T} \xi_i^a x_{i,T}^a \mathcal{R}_{T,t+1}^a + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{7}$$

Next-period wealth adds after-tax cash income, after-tax bond returns and the change in outside wealth to current funding. Investors choose portfolios each period, taking prices as given and conditioning on today's rate state. Nonnegative positions rule out short sales and borrowing (Section 3.5).

Required returns. The portfolio problem gives the Karush–Kuhn–Tucker conditions

$$\underbrace{\xi_i^a \mu_T^a - \xi_i^C r}_{\text{after-tax return over cash}} \leq \underbrace{\gamma_i \xi_i^a \operatorname{Cov}_z\big(u_{T,t+1}^a, \widehat{W}_i'\big)}_{\text{cost of the risk the bond adds}} + \underbrace{\kappa_i,}_{\text{shorting constraint on cash}} \tag{8}$$

with equality whenever $x_{i,T}^a > 0$. The covariance measures the risk the bond adds to total wealth, conditional on today's state. The multiplier $\kappa_i \geq 0$ on the shorting constraint on cash is zero

[8] I follow the discrete-time return approximation in Greenwood et al. (2024), Section 2.1. See also Greenwood et al. (2023), Section III.A, and Hanson and Stein (2015), footnote 18. Appendix A.1.1 gives the precise return convention and wealth equation.

when the investor holds cash and can invest another dollar without borrowing. A held bond must compensate for both its added risk and, when this constraint binds, the dollar it displaces from other bonds. If a bond offers less than an investor's required return in equilibrium, that investor holds none. Differences in taxes and balance sheets therefore allow the same bond to suit one investor but not another in equilibrium.

Equilibrium. Prices clear every market at every rate state:

$$\sum_i m_i x^a_{i,T}(z) = Q^a_T, \tag{9}$$

where m_i is the mass of type i, converting its positions and funding into aggregate holdings $m_i x^a_{i,T}$ and funding $m_i W_i$. In the calibration, one unit of mass represents one dollar of baseline portfolio assets (debt securities and cash). Thus m_i measures a type's portfolio size, and W_i is the share available for the bonds and cash included in the model. The rate process, supplies, taxes, funding, risk aversion and liability hedges are exogenous inputs. Equilibrium prices imply holding returns for which investors satisfy (8) and all markets clear. Who holds each bond is determined by these conditions. Appendix Table 6 collects the notation.

4.2 Taxes and the cost of bearing rate risk

To isolate the mechanism, I use a special case with two investors, two markets and rate risk alone. Consider a high-tax household H and a lower-tax investor I choosing municipal bonds M, taxable bonds C and cash. Municipal maturities begin at $T = 1$, while risky taxable bonds begin at $T = 2$. Maturity T counts periods, which last two years in the quantitative application. The two determinants of required returns are the investor's tax treatment and the cost of adding risk to its portfolio.

The relative after-tax share of Section 3.3 summarizes the tax comparison:

$$\xi_i = \frac{\xi^C_i}{\xi^M_i} = \frac{1-\tau^C_i}{1-\tau^M_i}, \qquad \xi_H < \xi_I.$$

The household's smaller ξ_H means a larger exemption benefit, $1 - \xi_H$. For portfolio choice, ξ_i is a sufficient statistic for an investor's tax treatment: dividing its after-tax payoffs by $1 - \tau^M_i$ keeps municipal returns in full and multiplies taxable returns and cash income by ξ_i, without changing its optimal portfolio.[9]

In the full model, prices are nonlinear in the rate state for three reasons: the short rate cannot fall below zero, the shorting constraint on cash can bind, and which investors hold which bonds can change as rates move. A bond's price today depends on its resale price next period, so these future nonlinearities also affect today's prices. For tractability, the illustration fixes participation: prices are computed as if the same investors held each bond at every future rate state. I call such

[9] With payoffs divided by ξ^M_i, risk aversion $\gamma_i \xi^M_i$ and liability hedge ℓ_i / ξ^M_i, the objective is $1/\xi^M_i$ times the original, up to a portfolio-independent constant. Below, γ_i and ℓ_i denote these normalized values.

a fixed holding pattern a participation branch. The illustration also removes the rate floor, setting $r = \bar{r} + z$, and the shorting constraint on cash, so $\kappa_i = 0$. Each log price is then a constant plus a constant multiple of the rate state, $\log P_T^a(z) = -A_T^a z - C_T^a$, where A_T^a is the bond's rate loading defined below. The quantitative model of Section 5 imposes none of these simplifications: participation changes with rates, and the rate floor and the shorting constraint on cash are kept. Fixing participation abstracts from the possibility that rate movements over a bond's remaining life change who holds it, and hence how strongly its price responds to rates (Appendix equation (45)).

Define a bond's rate loading as $A_T^a \equiv -\partial \log P_T^a / \partial z$, which measures its proportional price decline when the rate state rises and is the model's counterpart of duration. It is an equilibrium outcome, as Section 4.5 shows. Rate shocks affect the holding return through next period's resale price, when the bond has $T-1$ periods remaining. Define the investor's net rate exposure as

$$\widehat{D}_i \equiv -\frac{\partial \widehat{W}_i'}{\partial z_{t+1}} = \sum_T x_{i,T}^M A_{T-1}^M + \xi_i \sum_T x_{i,T}^C A_{T-1}^C - \ell_i. \tag{10}$$

The derivative measures how much next-period wealth falls when the rate state rises, holding today's portfolio fixed. Liabilities whose value also falls offset losses on bonds, so a larger liability hedge ℓ_i lowers this net rate exposure.

The investor's cost per unit of additional rate exposure is $\gamma_i \sigma_r^2 \widehat{D}_i$. The portfolio first-order conditions are

$$\begin{aligned} \mu_T^M - \xi_i r &\le \gamma_i \sigma_r^2 \widehat{D}_i \, A_{T-1}^M, \\ \mu_T^C - r &\le \gamma_i \sigma_r^2 \widehat{D}_i \, A_{T-1}^C, \end{aligned} \tag{11}$$

with equality for a positive position. The municipal condition compares μ_T^M with after-tax cash, $\xi_i r$. The taxable condition, divided by ξ_i, is on a pretax scale and compares μ_T^C with r. In both lines the right side is the return the investor requires for the rate risk added by the bond. The quantitative model uses common risk aversion in these normalized payoff units, so its investors differ in tax treatment ξ_i and net rate exposure $\widehat{D}_i$, which depends on portfolio choice and the liability hedge. The analytical benchmark allows γ_i to differ.

These conditions separate the two forces behind participation. The household's larger exemption benefit lowers its required municipal return. A lower unit cost of rate risk also lowers the required return, increasingly so for a bond with a larger loading.

4.3 Participation at longer maturities

At the riskless maturity, only the cash alternative matters. With $A_0^M = 0$, a municipal holder requires $\mu_1^M = \xi_i r$. The household accepts the lowest return because it keeps the smallest share of taxable interest. With a positive short rate and a slack shorting constraint on cash, it holds the municipal supply and

$$y_1^M = \mu_1^M = \xi_H r, \qquad y_1^C = r. \tag{12}$$

At the riskless short end the municipal yield equals the household benchmark exactly (Appendix Proposition A.2).

At a risky maturity, compare the returns the two investors require on the same municipal bond. The lower-tax investor requires less when its duration benefit exceeds the difference in exemption benefits:

$$\underbrace{\sigma_r^2 A_{T-1}^M(\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)}_{\text{duration benefit}} > \underbrace{(\xi_I - \xi_H)r}_{\text{difference in exemption benefits}}. \tag{13}$$

All quantities are evaluated at equilibrium. A positive $\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$ means that the lower-tax investor requires less return per unit of rate exposure. With the same γ_i, this happens when its liability hedge ℓ_I is large enough to give it a lower equilibrium $\widehat{D}_I$. Where the duration benefit is smaller than the difference in exemption benefits, the household requires less return and holds the bond. Where it is larger, the lower-tax investor requires less and holds the bond. Both can hold a maturity at which the two terms are equal (Appendix Theorem A.1).

In the illustration, the municipal rate loading A_{T-1}^M is endogenous and increases with maturity (Section 4.5). At a given equilibrium, the difference in unit risk costs is common across maturities. The duration benefit therefore grows with maturity, while $(\xi_I - \xi_H)r$ stays fixed. Lower-tax participation in municipal bonds can begin at the long end, from a threshold maturity T^*. For taxable bonds, bond returns and cash income receive the same tax treatment, so there is no difference in exemption benefits to overcome. The investor with lower $\gamma_i \widehat{D}_i$, the lower-tax investor in this illustration, requires less compensation for rate risk at every risky maturity and holds risky taxable bonds.

Figure 4 shows sharp sorting when rate risk is the only risk. The lower-tax investor holds all risky taxable bonds and the long municipal maturities, while the household holds the short municipal maturities. Adding a small bond-specific shock allows joint holdings at more than one maturity. In the otherwise identical illustration of Appendix A.2.9, both investors hold maturities 17 to 30, while municipal yields change by at most 0.33 basis points. Bond-specific risk thus matters for who holds each bond but has little effect on prices. The quantitative model includes this risk and multiple investor types, which generate overlapping holdings and partial segmentation.

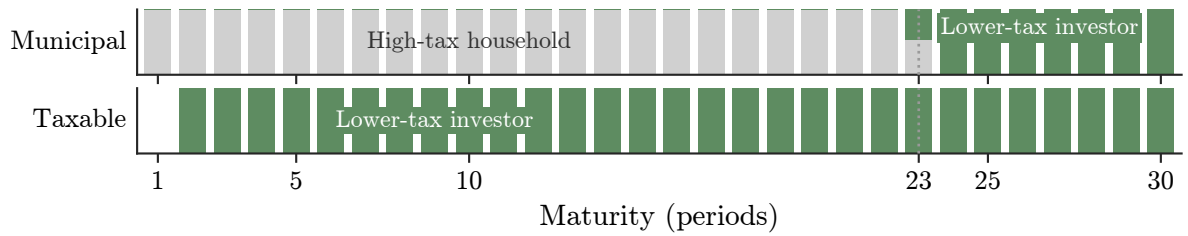


Figure 4: Participation in the two-investor illustration without bond-specific risk. Bars show supply shares at the mean rate state. Both investors hold municipal maturity $T^* = 23$ (dotted line). The high-tax household holds shorter municipal maturities, and the lower-tax investor holds longer municipal maturities and risky taxable bonds. The one-period taxable claim is cash. Appendix A.2.7 gives the parameters and checks.

4.4 Three channels in required returns

Given this allocation, I compare municipal and taxable bonds of the same maturity, the relevant comparison for the municipal puzzle. The municipal return above the household's after-tax taxable return is

$$g_T = \mu_T^M - \xi_H \mu_T^C. \tag{14}$$

The portfolio conditions separate the gap into three channels. Section 4.5 then translates holding returns into yields.

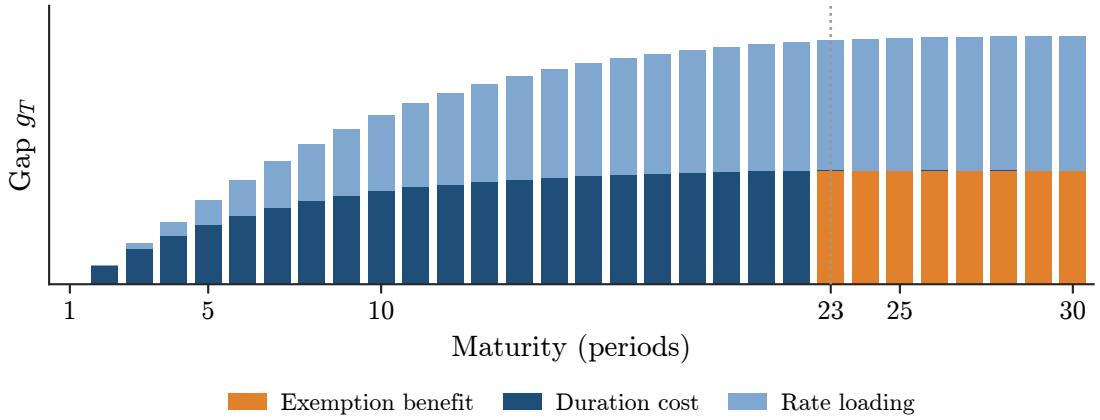


Figure 5: One-period return gap by channel. The allocation is that of Figure 4. Bars decompose g_T into the exemption-benefit, duration-cost and total rate-loading terms of (15). The vertical scale is omitted because levels depend on the illustrative parameters.

Proposition 1 (Three channels of the one-period return gap). In the two-investor affine allocation just described, the lower-tax investor alone holds risky taxable bonds. For municipal bonds, use the household's condition ($i = H$) below T^* and the lower-tax investor's condition ($i = I$) from T^* onward. Subtracting ξ_H times the taxable holder's condition from the municipal holder's condition in (11) gives

$$\begin{aligned} g_T = \; & \underbrace{(\xi_i - \xi_H) r}_{\text{exemption benefit}} + \underbrace{\sigma_r^2 A_{T-1}^M (\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I)}_{\text{duration cost}} \\ & + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I (A_{T-1}^M - \xi_H A_{T-1}^C)}_{\text{rate loading}}. \end{aligned} \tag{15}$$

Each term compares one element of the municipal holder's condition with the corresponding element of ξ_H times the taxable holder's condition. The exemption-benefit term compares the municipal holder's gain from tax exemption with the household's. The duration-cost term compares the two holders' costs per unit of rate exposure. The rate-loading term compares the municipal loading with the taxable loading scaled by ξ_H. Appendix A.2.4 states and proves the decomposition.

Exemption benefit. Below T^*, the household holds the municipal bond. The benchmark already uses its tax treatment, so this term is zero. From T^* onward, the lower-tax investor holds both

municipal and taxable bonds. Because it pays less tax on taxable interest, tax exemption saves it less than it saves the high-tax household. It therefore requires a higher municipal return than the household benchmark predicts, and the difference in exemption benefits contributes $(\xi_I - \xi_H)r$ to the gap.

Duration cost. At risky maturities below T^*, the household alone holds municipal bonds, while the lower-tax investor holds taxable bonds. Rate risk is more costly for the household, so it needs more return to bear the same rate exposure. This difference contributes $\sigma_r^2 A_{T-1}^M(\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)$ to the gap, even though the lower-tax investor holds no municipal bonds at these maturities. It is the duration benefit in (13), expressed as the household's higher risk cost. From T^* onward, the lower-tax investor holds both bonds and prices them with the same unit risk cost, so this term is zero.

Rate loading. When taxes scale returns, they also scale risk. The household benchmark scales taxable yields and rate exposure by ξ_H, predicting $A_{T-1}^M = \xi_H A_{T-1}^C$. In the illustration, lower-tax participation raises municipal rate loadings relative to taxable loadings, $A_{T-1}^M \geq \xi_H A_{T-1}^C$. The extra exposure contributes positively to the gap when the lower-tax investor's net rate exposure is positive, as in the illustration. The contribution equals the extra exposure times $\gamma_I \sigma_r^2 \widehat{D}_I$, the lower-tax investor's unit risk cost. The lower-tax investor holds the taxable bond at every risky maturity, so this term applies on both sides of T^*. I defer the derivation of the loading gap $A_{T-1}^M - \xi_H A_{T-1}^C$ to Section 4.5.

All three channels vanish at the riskless maturity, and $g_1 = 0$. Since $A_0^M = A_0^C = 0$, the duration-cost and rate-loading terms vanish. The exemption-benefit term is also zero because the household holds the riskless municipal bond. Figure 5 shows how the channels combine across maturities.

4.5 Rate loadings and the yield gap

Proposition 1 takes rate loadings as given and describes one-period returns. Two steps connect it to the municipal puzzle. The first derives the loading gap $A_{T-1}^M - \xi_H A_{T-1}^C$ in the rate-loading channel from the way required returns respond to rates. The second shows how the one-period gaps g_T accumulate into the yield gap $y_T^M - \xi_H y_T^C$, which defines the municipal puzzle (Section 3.6).

Loadings accumulate return responses. Because $\mu_T^a(z)$ is the log expected gross holding return, today's log price equals the log expected resale value less the required holding return:

$$\log P_T^a(z) = -\mu_T^a(z) + \log \mathbb{E}_z\big[P_{T-1}^a(z')\big].$$

Let $s_T^a = \partial \mu_T^a / \partial z$ denote the response of the required holding return to the rate state. With affine log prices and $\mathbb{E}_z[z'] = \rho z$, the log expected resale value has loading ρA_{T-1}^a. Taking the negative state derivative gives

$$A_T^a = \rho A_{T-1}^a + s_T^a, \qquad A_0^a = 0. \tag{16}$$

A bond's loading carries forward the loading of the bond it becomes next period, weighted by rate persistence, and adds the response of its current required return. Subtracting ξ_H times the taxable recursion from the municipal recursion gives

$$A_T^M - \xi_H A_T^C = \rho\big(A_{T-1}^M - \xi_H A_{T-1}^C\big) + \big(s_T^M - \xi_H s_T^C\big).$$

The loading gap therefore accumulates the response gaps $s_T^M - \xi_H s_T^C$. Under the household benchmark, municipal required returns are ξ_H times taxable required returns, so $s_T^M = \xi_H s_T^C$ and the loading gap is zero. A loading gap arises when municipal required returns respond to rates by more than ξ_H times taxable required returns.

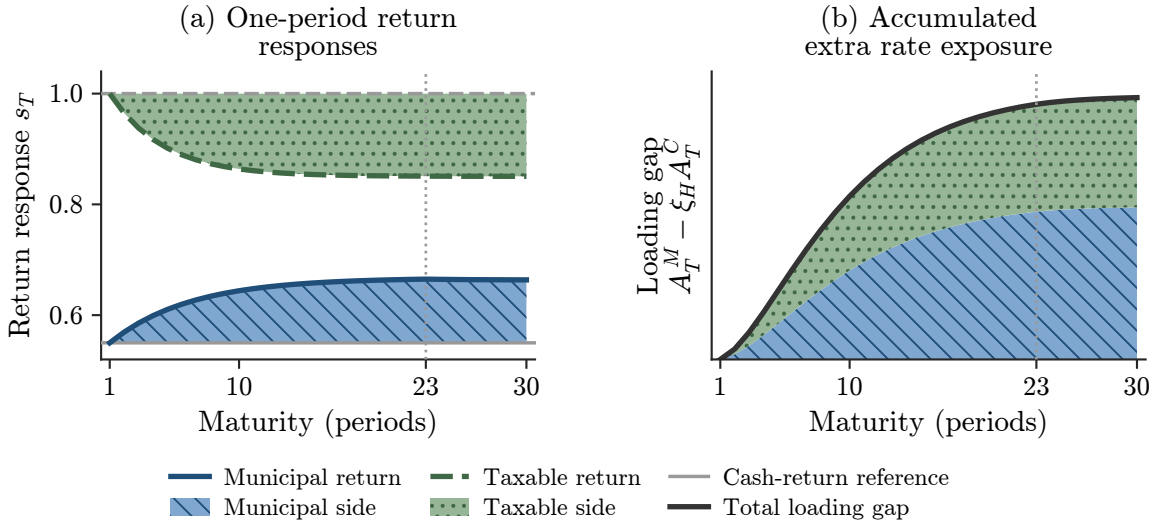


Figure 6: Rate responses in the illustration of Figure 4. Panel (a) shows municipal and taxable holding-return responses. Gray lines show the household's after-tax cash response, ξ_H, and the pretax cash response, 1. Blue shading combines the municipal tax mix and portfolio response into $s_T^M - \xi_H$. Green shading shows $1 - s_T^C$. Panel (b) accumulates $s_T^M - \xi_H$ and $\xi_H(1 - s_T^C)$ into the loading gap. The dotted line marks the shared municipal maturity $T^* = 23$.

How the return gap responds to rates. The response gap is the response of the return gap $g_T = \mu_T^M - \xi_H \mu_T^C$ of Section 4.4 to the rate state: $s_T^M - \xi_H s_T^C = \partial g_T / \partial z$. With the participation pattern fixed, the loadings A_{T-1}^a do not depend on the rate state, and the short rate moves one for one with it while $r > 0$. Differentiating (15), with i the municipal holder as in Proposition 1, gives

$$s_T^M - \xi_H s_T^C = \underbrace{\xi_i - \xi_H}_{\text{exemption benefit}} + \underbrace{\sigma_r^2\Big(\gamma_i A_{T-1}^M \frac{\partial \widehat{D}_i}{\partial z} - \xi_H \gamma_I A_{T-1}^C \frac{\partial \widehat{D}_I}{\partial z}\Big)}_{\text{portfolio reallocation}}. \tag{17}$$

The return gap thus responds to rates in two ways. First, the exemption benefit $(\xi_i - \xi_H)r$ grows with the short rate. Below T^*, the household holds the municipal bond and this term is zero. From T^* onward, the lower-tax investor holds it, so the required municipal return rises with rates by

its after-tax share ξ_I, more than the household's ξ_H. Second, $\partial \widehat{D}_i/\partial z$ measures how investor i's net rate exposure changes as rates rise. These exposures change because both investors reallocate their portfolios when rates move, and a changing exposure changes the investor's unit risk cost and hence the compensation it requires on every risky bond it holds.

The reallocation works as follows. As rates rise, the difference in exemption benefits $(\xi_I - \xi_H)r$ widens, so municipal bonds become less attractive to the lower-tax investor. It sells municipal bonds to the household, so the household's net exposure rises and the lower-tax investor's falls. As a result, $\partial \widehat{D}_H/\partial z > 0 > \partial \widehat{D}_I/\partial z$. Below T^*, both changes raise the response gap: the household requires more compensation on all its risky municipal bonds, including those it holds alone, and the lower-tax investor requires less on its taxable bonds. From T^* onward, the lower-tax investor holds both bonds, so its falling exposure lowers its required returns in both markets when rates rise. Intuitively, the household bears more rate risk when rates rise and therefore requires more compensation, which raises the rate loadings of the bonds it holds. The opposite holds for the lower-tax investor.

Proposition 2 (Participation and rate responses). On an affine participation branch of the two-investor benchmark with $r > 0$, suppose the household alone holds municipal maturities below T^*, both investors hold T^*, and the lower-tax investor alone holds longer municipal maturities and the risky taxable bonds, as in Figure 4. The municipal holders' tax mix equals ξ_H when the household holds the bond alone, ξ_I when the lower-tax investor holds it alone, and $\xi^* = \omega_H \xi_H + \omega_I \xi_I$ under joint holdings. Here ω_i is each investor's share of total risk tolerance.[10] Differentiating (11) and clearing (9) gives

$$s_T^M = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M \frac{\partial \widehat{D}_H}{\partial z}, & \text{household alone,} \\ \xi^*, & \text{both investors,} \\ \xi_I + \gamma_I \sigma_r^2 A_{T-1}^M \frac{\partial \widehat{D}_I}{\partial z}, & \text{lower-tax investor alone,} \end{cases} \tag{18}$$

$$s_T^C = 1 + \gamma_I \sigma_r^2 A_{T-1}^C \frac{\partial \widehat{D}_I}{\partial z}.$$

Figure 6(a) groups the same response gap by market:

$$s_T^M - \xi_H s_T^C = \underbrace{s_T^M - \xi_H}_{\text{municipal side}} + \underbrace{\xi_H (1 - s_T^C)}_{\text{taxable side}}.$$

The municipal side measures how much more the municipal required return responds than the household's after-tax cash return, and the taxable side how much less the taxable required return responds than the pretax cash return. Both sides are positive at every risky maturity in the illustration. Accumulated through (16), they give a positive loading gap, $A_T^M > \xi_H A_T^C$ for $T \geq 2$ (Figure 6(b)). Municipal bonds therefore carry more rate exposure than the household benchmark implies, and with the lower-tax investor's positive net exposure, the rate-loading channel raises

[10] The weights are $\omega_i = (m_i/\gamma_i)/(m_H/\gamma_H + m_I/\gamma_I)$.

g_T. The appendix proves the response formulas and the positive loading gap under condition (S) in Appendix equation (72), which keeps the lower-tax investor's falling risk compensation from reversing the sign of the loading gap (Proposition 2).

From return gaps to yield gaps. A bond with T periods to maturity earns a sequence of one-period returns as it ages: the return on a T-period bond today, on a $(T-1)$-period bond next period, and so on until it matures. Its yield averages these expected returns, less a convexity term (Appendix equation (54)). Applying this relation to both markets gives

$$y_T^M - \xi_H y_T^C = \frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t[g_{T-j}(z_{t+j})] - \text{convexity difference}. \tag{19}$$

The left side is the municipal yield above the household benchmark, the model counterpart of the measured puzzle. The right side averages the one-period gaps the bond earns over its remaining life, so the yield gap inherits the three channels of Proposition 1. A long municipal bond earns the exemption-benefit and rate-loading terms while its remaining maturity is at least T^*, and the duration-cost and rate-loading terms after it falls below T^*. Since $g_1 = 0$ and the one-period gaps rise with maturity in the illustration (Figure 5), longer bonds average larger gaps. The convexity difference is small, so the equilibrium municipal yield exceeds the household benchmark at every maturity beyond one period, by more at longer maturities.

4.6 Empirical predictions

The representative-tax benchmark predicts full pass-through of household tax changes at every maturity and no response to tax benefits specific to lower-tax institutions. In the model, household taxes determine the riskless short municipal yield, while longer yields also reflect lower-tax participants' tax treatment and portfolio responses. I first use tax variation specific to households to test whether household tax pass-through declines with maturity (P1). I then use tax variation specific to banks to test whether lower-tax institutions' benefits affect longer yields while leaving the riskless short yield unchanged (P2). Together, these tests distinguish the participation mechanism from a single household tax rate applied across the yield curve. P3 tests the associated prediction that municipal rate sensitivity rises with maturity relative to the taxable benchmark.

The outcome of interest is the yield-implied tax $\tau_T^{Y,\text{impl}} = 1 - y_T^M / y_T^C$ from (2). Investor i's pass-through measures how much this equilibrium implied tax changes when investor i's exemption benefit changes:

$$\mathcal{P}_i^Y(T) = \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad i \in \{H, I\}. \tag{20}$$

Here, pass-through of one means full transmission and zero means no effect.

By (12), the implied tax at the riskless shortest maturity is $1 - \xi_H$. Therefore

$$y_1^M = \xi_H y_1^C \quad \Longrightarrow \quad \mathcal{P}_H^Y(1) = 1, \qquad \mathcal{P}_I^Y(1) = 0. \tag{21}$$

Only the household's benefit enters this pricing relation.

At longer maturities, lower-tax participation changes whose exemption benefit affects returns. A larger benefit makes municipal bonds more attractive to its recipient and changes portfolios in both markets. The household's benefit therefore no longer has the exclusive role it has at the riskless short end. An institution's tax treatment affects returns where it holds bonds, while portfolio substitution transmits the effect to other risky maturities.

Pass-through is an equilibrium derivative: changing one investor's after-tax share re-solves portfolios, loadings and both yield curves, holding non-tax primitives fixed. At the long end, the lower-tax investor holds both municipal and taxable bonds, so the long-end municipal-to-taxable yield ratio equals its after-tax share ξ_I, up to a relative convexity term whose tax derivatives are small (Appendix Propositions A.6(iii) and A.8). Long-end pass-through is therefore close to zero for the household and close to one for the lower-tax investor:

$$\mathcal{P}_H^Y(\infty) < 1, \qquad \mathcal{P}_I^Y(\infty) > 0. \tag{22}$$

Between the two ends, household pass-through falls and the lower-tax investor's rises with maturity in the illustration (Appendix A.2.7). Appendix A.2.8 gives the counterpart when bond-specific risk extends joint holdings across long maturities. The empirical analysis tests the following three predictions, P1 to P3.

Household taxes (P1). A home-state exemption gives resident households an additional benefit on their state's bonds. That benefit is fully reflected in implied state taxes at the riskless short end and less strongly at longer maturities in the solved model. The representative-tax benchmark predicts full pass-through at every maturity.

Institutional taxes (P2). A benefit specific to a lower-tax institution has no effect at the riskless short end and matters more at longer maturities. Bank qualification tests this prediction. The measure is the share of the bank-only benefit reflected in yields (Section 6.2). The representative-tax benchmark gives these benefits no role.

Interest-rate responses (P3). Section 4.5 shows why municipal bonds carry more rate exposure than the household benchmark predicts. Since $y_T^a = -\log P_T^a/T$, yield sensitivity is A_T^a/T. Comparing bonds at the same maturity cancels this division by T. For the illustration's affine branch and its long-end continuation, this gives

$$\frac{\partial y_T^M/\partial z}{\partial y_T^C/\partial z} = \frac{A_T^M}{A_T^C}, \qquad \frac{A_1^M}{A_1^C} = \xi_H, \qquad \lim_{T\to\infty} \frac{A_T^M}{A_T^C} = \xi_I > \xi_H.$$

The lower-tax investor prices both markets at long maturities. Its changing rate exposure affects both municipal and taxable bond loading recursions in (18). This common adjustment cancels in the limiting ratio, leaving ξ_I (Appendix Proposition A.6). It still affects each bond's loading and return response. On the illustrated branch, the ratio rises with maturity from ξ_H toward ξ_I. P3 tests whether municipal yield sensitivity relative to same-maturity taxable bonds increases with maturity. The representative-tax benchmark predicts the constant ratio ξ_H.

4.7 Liability hedging and other motives

Liability hedging is one source of the advantage that leads lower-tax investors to hold long municipal bonds. A lower-tax investor can also hold long municipal bonds at lower cost if it trades less often or more cheaply than households, or diversifies default risk across markets. When these advantages grow with maturity, they can outweigh the maturity-invariant difference in exemption benefits and sustain participation in long municipal bonds.

The pricing mechanism extends to these motives when the relevant advantage is neutral across taxable and tax-exempt bonds. Neutrality means that the advantage itself creates no return gap relative to the household benchmark when the household holds both bonds. Lower-tax participation then brings the participant's smaller exemption benefit into the return comparison. Appendix A.3 states the result, and Appendix A.3.4 gives solved trading-cost and default-diversification examples. A preference for municipal bonds as such, including warm glow from financing local projects, lowers the return an investor requires on them. Such a preference alone does not explain yields above the household benchmark.

I use liability hedging in the quantitative application because rate risk is common to both markets and balance sheets provide a measurable source of differences in investors' risk-bearing costs. Observed differences in maturity holdings help discipline the outside exposures.

5 Quantitative Results

I now ask how much of the municipal puzzle the model of Section 4 explains when taxes, balance sheets and bond supplies take their measured values. The quantitative model keeps what the analytical benchmark set aside: many investor types, separate state municipal markets, the rate floor, the shorting constraint on cash and participation that changes with rates. Section 5.1 describes its markets and investors, Section 5.2 explains what is measured and what is estimated, and Section 5.3 compares its yield curves with the data and accounts for the puzzle.

5.1 The quantitative model

The model has 45 municipal markets and one taxable market. Each state with a positive personal income tax and the District of Columbia has its own municipal market. The seven states with no state income tax are combined into one market because investors treat their bonds alike in the model. The taxable market pools treasury and corporate supply. A model period is two years, giving fifteen maturities from two to thirty years in each municipal market. Cash and the two-year municipal claim are riskless.

Six investor groups hold these bonds: high-tax households, lower-tax households, P&C insurers, life insurers, banks and a tax-neutral residual group, which consists mainly of foreign investors, pensions, governments and nonprofits. They differ in three ways that matter for the mechanism of Section 4. The first is tax treatment, summarized by the exemption benefit $1 - \xi_i$, which follows from tax law (Table 3). High-tax households face a federal marginal rate of at least 35%.

Table 3: Investor groups in the quantitative model, 2015–2017

Group	Types	Types differ by	Gross exemption benefit $1-\xi_i$	Portfolio assets m_g
High-tax households	$7 \times 45 = 315$	Outside exposure, home state	0.43–0.50	6.97
Lower-tax households	$7 \times 45 = 315$	Outside exposure, home state	0.21–0.27	8.09
P&C insurers	2	Outside exposure	0.31	1.11
Life insurers	3	Outside exposure	0.21	3.05
Banks	1	None	0.35	4.83
Residual group	7	Outside exposure	0	26.47
Total	643			50.52

Notes. Each household group has seven outside-exposure types in each of the 45 markets and in the one-market model used to estimate the coefficients (Section 5.2). Estimated coefficients map each household type's measured outside assets and debt into exposure. Institutional exposures are estimated directly (Appendix Table 9). The gross exemption benefit $1-\xi_i$, with $\xi_i = (1-\tau_i^C)/(1-\tau_i^M)$, follows from tax law. Households multiply the federal and home-state after-tax shares of each SCF tax unit, with the range running across states, and institutions face the 35% corporate rate, with proration for insurers. Banks also pay a financing-tax cost of 0.11% a year on municipal holdings, which lowers their net benefit below this gross value (Table 2). Portfolio assets m_g, in trillions of dollars, are all of a group's debt securities and cash in the Financial Accounts, and they scale the group's risk-bearing capacity.

Households also differ by home state, because a resident's own state's bonds are exempt from state tax while other states' bonds and taxable bonds are not.[11] Insurers and banks face the corporate rate. Proration taxes part of insurers' municipal interest, and banks also pay a financing-tax cost on municipal holdings. The second is size. An investor's portfolio assets m_i are its debt securities and cash in the Financial Accounts, including holdings outside the model's markets, and they scale its risk-bearing capacity. A measured fraction W_i is available for the investments in the model: municipal bonds, corporate bonds, treasury notes and bonds, and cash. Its funding is therefore $m_i W_i$ (Appendix B.1). Positions $x^a_{i,T}$ are measured per dollar of portfolio assets, so aggregate bond holdings are $m_i x^a_{i,T}$.

The third difference is outside rate exposure: how the rest of an investor's balance sheet changes in value when rates move. Outside assets that lose value when rates rise add to the losses on bonds, while liabilities that lose value offset them. Outside rate exposure is the negative of the liability hedge ℓ_i of Section 4. As there, an investor whose outside exposure offsets bond losses bears rate risk more cheaply and can hold long bonds even when its exemption benefit is small. Within each group, types differ in outside rate exposure. Each household group has seven types in each of the 45 home markets, formed by sorting Survey of Consumer Finances (SCF) households on outside assets and debt relative to their bond and deposit holdings. P&C insurers have two types and life insurers three, which differ in exposure and in their shares of the sector's assets. Banks form a single type. The residual group has seven types whose exposures approximate a normal distribution. Appendix B.1 defines every type.

[11] Treasury interest is exempt from state income tax, while interest on corporate and other taxable bonds is not. The model taxes all interest in its taxable market at federal and state rates, as in the corporate comparison curve of Section 3.6.

The quantitative model keeps the structure of Section 4. Investors solve (7) with market-specific after-tax shares, and prices satisfy the covariance-based portfolio conditions (8) and clearing (9) in every market and rate state. Banks' measured financing-tax cost is subtracted from their municipal returns (Appendix B.1). The model retains the rate floor in (5) and allows the shorting constraint on cash to bind and participation to change across rate states. I solve $P_T^a(z)$ numerically through the general resale relation in Appendix equation (40), integrating over next-period states. Returns, portfolio risks, holdings and both yield curves are determined jointly.

The model of Section 4.1 has rate risk only. The quantitative model adds a small independent bond-specific return shock to every risky bond, which adds its own variance to the bond's log return, as in the general model of Appendix A.1 (equation (32)). The shock is a realistic force that makes bonds imperfect substitutes and lets several investor types hold the same maturities. In the two-investor benchmark, it extends joint holdings from one maturity to a block of maturities while changing municipal yields by at most 0.33 basis points (Appendix A.2.8). It vanishes at the riskless claim, and changing its variance by 20% moves the thirty-year gap by less than 0.1 basis points (Appendix B.5).

Measured inputs change over time, so I solve the model separately for six three-year periods, 2000–2002 through 2015–2017, each centered on an SCF wave. Each period has its own tax rules, portfolio assets, bond supplies, household balance sheets and observed rate states. Financial Accounts stocks give each sector's portfolio assets and holdings and the taxable supply. Mergent gives corporate maturity profiles and municipal supply by maturity and state. The SCF supplies household balance sheets and the income and filing status needed to apply statutory taxes. Historical short rates determine rate persistence and volatility, while two-year corporate yields set the rate level and observed states. Appendix B.1 details the construction and sources. I also solve the model for a seventh period, 2018–2020, with its own measured inputs and the tax rules of the 2017 TCJA. This period is not used in estimation: it keeps the coefficients estimated on the first six periods, and Section 5.3 includes it when comparing model and observed yields.

5.2 Calibration and fit

Everything the data measure directly is fixed before estimation: investor tax treatment, each group's portfolio assets and funding, bond quantities and maturity profiles, household outside assets and debt, and the short-rate process. These inputs differ across the six periods. The remaining seventeen coefficients cannot be measured directly and are common to all six periods and all states. One is the common risk aversion. Two set the size and maturity profile of the bond-specific shock. The other fourteen set outside rate exposures, which balance-sheet amounts alone cannot measure: three translate a household type's measured outside assets and debt into exposure, eight set the exposures and asset shares of the insurer types, one sets bank exposure, and two set the mean and dispersion of residual-group exposure.

I estimate the coefficients by fitting twenty-two targets. Nine are ownership stocks: municipal holdings of the two household groups, P&C insurers, life insurers and banks, and taxable holdings

of households and the same three institutional sectors. Four are the mean maturities of insurers' municipal and taxable portfolios. Six describe the composition of insurer portfolios. Two, one for each insurer sector, measure how unevenly municipal bonds are spread across insurers. The other four are the fractions of each insurer sector's municipal and taxable holdings that mature beyond twenty years. The last three are the taxable yields at ten, twenty and thirty years, measured as term spreads over the two-year yield.

Ownership provides the main information about outside exposure. Given a group's tax treatment and portfolio assets, how much it holds in each market and at which maturities reveals how much rate risk it is willing to bear. Insurer portfolio composition disciplines the differences among insurer types. The taxable term spreads and long-bond allocations discipline risk aversion and bond-specific risk. Each coefficient affects every target through equilibrium prices and portfolios, so the coefficients are estimated jointly. I evaluate the targets at each period's observed rate states, average them across the six periods and minimize their weighted squared deviations from the data.[12]

Calibration on the 45-market model is not computationally tractable, so I estimate the coefficients in a one-market version with a single national municipal market. In that market, each household type's municipal after-tax share averages its home-state shares, weighted by each state's municipal supply. I then restore the 45 municipal markets and their measured tax differences without re-estimating anything, since no coefficient is state-specific.[13] The one-market and 45-market models give similar relative municipal yields. Measured against each model's own taxable curve, their thirty-year gaps average 102.6 and 102.4 basis points across the six periods and differ by at most 2.5 basis points in any period, while their thirty-year taxable yields differ by at most 0.52 basis points. The gap is also insensitive to the individual coefficients. Changing one group of coefficients at a time, for example scaling an exposure by 20%, moves the one-market gap by at most 4.1 basis points (Appendix Table 11). The exception is the common risk aversion. It scales the compensation for rate risk in both markets, so it moves the taxable and municipal curves nearly in proportion. Because the mechanism works through the tax rate embedded in yields, the gap in basis points moves with the level of taxable yields. Lowering risk aversion by 20% lowers the thirty-year gap by 13.6 basis points, mostly because it lowers the thirty-year taxable yield by 70 basis points. The gap per unit of taxable yield, which is the shortfall of the implied tax below the household wedge and the quantity the comparison with data in Section 5.3 uses, falls only from 0.170 to 0.166. The same change also leaves taxable term spreads about 20% below their targets, so the data rule it out.[14]

Municipal yields, the municipal puzzle and empirical pass-through coefficients are not targeted. Ownership disciplines outside rate exposure, which combines with observed taxes to determine

[12] Each averaged deviation is scaled by 10% of its observed normalizer. Maturity targets and the fractions held beyond twenty years include model-portfolio weights. Appendix equation (113) gives the criterion, and Appendix Tables 9 and 10 report the coefficients and the observed and fitted target values.

[13] Fourteen of the twenty-two targets are matched to within 10% of their observed normalizers. The best small move away from the estimates improves the fit by less than 0.001% and changes the average thirty-year gap by 0.004 basis points. Appendices B.3 and B.4 report the estimation procedure and fit.

[14] It also raises the estimation criterion from 36.3 to 55.0 (Appendix Table 11).

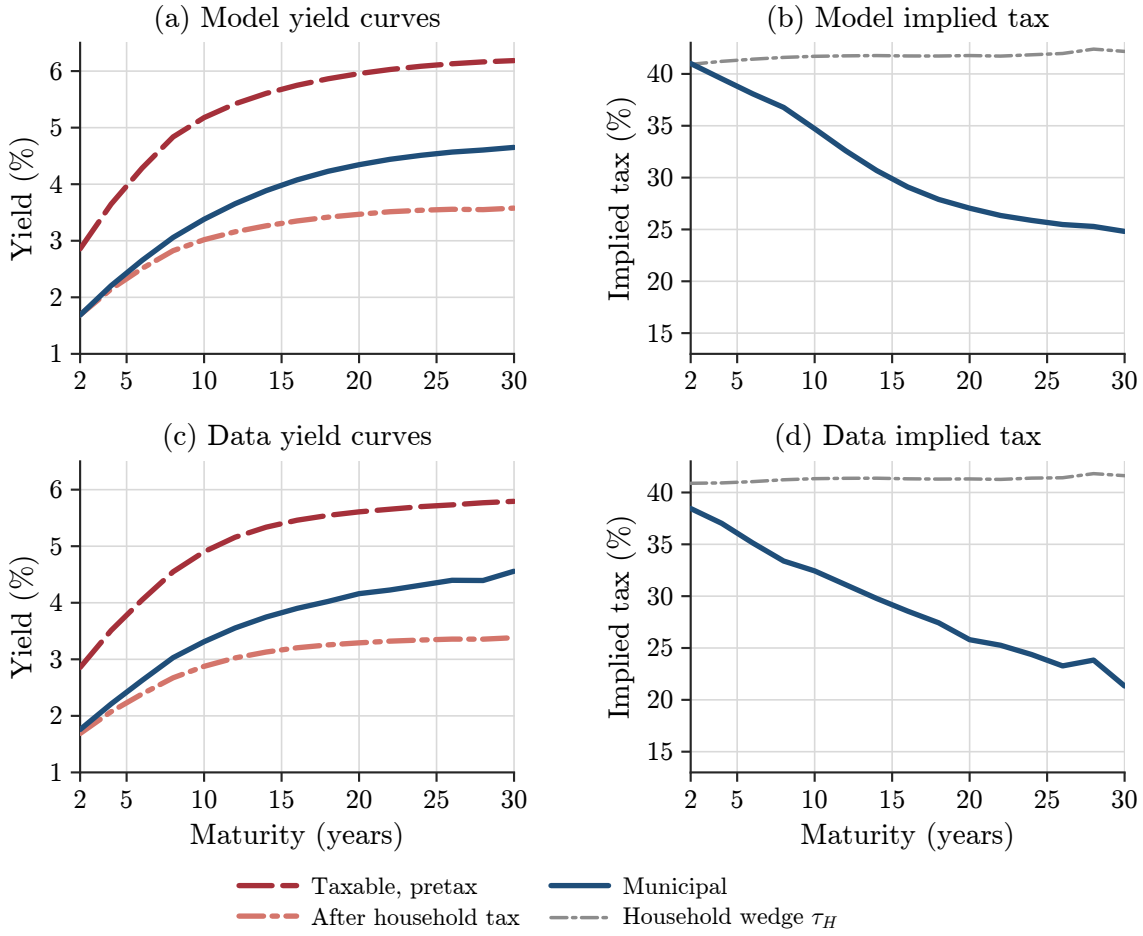


Figure 7: Model (top) and adjusted data (bottom), 2000–2020. Panels (a) and (c) plot the pretax taxable yield, the taxable yield after household tax (the household benchmark) and the municipal yield. Panels (b) and (d) plot the implied tax $1 - y_T^M / y_T^C$ and the household wedge, denoted τ_H in the legend. Corresponding panels share axis scales. The taxable curve is the model's pooled taxable market (top) and the observed corporate benchmark (bottom). The municipal data curve removes callability (Appendix C.1.2) and the issuance premium, and panel (d) uses this same adjusted curve. The model evaluates the estimated coefficients at observed issue-quarter rate states in the six calibration periods and the 2018–2020 period. Both rows use the empirical bond, rating and quarter weights and nominal semiannual yields. The household benchmark multiplies each observation's taxable yield by the high-tax household's relative after-tax share for that bond's state and date before averaging. Implied taxes and the household wedge are ratios of averaged yields.

municipal yields in equilibrium. The relative municipal yield curve and cross-state pass-through therefore test the mechanism (Sections 5.3 and 6.1).[15] In the headline comparison of Section 5.3, model and data share the same taxable curve and weights: I apply the model's implied tax at each maturity to the observed corporate yield and average both with the weights of the 2000–2020 bond sample.

[15] Appendix B describes the solution method, and Appendix B.7.2 reports accuracy.

5.3 Model yields and the municipal puzzle

Figure 7 sets the calibrated model (top) against the data (bottom). The left panels plot three yield curves: the pretax taxable yield, the household benchmark, which is that yield after the high-tax household's taxes, and the municipal yield. The right panels convert these curves into the implied tax $1 - y_T^M / y_T^C$ and compare it with the household wedge τ_H. In both rows, the municipal yield is close to the household benchmark at two years and rises above it with maturity, so the implied tax starts near the household wedge and falls well below it at long maturities. In nominal semiannual yields, the model's implied tax is 41.0% at two years, 34.7% at ten and 24.8% at thirty, a decline of 16.2 percentage points. In the adjusted data, it is 38.5%, 32.4% and 21.3%, a decline of 17.1 points. The household wedge stays between 40.9% and 42.4% in both rows. The model's thirty-year municipal and taxable yields are 4.651% and 6.185%, against 4.558% and 5.794% in the adjusted data.

The accounting below asks how much of the municipal puzzle the model explains, alongside the measured capital-gains, liquidity and default differentials. Because the model's thirty-year taxable yield is 39 basis points above the observed one, its gap in basis points is not directly comparable with the puzzle: a higher taxable yield enlarges the gap mechanically, even with the same implied tax. I therefore express all contributions on the same observed taxable yield curve.

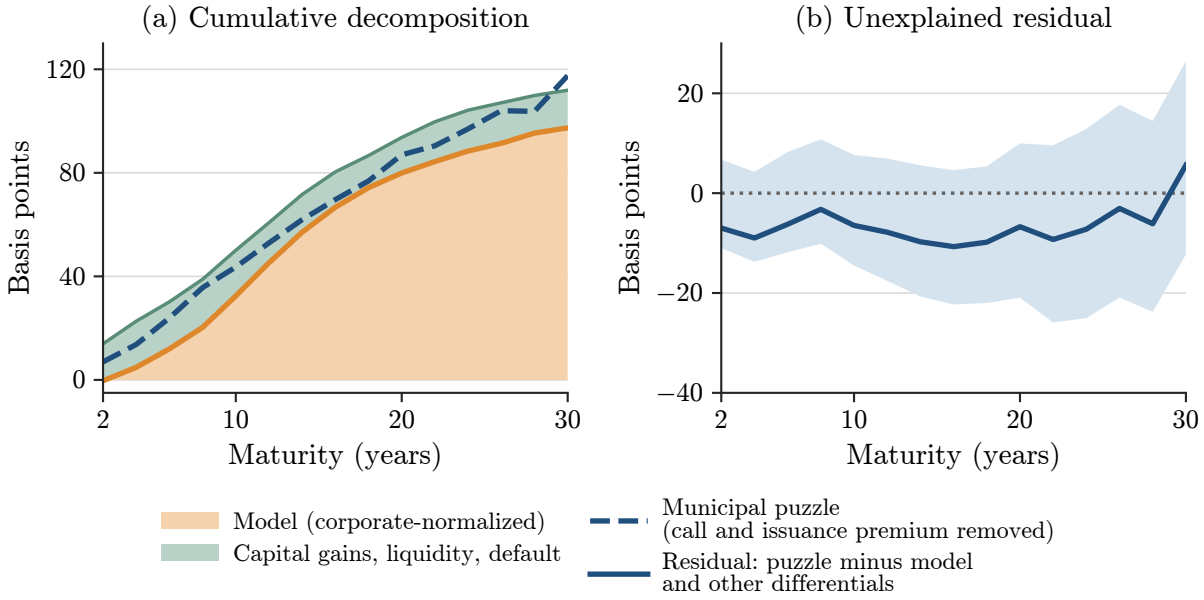


Figure 8: Cumulative decomposition of the municipal puzzle, 2000–2020, as in (24). The puzzle is the observed spread after removing callability and the issuance premium. Panel (a) stacks the model gap $\widetilde{\mathcal{G}}_T$ of (23) and the capital-gains, liquidity and default differentials against that puzzle. Panel (b) shows Residual(T), the puzzle less their sum. Shading spans the ranges of the five empirical adjustments (Appendix C.1) and excludes model uncertainty. Weights and yield conventions match Figure 7.

At each maturity, I keep the model's municipal-to-taxable yield ratio and multiply it by the observed taxable yield. I then subtract the household benchmark. The resulting gap, denoted $\widetilde{\mathcal{G}}_T$,

Table 4: Accounting for the municipal puzzle. Removing callability and the issuance premium from the raw spread over the household benchmark gives the puzzle. The model contribution applies its implied tax to observed corporate yields, as in equation (23).

		Comparability			Accounting for the puzzle			
Maturity (years)	Raw spread	Call	Issuance premium	Puzzle	Model	Model share	Bond-side adjustments	Unexplained
2	9.9	0.1	2.9	6.9	−0.3		14.3	−7.0
6	27.9	0.7	3.3	23.9	12.0	50%	18.2	−6.2
10	52.2	4.6	4.0	43.6	32.4	74%	17.6	−6.5
14	87.0	17.9	7.2	61.9	57.1	92%	14.6	−9.7
18	110.3	23.9	9.6	76.9	74.3	97%	12.4	−9.8
22	128.8	27.0	11.4	90.5	84.3	93%	15.4	−9.3
26	145.0	28.5	12.5	104.0	91.4	88%	15.7	−3.1
30	159.8	29.5	12.8	117.5	97.4	83%	14.5	5.6

Notes. Basis points, except maturity and shares. The accounting is puzzle = model + bond-side + unexplained, as in (24). The model column uses the corporate normalization, and the bond-side adjustments sum the capital-gains, liquidity and default differentials. The model share divides its contribution by the puzzle, using unrounded values, and is omitted where the puzzle is below 20 bp. Appendix C.1.7 reports the unnormalized gaps. Model and data both use the 2000–2020 sample, the averaging of Appendix C.1.1 and the nominal semiannual convention. The separately measured adjustments are not jointly estimated. Rows show every second model maturity. Figure 8 shows all fifteen.

measures the model's contribution on the data's taxable curve:

$$\begin{aligned} \text{Normalized municipal yield} &= \frac{\text{Model municipal yield}}{\text{Model taxable yield}} \times \text{Observed taxable yield}, \\ \widetilde{\mathcal{G}}_T &= \text{Normalized municipal yield} - \text{Household benchmark}. \end{aligned} \tag{23}$$

I call this the corporate normalization. It preserves the model's implied tax and changes only the taxable yield on which the gap is measured. The solved equilibrium and the curves in Figure 7 are unchanged. Appendix C.1.7 gives the averaging and normalization details.

The accounting has three steps at each maturity T:

$$\begin{aligned} \text{Spread}(T) - \underbrace{\big[\text{Call}(T) + \text{Prem}(T)\big]}_{\text{comparability adjustments}} &= \underbrace{\text{Puzzle}(T)}_{\text{municipal puzzle}}, \\ \text{Puzzle}(T) - \underbrace{\big[\text{CG}(T) + \text{Liq}(T) + \text{Def}(T)\big]}_{\text{bond-side adjustments}} &= \underbrace{\text{Puzzle}^{\text{adj}}(T)}_{\text{residual before the model}}, \\ \text{Puzzle}^{\text{adj}}(T) - \underbrace{\widetilde{\mathcal{G}}_T}_{\text{model}} &= \underbrace{\text{Residual}(T)}_{\text{unexplained}}. \end{aligned} \tag{24}$$

The first line removes the comparability adjustments of Section 3.7, callability and the issuance premium, from the raw spread and gives the municipal puzzle. The second line subtracts the capital-gains, liquidity and default differentials, CG, Liq and Def. These features affect municipal yields but are absent from the model, so $\text{Puzzle}^{\text{adj}}(T)$ is the part of the puzzle left for the model to explain. The third line subtracts the model gap and leaves the unexplained residual. The household

benchmark cancels in this last difference, so the residual equals the observed corporate yield times the difference between the model's implied tax and the implied tax of the fully adjusted data. A negative residual therefore means that the model's implied tax is below the data's and the model gap exceeds $\text{Puzzle}^{\text{adj}}(T)$. Appendix C.1 explains how each of the five adjustments is measured, in the order callability, issuance premium, capital gains, liquidity and default (Appendices C.1.2–C.1.6).

At thirty years (Figure 8 and Table 4), the model accounts for 97 basis points of the 118-basis-point puzzle, or 83%. The bond-side adjustments account for 15 basis points, or 12%, leaving about 6 basis points unexplained. The model's contribution rises with maturity, and its share of the puzzle is 74% at ten years, 97% at eighteen and 83% at thirty. The unexplained residual is negative at every maturity except thirty years: from twelve to twenty-four years, the model gap exceeds $\text{Puzzle}^{\text{adj}}(T)$ by 7 to 11 basis points, and the largest absolute residual is 11 basis points at sixteen years. A negative residual is consistent with the convenience-premium interpretation in Fleckenstein and Longstaff (2025).

6 Empirical Tests

I test the model's participation mechanism using variation in household taxes, institutional tax benefits and interest rates (Section 4.6). None of the estimates is a calibration target. P1 uses differences in residents' state income-tax rates. P2 uses bank qualification. These designs compare household and institutional tax effects across maturities. P3 compares changes in municipal yields with changes in treasury yields at the same remaining maturity.

The P1 and P3 tests use the AA-and-above municipal sample (agency score three or better) to limit credit differences that could affect both yields and trading. The curve accounting of Sections 3 and 5 uses the wider A-or-above sample (score seven or better), which matches the corporate benchmark's credit band, and Appendix C.10.2 repeats both tests on that sample, where the same maturity ordering holds. The bank-qualified design uses all new issues with rating-by-year fixed effects. Throughout, n_b denotes a bond's remaining maturity in years.

6.1 P1. State-tax pass-through

P1 asks whether resident households' state-tax benefit is reflected less in municipal yields at longer maturities. The representative-tax benchmark predicts full pass-through at every maturity. The model predicts full pass-through at the riskless short end and less at longer maturities, where institutions and investors from other states, who receive none of the state benefit, also participate. Most states exempt residents' interest on in-state municipal bonds and tax interest on out-of-state municipal bonds and taxable bonds. A household in state s has a state after-tax share of one on in-state municipal interest and $1 - \tau_s^{\text{st}}$ on out-of-state municipal and taxable interest.

State rates move resident households' exemption benefits, which institutions largely do not receive. Pensions and tax-deferred accounts owe no current state tax on interest, and insurer premium taxes generally do not distinguish municipal from taxable interest at the state level.

Under its stated conditions, Appendix Proposition A.8 gives household pass-through of one at the riskless short end and below one in the long-end limit, and Appendix Proposition A.18 gives the riskless anchor and the tax-mix part of the pass-through in the state-market application.

The yield ratio y^M/y^C contains both federal and state taxes. Under the pre-2018 federal deduction convention, the after-tax share of taxable interest is $\xi_s^{\text{st}}\xi_t^{\text{fed}}$, where $\xi_s^{\text{st}} = 1 - \tau_s^{\text{st}}$ is the state share and ξ_t^{fed} the federal share, while home-state municipal interest is exempt at both levels. Household indifference under the representative-tax benchmark gives $y_{s,t}^M = \xi_t^{\text{fed}}\xi_s^{\text{st}}\, y_t^C$. Dividing out the federal share defines the implied state tax

$$\mathit{ImpliedStateTax}_{s,t}(T) \;\equiv\; 1 - \frac{y_{s,t}^M(T)/y_t^C(T)}{\xi_t^{\text{fed}}}. \tag{25}$$

It equals the statutory rate τ_s^{st} under the benchmark, giving pass-through of one. I set ξ_t^{fed} to one minus the dated top federal ordinary-income rate, excluding the net investment income tax: 0.604 in 2000, 0.609 in 2001, 0.614 in 2002, 0.65 in 2003–2012 and 0.604 in 2013–2017. The sample ends before the TCJA capped state and local tax deductions at \$10,000 from 2018, and Appendix C.6 reports the later period under its tax mapping.

I estimate pass-through separately in five maturity bins ($[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$ and $[20,30]$ years):

$$\begin{aligned}\mathit{ImpliedStateTax}_{b,t} = &\sum_{g=1}^{5}\beta_g\big(\mathit{StateTax}_{s(b),t}\times\mathbf{1}\{n_b\in g\}\big) + \sum_{g=1}^{5}\delta_g'\big(Z_{b,t}\times\mathbf{1}\{n_b\in g\}\big)\\ &+\lambda' V_b + \alpha_{t\times n\times \mathit{callable}} + \varepsilon_{b,t},\end{aligned} \tag{26}$$

where $\mathit{StateTax}_{s,t}$ is the state's top statutory rate and the outcome (25) uses call-adjusted zero-coupon yields. The coefficient β_g is the change in the implied state tax per percentage point of statutory rate in maturity bin g, and the benchmark gives $\beta_g = 1$ in every bin. The controls $Z_{b,t}$, also interacted with the bins, are the rounded average rating and six lagged state variables: par-weighted outstanding maturity, the maturity of the preceding year's issuance, outstanding par relative to personal income, log income, log preceding-year issuance and the outstanding share in the bond's maturity bin. V_b contains bond-specific characteristics, and $\alpha_{t\times n\times \mathit{callable}}$ denotes issue-year–month–maturity–callability fixed effects.

The empirical sample excludes Illinois, Iowa and Wisconsin, which generally do not exempt residents' interest on in-state municipal bonds, and Oklahoma, whose exemption depends on the issuer and issue date. It retains bonds recorded as exempt from own-state tax and bonds issued in states with a zero rate. The 2000–2017 sample has 681,838 observations with complete controls. Standard errors are clustered by state, allowing dependence among a state's bonds across years and maturities. Cross-sectional variation in state tax rates identifies pass-through at each maturity. Because current tax policy could respond to current municipal yields, I also report an instrumental-variables (IV) comparison that instruments the current statutory rate with its 1990 counterpart,

following the policy-history approach of Babina et al. (2021). The IV interpretation requires that, conditional on the controls, historical rates affect current yields only through current tax rates. The results are robust to the choice of historical year, and Appendix C.6 reports other vintages with the same controls and sample.

Figure 9 compares the ordinary least squares (OLS) estimates of (26) with the cross-state slope in the full 45-market model. For the model, I compute the implied state tax (25) with its high-tax household's federal share ξ^{fed} and nominal semiannual yields. At each maturity, I regress it on each state's top rate, averaged within the three-year calibration period, weighting the 43 income-tax states and the District of Columbia equally, and I average the slopes across the six pre-2018 periods.

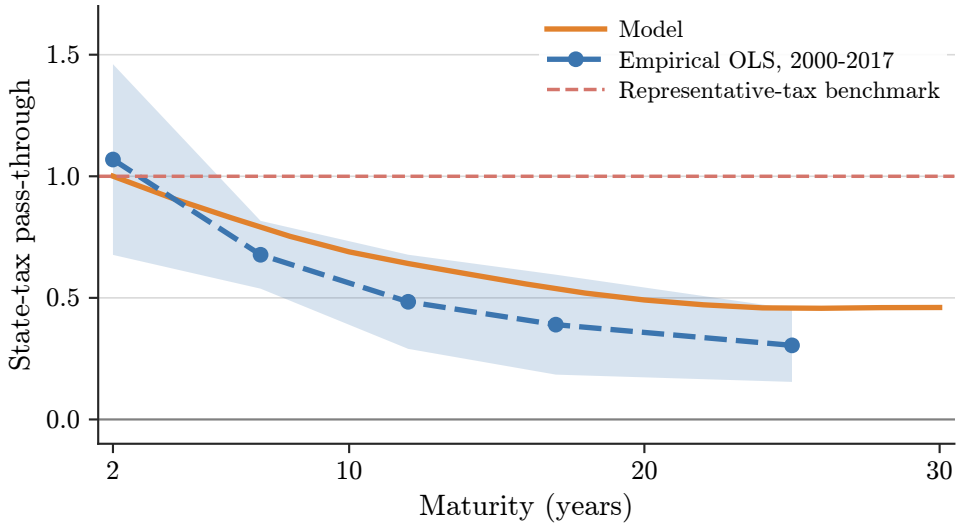


Figure 9: State-tax pass-through by maturity. Blue: OLS estimates for 2000–2017 call-adjusted zeros, with state-clustered 95% intervals using $t(46)$. The regression is unweighted and includes the controls in (26). Orange: the model's cross-state slope, using its own federal after-tax share and nominal semiannual yields at the mean rate state $z = 0$ and averaged equally across the six pre-2018 periods. Both use (25). The red dashed line is the representative-tax benchmark.

Estimated pass-through declines across all five bins, from 1.069 below five years to 0.305 beyond twenty (Appendix Table 22). A state top rate ten percentage points higher raises the implied tax by about 10.7 points at the short end and 3.0 points at the long end. The short-to-long decline is 0.764 (s.e. 0.235, $p = 0.002$). Unit pass-through cannot be rejected in the shortest bin and is strongly rejected in the longest, and every bin coefficient is positive and significant at one percent. The IV estimates also decline, from 1.088 to 0.229.

The model's untargeted profile declines from 1 at two years to 0.69 at ten and 0.46 at thirty. It lies within the 95% confidence intervals of the four bins below twenty years and just above the interval of the longest bin. The empirical decline survives alternative yield constructions, credit controls, weights, instruments, state fixed effects and single-state omissions (Appendices C.5 and C.6).

6.2 P2. Bank qualification

P2 uses a tax benefit that only banks receive. The representative-tax benchmark prices every maturity off household taxes and so gives this benefit no effect on yields. The model predicts no effect at the riskless short end and a larger effect at longer maturities, where banks and other lower-tax institutions participate. Banks ordinarily cannot deduct interest expense allocated to tax-exempt municipal holdings (Section 265(b) of the Internal Revenue Code). For bank-qualified (BQ) municipal bonds, Sections 265(b)(3) and 291 restore 80% of that deduction. A municipal issuer can designate its bonds as bank-qualified if its reasonably anticipated tax-exempt issuance for the year does not exceed $10 million ($30 million in 2009–2010 under the American Recovery and Reinvestment Act). Designation changes neither the bonds' payments nor any other investor's taxes.

Tax rules allocate a bank's interest expense across its assets in proportion to tax basis, so each dollar of municipal holdings carries the same annual financing cost c, whatever the bond's maturity. Designation makes 80% of that cost deductible and saves $X = 0.8\,\tau_B c$ a year per dollar of designated holdings, where τ_B is the federal bank tax rate. In any year, the benefit is therefore the same for short and long bonds. Measured from Call Report interest expense, X moves with banks' financing costs: it is about 65 basis points in 2005–2008 and 9 to 14 from 2011 to 2019 (Appendix C.7.2). A bond's benefit, however, accrues over its whole life, so the benefit relevant for its yield is $\bar{X}_b$, the expected annual benefit over bond b's life, which evaluates X at expected future financing costs (Appendix C.7.2). Its median is 45 basis points. The test measures how much of $\bar{X}_b$ is reflected in yields.

Three hypotheses differ in how much of the bank benefit is reflected in the designation premium $\varpi_k = y_k^N - y_k^Q$, the non-designated minus the designated municipal yield at maturity k. Under the representative-tax benchmark, households hold both bonds at the same yield, $y_k^N = y_k^Q = (1-\tau_H)y_k^C$, where $\tau_H = 1 - \xi_H$ is the household wedge. A bank then prefers a taxable bond, which pays $(1-\tau_B)y_k^C$ after tax, to a designated bond, which pays the household benchmark yield less the tax value of the disallowed fifth of its financing cost, whenever

$$(1-\tau_B)\,y_k^C > (1-\tau_H)\,y_k^C - 0.2\,\tau_B c \quad \Longleftrightarrow \quad \tau_H\,y_k^C > \tau_B\,y_k^C - 0.2\,\tau_B c,$$

that is, whenever the household's exemption benefit exceeds the bank's net exemption benefit on a designated bond. This holds when household rates are at least as high as banks', as they are for top-bracket households throughout the sample period. Banks then hold neither municipal bond, and $\varpi_k = 0$. Under a bank-only benchmark, banks' indifference to taxable bonds gives full capitalization: the premium equals the benefit at every maturity. Holding outside investment opportunities and other bond characteristics fixed, the premium is the benefit times the bank's capitalization share θ_k, the average bank share of the total demand response to the bond's yield

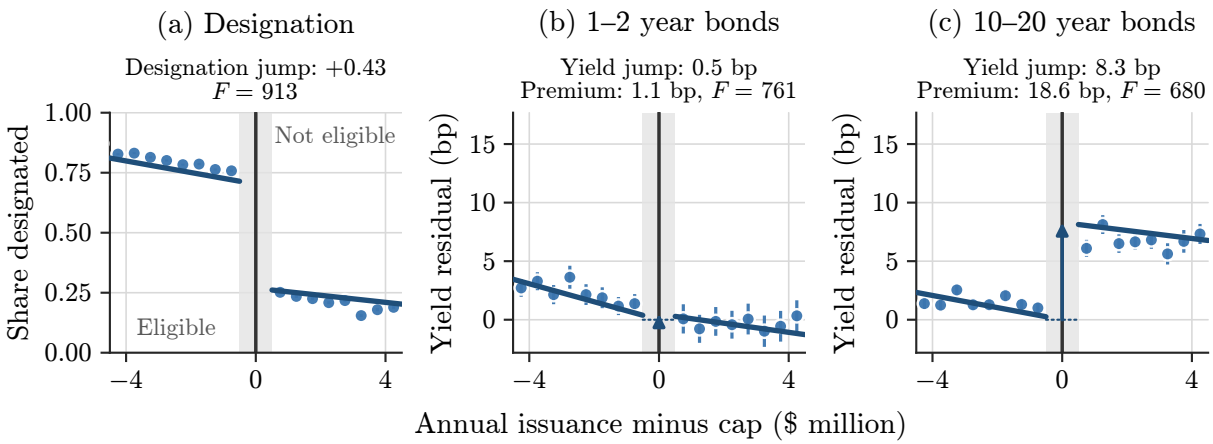


Figure 10: The bank-qualified discontinuity. The horizontal axis is an issuer's measured annual tax-exempt issuance minus that year's cap. Designation is more frequent to the left of zero. Dots are means in $0.5 million bins with 95% intervals: the share of bonds designated in panel (a), and offering-yield residuals after the fixed effects and log par of (28) for 1–2 and 10–20 year bonds in panels (b) and (c). Lines are the estimated local linear fits, which exclude the shaded donut. Yield fits are shifted to zero on the below-cap side at the cap, and the arrow marks the yield jump. Each premium divides the yield jump by its bin's designation jump. F is the first-stage Wald statistic with issuer-clustered standard errors. Appendix Figure 24 shows the other maturities.

(Appendix C.7):

$$\varpi_k = \theta_k X, \qquad \underbrace{\theta_k = 0}_{\text{representative-tax}}, \quad \underbrace{\theta_k = 1}_{\text{bank-only}}, \qquad \underbrace{\theta_{\text{short}} \simeq 0 < \theta_{\text{long}}}_{\text{model}}. \tag{27}$$

Dividing each yield by its bond's expected benefit $\bar{X}_b$ therefore measures θ_k, the share of the benefit reflected in yields: zero under the representative-tax benchmark, one under the bank-only benchmark and rising with maturity in the model. The capitalization share is the bank counterpart of $\mathcal{P}_I^Y(T)$ in Section 4.6. Household after-tax cash returns anchor short yields and banks hold few short bonds, so θ_k is near zero at short maturities. In the calibrated model, banks hold intermediate maturities, where their duration benefit offsets their smaller exemption benefit (Section 4.3), and yields there reflect more of the bank benefit.

The cap supports a regression-discontinuity comparison among issuers of similar size. Designation averages 0.79 below the cap and 0.21 above it across the estimation window. Measured annual issuance need not equal the issuance anticipated at each sale or counted under the statute, so some designated bonds appear above the measured cap. Below-cap status raises the probability of designation without determining it, giving a fuzzy discontinuity. The below-cap indicator instruments designation, and the premium is the local yield jump divided by the local designation jump.

The variable R_{jt} is issuer j's annual tax-exempt issuance minus that year's cap, in millions of dollars. The indicator $E_{jt} = \mathbf{1}\{R_{jt} \leq 0\}$ marks below-cap issuer-years, and B_b indicates whether bond b is bank-qualified. The sample covers tax-exempt new issues from 2000 to early 2021 with $0.5 \leq |R_{jt}| \leq 4$. The half-million exclusion around the cap, the donut, removes managed bunching

and issuer-years exactly at the cap. With bins g of maturity at issuance (1–2, 2–5, 5–10, 10–20 and more than 20 years), I estimate

$$y_b = \sum_g \beta_g B_b \mathbf{1}\{n_b \in g\} + Z_b'\delta + u_b, \tag{28}$$

by two-stage least squares (2SLS), instrumenting each designation interaction $B_b\mathbf{1}\{n_b \in g\}$ with the eligibility interaction $E_{jt}\mathbf{1}\{n_b \in g\}$ of the same bin. The first stage is the local designation jump conditional on controls, 0.42 to 0.45 across bins (Appendix Table 28). Here Z_b contains local linear controls in R_{jt} on each side of the cap, log tranche par, and fixed effects for maturity, state, offering month, rating, security type, use of proceeds, coupon and callability, with the interactions listed in Appendix C.7.2. The outcome is the offering yield in basis points, or that yield divided by $\bar{X}_b$, so $-\beta_g$ is the premium in basis points or, in expected-benefit units, an estimate of the capitalization share θ_k. Standard errors are clustered by issuer.

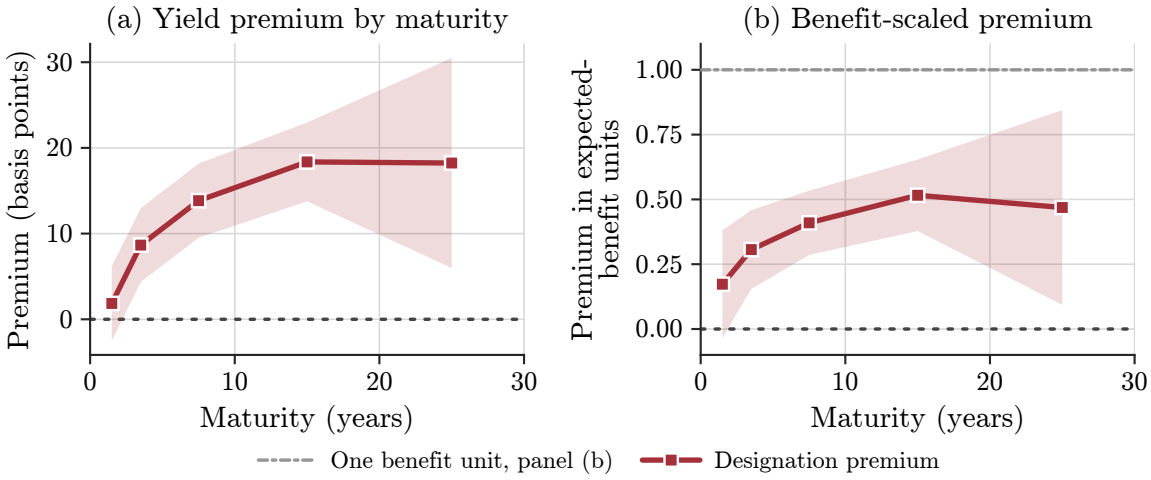


Figure 11: The bank-only benefit and municipal yields. Panel (a): 2SLS yield premium in basis points. Panel (b): the premium in expected-benefit units, with each yield divided by the expected annual benefit $\bar{X}_b$ and the same controls. Financing costs are municipal-holdings-weighted. Expected rates use the Kim–Wright expected-short-rate component, capped at ten years (Appendix C.7). Shading is the issuer-clustered 95% interval. The dashed line marks one benefit unit.

Identification requires that, given these controls, crossing the cap moves offering yields only through designation. Callability and the number of maturities per issue are balanced at the cap. A placebo cutoff at \$5 million shows essentially no designation effect, and its 10–20 year yield jump exceeds the 1–2 year jump by only 0.4 basis points, against 7.8 at the actual cap (Appendix C.7).

Figure 10 illustrates the discontinuity with separate local fits for each maturity bin. The 1–2 year yield barely moves at the cap, by 0.5 basis points, while the 10–20 year yield jumps by 8.3, implying bank-qualification premia of 1.1 and 18.6 basis points after dividing by the designation jumps. The main estimates come from the joint 2SLS regression (28), which estimates all five bins together. Figure 11 reports these estimates (Appendix Table 28). The premium is 1.8 basis points at 1–2 years, indistinguishable from zero, and 8.7, 13.8, 18.4 and 18.2 at 2–5, 5–10, 10–20 and more

than 20 years. The 10–20 year premium exceeds the 1–2 year premium by 16.5 basis points (s.e. 2.7). As a share of the expected benefit, the premium rises from 0.17 at 1–2 years to 0.52 at 10–20 years: the bank-only benefit is barely priced in short municipal yields, and about half of it is priced at 10–20 years.

The positive premia at 5–20 years reject the representative-tax benchmark, with t statistics of 6 to 8, and the rise in the share, 0.34 (s.e. 0.11), rejects full capitalization, which implies a share of one at every maturity. The gradient appears in every benefit regime (terciles of years ranked by X) and survives alternative donuts and bandwidths, credit-enhancement controls, a general-obligation sample, issuer effects, year clustering and the binned estimator of Dagostino (2025) (Appendix C.7).

The rising bank profile is the reverse of household pass-through in P1, which falls from 1.07 to 0.30 (Figure 9), as the model predicts for institutional and household tax benefits.

6.3 P3. Rate responses by maturity

P3 asks whether municipal yields respond more to a given treasury yield change at longer maturities. The representative-tax benchmark makes municipal zero yields a constant fraction ξ_H of treasury zero yields, so it predicts the same relative response at every maturity. The model predicts a relative response that rises with maturity, from ξ_H at the riskless short end toward the lower-tax investor's share ξ_I at the long end (Section 4.6). That section derives the pattern relative to the model's jointly priced taxable market C. The test measures rate movements with treasury yields, and for this reference the analytical illustration also gives a relative response that rises with maturity.[16] The observed response is about 0.35 at one year, below ξ_H. The test concerns how this relative response varies with maturity.

I compare consecutive trades in the same municipal bond, using its yield to maturity and the maturity-matched treasury par yield. With $\Delta y^{M}_{b,t}$ the change in its yield between the two trades and $\Delta y^{\text{Tsy}}_{n_b,t}$ the change over the same interval in the treasury yield matched to its remaining maturity n_b, I estimate

$$\Delta y^{M}_{b,t} = \beta_0 \, \Delta y^{\text{Tsy}}_{n_b,t} + \beta_1 \big(\Delta y^{\text{Tsy}}_{n_b,t} \times n_b \big) + \beta_2 \, n_b + \varepsilon_{b,t}. \tag{29}$$

The response to a treasury yield change is $\beta_0 + \beta_1 n_b$. A positive β_1 means that the same treasury yield change is associated with a larger municipal yield change at longer maturities. Comparing trades within a bond removes its time-invariant characteristics. The sample covers January 2005 to August 2020 and includes non-callable, federally tax-exempt municipal bonds rated AA and above. Trades are at most 90 days apart, and the treasury yield must move by at least 10 basis points in either direction to limit transaction noise. The treasury comparison follows the bond's remaining maturity at each trade, and standard errors are clustered by the second trade's date. Appendix C.9 details the construction.

[16] In the two-investor illustration, municipal sensitivity relative to a taxable reference whose one-period return responds one for one to the rate state increases over the thirty illustrated maturities. Appendix Proposition A.12(ii) establishes the maturity ordering for that reference when bond-specific risk extends joint holdings across long maturities.

The estimates are $\hat{\beta}_0 = 0.346$ and $\hat{\beta}_1 = 0.007$ (Appendix Table 32). A 100-basis-point rise in the treasury yield is associated with a municipal yield increase of about 35 basis points at one year and 42 at ten years.

All thirty combinations of alternative treasury-move cutoffs and trade windows give a positive maturity interaction, from 0.0047 to 0.0104, significant at five percent (Appendix Table 34). On the wider rating sample, the maturity pattern holds and the gradient is larger, 0.013 against 0.007 (Appendix Table 36).

7 Policy Counterfactuals

Tax reforms that change investors' exemption benefits change who holds municipal bonds. The Tax Reform Act of 1986 generally removed banks' interest deduction for new municipal purchases and was followed by a sharp decline in their holdings (Appendix D.5.1). The 2017 TCJA again changed banks' and insurers' relative exemption benefits. I study two reforms with existing policy precedents. The first removes a state's exemption for residents' interest on its own municipal bonds, which Illinois, Iowa and Wisconsin already generally tax. The second taxes municipal interest like taxable interest for every investor, as the 2009–2010 Build America Bonds program did for its taxable municipal bonds while subsidizing issuers' interest.

The two reforms differ in whether investors can retain the federal benefit by switching municipal markets. Home-state repeal leaves that benefit in place, so high-tax households mainly switch between municipal markets while their replacement investments generate state-taxable interest. Equal treatment removes the federal benefit as well. Investors who previously benefited most shorten their bond portfolios, while tax-neutral investors expand their long-bond holdings. The financing and revenue effects depend on both the interest these investors earn and the compensation they require for bearing rate risk.

7.1 Design and fiscal accounting

I solve each reform in each of the six three-year periods of 2000–2017 and report calendar-year-weighted averages. Each period uses the pre-2018 tax regime and period-specific investor and bond-market inputs (Section 5.2). I change the relevant tax rules and re-solve holdings and both yield curves, keeping bond supplies, investors' total funding, risk preferences and outside exposures at their dated baseline values.[17]

Annual financing costs equal debt outstanding at each maturity times its yield, summed across maturities. Their increase is the replacement-financing cost, the additional annual cost of financing the same debt at the new yields.[18] Each group's pretax interest on bonds and taxable cash divides

[17] Tax changes are applied in each investor's baseline payoff units, preserving preferences and liability hedges. Yield changes average each period's 36 monthly rate states. Dollar amounts, California's implied state taxes and Figure 12(a) use the mean rate state, $z = 0$. Ratios for 2000–2017 divide average dollar financing costs by average dollar receipts.

[18] The calculation uses continuously compounded yields to construct coupon-equivalent flows. Existing contractual payments and original-issue-discount schedules remain unchanged. The taxable market pools treasury and corporate

into tax receipts and after-tax interest income, with households taxed by residence and institutions under their own tax treatment. Because aggregate funding, bond supplies and the short rate are fixed, total cash interest does not change. Every additional dollar issuers pay therefore goes either to governments or to investors. Summing across both markets and all governments gives

$$\Delta(\text{Financing costs}) = \Delta(\text{Tax receipts}) + \Delta(\text{After-tax interest income}). \tag{30}$$

I report financing costs per dollar of added receipts. For home-state repeal, I compare both own-state and all-issuer costs with the repealing state's added revenue, excluding changes in other governments' receipts. For equal treatment, I compare total municipal and taxable issuer costs with the combined increase in federal and state revenue. The ratio exceeds one when investors' after-tax interest income rises: issuers then pay more than governments collect, and investors receive the difference.[19]

7.2 The home-state exemption

Most income-tax states exempt residents' interest on their own municipal bonds and tax interest on other states' bonds, which encourages residents to hold local debt. Illinois, Iowa and Wisconsin generally tax interest on their own bonds as well.[20]

Panel (a) of Figure 12 gradually removes California households' exemption on local municipal interest in the 2015–2017 period, keeping statutory state and federal tax rates and household composition fixed. The horizontal axis measures the remaining tax benefit of a local bond relative to an out-of-state bond for the high-tax household. The implied state tax is the percentage by which California's yield lies below the pooled municipal yield of the states without an income tax.[21] At the riskless two-year maturity, the implied state tax moves one for one with this benefit. Removing the exemption entirely lowers the implied state tax by 13.3 percentage points at two years but only 5.0 percentage points at thirty years. As in P1, the household's tax benefit is fully reflected in short yields and only partly in long yields, where lower-tax residents and institutions also hold the bonds.

debt. Other liability flows, issuer deductions, refinancing timing and fiscal feedback are outside this interest-income accounting (Appendix D.4).

[19] These ratios compare financing and interest-income flows. Welfare would also depend on changes in risk bearing, investment opportunities and public spending.

[20] Each of these three states exempts particular issues. The Supreme Court upheld the home-state preference in *Department of Revenue of Kentucky v. Davis* (2008). Home-state preferences also changed within the sample: Oklahoma exempts qualifying local obligations issued after July 1, 2001, and Utah has taxed out-of-state municipal interest on bonds acquired after January 1, 2003, except from states that do not tax Utah's bonds. Since 2018, the federal state-and-local-tax deduction cap has also increased the exemption's value for residents whose deductions it limits.

[21] The implied state tax is $1 - y^M_{\mathrm{CA},T}/y^0_T$, where y^0_T is the pooled no-income-tax-state municipal yield. It is the model counterpart of the P1 measure in (25), with y^0_T in place of $\xi^{\mathrm{fed}}_t y^C_T$. Both yields are solved at rate state $z = 0$. If a fraction f of local interest becomes state-taxable, the high-tax household's relative home-state tax benefit is $(1-f)\tau^{\mathrm{st}}_H/(1-f\tau^{\mathrm{st}}_H)$. This compares its after-tax shares on local and out-of-state interest and equals its state rate when the local exemption is full.

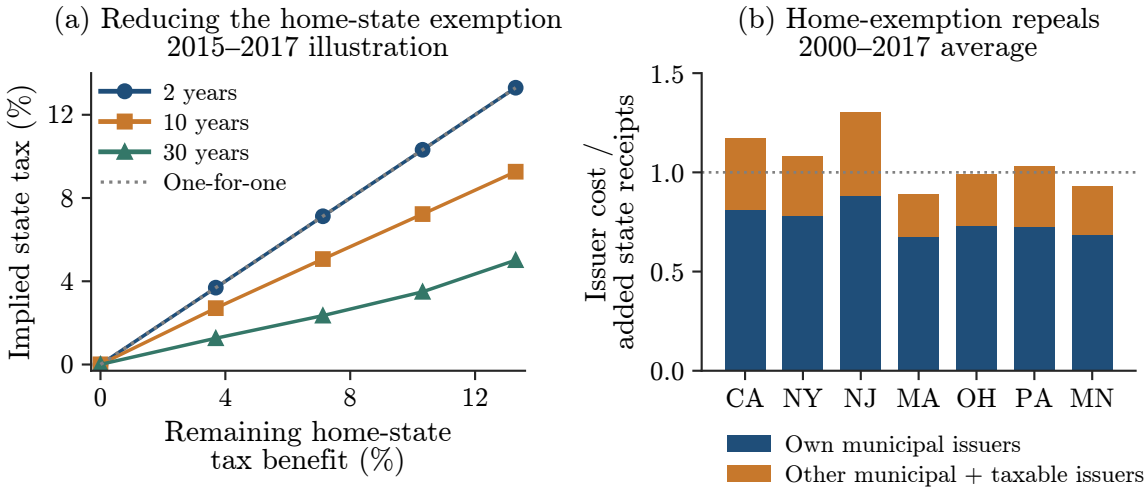


Figure 12: Home-state exemptions, municipal yields and financing costs. Panel (a) makes 0, 25, 50, 75 and 100% of local municipal interest state-taxable for both California household groups in the 2015–2017 period, keeping all statutory tax rates fixed. It plots $1 - y^M_{\text{CA},T}/y^0_T$ at two, ten and thirty years against the high-tax household's remaining relative home-state tax benefit. The comparison yield y^0_T pools the seven states without an income tax. Markers are solved equilibria at $z = 0$, and the dotted line is one-for-one. Panel (b) removes each listed state's home-state exemption at fixed tax rates in each of the six periods of 2000–2017. It divides average replacement-financing costs for own-state municipal issuers and other municipal and pooled taxable issuers by average added state receipts from residents' bond and taxable cash interest at $z = 0$.

Panel (b) removes residents' state exemption on home-state municipal bonds, separately in California, New York, New Jersey, Massachusetts, Ohio, Pennsylvania and Minnesota, seven states with high supply-weighted tax benefits. Residents then pay the same state tax on home-state as on out-of-state municipal interest, while the tax treatment of institutions, nonresidents, cash and taxable bonds is unchanged. Added state receipts exceed local issuers' added financing costs, which rise by \$0.68–\$0.88 per receipt dollar, below one in all seven six-period averages (Appendix Table 40). Including other municipal and taxable issuers raises the ratio to \$0.89–\$1.30, above one in California, New York, New Jersey and Pennsylvania. Both ratios use the same receipts, so their difference, \$0.21–\$0.42 per receipt dollar, is the cost transmitted to other markets: other states' municipal issuers bear \$0.08–\$0.11 and taxable issuers \$0.11–\$0.31.

The mechanism is clearest for one dollar of state s's maturity-T debt held by a resident. As the comparison yield, take y^0_T, the pooled municipal yield of the states without an income tax, whose bonds carry no home-state benefit for any investor. The implied state tax $\tau^{\text{impl}}_{s,T} = 1 - y^M_{s,T}/y^0_T$ then measures how much of the home-state exemption state s's yield reflects. Repeal has two effects on this dollar. The implied state tax falls, so with y^0_T fixed the issuer's yield rises by $-y^0_T\,\Delta\tau^{\text{impl}}_{s,T}$. The state collects its full rate τ^{st}_s on the resident's interest. For a dollar earning y^0_T, the annual

comparison is

$$\underbrace{\Delta y^{M}_{s,T} = -y^{0}_{T}\,\Delta\tau^{\text{impl}}_{s,T}}_{\text{added cost to the issuer}} \qquad \text{against} \qquad \underbrace{\tau^{\text{st}}_{s}\,y^{0}_{T}}_{\text{tax on the formerly exempt dollar}}.$$

Under the representative-tax benchmark, residents determine relative yields at every maturity, so repeal lowers the implied state tax by the full rate, $-\Delta\tau^{\text{impl}}_{s,T} = \tau^{\text{st}}_{s}$, and the two sides are equal. In the model, this equality holds only at the riskless short end. At long maturities, lower-tax residents and investors who do not receive the home-state exemption also hold the bonds, so long yields reflect only part of the high-tax household's benefit, consistent with the partial pass-through documented in P1. Repeal therefore raises long yields by less than the benefit it removes. Across the six periods, California's thirty-year implied state tax falls by 3.8 percentage points, against a mean high-tax household rate of 10.8%, so at thirty years the issuer's added cost is about a third of the tax a high-tax resident pays on the same dollar. Aggregate costs and receipts also depend on resident ownership: higher yields apply to all of the state's debt, while the state taxes only its residents' interest.

Repeal removes only the state exemption. Residents keep the federal exemption by buying other states' municipal bonds, so high-tax households mainly switch from home-state to other states' bonds, with modest changes in the maturities they hold. Across the seven experiments, residents' aggregate risky-bond share changes by less than two percentage points (Appendix Table 39). This switching does not erode the state's tax base, because after repeal the state taxes residents' interest on home-state bonds, other states' bonds and taxable bonds alike. What matters for receipts is the gross yield on residents' new holdings. Reinvestment at the same-maturity comparison yield y^{0}_{T} leaves the receipts in the comparison above unchanged. As a result, investors' reallocation does not erode the tax base. It in fact adds to receipts, mainly through substitution into higher-yielding taxable bonds (Appendix Table 40), and in equilibrium the added receipts exceed the increase in local issuers' borrowing costs.

Reallocation also spreads financing costs to other issuers because investors choose jointly across municipal and taxable markets. In the California experiment, other states' municipal issuers bear an additional \$0.11 and taxable issuers \$0.25 per dollar of added California receipts. These spillovers raise the cost from \$0.81 for California's own issuers to \$1.17 for all issuers. The state's receipts can therefore exceed its own borrowers' costs even when financing costs across all markets exceed those receipts.

7.3 Equal municipal and taxable treatment

The second reform taxes each investor's municipal interest as it taxes that investor's taxable interest. It removes federal and state exemption benefits for every investor group.[22] The Build America Bonds program of 2009–2010 tested a version of it: issuers could sell taxable bonds and receive a

[22] The reform is broader than the investor-specific changes in the 2017 TCJA.

federal subsidy of 35% of their interest. That program changed individual issues while the rest of the market stayed exempt. Equal treatment changes all municipal debt at once, so it also changes who holds long bonds and the yields they require.

I implement the reform by moving each municipal after-tax share 99.9% of the way to its taxable share and removing the same share of banks' interest-deduction disallowance, so that banks' municipal interest is taxed like their taxable interest. With exact equality, the riskless municipal bond and cash would be perfect substitutes, leaving investors' split between them indeterminate. The remaining 0.1% pins down that split and keeps yields and the allocation of bonds across maturities close to those of full removal (Appendix B.7.4). The tax treatment of cash and taxable interest does not change.

Equal treatment raises financing costs by more than it raises tax receipts. All amounts below are average annual changes across the six periods of 2000–2017. Per dollar of added federal and state receipts, municipal issuers bear about \$1.06 in added financing costs and taxable issuers bear another \$0.13. Total costs rise by about \$1.20, leaving roughly \$0.20 per receipt dollar as additional after-tax interest income for investors (Table 5 and Figure 13(a)). In annual dollars, municipal issuers pay an additional \$41.2 billion and taxable issuers \$5.1 billion, for a total of \$46.3 billion. Governments collect \$38.7 billion and investors retain the remaining \$7.6 billion after tax, by (30). Appendix Table 41 reports each period separately.

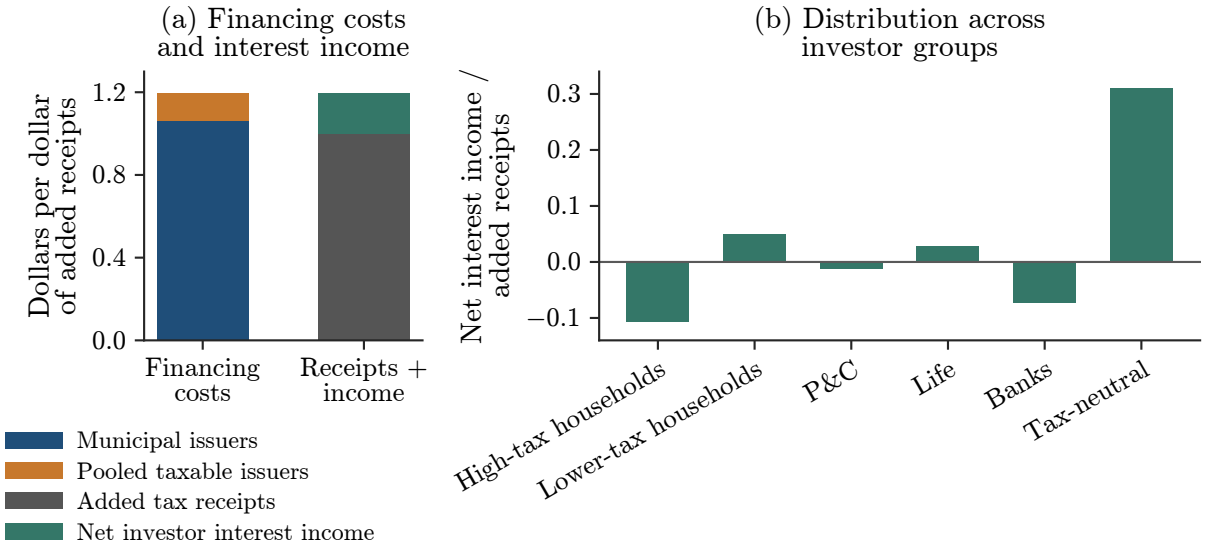


Figure 13: Average annual flows after removing 99.9% of each tax distinction in the six periods of 2000–2017. Panel (a) reports identity (30), dividing average financing costs and net investor interest income by average added federal and state receipts. Panel (b) allocates the income change across investor groups. Receipts and income include taxable cash interest. Each period uses continuous yields and equilibrium holdings at $z = 0$. Taxable issuers pool treasury and corporate quantities.

Unlike home-state repeal, equal treatment leaves investors no municipal market in which they can retain the federal exemption. High-tax households, banks and P&C insurers shorten the average maturity of their risky bonds by 1.8 to 3.7 years, and high-tax households also increase their riskless

Table 5: Average annual fiscal effects of equal municipal and taxable treatment, 2000–2017

Δ(Financing costs)			=	Δ(Tax receipts) + Δ(After-tax interest income)		
	$bn/ year	Per receipt dollar			$bn/ year	Per receipt dollar
Municipal issuers	41.16	1.063		Federal receipts	34.29	0.886
Pooled taxable issuers	5.15	0.133		Residence-state receipts	4.42	0.114
				Total receipts	38.71	1.000
			+	Net investor interest income	7.59	0.196
Total financing costs	46.30	1.196	=	Total receipts and income	46.30	1.196

Notes. Calendar-year-weighted changes across the six three-year periods, each relative to its own baseline, with 99.9% of the tax distinction removed. Dollar amounts use dated-period quantities without inflation adjustment and continuous yields at $z = 0$. Per-receipt-dollar columns divide each average dollar change by average total receipts. Receipts and income include taxable cash interest, and net investor interest income is the change in after-tax interest income. Before rounding, total issuer costs equal receipts plus net investor interest income.

holdings (Appendix Table 39). Shorter bonds and cash generate less interest than long bonds, limiting the revenue collected from the affected investors. Tax-neutral investors' share of bonds beyond ten years rises from 39% to 57%, while lower-tax households' and life insurers' shares decline slightly to 27% and 14%. Long-bond interest thus shifts toward the tax-neutral group, on whom the reform imposes no additional tax, limiting receipts.

At the same time, the tax-neutral group's larger long-bond holdings increase its rate exposure and the returns it requires in both markets. Municipal yields rise nearly to taxable yields, increasing by 170.6 basis points at thirty years, while thirty-year taxable yields rise by 23.1 basis points. The transfer of rate risk therefore raises borrowing costs in both markets.

The $7.6 billion net increase in annual after-tax interest income is distributed unevenly (Figure 13(b)). The tax-neutral group gains $12.0 billion a year, lower-tax households gain $1.9 billion and life insurers gain $1.1 billion. High-tax households lose $4.1 billion, banks lose $2.8 billion and P&C insurers lose $0.5 billion. Investors who benefited more from the exemption lose income, while those less affected by the exemption gain income.

A direct subsidy, as under Build America Bonds, can return the added receipts to municipal issuers. Returning all $38.7 billion of added annual receipts still leaves municipal issuers paying $2.4 billion more per year and taxable issuers paying $5.1 billion more.

8 Conclusion

I develop an equilibrium term-structure model in which heterogeneous investor taxes endogenously segment bond markets and maturities. Investors choose taxable and tax-exempt bonds jointly across maturities, weighing exemption benefits against the cost of bearing rate risk, which depends on their balance sheets. Under short-sale constraints, these choices determine who holds each bond

and therefore whose taxes its yield reflects. Portfolio substitution links markets and maturities, so an investor's tax treatment can affect even bonds it never holds.

The municipal application shows why one representative investor's taxes cannot explain an entire yield curve. Top-bracket households value the exemption most, yet investors with smaller exemption benefits hold substantial municipal portfolios. The model links this ownership to relative yields: household taxes anchor the riskless short end, while longer yields also reflect lower-tax investors' tax treatment and their lower cost of bearing rate risk. The puzzle is therefore one of relative pricing, not a mispricing of municipal bonds: overlapping investors price the taxable and tax-exempt markets jointly, and the allocation of rate risk among them shapes both yield curves. Without targeting municipal yields, the calibrated model accounts for most of the puzzle, and the empirical tests confirm that household tax effects weaken with maturity while those of lower-tax investors strengthen.

These results matter beyond the municipal bond puzzle. Under the U.S. tax code, investors face very different tax treatment of the same asset, and the code changes frequently. For bond pricing, a model that ignores differences in investors' taxes and balance sheets misses the resulting partial segmentation and misprices bonds. For tax policy, the incidence of a tax change extends beyond the market and the investors it targets. Because investors reallocate across markets and maturities, a change in the taxation of one market shifts financing costs across issuers in different markets and redistributes after-tax income among investors. Evaluations that ignore investor heterogeneity and this reallocation misstate the effects of such policies. The same mechanism operates in any setting where investors with different tax treatments choose portfolios under limits to arbitrage, including the state-tax difference between treasury and corporate bonds, tax-favored accounts and cross-border withholding. Wherever taxes differ across investors, who holds a bond shapes its yield, making tax heterogeneity a first-order determinant of the term structure.

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A The model: formal results and proofs

This appendix states and proves the model results used in Section 4. Appendix A.1 states portfolio choice, market clearing and the price recursion for the general economy, on which the later results build. It also shows when the riskless municipal yield equals the household benchmark. Appendix A.2 derives participation and pricing in the two-investor affine benchmark, then verifies the illustrations with and without bond-specific risk. Appendices A.3–A.6 consider other participation motives, multiple states and investor groups, and other bond markets.

Unless stated otherwise, the analytical solutions allow borrowing through the cash account. Wherever cash stays nonnegative, they also solve the problem with borrowing ruled out. Appendix B calibrates the quantitative model and solves it with the shorting constraint on cash, Appendix C constructs the empirical adjustments and tests, and Appendix D supplies the tax rules, institutional evidence, fiscal accounting and data sources. Table 6 collects the model notation.

Table 6: Notation for the model and the proofs

Symbol	Meaning
z, $r(z) = \max\{\bar{r} + z, 0\}$	Latent rate state and the one-period log gross cash return
τ_i^a, ξ_i^a, $\xi_i = \xi_i^C/\xi_i^M$	Investor i's tax rate and after-tax share in market a. In the two-market benchmark, ξ_i is its relative after-tax share and $1-\xi_i$ its exemption benefit
$\xi_H < \xi_I$, ξ^*	Benchmark relative after-tax shares of the high-tax household and lower-tax investor, and their tax mix $\omega_H\xi_H + \omega_I\xi_I$ with risk-tolerance shares ω_i
$p_T^a(z)$, $A_T^a = -\partial_z p_T^a$, $y_T^a = -p_T^a/T$	Log price of the maturity-T bond in market a, its rate loading (the model's duration), and its yield, with $A_0^a = 0$
$C_T^a(z) = -p_T^a(z) - A_T^a(z)z$	Level term, the constant intercept in the affine benchmark
$\mu_T^a(z) = \log \mathbb{E}_z[1 + R_T^a]$, $s_T^a = \partial_z \mu_T^a$	Mean adjusted holding return, equal to its holders' required return in the Karush–Kuhn–Tucker (KKT) conditions, and its response to the state
$R_{T,t+1}^a$, $\mathcal{R}_{T,t+1}^a = \mu_T^a + u_{T,t+1}^a$	Simple holding-period return and the return used in portfolio choice, with mean μ_T^a and centered log-return innovation $u_{T,t+1}^a$
$\sigma_{\eta,T}^a$, $v_T^a = (\sigma_{\eta,T}^a)^2$	Standard deviation of the bond-specific log-return shock and its variance, zero for the one-period claim
$\nu_{i,T}^a$	KKT slack: required return minus offered return, zero for a positive holding
$x_{i,T}^a$, Q_T^a, m_i, γ_i	Holdings per unit of mass, total supply, investor mass, risk aversion. In the calibration, m_i measures baseline portfolio assets
W_i, W_i', $\widehat{W}_i'$, κ_i	Funding per unit of mass, next-period net wealth, its discrete-time approximation, and the multiplier on the shorting constraint on cash
$\widehat{D}_i$, ℓ_i	Net rate exposure and the signed liability hedge. Appendix B.1 uses the outside rate exposure $h_i = -\ell_i$
$\widehat{\Lambda}$	Net-exposure difference weighted by risk aversion, $\gamma_H\widehat{D}_H - \gamma_I\widehat{D}_I$, at the equilibrium allocation
$\omega_{i,T}^a$, $\mathcal{U}_T^a$	Risk-tolerance share of an active holder and the set of active holders of the maturity-T bond in market a
$g_T = \mu_T^M - \xi_H\mu_T^C$	Gap between the municipal one-period return and the high-tax household's after-tax taxable return
$\tau_T^{Y,\text{impl}} = 1 - y_T^M/y_T^C$	Model yield-implied tax, the counterpart of *ImpliedTax* in (2)

A.1 The general model

This subsection sets out the general economy, which adds bond-specific return risk and trading exclusions to the model of Section 4.1. As in that model, investors with different tax rates and liability hedges choose jointly across bond markets and maturities. Their KKT conditions give the return each investor requires on each bond, market clearing determines who holds it, and a price recursion turns one-period returns into prices and yields. The quantitative model of Section 5 solves this economy with several investor groups, state municipal markets, shorting constraints on cash that can bind and the rate floor.

A.1.1 Assets and portfolio choice

The rate state. Time is discrete and indexed by t. One aggregate rate state z_t follows a stationary first-order autoregression, AR(1), and the short rate, the pretax log gross return on cash, cannot fall below zero:

$$z_{t+1} \;=\; \rho z_t \;+\; \sigma_r \varepsilon_{t+1}, \qquad r_t = r(z_t) := \max\{\bar{r} + z_t, 0\}, \qquad \rho \in (0,1), \quad \varepsilon_{t+1} \sim \mathcal{N}(0,1). \tag{31}$$

Here ρ measures the persistence of the rate state and σ_r the size of its shocks. The parameter $\bar{r}$ is the cash rate before the floor at the stationary mean $z = 0$. The floor applies to the cash return and leaves the state z_{t+1} Gaussian. Appendix A.1.2 describes how the floor affects continuation prices, the next-period resale prices of bonds.

Bonds and holding returns. Let $\mathcal{A}$ be a finite set of bond markets, indexed by a and distinguished by asset class or issuer. In the two-investor benchmark, $\mathcal{A}$ contains a taxable market C and a municipal market M. The quantitative model has 45 municipal markets, pooling the seven states without an income tax. Each market contains zero-coupon bonds indexed by their remaining maturity $T \in \{1, \ldots, T_{\max}\}$, the number of periods until the bond pays one unit, so $P^a_0 = 1$. The one-period claim of the taxable market C is the cash account, with log gross return $r(z)$, so risky taxable bonds begin at $T = 2$. A bond with T periods remaining has price $P^a_T(z_t)$ and holding-period return

$$1 + R^a_{T,t+1} \;=\; \frac{P^a_{T-1}(z_{t+1})}{P^a_T(z_t)} \exp\!\big(\sigma^a_{\eta,T}\, \eta^a_{T,t+1}\big). \tag{32}$$

The standard normal shocks $\eta^a_{T,t+1}$ are independent of each other and of ε_{t+1}. They add bond-specific log-return variance $v^a_T := (\sigma^a_{\eta,T})^2$, and setting $v^a_T = 0$ gives the main-text return. Every market has $\sigma^a_{\eta,1} = 0$, so a one-period claim is fully riskless. Bond-specific risk represents residual variation in holding returns, such as an issue-specific change in trading demand. It makes longer bonds imperfect substitutes and allows joint holdings at more than one maturity, without adding a state variable that affects future prices.

Investors. Investors belong to finitely many types, indexed by i. Type i has mass $m_i > 0$, risk aversion $\gamma_i > 0$, funding W_i and liability hedge ℓ_i against the rate shock from assets and liabilities beyond its bonds. It keeps the absolute after-tax share $\xi^a_i = 1 - \tau^a_i > 0$ of returns in market a, where τ^a_i is its tax rate on those returns. Investor i is eligible for a bond unless a predetermined trading exclusion fixes its holding at zero, and the KKT conditions below apply to eligible investor–bond pairs. Because after-tax shares differ across both investors and markets, the same structure covers municipal and taxable bonds, state taxes on treasury and corporate interest, tax-favored accounts and cross-border withholding (Appendix A.6).

Holdings and the return approximation. Investor i holds bonds $x^a_{i,T,t}$ and cash $x^0_{i,t} = W_i - \sum_{a,T} x^a_{i,T,t}$, in dollars per unit of mass. Aggregate holdings are $m_i x^a_{i,T,t}$, in the same units as supply. With simple holding-period returns $R^a_{T,t+1}$, exact next-period net wealth is

$$W_i' = W_i + \xi^C_i x^0_i \big(e^{r(z_t)} - 1\big) + \sum_{a,T} \xi^a_i x^a_{i,T} R^a_{T,t+1} + \ell_i \sigma_r \varepsilon_{t+1}.$$

Taking logs of (32) gives the change in log price plus the bond-specific shock. Following Greenwood et al. (2024), I approximate the simple bond return by this log return plus a Jensen adjustment, and $e^r - 1$ by r for cash. The adjusted bond return is

$$\mathcal{R}^a_{T,t+1} = \log(1 + R^a_{T,t+1}) + \underbrace{\log \mathbb{E}_t[1 + R^a_{T,t+1}] - \mathbb{E}_t[\log(1 + R^a_{T,t+1})]}_{\text{Jensen adjustment}}.$$

The Jensen adjustment is known at time t, so it shifts the mean but leaves log-return variances and covariances unchanged. As in Section 4.1, write $\mathcal{R}^a_{T,t+1} = \mu^a_T(z_t) + u^a_{T,t+1}$. Taking the conditional expectation gives the mean adjusted return, Jensen adjustment included, $\mathbb{E}_t[\mathcal{R}^a_{T,t+1}] = \mu^a_T(z_t) = \log \mathbb{E}_t[1 + R^a_{T,t+1}]$. Subtracting this mean cancels the adjustment and leaves the centered log-return innovation

$$u^a_{T,t+1} = \mathcal{R}^a_{T,t+1} - \mu^a_T(z_t) = \log(1 + R^a_{T,t+1}) - \mathbb{E}_t[\log(1 + R^a_{T,t+1})].$$

Thus μ sets expected income and u sets risk. In the affine benchmark, log returns are conditionally Gaussian, so

$$\mu^a_T(z_t) = \mathbb{E}_t[\log(1 + R^a_{T,t+1})] + \tfrac{1}{2} \operatorname{Var}_t[\log(1 + R^a_{T,t+1})].$$

This mean and the centered innovation u match equations (7)–(8) of Greenwood et al. (2024). The quantitative model uses $\mu = \log \mathbb{E}_t[1 + R]$ directly, retaining the full Jensen adjustment even when log returns are not Gaussian.

Exactly taxed simple returns. The price recursion (42) below is exact for the chosen mean $\mu = \log \mathbb{E}_t[1+R]$. Using this mean in the linear wealth equation (33) below approximates simple returns, whose conditional mean is $e^{\mu} - 1$. Appendix D.3 compares the two conventions for a deterministic return. For the solved model, I re-solve the one-market model used to estimate the coefficients in Section 5, now with exactly taxed simple returns, refit risk aversion to restore the targeted taxable yields and hold the other coefficients fixed. The thirty-year gap in that model rises from 102.6 to 107.7 basis points, so the approximation does not inflate the gap.

The portfolio problem. Substituting the adjusted returns into exact wealth gives approximated next-period wealth $\widehat{W}_i'$. Each investor maximizes the mean of $\widehat{W}_i'$ less a penalty for its variance:[23]

$$\begin{aligned} \max_{x_i \geq 0,\, x^0_{i,t} \geq 0} \quad & \mathbb{E}_t[\widehat{W}_i'] - \frac{\gamma_i}{2} \operatorname{Var}_t(\widehat{W}_i'), \\ \text{subject to} \quad & \widehat{W}_i' = W_i + \xi_i^C r(z_t) x^0_{i,t} \\ & \qquad + \sum_{a,T} \xi_i^a x^a_{i,T,t} \mathcal{R}^a_{T,t+1} + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{33}$$

Positions and cash are nonnegative, which rules out short sales and borrowing (Section 3.5). Funding and masses are fixed across states. The liability hedge ℓ_i can take either sign. A positive ℓ_i raises net wealth when rates rise, offsetting the rate exposure of the investor's bonds.

Collecting the cash and bond terms in (33) gives each bond's expected after-tax excess return over cash,

$$\widetilde{\mu}^a_{i,T}(z_t) := \underbrace{\xi_i^a \mu^a_T(z_t)}_{\text{after-tax bond return}} - \underbrace{\xi_i^C r(z_t)}_{\text{after-tax cash return}}. \tag{34}$$

The bond and outside innovations have conditional mean zero, so expected approximated wealth is initial funding grown at the after-tax cash rate plus the portfolio's excess returns:

$$\mathbb{E}_t[\widehat{W}_i'] = W_i\big(1 + \xi_i^C r(z_t)\big) + \sum_{a,T} x^a_{i,T,t} \widetilde{\mu}^a_{i,T}(z_t).$$

The variance of wealth is that of the after-tax portfolio innovation, the random part of (33):

$$\widetilde{\Pi}_i(t+1) = \sum_{a \in \mathcal{A}} \sum_T \xi_i^a x^a_{i,T,t} u^a_{T,t+1} + \underbrace{\ell_i \sigma_r \varepsilon_{t+1}}_{\text{liability hedge}}. \tag{35}$$

Dropping the portfolio-independent term $W_i(1 + \xi_i^C r(z_t))$ and eliminating cash gives the equivalent reduced problem:

$$\max_{\substack{x_i \geq 0 \\ \sum_{a,T} x^a_{i,T,t} \leq W_i}} \quad \underbrace{\sum_{a \in \mathcal{A}} \sum_T x^a_{i,T,t} \widetilde{\mu}^a_{i,T}(z_t)}_{\text{after-tax excess return}} - \underbrace{\frac{\gamma_i}{2} \operatorname{Var}_t\Big(\widetilde{\Pi}_i(t+1)\Big)}_{\text{whole-portfolio risk cost}}. \tag{36}$$

The KKT conditions. Let $\kappa_i(z) \geq 0$ denote the multiplier on the shorting constraint on cash, $\sum_{a,T} x^a_{i,T} \leq W_i$. The objective in (36) is concave and its constraints are linear, so the following KKT conditions are necessary and sufficient for optimality:

$$\begin{aligned} x^a_{i,T}(z) > 0 \quad &\Longrightarrow \quad \widetilde{\mu}^a_{i,T}(z) = \underbrace{\gamma_i \xi_i^a \operatorname{Cov}_z\Big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\Big)}_{\text{marginal portfolio-risk cost}} + \kappa_i(z), \\ x^a_{i,T}(z) = 0 \quad &\Longrightarrow \quad \widetilde{\mu}^a_{i,T}(z) \leq \gamma_i \xi_i^a \operatorname{Cov}_z\Big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\Big) + \kappa_i(z). \end{aligned} \tag{37}$$

[23] Appendices C.1.4 and D.3 give the correction for taxing resale gains separately from interest.

The multiplier satisfies complementary slackness, $\kappa_i(z)[W_i - \sum_{a,T} x^a_{i,T}(z)] = 0$. The right side is the investor's required after-tax excess return: the marginal cost of the bond's risk to the whole portfolio plus the multiplier. Since $\widehat{W}'_i - \widetilde{\Pi}_i(t+1)$ is known at t, this covariance equals $\mathrm{Cov}_z(u^a_{T,t+1}, \widehat{W}'_i)$ in (8).

Rate risk in the covariance comes from the next-period resale price. With log price $p^a_T := \log P^a_T$ and rate loading $A^a_T := -\partial_z p^a_T$, a bond with T periods remaining has rate innovation $-A^a_{T-1}(\rho z_t)\sigma_r \varepsilon_{t+1}$ to first order around the expected resale state. A one-period bond carries no rate innovation, since $A^a_0 = 0$, although its price can respond to today's state.

The multiplier $\kappa_i(z)$ is zero whenever the investor holds cash, since the investor can then buy more bonds without borrowing. It is positive only when the investor is fully invested, so that buying more of one bond means selling another, and it then raises the required pretax return on each held bond by κ_i/ξ^a_i.

Market clearing and equilibrium. Every bond other than cash clears at every rate state:

$$x^a_{i,T}(z) \geq 0, \qquad \sum_i m_i \, x^a_{i,T}(z) = Q^a_T, \tag{38}$$

where Q^a_T is the fixed dollar supply. An equilibrium consists of log-price functions with $p^a_0 \equiv 0$, holdings and multipliers $\kappa_i(z) \geq 0$ such that, at every state, returns follow (32), every investor satisfies $\sum_{a,T} x^a_{i,T}(z) \leq W_i$, (37) and complementary slackness, and (38) holds. These conditions determine participation, that is, which investors hold each bond.

Riskless one-period claims. Consider a riskless one-period claim in any market $a \neq C$. Its bond-specific variance $v^a_1 = (\sigma^a_{\eta,1})^2$ is zero and $p^a_0 = 0$, so its return has no covariance with portfolios or liabilities. Retaining κ_i in (37), each eligible investor's KKT condition becomes

$$\mu^a_1 \leq \frac{\xi^C_i r + \kappa_i}{\xi^a_i}, \qquad \text{with equality for a positive holding.}$$

With positive supply, at least one investor holds the claim, so its return equals the smallest of these bounds:

$$\mu^a_1(z) = \min_{i \text{ eligible}} \frac{\xi^C_i r(z) + \kappa_i(z)}{\xi^a_i}. \tag{39}$$

The price recursion (42) then gives $p^a_1 = -\mu^a_1$, hence $y^a_1 = \mu^a_1$.

A.1.2 Bond prices and required returns

A bond's price depends on the one-period returns it earns as it ages to maturity, and its rate loading accumulates their responses to the rate state. Section 4.5 uses both relations on an affine branch, and this subsection derives them for the general economy.

The price recursion. Taking expectations of the gross holding return (32), with next-period state $Z' = \rho z + \sigma_r \varepsilon'$, gives

$$P^a_T(z) = e^{-\mu^a_T(z) + v^a_T/2} \, \mathbb{E}_z\big[P^a_{T-1}(Z')\big], \qquad P^a_0 = 1. \tag{40}$$

A higher required return lowers today's price for a given expected resale value. The factor $e^{v^a_T/2}$ accounts for the mean of the bond-specific shock, and its yield effect is small (Appendix B.5).

With fixed supplies and funding, I seek an equilibrium log-price function for each market and maturity:

$$p^a_{T,t} = p^a_T(z_t), \qquad p^a_0(z) \equiv 0, \qquad a \in \mathcal{A}. \tag{41}$$

Since μ^a_T is the log of the bond's conditional expected gross return (32), solving for today's log price gives

$$p^a_T(z) = \underbrace{\log \mathbb{E}_z\big[\exp\{p^a_{T-1}(Z')\}\big]}_{\text{log expected resale price}} + \underbrace{\tfrac{1}{2}(\sigma^a_{\eta,T})^2}_{\text{bond-specific payoff term}} - \underbrace{\mu^a_T(z)}_{\text{one-period return}}. \tag{42}$$

This recursion links adjacent maturities, while the KKT conditions (37) link markets through investors' holdings. At every state, prices and portfolios jointly solve these equations and clearing (38). Appendix B gives the numerical implementation.

Sources of nonlinearity. The rate floor, changes in participation and binding shorting constraints on cash make prices nonlinear in the state. Resale carries these nonlinearities across maturities, because today's continuation value averages future prices across states that differ in these respects.

Prices and yields through one-period returns. Iterating (42) from $p_0^a = 0$ expresses the price through the one-period returns the bond earns as it ages through maturities $T, T-1, \ldots, 1$:

$$P_T^a(z_t) = e^{p_T^a(z_t)} = \mathbb{E}_t\Big[\exp\Big\{-\sum_{j=0}^{T-1}\Big(\mu_{T-j}^a(z_{t+j}) - \tfrac{1}{2}\big(\sigma_{\eta,T-j}^a\big)^2\Big)\Big\}\Big]. \tag{43}$$

Taking the negative log of (43) and dividing by T gives the yield $y_T^a = -p_T^a/T$. In the affine benchmark, this yield is the explicit average of expected one-period returns less convexity in (54).

Rate loadings and level terms. In the general economy, I summarize the yield's dependence on the state by writing the log price in loading form,

$$p_T^a(z) = -A_T^a(z)\, z - C_T^a(z), \qquad A_T^a(z) := -\frac{\partial p_T^a(z)}{\partial z}, \tag{44}$$

so that $P_T^a = \exp\{-A_T^a z - C_T^a\}$ and, at differentiability points,

$$y_T^a(z) = \frac{A_T^a(z)z + C_T^a(z)}{T}, \qquad \frac{\partial y_T^a(z)}{\partial z} = \frac{A_T^a(z)}{T}.$$

The rate loading A_T^a is the model's counterpart of duration. The level term C_T^a includes accumulated one-period returns and convexity, and the mean-state yield is $C_T^a(0)/T$. Outside the affine benchmark, both A_T^a and C_T^a can vary with the state. The yield derivative above still holds, because the definitions imply $\partial_z C_T^a = -z\,\partial_z A_T^a$. Same-maturity loading ratios therefore measure relative yield sensitivity, the model counterpart of P3.

The loading recursion. The loading recursion links A_T^a to the one-period return response $s_T^a := \partial_z \mu_T^a$. Above the rate floor, cash has response one and, by (39), a riskless one-period claim in another market has response ξ_i^C/ξ_i^a when investor i attains the minimum in (39) with $\kappa_i = 0$ at all nearby states. At longer maturities the response also includes the change in the holders' risk premia. Suppose that μ_T^a is differentiable at a given state, that continuation prices are locally absolutely continuous, and that the Gaussian exponential moment and its state derivative have an integrable envelope. Differentiating under the expectation in (42) then gives

$$A_T^a(z) = \underbrace{s_T^a(z)}_{\text{current return response}} + \underbrace{\rho\,\frac{\mathbb{E}_z\big[\exp\{p_{T-1}^a(Z')\}\, A_{T-1}^a(Z')\big]}{\mathbb{E}_z\big[\exp\{p_{T-1}^a(Z')\}\big]}}_{\text{continuation price response}}. \tag{45}$$

The fraction averages the continuation loading, giving more weight to states where the resale bond is more valuable. Through this average, future changes in who holds the bond affect today's loading. In the affine benchmark, (45) reduces to the loading recursion (53).

A.1.3 Market clearing and the riskless short yield

Proposition A.1 shows that, given continuation prices, clearing returns exist, and gives sufficient conditions for their uniqueness. Proposition A.2 shows that, with a slack household shorting constraint on cash and $r > 0$, the riskless one-period municipal yield equals the household benchmark, $\xi_H r$, when the high-tax household has the smallest relative after-tax share.

Supports and participation branches. An allocation's support is the set of investor–bond pairs with positive holdings. A participation branch fixes the support, and a bond's active set is its holders. Affine results use floorless rates and a fixed participation branch, whereas the numerical model lets participation change with rates and retains the floor and the shorting constraint on cash.

One KKT system for all markets. Index market–maturity coordinates other than cash by $j = (a, T)$ and write $v_j = (\sigma_{\eta,T}^a)^2$, $K_i = \mathrm{diag}(\xi_i^a)$ and the relative after-tax share $\xi_{ij} = \xi_i^C/\xi_i^a$, collected in the vector $\boldsymbol{\xi}_i$. In the two-market benchmark normalized as in Section 4.2, $\xi_{ij} = \xi_i$ at municipal coordinates and $\xi_{ij} = 1$ at taxable coordinates. With resale log price $p_{j,-} = p_{T-1}^a$, the exact covariance and liability credit are

$$\Sigma_{jk}(z) = \mathrm{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j, \qquad c_{ij}^L(z) = -\mathrm{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r \varepsilon'). \tag{46}$$

With a slack shorting constraint on cash, dividing each first-order condition by the positive after-tax share of its market gives

$$\nu_{ij} := \xi_{ij} r + \gamma_i [\Sigma K_i x_i - c_i^L]_j - \mu_j \geq 0, \qquad x_{ij} \geq 0, \qquad x_{ij}\nu_{ij} = 0, \\ \sum_i m_i x_{ij} = Q_j. \tag{47}$$

Here the slack ν_{ij} is the required minus the offered return. Concavity of (36) makes these conditions sufficient for investor optimality. With future resale prices held fixed, return covariances and liability credits are given, and current returns must adjust until investors' optimal holdings match bond supplies. The next proposition treats this clearing step.

Proposition A.1 (Market clearing with given continuation prices)**.** *Fix the state and continuation prices, with the shorting constraint on cash dropped, $\kappa_i \equiv 0$, finite nonnegative supplies, finite $\Sigma \succeq 0$, finite liability credits and at least one eligible investor for every positive-supply coordinate.*

(i) The KKT-and-clearing system (47) has a current-state solution. Holdings that also satisfy $\sum_{a,T} x^a_{i,T} \leq W_i$ for every i, one-period claims included, solve the problem with that constraint.

(ii) If, in addition, $\Sigma = V + \sigma_r^2 AA'$ with $V = \mathrm{diag}(v_j)$, and every risky coordinate has $v_j > 0$, predecessor loading $A_j = A^a_{T-1} > 0$ and supply $Q_j > 0$, the current-state clearing returns and risky holdings are unique, even with endogenous support changes. Deterministic one-period claims are treated separately. Payoff-equivalent holdings in tied one-period boundary claims need not be unique.

Neither part establishes the dynamic fixed point of continuation prices.

Proof. *(i) Existence.* Clearing returns are read off a fixed point on the set of clearing allocations. Let $X = \{x \geq 0 : \sum_i m_i x_i = Q,\ x_{ij} = 0 \text{ for ineligible } (i,j)\}$ be that set, with inner product $\langle x, y\rangle_m = \sum_i m_i x_i' y_i$. Positive masses, finite supplies and an eligible investor for each positive-supply coordinate make X nonempty, compact and convex. Let $\Upsilon_i(x_i) = \boldsymbol{\xi}_i r + \gamma_i(\Sigma K_i x_i - c_i^L)$ be investor i's required return on each coordinate. Brouwer's theorem gives a fixed point of the continuous map $x \mapsto \mathrm{Proj}_X(x - \Upsilon(x))$, with Proj_X the projection in $\langle\cdot,\cdot\rangle_m$, and the projection characterization implies

$$\sum_i m_i \langle \Upsilon_i(x_i), y_i - x_i\rangle \geq 0 \qquad (y \in X).$$

Transfers from positive holders to other eligible investors show that each positive holder minimizes Υ_{ij}. Thus $\mu_j = \min_{i \text{ eligible}} \Upsilon_{ij}$ and $\nu_{ij} = \Upsilon_{ij} - \mu_j$ satisfy (47). Conversely, complementarity implies the displayed inequality for every feasible transfer.

(ii) Uniqueness. Bonds are substitutes in demand, and own-return effects dominate in loading weights, so the averaged demand Jacobian is nonsingular. Strict concavity makes individual demand $x_i(\mu)$ continuous, Lipschitz and piecewise affine. On a region where investor i holds exactly the coordinates in a set B_i, its derivative is

$$\mathcal{J}_i = \gamma_i^{-1} \mathbf{E}_{B_i} K_{i,B_i}^{-1} \Sigma_{B_iB_i}^{-1} \mathbf{E}_{B_i}',$$

where $\mathbf{E}_B$ embeds the held coordinates. Sherman–Morrison gives

$$\Sigma_{BB}^{-1} = V_B^{-1} - \frac{\sigma_r^2 V_B^{-1} A_B A_B' V_B^{-1}}{1 + \sigma_r^2 A_B' V_B^{-1} A_B}, \qquad \Sigma_{BB}^{-1} A_B = \frac{V_B^{-1} A_B}{1 + \sigma_r^2 A_B' V_B^{-1} A_B} \gg 0.$$

Every selection of $\mathcal{J} = \sum_i m_i \mathcal{J}_i$ therefore has nonpositive off-diagonal entries and $\mathcal{J}A \geq 0$, strictly on held coordinates. Now suppose μ^0 and μ^1 both cleared. Piecewise-affine continuity gives $0 = \overline{\mathcal{J}}(\mu^1 - \mu^0)$, with $\overline{\mathcal{J}}$ the segment-average selection matrix. Positive supply at μ^0 and demand continuity imply positive holdings on an initial segment for each asset. Finitely many selections then give $(\overline{\mathcal{J}}A)_j > 0$ for every j. Hence $\overline{\mathcal{J}}\,\mathrm{diag}(A)$ is strictly diagonally dominant by rows and nonsingular, so returns coincide, and strict concavity gives identical risky holdings. On support-boundary segments, adjacent selections have the same directional derivative. □

Regularity for derivatives. Derivatives below are taken at regular states, at which the cash rate and price moments are continuously differentiable, complementarity is strict on risky coordinates and the active KKT-and-clearing Jacobian is nonsingular. Each investor's strict concavity does not imply this nonsingularity, because with heterogeneous after-tax shares the normalized operator contains ΣK_i, which need not be symmetric

or strongly monotone. At such states, the implicit-function theorem gives a locally fixed participation branch and differentiable holdings and returns. Fixed supplies imply, for each bond's active set $\mathcal{U}_j$,

$$\sum_{i \in \mathcal{U}_j} m_i \frac{\partial x_{ij}}{\partial z} = 0. \tag{48}$$

The one-period boundary. Specializing the riskless-claim return (39) to the normalized municipal–taxable benchmark gives the one-period yields (12) and tax pass-throughs (21).

Proposition A.2 (Deterministic one-period boundary)**.** *In the normalized municipal–taxable benchmark, suppose the household's shorting constraint on cash is slack, $\kappa_H = 0$, the one-period municipal claim has positive supply, and $r(z) > 0$. Suppose the high-tax household has the minimal relative after-tax share, $\xi_H = \min_i \xi_i$. Then only investors with the minimal relative after-tax share can hold the claim, and*

$$y_1^M = \mu_1^M = \xi_H r(z), \qquad y_1^C = r(z), \qquad \tau_1^{Y,\text{impl}} = 1 - \xi_H. \tag{49}$$

If the household is the unique minimizer and $\kappa_H = 0$ at nearby after-tax shares, then $\mathcal{P}_H^Y(1) = 1$ and $\mathcal{P}_i^Y(1) = 0$ for every investor with $\xi_i > \xi_H$, which is (21). At a tie the household derivative requires perturbing all tied minimal after-tax shares together. At $r = 0$ the yield ratio is undefined.

Proof. With $\xi_i^M = 1$ and $\xi_i^C = \xi_i$, the riskless-claim condition preceding (39) is $\mu_1^M \le \xi_i r + \kappa_i$, with equality for positive holdings. The household's condition gives $\mu_1^M \le \xi_H r$. Positive supply gives a holder i, and $\mu_1^M = \xi_i r + \kappa_i \ge \xi_H r$. Hence $\mu_1^M = \xi_H r$, $\kappa_i = 0$ and, since $r > 0$, $\xi_i = \xi_H$. The recursion gives $p_1^M = -\mu_1^M$, so $y_1^M = \xi_H r$. Taxable cash has $p_1^C = -r$, so $y_1^C = r$ and $\tau_1^{Y,\text{impl}} = 1 - y_1^M / y_1^C = 1 - \xi_H$. Only the minimal after-tax share appears in this ratio, so differentiating it proves the tax statements. □

A.2 The two-investor affine benchmark

This subsection specializes the general model to two investors and two jointly priced bond markets. It proves the participation and pricing results of Sections 4.3–4.6 in the setting used there: risky bonds carry rate risk only, with no bond-specific return shock. Without bond-specific risk, investors sort by maturity whenever $r > 0$ and rate loadings rise with maturity. When the lower-tax investor bears rate risk more cheaply, the high-tax household holds the shorter municipal bonds, the lower-tax investor holds the longer ones, and at most one maturity is shared (Theorem A.1). Appendices A.2.1–A.2.7 work on this sorting branch. Appendix A.2.8 then adds a small bond-specific shock, as in the quantitative model. The shock lets both investors hold a block of maturities, and the pricing results carry over with the shared maturity replaced by that block.

For the riskless one-period claim, a slack shorting constraint on cash gives $\mu_1^M = \xi_H r$ whenever $r > 0$, and the household, which keeps the smallest share of taxable interest, holds the riskless municipal claim (Proposition A.2). At longer maturities, participation determines whose taxes and risk costs shape equilibrium returns.

A.2.1 Assumptions and portfolio conditions

This part sets out the economy of Section 4.2, the simplifications that make prices affine and the net rate exposure behind the portfolio conditions (11).

Investors, assets and units. The high-tax household H and the lower-tax investor I choose among municipal maturities $T = 1, \dots, T_{\max}$, taxable maturities $T = 2, \dots, T_{\max}$ and cash, the one-period taxable claim. Bond supplies Q_T^a are fixed. Risk aversions γ_i can differ, and the relative after-tax share $\xi_i = \xi_i^C / \xi_i^M$ is that of Section 4.2. Payoffs are in units of after-tax municipal income, so taxable returns and cash income are multiplied by ξ_i, with γ_i and ℓ_i normalized to match. All two-investor results assume $0 < \rho < 1$, $\sigma_r > 0$, $m_i, \gamma_i > 0$, $0 < \xi_H < \xi_I$ and positive traded supplies. Bond-specific variance $v_T^a = (\sigma_{\eta,T}^a)^2$ is zero through Appendix A.2.7. The general formulas below keep the bond-specific terms, which Appendix A.2.8 uses.

Participation branches and three simplifications. A participation branch specifies which investors hold each bond. I fix a branch, solve its portfolio conditions and market clearing, and check that no investor benefits from changing its holdings. Closed forms use three simplifications:

(i) the shorting constraint on cash is dropped;

(ii) the floored cash rate is replaced by the linear rule $r(z) = \bar{r} + z$; and

(iii) prices are computed as if the same participation branch applied at every state, which makes log prices affine, and the branch is verified at the states studied.

The quantitative model lets participation change across states and keeps the shorting constraint on cash and the rate floor.

On a fixed branch, holdings, one-period returns and log prices are affine in z, and the price coefficients solve a fixed-point problem whose feasibility and uniqueness conditions are stated below. One-period returns take the form $\mu_T^{a,\mathrm{aff}}(z) = \mu_T^{a,\mathrm{aff}}(0) + s_T^{a,\mathrm{aff}} z$, so $s_T^{a,\mathrm{aff}}$ is the return's state response, and the rate loading $A_T^{a,\mathrm{aff}}$ is the coefficient on z in the negative log price. The superscript "aff", where shown, marks this branch.

Net rate exposure. Investor i's net rate exposure is the after-tax exposure of its whole balance sheet to the rate shock:

$$\widehat{D}_i(z) := \underbrace{\sum_T x_{i,T}^M(z) A_{T-1}^M + \xi_i \sum_T x_{i,T}^C(z) A_{T-1}^C}_{\text{tax-weighted asset exposure}} - \underbrace{\ell_i}_{\text{liability hedge}}. \tag{50}$$

With today's positions fixed, $\widehat{D}_i = -\partial \widehat{W}_i' / \partial z_{t+1}$, as in (10). It is dollar exposure before division by wealth. The loadings in (50) are predecessor loadings A_{T-1}^a, the bonds' loadings next period, when the rate shock moves their resale prices. The rate component of the wealth innovation is $-\widehat{D}_i \sigma_r \varepsilon_{t+1}$, and a larger liability hedge ℓ_i lowers this exposure. A bond's rate-risk compensation equals its loading times the investor's unit cost of rate risk, $\gamma_i \sigma_r^2 \widehat{D}_i$, and the portfolio conditions are those of (11). Loadings are constant on an affine branch, so changes in $\widehat{D}_i$ across rate states reflect trading alone.

Uniqueness and illustrations. On the sorting branch, the response of net rate exposures to the rate state has a unique solution (Proposition 2), and the holding split at the shared maturity is then pinned by a linear level condition. Proposition A.1 gives sufficient conditions for unique current-state clearing when every risky bond carries bond-specific risk. The main-text illustration shares only municipal maturity 23 (Appendix A.2.7). Adding small bond-specific risk gives overlapping holdings from maturity 17 (Appendix A.2.9). Both appendices report the parameters and check holdings and pricing. These are mechanism illustrations, so their level panels omit the vertical scale.

A.2.2 Affine price loadings and yields

This part shows how a bond's rate loading and yield build up from its one-period returns, deriving the loading recursion (16) and the yield average behind (19) of Section 4.5. It then constructs and checks affine solutions.

Prices, loadings and responses. Write $p_T^a = \log P_T^a$ for the log price and $y_T^a = -p_T^a/T$ for the yield. Under the affine assumptions above,

$$\log P_T^a(z) = -A_T^a z - C_T^a, \qquad y_T^a(z) = \frac{A_T^a z + C_T^a}{T}. \tag{51}$$

The coefficient A_T^a, the bond's rate loading, is the proportional fall in its price per unit rise in the rate state and the model's counterpart of duration. Thus A_T^a/T is the yield's response to the rate state and C_T^a/T is its level at the mean state. The response of the one-period return to the rate state is

$$s_T^a(z) := \frac{\partial \mu_T^a(z)}{\partial z}. \tag{52}$$

The loading recursion. Substituting affine prices into the price recursion (42) and matching its state and constant terms gives

$$A_T^a = \underbrace{\rho A_{T-1}^a}_{\text{inherited loading}} + \underbrace{s_T^a}_{\text{current response}}, \qquad C_T^a = C_{T-1}^a + \mu_T^a(0) - \tfrac{1}{2}\left[\sigma_r^2 \left(A_{T-1}^a\right)^2 + \left(\sigma_{\eta,T}^a\right)^2\right]. \tag{53}$$

The level term accumulates mean-state one-period returns less half-variance terms. Starting from $A_0^a = C_0^a = 0$, the loading accumulates the responses, weighted by rate persistence, as in (56) below. With a riskless one-period

claim, $\sigma^a_{\eta,1} = 0$, the same recursion gives the yield as the average of the expected one-period returns along the bond's life less convexity:

$$y^a_T(z_t) = \underbrace{\frac{1}{T}\sum_{j=0}^{T-1}\mathbb{E}_t\left[\mu^a_{T-j}(z_{t+j})\right]}_{\text{one-period returns as the bond ages}} - \underbrace{\frac{1}{2T}\sum_{q=2}^{T}\left[\sigma_r^2\left(A^a_{q-1}\right)^2 + \left(\sigma^a_{\eta,q}\right)^2\right]}_{\text{convexity}}. \tag{54}$$

On the sorting branch, the bond-specific terms are zero. Appendix A.2.8 makes them positive for risky municipal bonds.

Constructing affine solutions. *Step 1: candidate prices.* Fix a participation branch, set inactive positions to zero, set $r(z) = \bar{r} + z$ and use affine continuation prices on $\mathbb{R}$. For affine candidates $p^a_T(z) = -A^a_T z - C^a_T$ and next-period state $Z' = \rho z + \sigma_r \varepsilon'$, Gaussian integration gives exactly

$$\log \mathbb{E}_z e^{p^a_{T-1}(Z')} = -\rho A^a_{T-1} z - C^a_{T-1} + \tfrac{1}{2}\sigma_r^2 (A^a_{T-1})^2. \tag{55}$$

Step 2: current-state solution. The current-state covariance $\Sigma = \sigma_r^2 AA' + \mathrm{diag}(v)$ and liability credit $c^L_i = \ell_i \sigma_r^2 A$ of (46) are then constant, where A collects the predecessor loadings and v the bond-specific variances v_j. The active equalities are solved jointly across maturities, retaining whole-portfolio risk. If the active KKT-and-clearing matrix is nonsingular, holdings and μ are affine in z, and the price recursion returns affine prices with updated loadings $\rho A^a_{T-1} + s^a_T$.

Step 3: fixed point. At a coefficient fixed point, all active equalities, clearing equations and exact continuation equations hold identically, and, with $A^a_0 = 0$,

$$A^a_T = \rho A^a_{T-1} + s^a_T = \sum_{j=1}^{T} \rho^{T-j} s^a_j, \qquad s^a_T = \frac{\partial \mu^a_T}{\partial z}. \tag{56}$$

Feasibility. The candidate must have nonnegative holdings, nonnegative inactive slacks and, for finite funding, $W_i - \sum_j x_{ij}(z) \geq 0$. With affine holdings and slacks, the states where these KKT conditions hold form an interval, possibly empty, given globally affine continuation prices. Gaussian transitions leave any proper interval with positive probability, so feasibility on an interval does not establish global equilibrium feasibility. For example, on any branch on which the household holds the one-period municipal claim, the lower-tax investor's slack on that claim, $(\xi_I - \xi_H)(\bar{r} + z)$, turns negative at negative rates.

Existence and uniqueness. For a fixed participation branch, a continuous coefficient update on a nonempty compact convex invariant set has a fixed point by Brouwer. A contraction gives uniqueness within that set. Nonsingularity, invariance and KKT feasibility with the rate floor restored must be checked separately. Appendices A.2.7 and A.2.9 verify the reported analytical illustrations, and their checks supply no global existence argument.

A.2.3 Participation across maturities

Theorem A.1 proves the sorting described in Section 4.3. Without bond-specific risk, the two investors sort by maturity in equilibrium, with the lower-tax investor's municipal holdings at the long end. As loadings rise with maturity, that investor's duration benefit grows, while the difference in exemption benefits it must overcome is flat.

The participation comparison. At an equilibrium allocation, the household's and the lower-tax investor's municipal conditions are

$$\underbrace{\mu^M_T - \xi_H r}_{\text{offered to the household}} \leq \underbrace{\gamma_H \sigma_r^2 A^M_{T-1} \widehat{D}_H}_{\text{household's marginal risk cost}}, \tag{57}$$

$$\underbrace{\mu^M_T - \xi_I r}_{\text{offered to the lower-tax investor}} \leq \underbrace{\gamma_I \sigma_r^2 A^M_{T-1} \widehat{D}_I}_{\text{lower-tax investor's marginal risk cost}}, \tag{58}$$

each with equality for a positive holding. Both face the same offered return μ^M_T, so the lower-tax investor's required return is no higher than the household's exactly when

$$\underbrace{\sigma_r^2\, A^M_{T-1}\, \widehat{\Lambda}}_{\text{duration benefit (grows with maturity)}} \geq \underbrace{(\xi_I - \xi_H)\, r}_{\text{difference in exemption benefits (flat in maturity)}}, \quad \widehat{\Lambda} := \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I. \tag{59}$$

This is the comparison (13) of Section 4.3. The duration benefit, the lower-tax investor's saving in rate-risk cost on the bond, is the bond's predecessor loading times $\sigma_r^2\widehat{\Lambda}$, the difference between the two investors' unit costs of rate risk at the equilibrium allocation. Net rate exposures are common to all of an investor's bonds, so $\widehat{\Lambda}$ is the same at every maturity, while the loading rises with maturity in the illustration. A positive $\widehat{\Lambda}$ means that the household's risk-weighted net rate exposure exceeds the lower-tax investor's, as when the lower-tax investor's outside liabilities hedge more of its bond risk. In the main-text illustration, $\widehat{\Lambda} \approx 4.50$ at the mean rate state.

Theorem A.1 (Sorting without bond-specific risk)**.** *At any state of a two-investor affine solution without bond-specific risk, define*

$$\Psi_T^0 = (\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda}. \tag{60}$$

The household alone holds municipal maturity T if $\Psi_T^0 > 0$, the lower-tax investor alone holds it if $\Psi_T^0 < 0$, and both hold it only if $\Psi_T^0 = 0$. Suppose $r > 0$, $\widehat{\Lambda} > 0$ and the predecessor loadings A_{T-1}^M increase strictly with maturity, and put

$$T^*(z) \;=\; \min\Big\{T \geq 1 :\; A_{T-1}^M \geq \frac{(\xi_I - \xi_H)\, r}{\sigma_r^2\, \widehat{\Lambda}}\Big\}, \tag{61}$$

with $\min \varnothing = +\infty$. Then the household alone holds every municipal maturity below T^*, the lower-tax investor alone holds every maturity above T^*, and both can hold T^* only if $\Psi_{T^*}^0 = 0$. In particular, the household's municipal holdings form a lower interval of maturities, the lower-tax investor's form an upper interval, and at most one maturity is held by both.

Proof. *Step 1: holding cases.* Each investor's municipal condition is $\mu_T^M - \xi_i r \leq \gamma_i \sigma_r^2 A_{T-1}^M \widehat{D}_i$, with equality for a positive holding. If the lower-tax investor holds maturity T, its equality and the household's inequality give $\Psi_T^0 \leq 0$. If the household holds it, its equality and the lower-tax investor's inequality give $\Psi_T^0 \geq 0$. Positive supply means that someone holds every maturity. Hence $\Psi_T^0 > 0$ excludes the lower-tax investor, $\Psi_T^0 < 0$ excludes the household, and joint holdings require $\Psi_T^0 = 0$.

Step 2: monotonicity. With $\widehat{\Lambda} > 0$ and strictly increasing predecessor loadings, Ψ_T^0 decreases strictly with maturity. By (61), it is positive below T^*, negative above T^* and zero at most at T^*. Since $A_0^M = 0$ and $r > 0$, $\Psi_1^0 > 0$, so $T^* \geq 2$ and the household holds the one-period claim. □

What sets the threshold. Taxes, balance sheets and loadings set the threshold T^*, and no investor is assigned a maturity habitat. At given loadings and $\widehat{\Lambda}$, a larger difference in exemption benefits weakly raises the threshold, and a larger $\widehat{\Lambda}$ weakly lowers it. In equilibrium, $\widehat{\Lambda}$ and the loadings are determined jointly with holdings, so Theorem A.1 characterizes participation within a given equilibrium.

The shared maturity. At a regular state (Appendix A.1.3), strict complementarity makes $\Psi_{T^*}^0 = 0$ equivalent to joint holdings, so the threshold T^* is the first maturity the lower-tax investor holds. Without bond-specific risk, a maturity is shared only when the two investors' difference in rate-risk compensation exactly offsets their difference in exemption benefits, $\Psi_T^0 = 0$. At that maturity, holdings are not fixed by clearing alone: the split between the investors adjusts their net rate exposures until the condition holds, and it absorbs the trading as rates move (Proposition 2). In the main-text illustration, this maturity is 23, and the lower-tax investor holds about 47% of it (Appendix A.2.7). Appendix A.2.8 shows how bond-specific risk extends joint holdings to a block of maturities.

A.2.4 Return-gap decomposition and rate loadings

This part proves the results behind Sections 4.4 and 4.5: the split of the one-period return gap into the exemption benefit, the duration cost and the rate loading, and the tax-mix and portfolio responses that determine the loadings in the rate-loading channel.

The three channels of the one-period return gap. Subtracting ξ_H times the taxable equality from a municipal holder's equality separates the one-period return gap $g_T = \mu_T^M - \xi_H \mu_T^C$ into three channels. The restatement of Proposition 1 below states the conditions for (15) and the exemption-benefit and duration-cost contributions in each holding case.

Proposition 1 (Three channels of the one-period return gap)**.** *In the two-investor affine benchmark without bond-specific risk, consider a state at which the lower-tax investor alone holds the risky taxable bonds. Let i denote a holder of the maturity-T municipal bond, with $i = I$ whenever the lower-tax investor holds it. Then*

$$g_T = \underbrace{(\xi_i - \xi_H)r}_{\text{exemption benefit}} + \underbrace{\sigma_r^2 A_{T-1}^M(\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I)}_{\text{duration cost}} + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I(A_{T-1}^M - \xi_H A_{T-1}^C)}_{\text{rate loading}}. \tag{62}$$

At $T = 1$ the duration-cost and rate-loading contributions are zero, and so is the exemption-benefit contribution when $r > 0$. Where the household holds the bond alone, the exemption-benefit contribution is zero and the duration-cost contribution is at most $(\xi_I - \xi_H)r$. Where the lower-tax investor holds it, the exemption-benefit contribution is $(\xi_I - \xi_H)r$ and the duration-cost contribution is zero.

Proof. *Step 1: the identity.* Investor i's active municipal equality gives $\mu_T^M = \xi_i r + \gamma_i \sigma_r^2 A_{T-1}^M \widehat{D}_i$. The lower-tax investor's taxable equality, divided by its after-tax share, gives $\mu_T^C = r + \gamma_I \sigma_r^2 A_{T-1}^C \widehat{D}_I$, which also covers cash at $T = 1$ because $A_0^C = 0$. Subtracting ξ_H times the second equality from the first, then adding and subtracting $\gamma_I \sigma_r^2 A_{T-1}^M \widehat{D}_I$, gives (62).

Step 2: the one-period claim. At $T = 1$, $A_0^a = 0$, so the duration-cost and rate-loading terms vanish. When $r > 0$, the household alone holds the claim (Proposition A.2), so $i = H$ and the exemption-benefit term also vanishes.

Step 3: holding cases. If the household alone holds maturity T, then $i = H$, and Theorem A.1 gives $\Psi_T^0 \geq 0$, that is, $\sigma_r^2 A_{T-1}^M \widehat{\Lambda} \leq (\xi_I - \xi_H)r$. The duration-cost contribution is $\sigma_r^2 A_{T-1}^M \widehat{\Lambda}$ and is therefore at most $(\xi_I - \xi_H)r$. If the lower-tax investor holds the bond, the identity with $i = I$ has a zero duration-cost contribution and an exemption-benefit contribution of $(\xi_I - \xi_H)r$. □

At the shared maturity, $\Psi_T^0 = 0$, so either investor's equality gives the same total gap. Figure 5 plots this decomposition for the main-text illustration. At $T = 30$, the rate-loading term is 54% of the return gap, and the exemption-benefit term is the remainder. Figure 14 gives the counterpart with bond-specific risk, with the same channel names.

How the rate loading links maturities. Iterating the loading recursion (53) from $A_0^a = 0$ gives

$$A_{T-1}^M - \xi_H A_{T-1}^C = \sum_{j=1}^{T-1} \rho^{T-1-j}\left(s_j^M - \xi_H s_j^C\right),$$

so the predecessor-loading gap accumulates the response gaps $s_j^M - \xi_H s_j^C$ of the shorter maturities the bond passes through. This gap is zero at $T = 2$, since the one-period responses are $s_1^M = \xi_H$ and $s_1^C = 1$ when $r > 0$. The tax mix and portfolio responses derived below determine this gap in the rate-loading channel.

Tax mix and portfolio responses. One-period returns respond to the rate state for two reasons. Each holder's after-tax return on cash rises by its after-tax share per unit of the rate state. Trading also changes how much net rate exposure each holder carries, and with it the compensation each requires. Aggregating the holders' equalities separates the two. Multiplying investor i's active municipal equality by m_i/γ_i, summing over the active holders and differentiating with respect to z gives

$$\left(\sum_{i \in \mathcal{U}_T^M} \frac{m_i}{\gamma_i}\right) s_T^M - \sum_{i \in \mathcal{U}_T^M} \frac{m_i \xi_i}{\gamma_i} = \sigma_r^2 A_{T-1}^M \sum_{i \in \mathcal{U}_T^M} m_i \frac{\partial \widehat{D}_i(z)}{\partial z}, \tag{63}$$

where the active set $\mathcal{U}_T^M \subseteq \{H, I\}$ collects the investors who hold the maturity-T municipal bond. With bond-specific risk, the same aggregation removes the bond-specific terms, because fixed supply implies $\sum_i m_i \, \partial x_{i,T}^M(z)/\partial z = 0$. Define the active holders' risk-tolerance shares and aggregate exposure response as

$$\omega_{i,T}^M := \frac{m_i/\gamma_i}{\sum_{j \in \mathcal{U}_T^M} m_j/\gamma_j}, \qquad E_T^M := \frac{\sum_{i \in \mathcal{U}_T^M} m_i \, \partial \widehat{D}_i(z)/\partial z}{\sum_{i \in \mathcal{U}_T^M} m_i/\gamma_i}.$$

The weight $\omega_{i,T}^M$ is investor i's share of the active holders' risk tolerance, and E_T^M is the response of their aggregate net rate exposure to the rate state, divided by that total risk tolerance. The taxable counterpart E_T^C is defined in the same way from the holders of the taxable bond. Dividing (63) by total risk tolerance gives Proposition A.3, the general form of the response formulas (18) in Section 4.5.

Proposition A.3 (Holding-return response to the rate state)**.** *Consider a regular state of the two-investor affine benchmark: price moments are differentiable, active risky holdings and inactive KKT slacks are strictly positive, and the active KKT-and-clearing Jacobian is nonsingular. The active set $\mathcal{U}_T^M \subseteq \{H, I\}$ is therefore locally fixed. Then*

$$s_T^M = \underbrace{\sum_{i \in \mathcal{U}_T^M} \omega_{i,T}^M \xi_i}_{\text{holders' tax mix}} + \underbrace{\sigma_r^2 A_{T-1}^M E_T^M}_{\text{portfolio-risk response}} \tag{64}$$

exactly. The tax-mix term depends on who holds maturity T. The portfolio-risk response depends on how equilibrium trading changes those holders' exposure across their whole portfolios.

Proof of Proposition A.3. Divide (63) by the active holders' total risk tolerance. When bond-specific risk is present, fixed supply cancels its derivative by (48). The definitions of $\omega_{i,T}^M$ and E_T^M then give (64). □

Exact aggregation with nonlinear prices. Proposition A.4 extends the same aggregation to nonlinear prices by collecting the active holders' tax treatment and covariance-based risk compensation before taking the state derivative. The notation Σ, K_i, c_i^L and ξ_{ij} is that of (46)–(47). Appendix A.4 applies the proposition market by market in the multi-state model.

Proposition A.4 (Required returns aggregated across a bond's holders)**.** *At a regular state with slack shorting constraints on cash for the active holders, let $\Gamma_j = \sum_{i \in \mathcal{U}_j} m_i/\gamma_i$ be their total risk tolerance, $\bar{\xi}_j = \Gamma_j^{-1} \sum_{i \in \mathcal{U}_j} m_i \xi_{ij}/\gamma_i$ and $\Omega_j = \Gamma_j^{-1} \sum_{i \in \mathcal{U}_j} m_i [\Sigma K_i x_i - c_i^L]_j$. Here $\bar{\xi}_j$ is their tax mix and Ω_j their average risk compensation, weighted by risk tolerance. Then*

$$\begin{aligned} \mu_j &= \bar{\xi}_j r + \Omega_j, \\ s_j &= \bar{\xi}_j \, \partial_z r + \frac{1}{\Gamma_j} \sum_{i \in \mathcal{U}_j} m_i \left[(\partial_z \Sigma) K_i x_i + \Sigma K_i \partial_z x_i - \partial_z c_i^L \right]_j . \end{aligned} \tag{65}$$

In the normalized municipal–taxable benchmark, with $\bar{\xi}_T^a = \bar{\xi}_j$ and $\Omega_T^a = \Omega_j$ at $j = (a, T)$, above the rate floor the response difference is

$$s_T^M - \xi_H s_T^C = (\bar{\xi}_T^M - \xi_H) + \partial_z \Omega_T^M - \xi_H \partial_z \Omega_T^C. \tag{66}$$

Proof. Weighting the active equalities $\mu_j - \xi_{ij} r = \gamma_i [\Sigma K_i x_i - c_i^L]_j$ by m_i/γ_i, summing and dividing by Γ_j gives the first line. Differentiating on the locally fixed participation branch gives the second line. In the two-market normalization, $\bar{\xi}_j = \bar{\xi}_T^M$ at municipal and $\bar{\xi}_j = 1$ at taxable coordinates. Subtraction gives (66). □

When the shorting constraint on cash binds, each active equality also contains κ_i/ξ_i^a by (37). The return identity in (65) then gains $\Gamma_j^{-1} \sum_{i \in \mathcal{U}_j} m_i \kappa_i / (\gamma_i \xi_i^a)$, and the response identity gains its state derivative. A per-dollar charge, such as the bank financing-tax cost, appears in the return identity in the same units. When it does not vary with the rate state, it leaves the response identity unchanged on a given participation branch.

Responses in the affine benchmark. Substituting the affine covariance and liability credits into the active KKT equalities gives each bond's response

$$s_j = \bar{\xi}_j + \frac{\sigma_r^2 A_j}{\Gamma_j} \sum_{i \in \mathcal{U}_j} m_i \frac{\partial \widehat{D}_i}{\partial z} + \frac{v_j}{\Gamma_j} \sum_{i \in \mathcal{U}_j} m_i \xi_i^a \frac{\partial x_{ij}}{\partial z}, \tag{67}$$

where $A_j = A_{T-1}^a$ is the predecessor loading, the bond's loading one period later. The final term is zero without bond-specific risk. For municipal bonds, whose after-tax shares are normalized to one, (48) also cancels it when it is present. The tax-mix term is ξ_H under household-only holdings, ξ_I under lower-tax-only holdings and, under joint holdings, the tax mix

$$\xi^* := \omega_H \xi_H + \omega_I \xi_I, \qquad \omega_i := \frac{m_i/\gamma_i}{m_H/\gamma_H + m_I/\gamma_I},$$

where ω_i is investor i's share of total risk tolerance. The lower-tax investor's after-tax share therefore appears in this term only at maturities it holds.

Taxable counterpart. Proposition 2 also uses the taxable version of (67). With $\xi_{ij} = 1$ at taxable coordinates, its tax term is one. Its bond-specific term vanishes in three cases: if $v_T^C = 0$, if one investor holds the taxable bond, because clearing then fixes its holding, or if all holders have equal after-tax shares, because these then factor out of (48). Otherwise, unweighted clearing does not cancel the term. With the lower-tax investor the sole holder of the taxable bond,

$$s_T^C = 1 + \gamma_I \sigma_r^2 A_{T-1}^C \partial_z \widehat{D}_I. \tag{68}$$

Signing the responses. Proposition 2 signs these responses on the sorting branch of Theorem A.1 with a shared maturity, as in the main-text illustration. A higher rate state increases the difference in exemption benefits. At the shared maturity, the lower-tax investor sells municipal bonds to the household, whose required compensation therefore rises with rates, even on shorter bonds it holds alone. The lower-tax investor sheds net exposure, which lowers the response of the returns it requires on taxable and long municipal bonds. At the shared maturity, clearing cancels the holders' aggregate exposure response and leaves the tax mix ξ^*. The proposition supplies the responses and signs in (18) and the positive loading gap behind (76). Figure 6(a) plots this case.

Proposition 2 (Participation and rate responses)**.** *At a regular state with $r(z) > 0$ in the two-investor affine benchmark without bond-specific risk, suppose the lower-tax investor alone holds risky taxable bonds, the household alone holds the municipal maturities below $k \geq 2$, both investors hold maturity $k = T^*$, and the lower-tax investor alone holds the municipal maturities above k. Assume $A_{T-1}^M, A_{T-1}^C > 0$ for every $T \geq 2$, and put $d := \partial \widehat{D}_H / \partial z$. Then*

$$d = \frac{(\xi_I - \xi_H) m_I}{(\gamma_H m_I + \gamma_I m_H)\sigma_r^2 A_{k-1}^M} > 0, \qquad \frac{\partial \widehat{D}_I}{\partial z} = -\frac{m_H}{m_I} d. \tag{69}$$

With $\xi^* = \omega_H \xi_H + \omega_I \xi_I$ and the feedback $b := \gamma_I \sigma_r^2 (m_H/m_I) d = \omega_H (\xi_I - \xi_H)/A_{k-1}^M$, the responses are

$$s_T^M = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M d, & T < k, \\ \xi^*, & T = k, \\ \xi_I - b A_{T-1}^M, & T > k, \end{cases} \qquad s_T^C = 1 - b A_{T-1}^C. \tag{70}$$

The exposure response d is the unique positive solution of $d\, A_{k-1}^M(d) = (\xi_I - \xi_H) m_I / [(\gamma_H m_I + \gamma_I m_H)\sigma_r^2]$, where $A_{k-1}^M(d) = \xi_H \sum_{q=0}^{k-2} (\rho + \gamma_H \sigma_r^2 d)^q$ is the loading generated at household-held maturities. The response gap $s_T^M - \xi_H s_T^C$ is positive at every risky maturity $T \leq k$. If $b \leq \rho$, then $A_T^M - \xi_H A_T^C > 0$ at every risky maturity $T \geq 2$, with equality at $T = 1$.

Proof. *Step 1: exposure responses.* Clearing fixes each sole holder's position: Q_T^M / m_H for the household below k, Q_T^M / m_I for the lower-tax investor above k and Q_T^C / m_I for its risky taxable bonds. Only the shared position at k varies, so $\partial_z \widehat{D}_i = A_{k-1}^M \partial_z x_{i,k}^M$, and differentiating clearing at k gives

$$m_H \partial_z \widehat{D}_H + m_I \partial_z \widehat{D}_I = 0. \tag{71}$$

Both investors' municipal equalities hold at k. Subtracting the lower-tax investor's differentiated equality from the household's gives $\xi_I - \xi_H = \sigma_r^2 A_{k-1}^M (\gamma_H \partial_z \widehat{D}_H - \gamma_I \partial_z \widehat{D}_I)$. Substituting (71) gives (69). The identity for b follows from $\gamma_I m_H / (\gamma_H m_I + \gamma_I m_H) = \omega_H$.

Step 2: return responses. A sole holder's differentiated equality gives $s_T^M = \xi_i + \gamma_i \sigma_r^2 A_{T-1}^M \partial_z \widehat{D}_i$, which is the first line below k and, with (69), the third line above k. At k, weighting both equalities by risk tolerance as in (63) cancels the exposure responses by (71) and leaves ξ^*. The household's own equality at k gives the same value, $\xi_H + \gamma_H \sigma_r^2 A_{k-1}^M d = \xi_H + \omega_I (\xi_I - \xi_H) = \xi^*$. Equation (68) gives the taxable formula.

Step 3: uniqueness. Below k, the municipal recursion is $A_T^M = (\rho + \gamma_H \sigma_r^2 d) A_{T-1}^M + \xi_H$, so $A_{k-1}^M(d) = \xi_H \sum_{q=0}^{k-2} (\rho + \gamma_H \sigma_r^2 d)^q$ is positive and nondecreasing in d. The product $d\, A_{k-1}^M(d)$ therefore rises strictly from zero without bound, and it equals the positive constant in the statement exactly once.

Step 4: response and loading gaps. By (70), the response gap is

$$s_T^M - \xi_H s_T^C = \begin{cases} \sigma_r^2 d \left[\gamma_H A_{T-1}^M + \xi_H \gamma_I \dfrac{m_H}{m_I} A_{T-1}^C \right] > 0, & 2 \leq T < k, \\ \omega_I (\xi_I - \xi_H) + \xi_H b A_{k-1}^C > 0, & T = k, \end{cases}$$

and it is zero at $T=1$. Let $L_T = A_T^M - \xi_H A_T^C$. Then $L_1 = 0$ and $L_T = \rho L_{T-1} + (s_T^M - \xi_H s_T^C) > 0$ for $2 \le T \le k$. Above k, the responses give

$$L_T = (\rho - b)L_{T-1} + (\xi_I - \xi_H), \qquad T > k,$$

which is positive by induction when $b \le \rho$. □

If no maturity satisfies $\Psi_T^0 = 0$, clearing fixes every holding, both exposure responses are zero, and the responses are ξ_H below the threshold and ξ_I from it on. The loading gap is then zero below the threshold and positive from it on.

Condition (S). The positive loading gap in Section 4.5 and the pricing results below use one restriction on taxes, risk-tolerance shares and rate persistence:

$$\text{(S)} \qquad \omega_H \frac{\xi_I - \xi_H}{\xi_H} \le \rho, \tag{72}$$

written in masses and risk aversions as

$$\text{(S)} \qquad \frac{\gamma_I m_H (\xi_I - \xi_H)}{\xi_H(\gamma_H m_I + \gamma_I m_H)} \le \rho. \tag{73}$$

On the sorting branch, the loading at the shared maturity's predecessor is at least $A_1^M = \xi_H$, so (S) gives $b = \omega_H(\xi_I - \xi_H)/A_{k-1}^M \le \rho$. The feedback from the lower-tax investor's falling exposure into its required returns is then too weak to reverse the rise of taxable loadings with maturity or the sign of the loading gap. In the main-text illustration, the left side of (S) is about 0.209, against $\rho = 0.8$.

From responses to the rate-loading channel. The response gap $s_T^M - \xi_H s_T^C = \partial_z g_T$ accumulates into the loading difference of the rate-loading channel in (62). Applying Proposition A.3 in both markets, with the lower-tax investor the sole holder of risky taxable bonds, splits it into tax-mix and portfolio parts:

$$s_T^M - \xi_H s_T^C = \underbrace{\left(\bar{\xi}_T^M - \xi_H\right)}_{\text{difference in tax mix}} + \underbrace{\sigma_r^2\left(A_{T-1}^M E_T^M - \xi_H A_{T-1}^C E_T^C\right)}_{\text{difference in portfolio-risk responses}}, \tag{74}$$

where $\bar{\xi}_T^M := \sum_i \omega_{i,T}^M \xi_i$ is the active holders' tax mix (ξ_H, ξ^* or ξ_I). With one municipal holder, these are the two terms of (17). Below the shared maturity, trading raises the household's municipal response and lowers the taxable response. At the shared maturity, the holders' exposure responses cancel in aggregate and the tax mix adds $\xi^* - \xi_H$. Above it, the tax mix adds $\xi_I - \xi_H$, and the lower-tax investor's falling exposure lowers the municipal and taxable responses by bA_{T-1}^M and bA_{T-1}^C, which changes the response gap by $-b(A_{T-1}^M - \xi_H A_{T-1}^C)$. All contributions vanish at the riskless one-period claim, and (66) gives the exact counterpart with nonlinear covariances. Figure 6 combines the municipal tax and portfolio terms and shows the taxable contribution separately.

The rate-loading channel at maturity T uses the predecessor loadings, which accumulate only the response gaps below T. This channel of Proposition 1 therefore incorporates portfolio-risk responses from maturity three on and the change in tax mix from maturity $k+1$ on.

Comparison with a treasury reference. The empirical treasury comparison in Section 6.3 divides municipal loadings by $G_T := \sum_{j=1}^T \rho^{T-j} = A_T^{\text{Tsy}}$, the loading of a reference whose one-period return responds one-for-one to the rate state. Since $A_T^M/G_T = \sum_{j=1}^T (\rho^{T-j}/G_T) s_j^M$, the relative yield sensitivity averages municipal responses with positive weights and equals ξ_H at the allocation of Proposition A.5. Its difference from ξ_H reflects municipal holders' tax treatment and portfolio responses, both of which depend on participation. Figure 6(b) uses the equilibrium taxable market C, whose own portfolio response also contributes to $A_T^M - \xi_H A_T^C$.

State-independent holding benefits. Suppose an investor's objective includes $\sum_{a,T} \vartheta_{i,T}^a x_{i,T}^a$, with each $\vartheta_{i,T}^a$ independent of the rate state, for example a trading-cost advantage (Section 4.7) or the calibration's fixed bank financing-tax cost. Its active municipal condition becomes

$$\mu_T^M - \xi_i r + \vartheta_{i,T}^M = \gamma_i \left[\sigma_r^2 A_{T-1}^M \widehat{D}_i + v_T^M x_{i,T}^M\right],$$

and the taxable condition similarly gains $\vartheta_{i,T}^C/\xi_i$. With covariance, financing and constraints fixed, differentiation removes state-independent benefits and costs. The response formulas and loading equations therefore hold on the same participation branch, so a given loading solution is unchanged. Such benefits and costs still change

which maturities an investor holds and the levels of one-period returns: they affect return intercepts, gap levels (Appendix A.3) and the feasible support. State-dependent benefits add their derivatives. Fixed participation fees affect global participation without a per-unit KKT term.

Household-only loadings. The last result covers the allocation in which the household holds every municipal bond. In that allocation, the loading gap of Section 4.5 is zero and the municipal-to-taxable loading ratio of P3 equals ξ_H at every maturity.

Proposition A.5 (Household-only loadings)**.** *In the two-investor affine benchmark, if the high-tax household alone holds every municipal bond and the lower-tax investor alone holds every taxable bond, then* $s_T^M = \xi_H$, $s_T^C = 1$ *and* $A_T^M = \xi_H A_T^C$ *at every maturity.*

Proof. Clearing fixes every holding, so both net rate exposures have zero derivatives. Equations (67) and (56) give the result. The identity concerns loadings, and different intercept risk premia can still affect the yield ratio. □

A.2.5 Yield gaps and implied taxes

The three channels act on one-period returns, while the municipal puzzle concerns yields to maturity. This subsection carries the channels into yields, formalizing (19) of Section 4.5. Proposition A.6 then orders the municipal-to-taxable loading ratio and the yield-implied tax by maturity and gives the long-end implied tax. These results underlie the long-end pass-throughs and prediction P3 of the main text.

From one-period gaps to yield gaps. Applying (54) to both markets turns the one-period gap g_T of Proposition 1 into the yield gap:

$$y_T^M - \xi_H y_T^C = \underbrace{\frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t\big[g_{T-j}(z_{t+j})\big]}_{\text{average gap along the bond's life}} \underbrace{- \frac{1}{2T}\sum_{q=2}^{T}\Big[\sigma_r^2\Big(\big(A_{q-1}^M\big)^2 - \xi_H\big(A_{q-1}^C\big)^2\Big) + \big(\sigma_{\eta,q}^M\big)^2 - \xi_H\big(\sigma_{\eta,q}^C\big)^2\Big]}_{\text{convexity difference}}. \tag{75}$$

The last term of the average, the riskless one-period gap, is zero. The yield-implied tax is

$$\tau_T^{Y,\text{impl}} := 1 - \frac{y_T^M}{y_T^C}, \qquad \text{the model counterpart of } \textit{ImpliedTax} \text{ in (2).}$$

It equals the household wedge less the yield gap per unit of taxable yield, $\tau_T^{Y,\text{impl}} = (1-\xi_H) - (y_T^M - \xi_H y_T^C)/y_T^C$, as in the identity (3) of Section 3.6. A positive gap accumulated in (75) therefore pulls the implied tax below the household wedge, more so at longer maturities when the gap grows proportionally faster than the taxable yield (Proposition A.6(ii)).

The relative yield response. The relative yield response is the loading gap divided by maturity, and the loading gap accumulates the response gaps in (74). Under condition (S), the loading gap is positive at every risky maturity (Proposition 2). It reflects portfolio responses below the shared maturity, and both the tax mix and portfolio responses from that maturity on. Hence, for $T \geq 2$,

$$\underbrace{\frac{\partial y_T^M}{\partial z} - \xi_H \frac{\partial y_T^C}{\partial z}}_{\text{relative yield response}} = \frac{1}{T}\sum_{j=2}^{T} \rho^{T-j} \underbrace{\left(\frac{\partial \mu_j^M}{\partial z} - \xi_H \frac{\partial \mu_j^C}{\partial z}\right)}_{\text{holding-return response gap}} > 0, \tag{76}$$

in model-period units, and both sides are zero for the riskless one-period bond. Long municipal yields therefore move more with the short rate than the household benchmark implies. Through the household's exposure response, the lower-tax investor's tax affects these responses even at maturities it does not hold.

Forward rates. Parts (ii) and (iii) of Proposition A.6 use forward rates. Define the forward rate implied by today's prices for the period from $t+T-1$ to $t+T$:

$$f_T^a(z) = \log \frac{P_{T-1}^a(z)}{P_T^a(z)}, \qquad y_T^a(z) = \frac{1}{T}\sum_{j=1}^{T} f_j^a(z). \tag{77}$$

With positive taxable forwards, the yield ratio is therefore a weighted average of municipal-to-taxable forward-rate ratios.

Proposition A.6 (Maturity ordering of the loading ratio and the yield-implied tax)**.** *In the two-investor affine benchmark without bond-specific risk, consider the sorting branch of Proposition 2. The high-tax household alone holds municipal bonds shorter than k, both investors hold maturity k, and the lower-tax investor alone holds longer municipal bonds and all risky taxable bonds. Assume condition (S) in (72).*

(i) Rate sensitivity. *The municipal-to-taxable loading ratio is nondecreasing in maturity. It starts at the household's after-tax share. When the recursions are extended to infinite maturity with the household's exposure response d held at its solved value, the ratio converges to the lower-tax investor's after-tax share:*

$$\frac{A_1^{M,\mathrm{aff}}}{A_1^{C,\mathrm{aff}}} = \xi_H \qquad \textit{and} \qquad \lim_{T\to\infty} \frac{A_T^{M,\mathrm{aff}}}{A_T^{C,\mathrm{aff}}} = \xi_I. \tag{78}$$

(ii) Yield levels. *Suppose also that taxable forward rates are positive and the municipal-to-taxable forward-rate ratio is nondecreasing in maturity at the stationary mean $z=0$. Then the yield ratio y_T^M/y_T^C is nondecreasing at $z=0$ (Proposition A.7), so the yield-implied tax $\tau_T^{Y,\mathrm{impl,aff}}$ is nonincreasing.*

(iii) Long end. *With the recursions extended to infinite maturity and the lower-tax investor's risk-premium coefficient $\pi_I = \gamma_I \sigma_r^2 \widehat{D}_I(0)$ held at its solved value, assume the limiting taxable forward is positive. Then*

$$\lim_{T\to\infty} \tau_T^{Y,\mathrm{impl,aff}} = 1 - \frac{\xi_I \bar{r} + A_\infty^M \pi_I - \frac{1}{2}\sigma_r^2 (A_\infty^M)^2}{\bar{r} + A_\infty^C \pi_I - \frac{1}{2}\sigma_r^2 (A_\infty^C)^2} = (1-\xi_I) - \frac{\frac{1}{2}\sigma_r^2 \xi_I (1-\xi_I)(A_\infty^C)^2}{y_\infty^C}, \tag{79}$$

where $A_\infty^C = 1/(1-\rho+b)$, $A_\infty^M = \xi_I A_\infty^C$, y_∞^C is the denominator of the first fraction, the limiting taxable yield, and b is the feedback in (70).

Proof of Proposition A.6(i). Step 1 (recursions). Put $\alpha = \rho - b$ and $\beta = \rho + \gamma_H \sigma_r^2 d$. Condition (S) and $b > 0$ give $0 \le \alpha < \rho$. By (70), the loadings satisfy

$$A_T^C = \alpha A_{T-1}^C + 1, \qquad A_T^M = \begin{cases} \beta A_{T-1}^M + \xi_H, & T \le k, \\ \alpha A_{T-1}^M + \xi_I, & T > k. \end{cases}$$

The first municipal recursion holds at k as well, because the household's response there equals ξ^* (Proposition 2, Step 2).

Step 2 (up to the shared maturity). For $T \le k$, $A_T^M/\xi_H = \sum_{j=0}^{T-1} \beta^j$ and $A_T^C = \sum_{j=0}^{T-1} \alpha^j$. Since $\beta > \alpha \ge 0$, $\beta^T \alpha^j \ge \alpha^T \beta^j$ for $j < T$, so adding the next terms cannot lower the ratio. It equals ξ_H at $T = 1$.

Step 3 (from the shared maturity on). Put $F_T = A_T^M - \xi_I A_T^C$. By (69), $\beta - \alpha = (\xi_I - \xi_H)/A_{k-1}^M$, so for $T \le k$

$$F_T = \alpha F_{T-1} + (\xi_I - \xi_H)\left(\frac{A_{T-1}^M}{A_{k-1}^M} - 1\right) \le \alpha F_{T-1},$$

because household-only loadings rise with maturity. Since $F_1 = \xi_H - \xi_I < 0$, $F_T \le 0$ through k. For $T > k$, $F_T = \alpha F_{T-1}$. The ratio is $A_T^M/A_T^C = \xi_I + F_T/A_T^C$. From k on, $|F_T|$ is nonincreasing and A_T^C is nondecreasing, so the ratio is nondecreasing.

Step 4 (limit). As $T \to \infty$, $A_T^C \to 1/(1-\alpha) = 1/(1-\rho+b) > 0$ and $F_T \to 0$, so the ratio converges to ξ_I. When $b < \rho$, A_T^C rises strictly, and so does A_T^M, the product of A_T^C and a positive nondecreasing ratio. The sorting branch therefore has the rising loadings that Theorem A.1 uses. □

Part (ii): the forward-ratio bridge. Part (ii) rests on the following bridge.

Proposition A.7 (Exact forward-ratio bridge)**.** *For any price functions, affine or nonlinear, let $f_T^a(z) = -p_T^a(z) + p_{T-1}^a(z)$. If, over the maturities considered, $f_T^C(z) > 0$ and $f_T^M(z)/f_T^C(z)$ is nondecreasing, then $y_T^M(z)/y_T^C(z)$ is nondecreasing and $\tau_T^{Y,\mathrm{impl}}(z)$ is nonincreasing.*

Proof. Telescoping gives $Ty_T^a = -p_T^a = \sum_{j \le T} f_j^a$. The yield ratio averages forward ratios with positive weights $f_j^C > 0$. Adding a ratio at least as large as its predecessors cannot lower the average. On a two-year grid, T counts steps and annualization cancels. □

The forward rate compares adjacent-maturity prices at the same date, whereas the holding return uses next period's resale price. The affine recursions (53) link the two:

$$f_T^a(z) = \mu_T^a(z) - (1-\rho)A_{T-1}^a z - \tfrac{1}{2}\left[\sigma_r^2 (A_{T-1}^a)^2 + (\sigma_{\eta,T}^a)^2\right]. \tag{80}$$

The premise of the bridge concerns full forward rates, including convexity terms, so monotone loadings alone do not establish it. Section 5 checks the nonlinear model's maturity profiles numerically.

Part (iii): long-end forward levels. Part (iii) concerns long-end yield levels, which add term premia and convexity to the long-end holders' tax treatment. For taxable bonds, the term premium is the yield less the average expected future short rate. Equation (54) expresses it as average expected bond risk premia less convexity. For coordinate $j = (a, T)$, put $\Gamma_T^a := \sum_{i \in \mathcal{U}_j} m_i/\gamma_i$ and

$$\bar{\xi}_T^a = \bar{\xi}_j = \frac{\sum_i (m_i/\gamma_i)\xi_{ij}}{\Gamma_T^a}, \quad \pi_T^a = \frac{\sigma_r^2}{\Gamma_T^a}\sum_i m_i \widehat{D}_i(0), \quad \iota_T^a = v_j\left[\frac{\sum_i m_i \xi_i^a x_{i,T}^a(0)}{\Gamma_T^a} - \tfrac{1}{2}\right],$$

with sums over active holders and ξ_i^a the after-tax share in normalized units, which is one for municipal bonds. Here π_T^a is the holders' cost of rate risk per unit of loading, weighted by risk tolerance, and ι_T^a is their compensation for bond-specific risk net of its convexity term. Aggregating the holders' level KKT equalities at $z = 0$ and subtracting the two Jensen terms in the exact Gaussian mean gives

$$C_T^a - C_{T-1}^a = \bar{\xi}_T^a \bar{r} + A_{T-1}^a \pi_T^a - \tfrac{1}{2}\sigma_r^2 (A_{T-1}^a)^2 + \iota_T^a. \tag{81}$$

For municipal bonds the weighted holding sum in ι_T^a is Q_T^M, while for taxable bonds heterogeneous after-tax shares remain inside it. The increment $C_T^a - C_{T-1}^a$ is the forward rate at $z = 0$. If the displayed coefficients and loadings converge and the limiting taxable forward f_∞^C is positive, Cesàro convergence gives

$$y_T^a(z) \longrightarrow f_\infty^a = \bar{\xi}_\infty^a \bar{r} + A_\infty^a \pi_\infty^a - \tfrac{1}{2}\sigma_r^2 (A_\infty^a)^2 + \iota_\infty^a \tag{82}$$

at fixed z, because C_T^a/T converges to the forward limit and $A_T^a z/T \to 0$. On the sorting branch without bond-specific risk, the lower-tax investor alone holds both long bonds, so $\bar{\xi}_\infty^M = \xi_I$, $\bar{\xi}_\infty^C = 1$, $\pi_\infty^M = \pi_\infty^C = \pi_I$ and $\iota_\infty^a = 0$. Steps 1 and 4 of part (i) give $A_\infty^C = 1/(1-\rho+b)$ and $A_\infty^M = \xi_I A_\infty^C$. The implied-tax limit $1 - f_\infty^M/f_\infty^C$ is therefore the first form in (79). Since $A_\infty^M = \xi_I A_\infty^C$, the interest and rate-risk terms of the municipal yield are ξ_I times those of the taxable yield, and

$$y_\infty^M - \xi_I y_\infty^C = \tfrac{1}{2}\sigma_r^2 \xi_I (1-\xi_I)(A_\infty^C)^2,$$

which gives the second form. The long-end implied tax is thus the lower-tax investor's wedge, $1 - \xi_I$, less a relative convexity term, which proves part (iii).

A.2.6 Tax pass-through and relative rate sensitivity

The mechanism predicts whose taxes are priced at each maturity. This subsection proves the pass-through contrasts and the rate-sensitivity ordering behind predictions P1–P3 of Section 4.6 on the sorting branch. Appendix A.2.8 gives their counterparts when bond-specific risk extends joint holdings across long maturities.

Two implied taxes and their pass-throughs. P1 and P2 concern yield levels, while P3 concerns rate sensitivities. To keep these objects distinct, I pair the yield-implied tax of Appendix A.2.5 with a loading-implied tax:

$$\tau_T^{Y,\text{impl}} = 1 - \frac{y_T^M}{y_T^C}, \qquad \tau_T^A = 1 - \frac{A_T^M}{A_T^C}.$$

To measure pass-through, I vary one investor's after-tax share at a fixed rate state and re-solve the equilibrium. Holdings, risk costs and both yield curves adjust while non-tax primitives stay fixed. The pass-throughs of investor i's exemption benefit $1 - \xi_i$ into the two implied taxes are

$$\mathcal{P}_i^Y(T) := \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad \mathcal{P}_i^A(T) := \frac{d\tau_T^A}{d(1-\xi_i)}, \qquad i \in \{H, I\},$$

where an argument ∞ denotes the limit as $T \to \infty$. Equivalently, they are the derivatives of y_T^M/y_T^C and A_T^M/A_T^C with respect to ξ_i.

Under the representative-tax benchmark of Section 3.3, $y_T^M = \xi_H y_T^C$ and $A_T^M = \xi_H A_T^C$, so in both measures household pass-through is one and the lower-tax investor's is zero at every maturity. With heterogeneous taxes, the household alone holds the riskless short bond, whose return is $\xi_H r$. The lower-tax investor holds the long bonds, whose returns depend on its after-tax share directly and on the household's through the exposure response at the shared maturity. For the empirical treasury comparison of Section 6.3, I also use a taxable reference whose one-period return moves one-for-one with the rate state, giving loading $A_T^{\text{Tsy}} = G_T$ (Appendix A.2.4).

Proposition A.8 (Tax pass-through at the short and long ends)**.** *Consider the sorting branch of Proposition 2 without bond-specific risk, under condition (S) of (72), with a positive cash rate and a slack household shorting constraint on cash. With its recursions extended to infinite maturity, suppose long yields, loading ratios and their tax derivatives converge locally uniformly, with a positive long taxable yield.*

(i) (Short end) Both implied taxes equal the household wedge, $\tau_1^{Y,\text{impl}} = \tau_1^A = 1 - \xi_H$, so $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.

(ii) (Long end, P1 and P2) The long-end loading-implied tax is $1 - \xi_I$, so

$$\mathcal{P}_H^A(\infty) = 0, \qquad \mathcal{P}_I^A(\infty) = 1.$$

Yield pass-throughs equal these values plus the tax derivatives of the relative convexity term $\frac{1}{2}\sigma_r^2 \xi_I(1-\xi_I)(A_\infty^C)^2/y_\infty^C$ in (79), which are $O(1/\bar{r})$ along the branch's algebraic continuation, the solution of the branch's equations as $\bar{r}$ varies with participation held fixed. Whenever these derivatives are below one in absolute value,

$$\mathcal{P}_H^Y(\infty) < 1, \qquad \mathcal{P}_I^Y(\infty) > 0.$$

(iii) (Response ratios, P3) The municipal-to-taxable zero-coupon response ratio A_T^M/A_T^C is nondecreasing in maturity, equals ξ_H at the short end and converges to ξ_I.

On the sorting branch, the lower-tax investor alone holds the long bonds of both markets, so their returns are its required returns. Its exposure response appears in both loading recursions through the same feedback b, which cancels from the loading ratio. Long-end loading pass-throughs are therefore exactly zero for the household and one for the lower-tax investor, and yield pass-throughs differ from them only through relative convexity. At finite maturities, the tax-mix part of the pass-throughs averages household-only, shared and lower-tax-only maturities, weighted by rate persistence (Proposition A.9), and exposure responses add to it. In both illustrations, household yield pass-through falls and lower-tax pass-through rises with maturity (Appendices A.2.7 and A.2.9). With more clienteles, each has its own participation threshold and risk-tolerance share (Appendix A.5). Section 6 tests P1–P3.

Proof of Proposition A.8. *(i) The riskless short end.* The one-period claims have zero variance. With a slack household shorting constraint on cash, Proposition A.2 gives $y_1^M = \xi_H r$ and $y_1^C = r$ at every rate state with $r > 0$. The one-period loadings equal the responses $s_1^M = \xi_H$ and $s_1^C = 1$ by (53), so $y_1^M/y_1^C = A_1^M/A_1^C = \xi_H$. Differentiating the two ratios gives $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.

(ii) The long end. Proposition A.6 gives $A_\infty^M = \xi_I A_\infty^C$ on the branch, so the long-end loading ratio equals ξ_I at every pair of after-tax shares at which the branch is feasible. Differentiating it gives $\mathcal{P}_H^A(\infty) = 0$ and $\mathcal{P}_I^A(\infty) = 1$.

For yields, part (iii) of the same proposition gives

$$\frac{y_\infty^M}{y_\infty^C} = \xi_I + \frac{K_\infty}{y_\infty^C}, \qquad K_\infty = \tfrac{1}{2}\sigma_r^2 \xi_I(1-\xi_I)(A_\infty^C)^2,$$

so $\mathcal{P}_i^Y(\infty) = \mathcal{P}_i^A(\infty) + d(K_\infty/y_\infty^C)/d\xi_i$, with holdings, loadings and both yields adjusting. The feedback b is independent of $\bar{r}$: by Proposition 2, d solves $d\, A_{k-1}^M(d) = (\xi_I - \xi_H)m_I/[(\gamma_H m_I + \gamma_I m_H)\sigma_r^2]$, which contains no level terms. Hence K_∞ and its tax derivatives are independent of $\bar{r}$.

The mean rate affects y_∞^C directly and through π_I. At $z = 0$, both investors' level equalities hold at the shared maturity, so $(\xi_I - \xi_H)\bar{r} = \sigma_r^2 A_{k-1}^M(\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)$. The aggregate exposure $m_H \widehat{D}_H + m_I \widehat{D}_I = \sum_T A_{T-1}^M Q_T^M + \xi_I \sum_T A_{T-1}^C Q_T^C - m_H \ell_H - m_I \ell_I$ does not depend on how the shared maturity is split. Solving the two equations gives $\pi_I = \gamma_I \sigma_r^2 \widehat{D}_I = \pi_0 - b\bar{r}$, with π_0 independent of $\bar{r}$. Since $1/A_\infty^C = 1 - \rho + b$,

$$y_\infty^C = \bar{r} + A_\infty^C(\pi_0 - b\bar{r}) - \tfrac{1}{2}\sigma_r^2 (A_\infty^C)^2 = (1-\rho)A_\infty^C \bar{r} + O(1),$$

and $dy^C_\infty/d\xi_i = O(\bar{r})$. Therefore $d(K_\infty/y^C_\infty)/d\xi_i = O(1/\bar{r})$. When both derivatives are below one in absolute value, $\mathcal{P}^Y_H(\infty) < 1$ and $\mathcal{P}^Y_I(\infty) > 0$. In the illustration of Appendix A.2.7, they are 0.001 and -0.052.

(iii) The yield-response ratios. At a common maturity, dividing both price loadings by T leaves their ratio unchanged. Proposition A.6(i) gives the ordering and endpoints: the ratio starts at ξ_H and converges to ξ_I. □

The tax-mix pass-through across maturities. The next result gives the tax-mix part of the loading pass-through at every maturity, the intermediate-maturity profile used above and in Appendix A.4.

Proposition A.9 (Tax-mix pass-through)**.** *Switch off exposure-response terms in the affine pricing recursions, fixing each maturity's holders and risk-tolerance weights, and let $\bar{\xi}_j$ be the holders' tax mix at municipal maturity j. Put $G_T = \sum_{j=1}^T \rho^{T-j}$. The tax-mix pass-throughs $\mathcal{D}_i(T) := -\partial_{\xi_i}\tau^A_T$ are the part of $\mathcal{P}^A_i(T)$ that remains with exposure responses switched off, and*

$$\mathcal{D}_i(T) = \sum_{j=1}^{T} \frac{\rho^{T-j}}{G_T}\,\frac{\partial\bar{\xi}_j}{\partial\xi_i}. \tag{83}$$

On the sorting branch with shared maturity k, put

$$\Theta_S(T) = \frac{\sum_{j=1}^{\min(T,k-1)}\rho^{T-j}}{G_T}, \qquad \Theta_k(T) = \frac{\rho^{T-k}}{G_T}\mathbf{1}_{\{T\geq k\}}, \qquad \Theta_U(T) = 1-\Theta_S(T)-\Theta_k(T).$$

Then

$$\mathcal{D}_H(T) = \Theta_S(T) + \omega_H\Theta_k(T), \qquad \mathcal{D}_I(T) = \omega_I\Theta_k(T) + \Theta_U(T), \tag{84}$$

which equal $(1,0)$ before participation and converge monotonically to $(0,1)$. If both investors hold every municipal maturity from T^* on, as with bond-specific risk in Appendix A.2.8, define Θ_S with T^* in place of k and put $\Theta_L(T) = 1-\Theta_S(T)$. Then

$$\mathcal{D}_H(T) = \Theta_S(T) + \omega_H\Theta_L(T), \qquad \mathcal{D}_I(T) = \omega_I\Theta_L(T), \tag{85}$$

which equal $(1,0)$ before participation and converge monotonically to (ω_H,ω_I).

Proof. With exposure responses switched off, $s^C_j = 1$ and $s^M_j = \bar{\xi}_j$, so the recursions give $A^C_T = G_T$ and $A^M_T = \sum_{j=1}^T \rho^{T-j}\bar{\xi}_j$. Differentiating A^M_T/G_T gives (83). The derivative $\partial\bar{\xi}_j/\partial\xi_H$ is 1, ω_H and 0 at household-only, jointly held and lower-tax-only maturities, and $\partial\bar{\xi}_j/\partial\xi_I$ is 0, ω_I and 1, which gives (84) and (85). From the first jointly held maturity on, $G_{T+1} = \rho G_T + 1$ gives $\Theta_S(T+1) = \Theta_S(T)\rho G_T/(\rho G_T+1) < \Theta_S(T)$ and, from $T = k$ on, the same recursion for Θ_k, so both fall to zero. At $T = k$, $\mathcal{D}_H$ falls from 1 to $1-\omega_I/G_k$. On the sorting branch, $\mathcal{D}_H$ therefore falls to zero and $\mathcal{D}_I = 1-\Theta_S-\omega_H\Theta_k$ rises to one. On the jointly held branch, $\mathcal{D}_H = \omega_H + \omega_I\Theta_S$ falls to ω_H and $\mathcal{D}_I = \omega_I(1-\Theta_S)$ rises to ω_I. □

On the sorting branch, the tax-mix pass-throughs average $(1,0)$ at household-only maturities, (ω_H,ω_I) at the shared maturity and $(0,1)$ at lower-tax-only maturities. The weights are the persistence-weighted shares of the bond's remaining life spent at each type of maturity.

A.2.7 Illustration without bond-specific risk

The main-text figures use this rate-risk-only illustration. This subsection gives its parameters, checks the sorting branch and the conditions of the results above, and verifies the values reported in the main text.

Parameters and allocation. Both investors have risk aversion $\gamma_H = \gamma_I = 5$, and their liability hedges differ, $\ell_H \approx 1.053$ and $\ell_I \approx 1.452$. Values below are rounded, but calculations use unrounded inputs. The remaining parameters are $\rho = 0.8$, $\sigma_r = 0.025$, $(\xi_H,\xi_I) = (0.55, 0.78)$, equal unit masses, $\bar{r} = 0.04$, municipal supply 0.1 per maturity and taxable supply 0.03 at each $T \geq 2$. With common risk aversion, the lower-tax investor's balance-sheet advantage comes entirely from its greater liability hedge, which leaves it the smaller net exposure and the lower unit cost of rate risk. In the solved allocation, the household alone holds municipal maturities $T = 1,\ldots,22$, both investors hold maturity $k = 23$, the threshold T^* of Theorem A.1 and Figure 4, and the lower-tax investor alone holds $T = 24,\ldots,30$. The lower-tax investor holds about 47% of maturity 23 and all risky taxable bonds. The analytical price recursions use an affine continuation on this allocation branch, and positive holdings and the inactive KKT inequalities are checked locally around the reference state $z = 0$.

Solution checks and return-gap channels. The solved branch has $d \approx 11.24$, $\widehat{D}_H \approx 4.47$, $\widehat{D}_I \approx 3.57$ and $\rho - b \approx 0.765$. The sorting index (60) is positive below maturity 23, zero at 23 and negative above it, as Theorem A.1 requires. The left side of condition (S) is about 0.209, below $\rho = 0.8$. The municipal yield exceeds the household benchmark at every maturity beyond one period, by at least 8.1 basis points, and the excess rises with maturity. Both $s_T^M - \xi_H$ and $1 - s_T^C$ are positive at every risky maturity. The active portfolio equalities and market clearing hold to numerical precision, and inactive inequalities are strict. Independent portfolio optimization reproduces the return responses (70), and direct finite sums reproduce the loading recursion. At $T = 30$, the return gap is about 201 basis points, comprising 92 of exemption benefit and 109 of rate loading. The duration-cost contribution is about 92 basis points at $T = 22$ and zero from $T = 23$, where the lower-tax investor's conditions are used for both bonds.

Tax pass-through. For P1 and P2, I differentiate the exposure fixed point, loading recursions, portfolio conditions and yield ratio with respect to one after-tax share at a time, fixing the remaining primitives. Re-solving after small tax changes confirms these derivatives, with the shared maturity and portfolio feasibility preserved. Household yield pass-through falls from 1 at $T = 1$ to 0.581 at $T = 10$, 0.301 at 23 and 0.230 at 30. Lower-tax pass-through rises from 0 to 0.402, 0.668 and 0.736 at those maturities. Both profiles are monotone in the solved illustration.

Long-end continuation. The continuation keeps the risk-premium coefficients of the solved finite-supply economy. Since the lower-tax investor's conditions bind in both markets, its exposure response appears in both loading recursions through $-b = \gamma_I \sigma_r^2 \partial \widehat{D}_I / \partial z$. Thus $A_\infty^C = 1/(1 - \rho + b)$ and $A_\infty^M = \xi_I/(1 - \rho + b)$. The feedback b changes both loadings and cancels from their ratio. The individual return responses are $s_\infty^C = (1 - \rho)/(1 - \rho + b) = 0.851$ and $s_\infty^M = \xi_I s_\infty^C = 0.663$, so both responses include the portfolio adjustment. With $\pi_I = \gamma_I \sigma_r^2 \widehat{D}_I$, the limiting yields are

$$y_\infty^C = \bar{r} + \pi_I A_\infty^C - \tfrac{1}{2}\sigma_r^2 (A_\infty^C)^2, \qquad y_\infty^M = \xi_I \bar{r} + \pi_I A_\infty^M - \tfrac{1}{2}\sigma_r^2 (A_\infty^M)^2.$$

Re-solving the finite economy under the same tax perturbations and differentiating these limits gives $\mathcal{P}_H^Y(\infty) = 0.001$ and $\mathcal{P}_I^Y(\infty) = 0.948$, compared with the loading-implied limits of 0 and 1 in Proposition A.8(ii). For the unit-response reference of the P3 treasury comparison, the ratio A_T^M / G_T increases over the thirty illustrated maturities. Its infinite-maturity limit is $(1 - \rho) A_\infty^M = s_\infty^M$, below ξ_I, because this reference lacks the portfolio response of the jointly priced taxable market C.

A.2.8 Bond-specific risk and overlapping holdings

The quantitative model gives every risky bond a small bond-specific return shock. This subsection adds that shock to the two-investor benchmark. It changes one thing qualitatively: both investors can now hold a block of maturities, because the shock adds a risk cost that rises with each investor's holding of that particular bond. The pricing results of Appendices A.2.3–A.2.6 carry over, with the shared maturity replaced by the jointly held block. In the illustration of Appendix A.2.9, adding the shock changes municipal yields by at most 0.33 basis points. Bond-specific risk thus changes who holds each bond with little effect on prices.

Portfolio conditions with bond-specific risk. Fix maturity T, and let the active set $\mathcal{U}_T^M \subseteq \{H, I\}$ be the investors who hold that municipal bond. An active holder's portfolio condition (37) becomes the main-text condition (11) with bond-specific risk added:

$$\underbrace{\mu_T^M - \xi_i r}_{\text{after-tax excess return offered to } i} = \gamma_i \left[\underbrace{\sigma_r^2 A_{T-1}^M \widehat{D}_i}_{\text{whole-portfolio rate risk}} + \underbrace{(\sigma_{\eta,T}^M)^2 x_{i,T}^M}_{\text{bond-specific risk}} \right]. \tag{86}$$

On the branches studied here, the lower-tax investor alone holds the risky taxable bonds. Dividing its taxable condition by its after-tax share gives, for $T \geq 2$,

$$\mu_T^C - r = \gamma_I \left[\sigma_r^2 A_{T-1}^C \widehat{D}_I + \xi_I (\sigma_{\eta,T}^C)^2 x_{I,T}^C \right], \tag{87}$$

and $\mu_1^C = r$ for cash.

The return gap. The subtraction in the proof of Proposition 1 now carries the bond-specific terms, which add bond-specific compensation to the three channels:

$$g_T = (\xi_i - \xi_H)r + \sigma_r^2 A_{T-1}^M\big(\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I\big) + \gamma_I \sigma_r^2 \widehat{D}_I\big(A_{T-1}^M - \xi_H A_{T-1}^C\big) + \underbrace{\gamma_i (\sigma_{\eta,T}^M)^2 x_{i,T}^M - \xi_H \gamma_I \xi_I (\sigma_{\eta,T}^C)^2 x_{I,T}^C}_{\text{bond-specific compensation}}. \tag{88}$$

The holding-case statements of Proposition 1 hold with the index Ψ_T of (89) below in place of Ψ_T^0.

Participation. Bond-specific risk adds the incumbent's bond-specific risk cost to the participation comparison. A new buyer bears no bond-specific risk cost at a zero position, so this cost adds to the lower-tax investor's participation incentive. The equilibrium index (89) makes this explicit.

Theorem A.2 (Upper-interval participation by the lower-tax investor on an affine branch)**.** *At a regular state of a two-investor affine solution, suppose $v_T^M > 0$ on the risky maturities considered. With the risk-weighted net-exposure difference $\widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$ at the equilibrium allocation, define*

$$\Psi_T = (\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} - \gamma_H v_T^M Q_T^M / m_H. \tag{89}$$

Then $x_{I,T}^M > 0$ if and only if $\Psi_T < 0$. If $\widehat{\Lambda} > 0$ and the predecessor loadings and $v_T^M Q_T^M$ are nondecreasing, the lower-tax investor's positive support is an upper interval, possibly empty. This identifies the cutoff within a given equilibrium and does not establish uniqueness across equilibria.

Proof. *Step 1: holding cases.* If the lower-tax investor is inactive, $x_{H,T}^M = Q_T^M / m_H$. Substituting the household equality into the lower-tax investor's inequality gives $\Psi_T = \nu_{I,T} \geq 0$, the lower-tax investor's KKT slack. If both investors hold the bond, subtracting their equalities and using clearing gives

$$\Psi_T = -v_T^M \left(\gamma_I + \gamma_H m_I / m_H\right) x_{I,T}^M < 0.$$

If the lower-tax investor alone is active, its equality and the household inequality imply $(\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} \leq -\gamma_I v_T^M Q_T^M / m_I < 0$, which also gives $\Psi_T < 0$. These cases prove the equivalence.

Step 2: monotonicity. Adjacent indices differ by

$$\Psi_{T+1} - \Psi_T = -\sigma_r^2 (A_T^M - A_{T-1}^M)\widehat{\Lambda} - \frac{\gamma_H}{m_H}(v_{T+1}^M Q_{T+1}^M - v_T^M Q_T^M) \leq 0.$$

With $\widehat{\Lambda} > 0$ and nondecreasing predecessor loadings and $v_T^M Q_T^M$, the index is nonincreasing in maturity, and the first maturity with a strictly negative index is the participation threshold T^*. Equality is a zero-holding boundary. □

Joint holdings. A small bond-specific variance adds a position-dependent cost, so both investors' portfolio conditions can bind over several maturities. At the equilibrium net rate exposures, define $N_T = (\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda}$, the difference in exemption benefits less the duration benefit. For $v_T := v_T^M > 0$, subtracting the two active KKT equalities and imposing clearing gives

$$x_{H,T}^M = \frac{\gamma_I Q_T^M + m_I N_T / v_T}{\gamma_H m_I + \gamma_I m_H}, \qquad x_{I,T}^M = \frac{\gamma_H Q_T^M - m_H N_T / v_T}{\gamma_H m_I + \gamma_I m_H}. \tag{90}$$

Both holdings are nonnegative exactly when $-\gamma_I Q_T^M / m_I \leq N_T / v_T \leq \gamma_H Q_T^M / m_H$, and strict inequalities give positive holdings for both investors. Each maturity's holdings depend on the net rate exposures, which are common to all maturities and must match the resulting portfolios. Two maturities $T < T'$ can both be jointly held only if $\sigma_r^2 \widehat{\Lambda}(A_{T'-1}^M - A_{T-1}^M) < v_T \gamma_H Q_T^M / m_H + v_{T'} \gamma_I Q_{T'}^M / m_I$, because both investors' holdings must be positive at both maturities. When the bond-specific variances are small relative to $\sigma_r^2 \widehat{\Lambda}$ times the loading differences, at most one maturity is jointly held, as in Theorem A.1.

Responses on a jointly held block. The next proposition is the counterpart of Proposition 2 on the branch that the illustration with bond-specific risk selects: the household alone holds municipal bonds below T^*, and both investors hold them from T^* on.

Proposition A.10 (Responses with a jointly held block)**.** *At a regular state with $r(z) > 0$ in the two-investor affine benchmark, suppose the lower-tax investor alone holds risky taxable bonds. Let $\varnothing \neq J \subseteq \{2, \ldots, T_{\max}\}$ be*

the fixed set of municipal maturities held by both investors. The high-tax household alone holds every municipal maturity outside J, including the riskless one-period claim. Assume $A^M_{T-1}, A^C_{T-1} > 0$ for every $T \geq 2$ and $v^M_T > 0$ on J. Define

$$P_J = \sum_{T\in J} \frac{A^M_{T-1}}{v^M_T}, \quad S_J = \sum_{T\in J} \frac{(A^M_{T-1})^2}{v^M_T}, \quad d := \frac{\partial \widehat{D}_H}{\partial z}.$$

Then

$$d = \frac{(\xi_I - \xi_H) m_I P_J}{(\gamma_H m_I + \gamma_I m_H)(1 + \sigma_r^2 S_J)} > 0, \qquad \frac{\partial \widehat{D}_I}{\partial z} = -\frac{m_H}{m_I} d. \tag{91}$$

With $b = \gamma_I \sigma_r^2 (m_H/m_I) d$, the responses are

$$s^M_T = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A^M_{T-1} d, & T \notin J, \\ \xi^*, & T \in J, \end{cases} \qquad s^C_T = 1 - b A^C_{T-1}. \tag{92}$$

Hence $s^M_T - \xi_H s^C_T > 0$ and $A^M_T - \xi_H A^C_T > 0$ at every risky maturity $T \geq 2$, with equality in both comparisons at $T = 1$.

Proof. *Step 1: exposure responses.* Clearing fixes each sole holder's position: Q^M_T/m_H for the household at maturities outside J and Q^C_T/m_I for the lower-tax investor's risky taxable bonds. Only jointly held municipal positions vary. With constant liabilities and fixed taxable holdings, differentiating clearing, weighting by municipal predecessor loadings and summing gives

$$m_H \partial_z \widehat{D}_H + m_I \partial_z \widehat{D}_I = 0. \tag{93}$$

At each $T \in J$, subtracting the lower-tax investor's differentiated municipal equality from the household's gives

$$\xi_I - \xi_H = \sigma_r^2 A^M_{T-1} (\gamma_H \partial_z \widehat{D}_H - \gamma_I \partial_z \widehat{D}_I) + v^M_T (\gamma_H \partial_z x^M_{H,T} - \gamma_I \partial_z x^M_{I,T}).$$

With $k_0 = (\gamma_H m_I + \gamma_I m_H)/m_H$, clearing reduces this to $\partial_z x^M_{I,T} = -(\xi_I - \xi_H)/(k_0 v^M_T) - \sigma_r^2 A^M_{T-1} \partial_z \widehat{D}_I / v^M_T$. Since $\partial_z \widehat{D}_I = \sum_{T\in J} A^M_{T-1} \partial_z x^M_{I,T}$, multiplying by A^M_{T-1} and summing over J gives

$$(1 + \sigma_r^2 S_J) \partial_z \widehat{D}_I = -\frac{\xi_I - \xi_H}{k_0} P_J < 0.$$

Equation (93) then gives (91).

Step 2: return responses. The responses follow from (67). Household-only quantities have zero derivatives, giving the first municipal response. At jointly held maturities, risk-tolerance aggregation cancels the aggregate exposure and bond-specific derivatives and leaves ξ^*, and (68) gives the taxable formula.

Step 3: response and loading gaps. Equation (92) gives the response gap $\partial_z g_T = s^M_T - \xi_H s^C_T$ as

$$\partial_z g_T = \begin{cases} \sigma_r^2 d \left[\gamma_H A^M_{T-1} + \xi_H \gamma_I \dfrac{m_H}{m_I} A^C_{T-1} \right], & T \notin J, \\ \omega_I (\xi_I - \xi_H) + \xi_H b A^C_{T-1}, & T \in J. \end{cases}$$

Thus $\partial_z g_T > 0$ for every $T \geq 2$, while $\partial_z g_1 = 0$ because both predecessor loadings vanish. Iterating (56), with $0 < \rho < 1$, yields

$$A^M_T - \xi_H A^C_T = \sum_{j=2}^{T} \rho^{T-j} \partial_z g_j > 0, \qquad T \geq 2.$$

□

Step 1 uses only that clearing fixes the positions outside J, so (91) also holds when the lower-tax investor alone holds the maturities above J, which then respond by $\xi_I - b A^M_{T-1}$. With $J = \{k\}$, the factor $P_J/(1+\sigma_r^2 S_J) = A^M_{k-1}/(v^M_k + \sigma_r^2 (A^M_{k-1})^2)$ tends to $1/(\sigma_r^2 A^M_{k-1})$ as $v^M_k \to 0$, so (91) tends to the exposure response (69), and every response converges to its value on the sorting branch, which is the $v^M_k \to 0$ limit of this configuration. On a block that extends to the longest maturity, long maturities respond by the tax mix ξ^*.

Uniqueness of the loading fixed point. Appendix A.2.9 verifies the following sufficient condition for a unique loading fixed point in the illustration. For given d, let $\mathcal{F}_J(d)$ be the right side of (91) at the municipal

loadings that (92) and the loading recursion (56) generate. If $\mathcal{F}_J$ maps $[0,\infty)$ into a bounded interval $[0, d_{\max}]$ and $\sup_{[0,d_{\max}]} |\mathcal{F}_J'| < 1$, every nonnegative exposure fixed point lies in $[0, d_{\max}]$, and Banach's fixed-point theorem gives exactly one, with its loading path on this branch. Holdings and non-holders' inequalities must still be checked separately, and equilibria with other feasible participation thresholds remain possible. The derivative of $\mathcal{F}_J$ is

$$\mathcal{F}_J'(d) = c\frac{P_J'(1+\sigma_r^2 S_J) - \sigma_r^2 P_J S_J'}{(1+\sigma_r^2 S_J)^2}, \qquad c = \frac{(\xi_I - \xi_H)m_I}{\gamma_H m_I + \gamma_I m_H},$$

with the derivatives of P_J, S_J obtained by differentiating the same loading recursions.

Long-end pricing with a jointly held block. For the upper interval $J = \{T^*, \dots, T_{\max}\}$, Proposition A.11 below uses condition (S) and the bound (M) on the threshold loading:

$$\textbf{(M)} \quad A^M_{T^*-1} < \frac{\xi^*}{1-\rho}. \tag{94}$$

Condition (S) keeps the feedback b at or below ρ, so taxable loadings are nondecreasing in maturity. The bound (M) places the threshold loading below its long-end value, so municipal loadings also rise beyond the participation threshold. The long-end taxable response $s^{C,\text{aff}}_\infty := \lim_{T\to\infty} s^{C,\text{aff}}_T$ lies in $(0,1)$, because the lower-tax investor's net rate exposure falls as the rate state rises.

The bound (M) holds at every solution of (91) on this branch. To prove this, put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$, so that (91) reads $\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 P_J/(1+\sigma_r^2 S_J)$. The recursions at household-only and jointly held maturities give $A^M_n = \xi_H \sum_{q=0}^{n-1}(\rho+\lambda)^q$ and $A^M_{n+l} = \rho^l A^M_n + (1-\rho^l)\xi^*/(1-\rho)$ for $l \geq 0$. Suppose $A^M_n \geq \xi^*/(1-\rho)$. Every loading on J is then at least $\xi^*/(1-\rho)$, so $S_J/P_J \geq \xi^*/(1-\rho)$. Using $\xi^* - \xi_H = \omega_I(\xi_I - \xi_H)$, this gives $\lambda < (1-\rho)(\xi^* - \xi_H)/\xi^* < 1-\rho$. Then $A^M_n < \xi_H/(1-\rho-\lambda) < \xi^*/(1-\rho)$, a contradiction.

Proposition A.11 (Loading ratio and long-end implied tax with a jointly held block)**.** *Consider the branch of Proposition A.10 with $J = \{T^*, \dots, T_{\max}\}$: the household alone holds municipal bonds shorter than T^*, both investors hold municipal bonds from T^* to $T_{\max}$, and the lower-tax investor alone holds the risky taxable bonds. Assume condition (S).*

(i) The municipal-to-taxable loading ratio is nondecreasing in maturity, starts at ξ_H and, with the recursions extended to infinite maturity and d held at its solved value, converges to $\xi^*/s^{C,\text{aff}}_\infty$. The ratio to the unit-response reference, A^M_T/G_T, is also nondecreasing, starts at ξ_H and converges to $\xi^* > \xi_H$.

(ii) With the recursions extended to infinite maturity, suppose the loadings, the risk-premium coefficients π^a_T and the bond-specific compensation terms ι^a_T of (81) converge, and the limiting taxable forward is positive. Then

$$\lim_{T\to\infty} \tau^{Y,\text{impl,aff}}_T = 1 - \frac{\xi^*\bar{r} + A^M_\infty \pi^M_\infty - \frac{1}{2}\sigma_r^2 (A^M_\infty)^2 + \iota^M_\infty}{\bar{r} + A^C_\infty \pi^C_\infty - \frac{1}{2}\sigma_r^2 (A^C_\infty)^2 + \iota^C_\infty}. \tag{95}$$

Proof. Step 1 (municipal loadings). At household-only maturities, $A^M_T = (\rho + \gamma_H \sigma_r^2 d)A^M_{T-1} + \xi_H$, so A^M_T is increasing from zero. At jointly held maturities, $A^M_T = \rho A^M_{T-1} + \xi^*$, whose increments have the sign of $\xi^* - (1-\rho)A^M_{T^*-1}$. Condition (M) makes them positive.

Step 2 (taxable loadings). The bound $A^M_{T-1} \geq \xi_H$ on J gives $P_J/S_J \leq 1/\xi_H$, and (91) and (S) give $0 \leq b \leq \rho$. Therefore

$$A^C_T = \sum_{q=0}^{T-1} (\rho - b)^q, \qquad s^C_T = 1 - bA^C_{T-1} \downarrow \frac{1-\rho}{1-\rho+b} > 0.$$

Step 3 (return responses). The municipal response rises at household-only maturities, is constant at jointly held maturities and does not jump down when the lower-tax investor begins to participate. For the last claim, the response at $T^* - 1$ must not exceed ξ^*. Substituting (91) shows that the required inequality $\omega_I(\xi_I - \xi_H) \geq \gamma_H \sigma_r^2 A^M_{T^*-2} d$ is equivalent to $\sigma_r^2 A^M_{T^*-2} P_J \leq 1 + \sigma_r^2 S_J$. This inequality follows from $A^M_{T-1} > A^M_{T^*-2}$ for every $T \in J$, established in Step 1 using (M). Since Step 2 makes s^C_j positive and nonincreasing, the return-response ratio s^M_j/s^C_j is nondecreasing in maturity.

Step 4 (averaging). For any positive e_j and nondecreasing ratios $q_j = a_j/e_j$, the ratio weighted by rate persistence,

$$\bar{q}_T = \frac{\sum_{j\leq T} \rho^{T-j} a_j}{\sum_{j\leq T} \rho^{T-j} e_j}$$

is nondecreasing: as a positive-weight average of $q_1, \ldots, q_T$, it is at most $q_T \le q_{T+1}$, and

$$\bar{q}_{T+1} - \bar{q}_T = \frac{e_{T+1}(q_{T+1} - \bar{q}_T)}{\rho \sum_{j \le T} \rho^{T-j} e_j + e_{T+1}} \ge 0.$$

By (56), A_T^M / A_T^C has this form. Applying the argument with $a_j = s_j^M$ and $e_j = s_j^C$ proves the loading-ratio ordering, starting from ξ_H. With $e_j = 1$, for which (M) suffices, the same argument proves that A_T^M / G_T is nondecreasing.

Step 5 (limits). Extending the same branch recursions to arbitrarily long maturities with fixed d gives $A_\infty^M = \xi^*/(1-\rho)$, $A_\infty^C = 1/(1-\rho+b)$ and $G_\infty = 1/(1-\rho)$, hence both endpoints in part (i). For part (ii), the level recursion (81) and its Cesàro limit (82) apply with $\bar{\xi}_\infty^M = \xi^*$ at jointly held long maturities and $\bar{\xi}_\infty^C = 1$, which gives (95). □

The long-end implied tax to first order. Aggregating by risk tolerance over both investors for long municipal bonds and over the lower-tax investor alone for long taxable bonds gives

$$\pi_\infty^M - \pi_\infty^C = \omega_H \, \sigma_r^2 \, \widehat{\Lambda}, \qquad \widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I, \tag{96}$$

where $\widehat{D}_i = \widehat{D}_i(0)$ is the net rate exposure at the stationary mean. The approximation below treats the convexity and bond-specific terms as negligible. With small taxable loading feedback ($b \ll 1-\rho$) and a positive denominator, substituting $A_\infty^M = \xi^*/(1-\rho)$, $A_\infty^C \approx 1/(1-\rho)$ and (96) into (82) gives the first-order endpoint

$$\tau_\infty^{Y,\text{impl,aff}} \approx \underbrace{(1-\xi^*)}_{\text{one minus tax mix}} - \underbrace{\frac{\xi^* \, \omega_H \, \sigma_r^2 \, \widehat{\Lambda}}{(1-\rho)\bar{r} + \pi_\infty^C}}_{\text{municipal term-premium excess}}. \tag{97}$$

With $\widehat{\Lambda} > 0$, this first-order endpoint lies below one minus the tax mix, $1-\xi^*$, and hence below the one-period implied tax $1-\xi_H$. The household's higher unit risk cost affects long municipal yields here because the household also holds them.

Pass-through with a jointly held block. On this branch the long-end loading ratio is ξ^*/s_∞^C, and both after-tax shares affect it through the tax mix and the feedback b. Its pass-throughs are bounded by the risk-tolerance shares under a condition on the relative tax difference.

Proposition A.12 (Tax pass-through with a jointly held block)**.** *Consider the branch of Proposition A.11, under condition (S), with a positive cash rate and a slack household shorting constraint on cash. With its recursions extended to infinite maturity, suppose long yields, loading ratios and their tax derivatives converge locally uniformly, with a positive long taxable yield. The short-end pass-throughs are those of Proposition A.8(i).*

(i) (Long end, P1 and P2) Suppose the after-tax shares satisfy the tax-difference condition

$$2\omega_H e + \omega_I e^2 \le 1, \qquad e := \frac{\xi_I - \xi_H}{\xi_H}.$$

Then

$$\mathcal{P}_H^A(\infty) < \omega_H, \qquad \mathcal{P}_I^A(\infty) > \omega_I.$$

Yield pass-throughs add to these values the correction terms in (98) below, which vanish at rate $1/\bar{r}$ along the branch's algebraic continuation. Whenever the corrections are smaller in absolute value than the margins of Lemma A.1,

$$\mathcal{P}_H^Y(\infty) < \omega_H < 1, \qquad \mathcal{P}_I^Y(\infty) > \omega_I > 0.$$

(ii) (Response ratios, P3) The municipal-to-taxable zero-coupon response ratio A_T^M / A_T^C is nondecreasing in maturity, equals ξ_H at the short end and converges to ξ^*/s_∞^C, with $0 < s_\infty^C < 1$. For the unit-response reference, A_T^M / A_T^{Tsy} has the same ordering and short endpoint, and converges to $\xi^* > \xi_H$.

With the tax mix alone, each investor's long-end loading pass-through on this branch equals its risk-tolerance share (Proposition A.9). The tax-difference condition holds whenever $\xi_I < \frac{3}{2}\xi_H$, regardless of risk-tolerance shares or the participation threshold. Under it, portfolio adjustment offsets the household effect and reinforces the lower-tax investor's effect, as Lemma A.1 proves with both portfolios and loadings adjusting. Yield pass-

through also includes relative risk costs, convexity and bond-specific compensation, whose correction to loading pass-through shrinks as $1/\bar{r}$ along the fixed-branch continuation.

Proof of Proposition A.12. *(i) The long end.* Let d be the household's exposure response in (91) and $b = \gamma_I \sigma_r^2 (m_H/m_I) d$ the feedback into taxable responses, as in (92). The solved long responses and loadings satisfy

$$s_\infty^M = \xi^*, \qquad s_\infty^C = \frac{1-\rho}{1-\rho+b}, \qquad A_\infty^a = \frac{s_\infty^a}{1-\rho}.$$

Thus $R_\infty = A_\infty^M / A_\infty^C = \xi^* / s_\infty^C$. At the mean state, the solved yield formula (82) gives

$$y_\infty^M = \xi^* \bar{r} + A_\infty^M \pi_\infty^M - \tfrac{1}{2}\sigma_r^2 (A_\infty^M)^2 + \iota_\infty^M,$$
$$y_\infty^C = \bar{r} + A_\infty^C \pi_\infty^C - \tfrac{1}{2}\sigma_r^2 (A_\infty^C)^2 + \iota_\infty^C.$$

The four terms are interest income, portfolio rate-risk compensation, convexity and bond-specific compensation. Dividing the two yields separates the loading ratio from the remaining yield-level difference:

$$\frac{y_\infty^M}{y_\infty^C} = R_\infty + \frac{y_\infty^M - R_\infty y_\infty^C}{y_\infty^C}.$$

All derivatives below are with respect to ξ_i, with holdings, loadings and both yields adjusting. The assumed convergence permits differentiation of the limiting ratio. The quotient rule gives

$$\mathcal{P}_i^Y(\infty) = \mathcal{P}_i^A(\infty) + \frac{1}{y_\infty^C} \frac{d(y_\infty^M - R_\infty y_\infty^C)}{d\xi_i} - \frac{y_\infty^M - R_\infty y_\infty^C}{(y_\infty^C)^2} \frac{dy_\infty^C}{d\xi_i}. \tag{98}$$

The three terms are the responses of the loading ratio, the relative yield compensation and the taxable denominator. The first retains the endogenous taxable response:

$$\mathcal{P}_i^A(\infty) = \frac{\omega_i}{s_\infty^C} - \frac{\xi^*}{(s_\infty^C)^2} \frac{ds_\infty^C}{d\xi_i}.$$

Lemma A.1 below proves $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$ under the tax-difference condition, with uniform positive margins.

For the two correction terms, the calculation after the lemma shows that $y_\infty^M - R_\infty y_\infty^C$ and its tax derivatives are independent of $\bar{r}$. Also, $y_\infty^C = A_\infty^C(1-\rho)\bar{r} + O(1)$ and $dy_\infty^C/d\xi_i = O(\bar{r})$. Both correction terms in (98) therefore vanish at rate $1/\bar{r}$ along the branch's algebraic continuation, and the lemma's margins are uniform over the finitely many cutoffs. When the corrections are smaller in absolute value than these margins, $\mathcal{P}_H^Y(\infty) < \omega_H < 1$ and $\mathcal{P}_I^Y(\infty) > \omega_I > 0$. In the illustration of Appendix A.2.9, the corrections are 0.006 and -0.054, against margins above one half.

(ii) The yield-response ratios. At a common maturity, dividing both price loadings by T leaves their ratio unchanged. Proposition A.11(i) gives the ordering and endpoints for the equilibrium taxable market C and for the unit-response reference, for which $A_T^{\mathrm{Tsy}} = G_T$, $A_1^{\mathrm{Tsy}} = 1$ and $A_\infty^{\mathrm{Tsy}} = 1/(1-\rho)$. □

Lemma A.1 (The loading response)**.** *Under the tax-difference condition in Proposition A.12, $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$. With other primitives fixed, both strict margins have positive lower bounds over all feasible upper-interval cutoffs.*

Proof. The proof differentiates the exposure fixed point and bounds its feedback into the loading ratio.

Step 1 (the fixed point and two bounds). Put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$, the response of the household's unit risk cost to the rate state. Equation (91) gives

$$\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 f, \qquad f = \frac{P_J}{1+\sigma_r^2 S_J}, \qquad R_\infty = \xi^* \left[1 + \frac{\omega_H \lambda}{\omega_I(1-\rho)}\right].$$

The sums P_J and S_J are those of Proposition A.10. With $G_n(p) = \sum_{q=0}^{n-1} p^q$, the loading recursion gives

$$A_n^M = \xi_H G_n(\rho+\lambda), \qquad A_{n+l}^M = \rho^l A_n^M + (1-\rho^l) A_\infty^M, \quad l \geq 0.$$

By (94), $A_n^M < A_\infty^M$, so loadings rise along the jointly held block. Thus $S_J/P_J \geq A_n^M$ and

$$\lambda G_n(\rho+\lambda) < \omega_I e, \qquad \lambda \frac{G_n'(\rho+\lambda)}{G_n(\rho+\lambda)} < \omega_I e,$$

where e is the relative tax difference of Proposition A.12. The first bound uses $\lambda < \omega_I(\xi_I - \xi_H)P_J/S_J$, and the second follows from $G_n' \leq G_n^2$. The tax-difference condition implies $e < 1$ and hence $b = (\omega_H/\omega_I)\lambda < \omega_H e < 1$, ensuring convergence of the taxable loadings.

Step 2 (derivatives at fixed λ). To differentiate the loading sums, let $t_j = \rho^l A_n^M / A_{j-1}^M$ for $j = T^* + l$ and

$$X = \bar{t}_P - \frac{2\sigma_r^2 S_J}{1+\sigma_r^2 S_J}\bar{t}_S.$$

Here $\bar{t}_P$ is the mean of t_j with weights proportional to A_{j-1}^M/v_j^M, and $\bar{t}_S$ uses weights proportional to $(A_{j-1}^M)^2/v_j^M$. Since $0 \leq t_j \leq 1$, $-2 < X \leq 1$. Let ∂_i denote differentiation with respect to ξ_i holding λ fixed. Differentiating the loadings gives

$$\partial_H \ln f = \left(\frac{1}{\xi_H} - \frac{\omega_H}{\xi^*}\right) X + \frac{\omega_H}{\xi^*}\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J},$$
$$\partial_I \ln f = \frac{\omega_I}{\xi^*}\left(\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J} - X\right).$$

Step 3 (the response of λ). To include the equilibrium response of λ, differentiate its fixed point. With G_n and G_n' evaluated at $\rho+\lambda$,

$$\frac{d\lambda}{d\xi_i} = \frac{\lambda\,\partial_i \ln[(\xi_I - \xi_H)f]}{D}, \qquad D = 1 - \lambda\frac{G_n'}{G_n}X > 1 - \omega_I e > 0.$$

Step 4 (the margins). Substitute into $dR_\infty/d\xi_i = \mathcal{P}_i^A(\infty)$. The coefficient of X is positive in the household derivative and negative in the lower-tax derivative. Using $X \leq 1$ and $\lambda G_n'/G_n < \omega_I e$ gives

$$\mathcal{P}_H^A(\infty) - \omega_H \leq \frac{\omega_H \lambda}{\omega_I(1-\rho)D}\left(\frac{2\omega_H}{1+\sigma_r^2 S_J} - \frac{1}{e} + \omega_I e\right) < 0,$$
$$\mathcal{P}_I^A(\infty) - \omega_I > \frac{\omega_H \lambda}{\omega_I(1-\rho)D}(1/e - \omega_I^2 e) > 0.$$

The tax-difference condition makes the household's parenthesized term at most $-2\omega_H\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$, giving a positive margin for each investor.

Step 5 (uniform margins). Consider $n \in \{1, \ldots, T_{\max} - 1\}$ in the finite loading system and trial feedbacks $x \geq 0$ with $xG_n(\rho+x) \leq \omega_I e$. This compact domain, defined by primitives, contains every equilibrium pair. Its positive loading sums give positive minima for f and $\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$, while $D \geq 1 - \omega_I e$ keeps full tax derivatives bounded. At a fixed point, $\lambda \geq \omega_I(\xi_I - \xi_H)\sigma_r^2 \min f > 0$ and $D < 1 + 2\omega_I e$. The displayed margins consequently have uniform positive lower bounds. □

The relative yield compensation. This calculation completes the proof of Proposition A.12(i). Subtract the two solved long yields, weighting the taxable yield by $R_\infty = A_\infty^M/A_\infty^C$. Equation (96) gives $\pi_\infty^M - \pi_\infty^C = \omega_H\sigma_r^2\widehat{\Lambda}$, while $\xi^* - R_\infty = -bA_\infty^M$. Therefore

$$y_\infty^M - R_\infty y_\infty^C = A_\infty^M(\omega_H\sigma_r^2\widehat{\Lambda} - b\bar{r}) + \tfrac{1}{2}\sigma_r^2 A_\infty^M(A_\infty^C - A_\infty^M) + \iota_\infty^M - R_\infty\iota_\infty^C.$$

The three terms are the difference in risk costs on the bonds each investor holds alone, relative convexity and bond-specific compensation. On the jointly held block, subtract the lower-tax investor's municipal equality from the household's and use clearing:

$$\omega_H\sigma_r^2\widehat{\Lambda} - b\bar{r} = \frac{\omega_H\sigma_r^2}{1+\sigma_r^2 S_J}\left[\gamma_H\left(\sum_{T\notin J}\frac{A_{T-1}^M Q_T^M}{m_H} - \ell_H\right) - \gamma_I\left(\frac{\xi_I}{m_I}\sum_T A_{T-1}^C Q_T^C - \ell_I\right)\right].$$

Every term, and hence its tax derivative, is independent of $\bar{r}$ on the fixed branch. Common rate-risk compensation per unit of loading cancels from $y^M_\infty - R_\infty y^C_\infty$.

The joint holders' risk-premium coefficient is also independent of $\bar{r}$. With $\Gamma := m_H/\gamma_H + m_I/\gamma_I$, the value of Γ^M_T on the jointly held block, clearing gives

$$\pi^M_\infty = \frac{\sigma_r^2}{\Gamma}\left(\sum_T A^M_{T-1}Q^M_T + \xi_I \sum_T A^C_{T-1}Q^C_T - m_H\ell_H - m_I\ell_I\right).$$

By (96) and the two previous displays, the mean-rate term in π^C_∞ is $-b\bar{r}$. Since $1/A^C_\infty = 1-\rho+b$, dividing the taxable yield by its loading cancels $b\bar{r}$ against this term. This establishes $y^C_\infty = A^C_\infty(1-\rho)\bar{r} + O(1)$. The bond-specific terms are $\iota^M_\infty = v^M_\infty(Q^M_\infty/\Gamma - \frac{1}{2})$ and $\iota^C_\infty = v^C_\infty(\gamma_I\xi_I Q^C_\infty/m_I - \frac{1}{2})$, where Q^a_∞ and v^a_∞ are the limiting supply and bond-specific variance in market a.

Convexity in the long-end correction. Dividing by y^C_∞ isolates relative convexity, the only correction on the sorting branch (Proposition A.6(iii)):

$$\Delta\tau_\infty := \frac{y^M_\infty}{y^C_\infty} - R_\infty = R_\infty(1-R_\infty)\chi_\infty + \psi_\infty,$$
$$\chi_\infty = \frac{\frac{1}{2}\sigma_r^2(A^C_\infty)^2}{y^C_\infty}, \qquad \psi_\infty = \frac{A^M_\infty(\omega_H\sigma_r^2\widehat{\Lambda} - b\bar{r}) + \iota^M_\infty - R_\infty\iota^C_\infty}{y^C_\infty}.$$

Since the implied tax is one minus the yield ratio, the yield-implied tax equals the loading-implied tax less the correction $\Delta\tau_\infty$ to the yield ratio. The signed convexity contribution is at most $\chi_\infty/4$, since $R_\infty(1-R_\infty) \le 1/4$. The derivatives of the two terms sum to $\mathcal{P}^Y_i(\infty) - \mathcal{P}^A_i(\infty)$ in (98).

What bond-specific risk changes. The short end, the signs of the exposure responses, the positive loading gap and the rising loading ratio are the same on both branches. What changes is who holds the longest bonds. On the sorting branch, the lower-tax investor alone holds them, so the long-end loading ratio is ξ_I and long-end loading pass-throughs are $(0,1)$. On a jointly held block that reaches the longest maturity, both investors hold them, the long-end response is the tax mix ξ^*, and the household's higher unit risk cost adds the term-premium excess in (97). The two branches give close long-end responses when the shared maturity's loading is near its long-end value: on the sorting branch, $bA^M_{k-1} = \xi_I - \xi^*$, so the long-end response $\xi_I - bA^M_\infty$ is near ξ^* when A^M_∞ is near A^M_{k-1}. In the illustrations, the long-end municipal response is 0.663 on the sorting branch against $\xi^* = 0.665$ with bond-specific risk, and the long-end loading pass-throughs are $(0,1)$ against $(-0.012, 1.013)$.

A.2.9 Illustration with bond-specific risk

Adding bond-specific risk to the main-text illustration broadens joint participation without materially changing the return gap. This subsection reports the holdings, return-gap channels and loading decomposition plotted in Figure 14, the tax pass-throughs, and checks of the conditions used in Appendix A.2.8.

Setup and intuition. I keep the other primitives of Appendix A.2.7 and add a municipal log-return standard deviation of $0.004[1 + 7\sqrt{(T-1)/29}]$ for $2 \le T \le 30$. Taxable bonds have no bond-specific shock, and the one-period claims remain riskless. Both investors now hold municipal maturities 17 through 30, while without the shock they share only maturity 23. With rate risk alone, joint holdings require both investors to demand exactly the same return. Bond-specific risk adds a cost that rises with each investor's holding of that bond and can offset a small difference in the returns the investors require for taxes and rate risk. In this illustration, those requirements are close across long maturities, where rate loadings flatten, so a small bond-specific cost supports joint holdings over a broad maturity range while adding little to required returns.

Return-gap channels. Bond-specific compensation is at most 2.7% of the return gap. At thirty periods, the gap is about 205 basis points, comprising 92 of exemption benefit, 109 of rate loading and 4 of bond-specific compensation. At the last household-only maturity, $T = 16$, the 182-basis-point gap comprises 88 of duration cost, 91 of rate loading and 3 of bond-specific compensation. Removing the shock changes the return gap by at most 3.91 basis points over all thirty maturities, municipal yields by at most 0.33 basis points and taxable yields by at most 0.07 basis points.

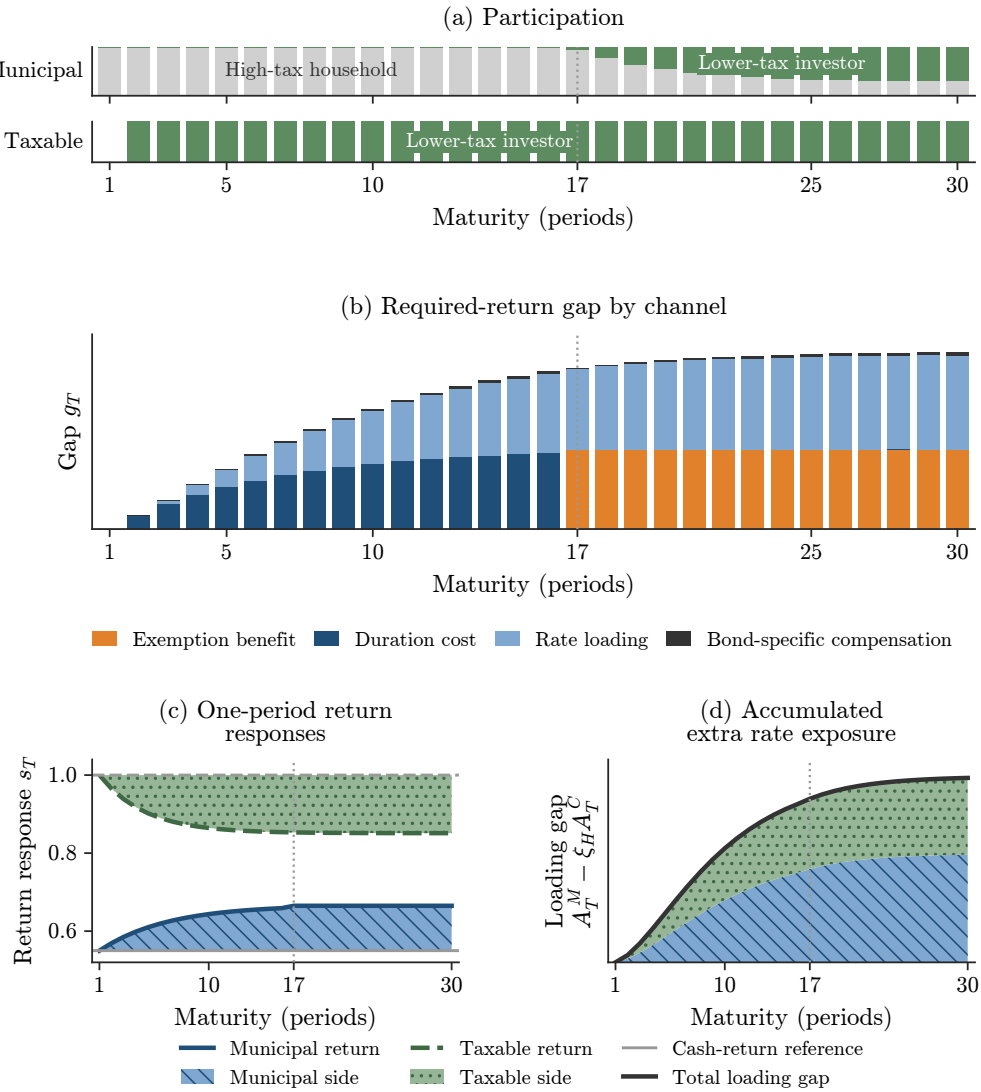


Figure 14: The analytical illustration with bond-specific risk. All other primitives match Appendix A.2.7. Panel (a) shows ownership. Panel (b) gives the three pricing channels of Figure 5, with small bond-specific compensation in gray. Panels (c) and (d) use the grouping of Figure 6: the municipal side combines the tax mix and municipal portfolio response, while the taxable side reflects the taxable portfolio response. Panel (c) shades $s_T^M - \xi_H$ and $1 - s_T^C$, measured from the cash-return references ξ_H and 1 (gray lines). Panel (d) accumulates $s_T^M - \xi_H$ and $\xi_H(1 - s_T^C)$ into the total loading gap. The dotted line marks joint municipal holdings from $T^* = 17$.

A unique loading solution. For $a_{\min} = \xi_H \sum_{q=0}^{15} \rho^q$, predecessor loadings on the jointly held block satisfy $A_{T-1}^M \geq a_{\min}$ and $P_J/S_J \leq 1/a_{\min}$. The exposure map $\mathcal{F}_J$ of Appendix A.2.8 therefore maps $[0, \infty)$ into $[0, d_{\max}]$, where

$$d_{\max} = \frac{\xi_I - \xi_H}{(\gamma_H + \gamma_I)\sigma_r^2 a_{\min}} < 13.77.$$

Interval bounds give $\sup |\mathcal{F}_J'| < 0.066 < 1$, so the map is a contraction and has a unique fixed point, $d^* \approx 11.20$. The resulting $A_{16}^M \approx 3.147$ is below $\xi^*/(1-\rho) = 3.325$, verifying condition (M) in (94). The left side of condition (S) in (72) is about 0.209, below $\rho = 0.8$.

Levels and the KKT conditions. The liability hedges above give $\widehat{\Lambda} \approx 4.53$, which satisfies the participation inequalities selecting $T^* = 17$ under Theorem A.2. At the mean rate state, the active KKT conditions,

market clearing and the return-gap identity hold to numerical precision, and all inactive portfolio inequalities are strict.

Pass-through across maturities. Re-solving the exposure fixed point, loadings and holdings for tax changes gives household yield pass-through declining from 1 to 0.230 and lower-tax pass-through rising from 0 to 0.739. Lower-tax pass-through is 0.06 at $T = 2$, a maturity the lower-tax investor does not hold, because its tax changes both investors' net rate exposures, which affect their required returns at every risky maturity. Interval bounds confirm that household pass-through falls and lower-tax pass-through rises at every successive maturity. Their sum ranges from about 0.969 to 1, and the participation threshold remains $T^* = 17$.

Long-end continuation. I continue the price recursions with the solved economy's exposure responses and risk-premium coefficients, holding municipal supply and bond-specific variance at their thirty-period values, $Q^M_\infty = 0.1$ and $v^M_\infty = 0.001024$. This gives $A^M_\infty = 3.325$ and $s^C_\infty = 0.851$. Loading-implied pass-throughs are $(-0.012, 1.013)$ and yield-implied pass-throughs $(-0.006, 0.959)$. The tax-difference condition of Proposition A.12 has left side $0.506 < 1$. Portfolio responses in both markets move both sets of pass-throughs away from the risk-tolerance weights, in the directions that part (i) of that proposition predicts. The yield pass-throughs also include relative risk costs, convexity and bond-specific compensation.

Equilibrium selection. To select each illustration's participation pattern, I enumerate affine configurations and keep a configuration only when its level KKT conditions and clearing, a linear system, have a unique feasible solution. This check applies to each configuration separately, and other feasible cutoffs remain possible (Appendix A.2.8). Full 45-market nonlinear solutions pass the independent grid checks in Appendix B.7.2, and their maturity patterns are numerical results.

A.3 Alternative motives for municipal participation

Lower trading costs or better default diversification can induce long-maturity participation without a liability-hedging advantage (Section 4.7). The pricing mechanism extends to any such motive that is neutral: by itself, it creates no gap relative to the household benchmark when the household holds both bonds. Participation then brings the lower-tax investor's smaller exemption benefit into the return comparison. If municipal and taxable default risks load on separate shocks, the comparison requires subtracting each market's holder-specific default compensation. A municipal-specific benefit has the opposite pricing implication. Throughout, $g_T = \mu^M_T - \xi_H \mu^C_T$ and $\xi_H < \xi_I$.

Proposition A.13 (Participation creates the gap)**.** *In the two-investor affine benchmark without bond-specific risk, suppose every non-tax characteristic other than rate risk is neutral and zero at the one-period claim. Each is a risk common to both markets, independent of rate risk, with municipal exposure ξ_H times the taxable exposure, or a cost that reduces after-tax wealth by the same amount in both markets.*

(i) If the household alone holds every municipal and taxable bond and its shorting constraint on cash is slack, $g_T = 0$ at every maturity.

(ii) Wherever the lower-tax investor holds the maturity-T municipal bond,

$$g_T \;\geq\; \underbrace{(\xi_I - \xi_H) r}_{\text{exemption benefit}} \;+\; \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I \big(A^M_{T-1} - \xi_H A^C_{T-1}\big)}_{\text{rate loading}}.$$

Part (i) is proved in Appendix A.3.2, and part (ii) follows from Proposition A.14. On the branch of Proposition 2 under condition (S) in (72), the rate-loading term is zero at $T \leq 2$ and positive at every $T \geq 3$ when the lower-tax investor's net rate exposure is positive.

A.3.1 Non-tax required returns and neutral characteristics

Add a differentiable concave term $\Phi_i(x_i)$, in the same after-tax units, to investor i's objective in (36). Dividing each KKT condition (37) by its after-tax share gives, under the normalized municipal–taxable benchmark,

$$\mu^M_T \leq \xi_i r + K^M_{i,T}, \qquad \mu^C_T \leq r + K^C_{i,T}, \qquad \text{with equality for a positive holding,} \tag{99}$$

where

$$K^a_{i,T} := \gamma_i \operatorname{Cov}_z\big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\big) + \frac{\kappa_i}{\xi^a_i} - \frac{1}{\xi^a_i} \frac{\partial \Phi_i}{\partial x^a_{i,T}}.$$

With additive shocks, $K^a_{i,T}$ splits into components $k^a_{ij,T}$ for rate risk, bond-specific risk, each additional shock or wealth characteristic, and the shorting constraint on cash. Additional shocks are independent of the rate and bond-specific shocks.

A common return characteristic is a unit-variance shock q with nonnegative maturity-T exposures e^M_T, e^C_T in both markets, taxed with returns. Its component is $k^a_{iq,T} = c_{iq}e^a_T$, where the cost per unit of exposure, $c_{iq} := \gamma_i \operatorname{Cov}_z(q_{t+1}, \widetilde{\Pi}_i(t+1))$, is the same in both markets. Affine rate risk has $q = -\varepsilon$, $e^a_T = \sigma_r A^a_{T-1}$, $c_{iq} = \gamma_i\sigma_r\widehat{D}_i$ and component $\gamma_i\sigma_r^2 A^a_{T-1}\widehat{D}_i$.

A wealth characteristic, such as a trading cost, reduces after-tax wealth by w^a_T per dollar, scaled by investor i's intensity $\varsigma_i \geq 0$. It is the state-independent holding benefit of Appendix A.2.4 with $\vartheta^a_{i,T} = -\varsigma_i w^a_T$. For $\Phi_i = -\varsigma_i \sum_{a,T} w^a_T x^a_{i,T}$, the component is $k^a_{iw,T} = \varsigma_i w^a_T/\xi^a_i$, and the shorting constraint on cash similarly gives κ_i/ξ^a_i.

Definition A.1 (Neutral characteristic)**.** *A common return characteristic is neutral if $e^M_T = \xi_H e^C_T$, and a wealth characteristic if $w^M_T = w^C_T$, at every maturity. For a household that bears it, a neutral characteristic satisfies $k^M_{Hj,T} = \xi_H k^C_{Hj,T}$ for its component j, so it creates no gap when the household holds both bonds.*

Neutrality for each investor. Neutrality is stated for the household because the puzzle is defined against the household benchmark. A characteristic that weighs more on municipal bonds would add a premium even there. A neutral common return characteristic gives $k^M_{iq,T} = c_{iq}e^M_T = c_{iq}\xi_H e^C_T = \xi_H k^C_{iq,T}$ for every investor i. A neutral wealth characteristic, with common cost w_T, gives $k^M_{iw,T} = \varsigma_i w_T = \xi_i\,(\varsigma_i w_T/\xi_i) = \xi_i k^C_{iw,T}$, so

$$k^M_{Iw,T} - \xi_H k^C_{Iw,T} = (\xi_I - \xi_H)k^C_{Iw,T}. \tag{100}$$

Rate risk is neutral in the affine benchmark where the household alone holds both bonds. Clearing fixes the household's exposure, so differentiation gives $s^M_T = \xi_H$ and $s^C_T = 1$. Starting at $A^M_1 = \xi_H$, $A^C_1 = 1$, recursion (53) then gives $A^M_{T-1} = \xi_H A^C_{T-1}$ at every maturity by induction, as in Proposition A.5.

A.3.2 The gap without and with participation

No one-period return gap when the household holds both bonds. Suppose the household alone holds both bonds at every maturity, its shorting constraint on cash is slack, there is no bond-specific risk and every additional characteristic is neutral and zero at the one-period claim. Both household conditions in (99) bind, so $g_T = \sum_j (k^M_{Hj,T} - \xi_H k^C_{Hj,T})$. At risky maturities, the terms κ_H and κ_H/ξ_H from a binding shorting constraint on cash would cancel. Definition A.1 removes the additional components. In the affine benchmark, the neutrality of rate risk makes the rate contribution $\gamma_H\sigma_r^2\widehat{D}_H(A^M_{T-1} - \xi_H A^C_{T-1})$ zero. With a constant cash rate, there is no rate component. At the riskless maturity $T = 1$, the municipal condition gives $\mu^M_1 = \xi_H r$ and cash pays $\mu^C_1 = r$, so $g_1 = 0$. A binding constraint would leave $g_1 = \kappa_H$. Hence

$$g_T = 0 \qquad \text{at every maturity.} \tag{101}$$

The yield gap also includes the convexity difference in (75), so it need not be zero.

Participation comparison for any motive. Evaluate the components at an equilibrium allocation. Both investors' municipal conditions in (99) share the offered return μ^M_T, so the lower-tax investor's required return is no higher than the household's exactly when

$$\Delta_T := K^M_{H,T} - K^M_{I,T} \;\geq\; (\xi_I - \xi_H)\, r. \tag{102}$$

The lower-tax investor can hold the maturity-T municipal bond only if (102) holds, and the household only if the reverse weak inequality holds. Rate risk gives $\Delta_T = \sigma_r^2 A^M_{T-1}\widehat{\Lambda}$, the duration benefit in (59), and Theorem A.1 gives the resulting sorting. Advantages from several motives add. If $r > 0$, $\Delta_1 = 0$ and Δ_T increases strictly in T, the lower-tax investor's municipal holdings form an upper interval of maturities, possibly empty, that excludes the one-period claim, and the household holds every shorter maturity. A strictly increasing Δ_T crosses the maturity-invariant tax difference at most once, and $\Delta_1 = 0 < (\xi_I - \xi_H)r$. Proposition A.15(i) applies this to the three motives below. With bond-specific risk, both equalities give $\Delta_T = (\xi_I - \xi_H)r$ at every jointly held maturity, so Δ_T is not strictly increasing there. For that case, Theorem A.2 uses the index Ψ_T, which adds the household's bond-specific compensation at its full holding, $\gamma_H v^M_T Q^M_T/m_H$.

Proposition A.14 (The gap when the lower-tax investor participates)**.** *In the normalized municipal–taxable benchmark of Appendix A.1.1, with one municipal market, suppose every additional characteristic is a neutral*

common return or wealth characteristic. If the lower-tax investor holds the maturity-T municipal bond at state z, then

$$g_T \;\geq\; (\xi_I - \xi_H)\, r + g_T^{\text{risk}} + (\xi_I - \xi_H)\Big(\sum_w k^C_{Iw,T} + \frac{\kappa_I}{\xi_I}\Big),$$

$$g_T^{\text{risk}} := \gamma_I \Big[\operatorname{Cov}_z\big(u^M_{T,t+1}, \widetilde{\Pi}_I(t+1)\big) - \xi_H \operatorname{Cov}_z\big(u^C_{T,t+1}, \widetilde{\Pi}_I(t+1)\big)\Big],$$

where the covariances run over the rate and bond-specific shocks and w runs over the wealth characteristics, with equality if this investor also holds the maturity-T taxable bond. In the affine benchmark without bond-specific risk, $g_T^{\text{risk}} = \gamma_I \sigma_r^2 \widehat{D}_I (A^M_{T-1} - \xi_H A^C_{T-1})$ and the shorting constraint on cash is slack. With nonnegative costs, this gives Proposition A.13(ii).

Proof. The active municipal condition gives $\mu_T^M = \xi_I r + K^M_{I,T}$. The taxable KKT condition gives $\mu_T^C \leq r + K^C_{I,T}$, with equality when taxable holdings are positive. Subtract ξ_H times this inequality:

$$g_T \;\geq\; (\xi_I - \xi_H) r + \sum_j \big(k^M_{Ij,T} - \xi_H k^C_{Ij,T}\big).$$

Rate and bond-specific components give g_T^{risk}. Neutral common-return components cancel. Neutral wealth costs give $(\xi_I - \xi_H) k^C_{Iw,T}$ by (100). The terms κ_I and κ_I/ξ_I from the shorting constraint on cash give $(\xi_I - \xi_H)\kappa_I/\xi_I$. □

The third term prices wealth costs and a binding shorting constraint on cash at the lower-tax investor's after-tax share. Nonnegative g_T^{risk} and nonnegative costs suffice for $g_T \geq (\xi_I - \xi_H) r$.

A.3.3 Three motives that produce the long-end gap

Proposition A.15 (Three motives that produce the long-end gap)**.** *Let $r > 0$ and consider one of the three cases below. Cases (b) and (c) are set in the two-investor economy with a constant cash rate, no bond-specific risk and a slack shorting constraint on cash, so the stated characteristic is the only non-tax component. In each case the characteristic is zero at the one-period claim. In (b), investors have no outside exposure to the default shocks.*

(a) Liability hedging and risk-bearing capacity: the two-investor affine benchmark at an equilibrium with strictly increasing municipal predecessor loadings, without bond-specific risk for parts (i) and (iv).

(b) Default diversification: municipal default risk loads on one unit-variance shock and taxable default risk on another, with correlation $\varrho \in [0,1]$ and exposures $\xi_H e_T$ and e_T, respectively, where e_T is strictly increasing for $T \geq 1$. Write $X_i^M := \xi_H \sum_T e_T x^M_{i,T}$ and $X_i^C := \xi_i \sum_T e_T x^C_{i,T}$ for investor i's municipal and taxable default exposures, and $k^M_{id,T} := \gamma_i \xi_H e_T (X_i^M + \varrho X_i^C)$ and $k^C_{id,T} := \gamma_i e_T (X_i^C + \varrho X_i^M)$ for its default components.

(c) Trading costs: the same round-trip cost $w_T \geq 0$ in both markets, strictly increasing for $T \geq 1$, and trading intensities $\varsigma_I < \varsigma_H$.

Then:

(i) at every equilibrium allocation, the participation advantage Δ_T in (102) is zero at $T = 1$. If the lower-tax investor holds a municipal bond, Δ_T increases strictly with maturity, and the municipal maturities it holds form an upper interval that excludes the one-period claim;

(ii) $g_1 = 0$: the household anchors the one-period claim;

(iii) at every maturity at which the lower-tax investor holds the municipal bond, $g_T \geq (\xi_I - \xi_H) r + g_T^{\text{risk}}$ in (a), with g_T^{risk} of Proposition A.14, $g_T \geq (\xi_I - \xi_H) r$ in (c), and in (b)

$$g_T = (\xi_I - \xi_H) r + k^M_{Id,T} - \xi_H k^C_{d,T},$$

where $k^C_{d,T}$ is the default component of any investor who holds the taxable bond, so the gap net of these default components is $(\xi_I - \xi_H) r$;

(iv) with the household alone holding both bonds, $g_T = 0$ in (a) and (c), and $g_T = k^M_{Hd,T} - \xi_H k^C_{Hd,T}$ in (b).

Proof. (i) At an equilibrium allocation, (a) gives $\Delta_T = \sigma_r^2 A^M_{T-1} \widehat{\Lambda}$. In (b), define investor i's unit municipal-default cost $c_{id} := \gamma_i (X_i^M + \varrho X_i^C)$, constant across maturities, so that $\Delta_T = \xi_H e_T (c_{Hd} - c_{Id})$. Case (c) gives

$\Delta_T = (\varsigma_H - \varsigma_I)w_T$. Each vanishes at $T = 1$, where the characteristic is zero. If the lower-tax investor holds some maturity T, its equality and the household's inequality give $\Delta_T \geq (\xi_I - \xi_H)r > 0$, so $T \geq 2$, where the loading, A^M_{T-1}, e_T or w_T, is positive, and the unit-cost difference, $\widehat{\Lambda}$, $c_{Hd} - c_{Id}$ or $\varsigma_H - \varsigma_I$, is positive. Strictly increasing loadings, A^M_{T-1}, e_T or w_T, then make Δ_T strictly increasing, so it exceeds $(\xi_I - \xi_H)r$ at every longer maturity. There the household's condition cannot bind, so the lower-tax investor holds every longer maturity, and $\Delta_1 = 0$ excludes the one-period claim.

(ii) Every non-tax component vanishes at the one-period claim, so the municipal conditions reduce to $\mu^M_1 \leq \xi_i r$. Positive supply gives $\mu^M_1 = \xi_H r$, while $\mu^C_1 = r$, as in Proposition A.2.

(iii) Apply Proposition A.14 to (a), with no additional characteristic, and (c), with $g^{\text{risk}}_T = 0$ and cost $(\xi_I - \xi_H)\varsigma_I w_T/\xi_I \geq 0$. In (b), active conditions give $\mu^M_T = \xi_I r + k^M_{Id,T}$ and $\mu^C_T = r + k^C_{d,T}$. Every taxable holder's equality has the cash term r and no other non-tax component, so every taxable holder has the same default component. Subtraction proves the expression.

(iv) In (a) and (c) this is (101). In (b), the household's conditions give $\mu^M_T = \xi_H r + k^M_{Hd,T}$ and $\mu^C_T = r + k^C_{Hd,T}$. □

The three cases. In (a), $\widehat{\Lambda}$ is positive when liability hedges or lower risk aversion reduce the lower-tax investor's unit cost of rate risk. In (b), the lower-tax investor participates at maturities where its saving on default compensation, $\xi_H e_T(c_{Hd} - c_{Id})$, covers the difference in exemption benefits. Its taxable holdings add to that cost only through ϱ, so it can take on municipal default risk more cheaply than households, which bear the default risk of the municipal bonds they hold. Raw gaps also contain holder-specific default-premium differences driven by relative risk supplies, which Appendix C.1.6 adjusts with credit default swap (CDS) spreads. In (c), trading costs reduce after-tax wealth (Appendix C.1.5). For matched long bonds at 26 to 30 years, institutional round-trip costs are 0.53% of price in both markets (Table 18), supporting equal costs. With combined motives, the decomposition of Proposition A.14 still applies, and its bound $g_T \geq (\xi_I - \xi_H)r$ holds whenever g^{risk}_T is nonnegative at the resulting equilibrium. Separate default shocks are treated as in (b).

A.3.4 Two solved examples without a duration motive

Consider two investors and three maturities with no bond-specific risk and a constant $r > 0$, which eliminates rate-risk advantages. Both investors have risk aversion γ, masses m_H, m_I, after-tax shares $\xi_H < \xi_I$ and slack shorting constraints on cash. Supplies are Q^M_T for $T = 1, 2, 3$ and Q^C_T for $T = 2, 3$. Write $M1$, $M2$, $M3$ for the municipal bonds and $C2$, $C3$ for the longer taxable bonds. Each example adds only its stated non-tax characteristic.

Proposition A.16 (Participation for diversification or lower trading costs)**.** *Consider the economy just described.*

(a) Default diversification. Municipal and taxable default risk load on independent unit-variance shocks, with exposures $\xi_H e_T$ and e_T and $0 = e_1 < e_2 < e_3$. Let $X^M_H := \xi_H e_2 Q^M_2/m_H$ and $X^M_I := \xi_H e_3 Q^M_3/m_I$ be the investors' municipal default exposures when the household holds $M1$ and $M2$ and the lower-tax investor holds $M3$, and let

$$X^C := \frac{e_2 Q^C_2 + e_3 Q^C_3}{m_H/\xi_H + m_I/\xi_I}.$$

There is an equilibrium with this municipal allocation if and only if

$$\gamma\,\xi_H e_2(X^M_H - X^M_I) \;\leq\; (\xi_I - \xi_H)r \;\leq\; \gamma\,\xi_H e_3(X^M_H - X^M_I).$$

Both investors then hold taxable bonds, each with taxable default exposure X^C, and

$$g_1 = 0, \qquad g_2 = \gamma\,\xi_H e_2(X^M_H - X^C), \qquad g_3 = (\xi_I - \xi_H)r + \gamma\,\xi_H e_3(X^M_I - X^C).$$

Net of the default components of the investors who hold each bond, the gaps are 0, 0 and $(\xi_I - \xi_H)r$.

(b) Trading costs. Each investor pays the after-tax cost $\varsigma_i w_T$ per period on either bond of maturity T, with $0 = w_1 < w_2 < w_3$ and $0 \leq \varsigma_I < \varsigma_H$. If $(\varsigma_H - \varsigma_I)w_2 < (\xi_I - \xi_H)r \leq (\varsigma_H - \varsigma_I)w_3$, there is an equilibrium in which the household holds $M1$ and $M2$, the lower-tax investor holds $M3$, $C2$ and $C3$, and

$$g_1 = 0, \qquad g_2 = \Big(\varsigma_H - \frac{\xi_H}{\xi_I}\varsigma_I\Big)w_2, \qquad g_3 = (\xi_I - \xi_H)\Big(r + \frac{\varsigma_I w_3}{\xi_I}\Big),$$

with $0 < g_2 < g_3$. The ratio μ_T^M/μ_T^C equals ξ_H at $T = 1$, lies strictly between ξ_H and ξ_I at $T = 2$ and equals ξ_I at $T = 3$.

Proof. (a) Independent default shocks separate market risks. Investor i's KKT conditions are $\mu_T^M \le \xi_i r + \gamma\xi_H e_T X_i^M$ and $\mu_T^C \le r + \gamma e_T X_i^C$, for municipal and taxable exposures X_i^M, X_i^C. Lower taxable exposure implies lower required return and profitable purchases, so both investors hold taxable bonds with equal exposures. Clearing $\sum_i m_i X_i^C/\xi_i = e_2 Q_2^C + e_3 Q_3^C$ gives $X_H^C = X_I^C = X^C$ and $\mu_T^C = r + \gamma e_T X^C$. Supply-proportional holdings, one of possibly several splits across the two maturities, implement these exposures and clear both maturities.

Active municipal conditions give $\mu_1^M = \xi_H r$, $\mu_2^M = \xi_H r + \gamma\xi_H e_2 X_H^M$ and $\mu_3^M = \xi_I r + \gamma\xi_H e_3 X_I^M$. The inactive investors' KKT conditions require $\xi_H r < \xi_I r$ at $M1$, the left threshold inequality at $M2$ and the right at $M3$.

Subtracting returns gives the gaps. Remove the holder-specific municipal component $\gamma\xi_H e_T X_i^M$ and ξ_H times the taxable component $\gamma e_T X^C$ to leave $\xi_H r - \xi_H r$ at $T = 2$ and $\xi_I r - \xi_H r$ at $T = 3$.

(b) Investor i requires $\xi_i r + \varsigma_i w_T$ on municipal bonds and $r + \varsigma_i w_T/\xi_i$ on taxable bonds. With slack shorting constraints on cash, each bond's return equals the minimum eligible requirement. Since $\varsigma_I/\xi_I < \varsigma_H/\xi_H$, the lower-tax investor holds all taxable bonds. It holds the maturity-T municipal bond when $\xi_I r + \varsigma_I w_T \le \xi_H r + \varsigma_H w_T$, here only at $T = 3$. Subtract returns to obtain the gaps and

$$g_2 = (\varsigma_H - \varsigma_I)w_2 + (\xi_I - \xi_H)\frac{\varsigma_I w_2}{\xi_I} < (\xi_I - \xi_H)\Big(r + \frac{\varsigma_I w_2}{\xi_I}\Big) \le g_3.$$

Hence $g_2 > 0$, so the $T = 2$ ratio exceeds ξ_H, and it is below ξ_I because the household's required municipal return is below the lower-tax investor's. At $T = 3$ it is $(\xi_I r + \varsigma_I w_3)/(r + \varsigma_I w_3/\xi_I) = \xi_I$. □

Case (a) of Proposition A.16 implements default diversification (Proposition A.15), requiring $X_H^M > X_I^M$. The intermediate raw gap $\gamma\xi_H e_2(X_H^M - X^C)$ is a default-premium difference. With $\varrho \in (0, 1)$, the same allocation, where it is an equilibrium, has default-net gaps 0, 0 and $(\xi_I - \xi_H)r$ (Proposition A.15(b)). Case (b)'s intermediate gap reflects higher household trading costs, independent of other holdings.

A.3.5 A municipal-specific benefit

A municipal-specific benefit $\zeta_T > 0$, from a mandate, issuer relationship or favorable regulation at matched maturity and credit, lowers the required municipal return without changing the taxable alternative and so violates neutrality. For the lower-tax investor, it gives the participation advantage $\Delta_T = \zeta_T$ but reduces the return gap by ζ_T. A benefit that grows with maturity can thus induce long-end participation while lowering municipal returns. Unlike default compensation, it has no taxable counterpart to subtract. A preference for municipal bonds as such is a benefit of this kind, so by itself it does not explain yields above the household benchmark.

Proposition A.17 (Same participation, opposite pricing)**.** *Consider the two-investor economy without rate risk or bond-specific risk, with a constant cash rate $r > 0$, a slack shorting constraint on cash and no additional household characteristic. A fixed per-dollar benefit lowers the lower-tax investor's required municipal return by $\zeta_T \ge 0$, with $\zeta_1 = 0$, ζ_T nondecreasing and $\zeta_T < \xi_H r$. Compare two benefits. A neutral benefit is paid and taxed with returns: ζ_T on the municipal bond and ζ_T/ξ_H before tax on the taxable bond. It also lowers the investor's required taxable return by the household-equivalent ζ_T/ξ_H. A municipal-specific benefit leaves the required taxable return unchanged.*

(i) In both cases the lower-tax investor holds the maturity-T municipal bond only if $\zeta_T \ge (\xi_I - \xi_H)r$, so participation begins at the same maturity.

(ii) With the neutral benefit, $\mu_T^C = r - \zeta_T/\xi_H$ and

$$g_T = \min\{\zeta_T,\ (\xi_I - \xi_H)r\} \ge 0,$$

so the gap equals the full difference in exemption benefits wherever the lower-tax investor participates. The ratio μ_T^M/μ_T^C rises with ζ_T.

(iii) With the municipal-specific benefit, $\mu_T^C = r$ and

$$g_T = \min\{0,\ (\xi_I - \xi_H)r - \zeta_T\} \le 0,$$

and the ratio $\mu_T^M/\mu_T^C = \min\{\xi_H,\ \xi_I - \zeta_T/r\}$ is weakly below ξ_H where the lower-tax investor participates and strictly below it wherever $\zeta_T > (\xi_I - \xi_H)r$.

Because the rate is constant, each yield averages the one-period returns $\mu_1^a, \ldots, \mu_T^a$. The yield gap $y_T^M - \xi_H y_T^C$ therefore has the sign of the g_j, and the yield-implied tax is nonincreasing in maturity in part (ii) and nondecreasing in part (iii).

Proof. As in Proposition A.16(b), minimum required returns determine holdings. The household requires $\xi_H r$ and r. The lower-tax investor requires $\xi_I r - \zeta_T$ and either $r - \zeta_T/\xi_H$ or r. This proves (i).

With neutrality, $g_T = \mu_T^M - \xi_H r + \zeta_T$ equals ζ_T when $\xi_H r$ is minimal, otherwise $(\xi_I - \xi_H)r$. The ratios $\xi_H r/(r - \zeta_T/\xi_H)$ below the threshold and $(\xi_I r - \zeta_T)/(r - \zeta_T/\xi_H)$ above increase in ζ_T since $\xi_I > \xi_H$. A municipal-specific benefit gives $g_T = \mu_T^M - \xi_H r$, with $\mu_T^M = \min\{\xi_H r,\ \xi_I r - \zeta_T\}$.

Since ζ_T is nondecreasing, μ_j^M/μ_j^C is nondecreasing in j in (ii) and nonincreasing in (iii). Adding a maturity whose ratio is at least (in (iii), at most) the running yield ratio cannot lower (raise) that ratio, which proves the yield statements. □

A.4 State-specific exemptions across municipal markets

The quantitative model of Section 5 has separate state municipal markets. This multi-state model gives households a home-state municipal exemption. A household resident in state s' pays its home state's income tax on interest from a bond issued by another state $s \neq s'$, while home-state municipal interest is exempt. Appendix D.2 distinguishes this common model convention from states' statutory exceptions. Institutional types have no modeled state-income-tax wedge on either bond, so state taxation is a household-specific channel. Statutory top state rates vary across states, with a supply-weighted mean of 5.8%. This cross-state variation is both a calibration input and P1's source of identification.

What changes. The multi-state model replaces ξ_i with $\xi_{i,s} = \xi_i^C/\xi_{i,s}^M$, which depends on the issuing state s and on the investor, whose type includes a household's state of residence. Here $\xi_{i,s}^M$ is investor i's after-tax share of state s's municipal interest. Each investor uses one payoff normalization across all markets, with after-tax shares ξ_i^a as in (33). Its net rate exposure in these units is $\widehat{D}_i = \sum_{a,T} \xi_i^a x_{i,T}^a A_{T-1}^a - \ell_i$, which reduces to (50) when the municipal after-tax shares all equal one.

What carries over. For state s's market M_s, put $\Gamma_T^{M_s} := \sum_{i \in \mathcal{U}_T^{M_s}} m_i/\gamma_i$ and $\omega_{i,T}^s := (m_i/\gamma_i)/\Gamma_T^{M_s}$, where $\mathcal{U}_T^{M_s}$ is the set of holders of the maturity-T bond. With slack shorting constraints on cash, the exact KKT aggregation (65) applies market by market, with $\xi_{ij} = \xi_{i,s}$. Applying (67) coordinate by coordinate gives, in the affine construction,

$$\begin{aligned} s_T^{M_s} = &\sum_{i \in \mathcal{U}_T^{M_s}} \omega_{i,T}^s \xi_{i,s} + \frac{\sigma_r^2 A_{T-1}^{M_s}}{\Gamma_T^{M_s}} \sum_{i \in \mathcal{U}_T^{M_s}} m_i \partial_z \widehat{D}_i \\ &+ \frac{v_T^{M_s}}{\Gamma_T^{M_s}} \sum_{i \in \mathcal{U}_T^{M_s}} m_i \xi_{i,s}^M \partial_z x_{i,T}^{M_s}. \end{aligned} \tag{103}$$

The final term vanishes under common active municipal after-tax shares, a single active holder or zero bond-specific variance, as for the taxable counterpart before (68). Unweighted clearing alone does not remove it.

The state-tax anchor and pass-through. Proposition A.18 is the state-tax counterpart of Proposition A.9. Its implied state tax is the model counterpart of (25), and T_s^* is the first maturity at which investors other than the resident household hold state s's bonds.

Proposition A.18 (State-tax anchor and tax-mix pass-through)**.** *If the household H_s resident in state s uniquely minimizes $\xi_{i,s}$, its shorting constraint on cash is slack and the one-period municipal boundary is deterministic with $r > 0$, then $y_1^{M_s}/y_1^C = \xi_{H_s,s}$. Under the multiplicative mapping $\xi_{H_s,s} = \xi^{\text{fed}}(1 - \tau_s^{\text{st}})$, with federal after-tax share ξ^{fed} and state income-tax rate τ_s^{st}, define*

$$\tau_{s,T}^{\text{st,impl}} = 1 - \frac{y_T^{M_s}/y_T^C}{\xi^{\text{fed}}}, \qquad \tau_{s,T}^{A,\text{st}} = 1 - \frac{A_T^{M_s}/A_T^C}{\xi^{\text{fed}}}. \tag{104}$$

Then $\tau_{s,1}^{\text{st,impl}} = \tau_s^{\text{st}}$, with unit pass-through. Switch off exposure responses in the affine recursions, and suppose only the resident household's effective after-tax share changes, with its active weight one below T_s^* and constant

at $\omega^s_{H_s}$ thereafter. Then the tax-mix derivative of the loading-implied state tax $\tau^{A,\mathrm{st}}_{s,T}$ with respect to τ^{st}_s is $\Theta^s_S(T)+\omega^s_{H_s}\Theta^s_L(T)$, where Θ^s_S,Θ^s_L are the weights in (85) with T^*_s in place of T^*.

Proof. With this household's constraint slack, (39) gives the boundary return $\min_i \xi_{i,s} r$, as in Proposition A.2. Equation (104) gives the anchor. Applying (85) with T^*_s in place of T^* gives the tax-mix derivative, because $\partial \xi_{H_s,s}/\partial \tau^{\mathrm{st}}_s = -\xi^{\mathrm{fed}}$ cancels the factor $1/\xi^{\mathrm{fed}}$ in (104). □

Changes to out-of-state after-tax shares or other clienteles add their derivatives. Other statutory mappings require a corresponding implied-tax transformation, and the primitive boundary remains $y_1^{M_s}/y_1^C = \xi_{H_s,s}$. Reported state-tax responses use the full multi-state model, which supplies P1's full-yield pass-through profile, including level and exposure corrections.

A.5 Participation and tax pass-through with multiple investors

To allow for more clienteles, as in the six investor groups of the quantitative model, extend the two-investor affine benchmark of Appendix A.2 to N investor types with slack shorting constraints on cash. Order after-tax shares $\xi_1 < \cdots < \xi_N$, so type 1 is the household with the largest exemption benefit. Without bond-specific risk, compare type $j>1$'s municipal condition with type 1's at an equilibrium allocation, as in Appendix A.2.3. Type j's required return on the maturity-T municipal bond is no higher than type 1's exactly when

$$\underbrace{\sigma_r^2 A^M_{T-1}\widehat{\Lambda}_j}_{\text{duration benefit}} \;\geq\; \underbrace{(\xi_j-\xi_1)r}_{\text{difference in exemption benefits}}, \qquad \widehat{\Lambda}_j := \gamma_1\widehat{D}_1 - \gamma_j\widehat{D}_j. \tag{105}$$

As in (59), net rate exposures are common to all maturities, so the duration benefit grows with the bond's rate loading, while the difference in exemption benefits is flat in maturity. Without bond-specific risk, type j requires $\xi_j r+\sigma_r^2 A^M_{T-1}\gamma_j\widehat{D}_j$, which is affine in the predecessor loading with a slope common to all maturities, and holders attain the lowest requirement. When predecessor loadings increase strictly with maturity, each participating type holds an interval of maturities, types with lower $\gamma_j\widehat{D}_j$ hold longer maturities, and two types share at most one maturity, as in Theorem A.1. Positive bond-specific risk adds the incumbent's bond-specific compensation to the comparison, as in the KKT system (47) and the participation index (89), and lets several clienteles hold the same maturities.

Assumption A.1 (Long-end participation, clientele by clientele)**.** *Bond-specific risk is positive, and type 1 is active at every maturity, so its first active maturity is $T^*_1=1$. For each type $j>1$, $\widehat{\Lambda}_j>0$, and its active municipal set is an upper maturity interval $[T^*_j,T_{\max}]$ with $T^*_j\geq 2$. The thresholds need not be ordered by tax advantage.*

Many-clientele tax-mix pass-through. To isolate the tax mix under the nested holding intervals of Assumption A.1, switch off exposure-response terms and fix each maturity's holders and risk-tolerance weights, as in Proposition A.9. Let $\omega_{n,j}$ be type n's share of the holders' risk tolerance at municipal maturity j, zero when it does not hold that bond. Then $A^C_T = G_T$ and $A^M_T = \sum_{j\leq T}\rho^{T-j}\sum_n \omega_{n,j}\xi_n$, where $G_T=\sum_{j=1}^T\rho^{T-j}$. Differentiating the loading-implied tax $\tau^A_T = 1-A^M_T/G_T$ in ξ_i gives its tax-mix pass-through $-\partial\tau^A_T/\partial\xi_i$,

$$\mathcal{D}_i(T)=\frac{\sum_{j=T^*_i}^T \rho^{T-j}\omega_{i,j}}{\sum_{j=1}^T\rho^{T-j}}, \tag{106}$$

with an empty numerator equal to zero. It is positive after participation begins and converges to ω_i if the active weight is eventually constant at ω_i, because the finitely many maturities before the active weight becomes constant receive vanishing weight at the long end.

Participation in order of increasing after-tax shares makes the tax mix nondecreasing in maturity: adding an after-tax share no smaller than the incumbents' does not lower their risk-tolerance-weighted mean. Such ordered participation does not by itself order total return responses, loading ratios or yields, since other terms in (67) and the taxable leg remain unrestricted. Clientele-specific pass-through and endpoint comparisons need no tax ordering.

Under Assumption A.1, the household anchors the riskless boundary, and (106) gives each clientele's tax-mix derivative. Full yield responses also include relative compensation and convexity (Appendix A.2.5). The portfolio-risk response can transmit a later clientele's tax to shorter risky bonds it never holds. P1 concerns the always-active household, and P2 a later-participating clientele, whose tax-mix pass-through is zero below its threshold. Assumption A.1 restricts only this affine extension. The calibration does not impose it (Section 5).

A.6 Applications to other bond markets

As the conclusion argues, the mechanism operates beyond municipal bonds. It requires heterogeneous relative taxes, a shorting friction and a neutral participation motive that grows with maturity. Common rate risk supplies such a motive and also carries participation into rate loadings. For a tax-favored bond and a comparable fully taxed bond, $\xi_i = (1 - \tau_i^{\text{tax}})/(1 - \tau_i^{\text{fav}})$, with investor i's tax rates on the taxed and favored bonds. Symmetric taxation gives $\xi_i = 1$ and removes the relative tax incentive. When after-tax shares differ, the response decomposition (65) applies. Positive taxable forwards and an increasing relative forward ratio imply a declining yield-implied tax (Proposition A.7). Condition (S) in (72) orders the affine loading ratio (Proposition A.6).

Applications include:

- state-exempt treasury versus corporate interest
- state-exempt Federal Home Loan Bank and Farm Credit issues versus Fannie Mae and Freddie Mac issues
- treaty-dependent withholding on cross-border sovereign debt
- the inflation accrual on treasury inflation-protected securities (TIPS), taxed before payment in taxable accounts
- taxable accounts, individual retirement accounts (IRAs) and pensions, which assign different after-tax payoffs to the same security at one market price

In each application, the maturity profile depends on which clienteles hold the bonds. Municipal bonds, the central application, combine a large federal wedge with tight shorting restrictions.

B Quantitative model, solution and calibration

This appendix explains the quantitative model, how I solve it and how I estimate the coefficients that are not directly measured. Measured taxes, portfolio assets, funding, bond supplies, household balance sheets and rate dynamics are fixed before estimation. Seventeen common coefficients are fitted to twenty-two ownership, insurer-portfolio and term-spread conditions, with municipal yields left untargeted. One model period is two years, and calendar maturities are stated in years. Later subsections show that the thirty-year gap changes little when the mean of the bond-specific shock is normalized, when bond supply responds to yields and, in 2015–2017, when the grid or barrier changes.

B.1 Markets, investors and measured inputs

Markets and maturities. The application has 45 municipal markets and one taxable market. Each of the 43 states with a personal income tax and the District of Columbia has its own municipal market. The seven states without one (Alaska, Florida, Nevada, South Dakota, Texas, Washington and Wyoming) share a market because investors treat their bonds alike in the model. Each municipal market has fifteen maturities in two-year steps from two to thirty years. The taxable market has fourteen risky maturities, four through thirty years, and cash is its one-period claim. Each rate state therefore has $45 \times 15 + 14 = 689$ bond positions to clear. Cash and two-year municipal bonds are riskless. The taxable market pools treasury and corporate supply, is taxed like corporate interest and has its yields compared with the corporate curve.

Time periods and the one-market estimation model. Measured inputs vary across six three-year periods, 2000–2002 through 2015–2017, centered on the 2001, 2004, 2007, 2010, 2013 and 2016 Survey of Consumer Finances (SCF) waves. One set of estimated coefficients applies to all six periods. For tractability, estimation uses a smaller one-market model with one national municipal market, in which household tax treatment is averaged across states with municipal supply weights. This model has twenty-seven investor types and twenty-nine bond positions. After estimation, I restore all 45 municipal markets and solve again with the coefficients fixed.

Investor groups, portfolio assets and funding. The six investor groups are high-tax households, lower-tax households, property-and-casualty (P&C) insurers, life insurers, banks and a tax-neutral residual group. High-tax households face a federal marginal rate of at least 35% under the law of the SCF year. Table 3 reports the six groups' portfolio assets and gross exemption benefits in 2015–2017. For type i, baseline portfolio assets m_i are its debt securities and cash in the Financial Accounts. A measured fraction W_i is available for the modeled investments, giving funding $m_i W_i$. These investments are municipal bonds, corporate bonds excluding

asset-backed securities, treasury notes and bonds, and cash excluding bank cash. Positions x_i are measured per dollar of portfolio assets, so aggregate bond holdings are $m_i x_i$ and cash is $m_i x_i^0$, where $x_i^0 = W_i - \mathbf{1}'x_i$. The amount m_i scales the investor's risk-bearing capacity, and W_i limits the fraction it invests in these markets. Neither is chosen to fit yields.

Outside rate exposure. Each type has an outside rate exposure $h_i = -\ell_i$, the negative of the liability hedge of Section 4, measured per dollar of portfolio assets on the same scale as a bond's rate loading. It summarizes how the rest of the balance sheet changes in value when rates move. Positive h_i adds to the losses on the bond portfolio when rates rise. Negative h_i offsets those losses, as when rising rates reduce the value of liabilities. Balance-sheet amounts are measured, and the coefficients that translate them into exposure are estimated (Appendix B.3).

Investor types. Investors with the same tax treatment can hold different portfolios because their other assets and liabilities expose them differently to rates. Types within each group represent this variation. Their asset shares divide each group's measured portfolio assets, so the types add no assets, and all types in a sector share the measured funding fraction W_i. The definitions below give $45 \times 2 \times 7 + 2 + 3 + 1 + 7 = 643$ types in the full 45-market model.

Households. High- and lower-tax households each have seven balance-sheet types in every municipal market. Each SCF wave supplies tax-unit income, filing status, outside financial assets and debt. Within each tax group, I sort households by outside assets less debt relative to their portfolio base of deposits, bonds and bond funds, and form seven bins, each holding about one seventh of the group's portfolio assets. Let Out_k and $Debt_k$ be bin k's outside-asset and debt ratios. Its outside rate exposure is

$$h_k = a_H + \frac{b_A Out_k - b_D Debt_k}{2}.$$

The intercept a_H gives common exposure, b_A measures the risk added by outside assets and b_D the hedge from debt. The factor of one half expresses the coefficients in the model's two-year units. These three coefficients are estimated jointly and apply to both tax groups. The balance-sheet ratios and bin weights are measured, while dated federal and state tax rules determine each group's after-tax shares. SCF direct and fund holdings divide household municipal ownership between the two tax groups. The national household composition is repeated across states, with portfolio assets allocated by municipal supply shares.

Property-and-casualty insurers. Two types share the sector's tax treatment but differ in outside rate exposure. Their exposures are a_P and $a_P + d_P$, and their shares of the sector's portfolio assets are w_P and $1 - w_P$. Thus a_P sets the lower exposure, d_P the difference between types and w_P their relative size. These three coefficients are estimated, while the Financial Accounts fix total sector portfolio assets and funding. Schedule D filings with the National Association of Insurance Commissioners (NAIC) supply the municipal and taxable portfolio composition and maturity moments used in estimation. I retain positive-value positions, match their tax status and remaining maturity, and construct cross-sectional moments separately each year before averaging equally within each period. Statutory proration rules determine tax treatment.

Life insurers. Three types have outside exposures a_L, $a_L + t_L d_L$ and $a_L + d_L$. Their portfolio-asset shares are w_L, $(1 - w_L)u_L$ and $(1 - w_L)(1 - u_L)$. The coefficients a_L and d_L set the exposure range, t_L locates the middle type within it, and w_L and u_L divide the sector's assets across types. All five are estimated. Financial Accounts stocks fix the sector total and funding, NAIC Schedule D supplies portfolio moments constructed in the same way as for P&C insurers, and statutory life-insurer proration fixes tax treatment. The types therefore share the sector's tax treatment and funding fraction and differ in outside rate exposure and portfolio-asset share.

Banks. One type represents the sector, with a single estimated outside exposure h_B. Financial Accounts stocks fix portfolio assets and funding. In addition to statutory corporate taxes, banks incur the tax cost of disallowed interest deductions on municipal holdings. Call Reports supply interest expense divided by assets, which I average across banks using municipal-holdings weights. Multiplying this ratio by the statutory corporate tax rate gives the financing-tax cost under the calibration's non-bank-qualified treatment. This cost is fixed before calibration, separately from the estimated exposure.

Tax-neutral residual investors. The residual group consists mainly of foreign investors, pensions, governments, money market funds and nonprofits. Seven types approximate a normal distribution of outside

Table 7: Measured household outside assets and debt by balance-sheet type

Bin	2000–02	2003–05	2006–08	2009–11	2012–14	2015–17
High-tax households						
1	0.93, 3.17	1.17, 4.21	0.83, 3.00	0.77, 2.85	0.63, 3.23	0.63, 2.93
2	0.39, 0.23	0.10, 0.07	0.19, 0.10	0.22, 0.18	0.26, 0.15	0.12, 0.08
3	0.62, 0.12	0.43, 0.12	0.45, 0.07	0.25, 0.06	0.57, 0.11	0.35, 0.12
4	1.18, 0.24	0.84, 0.12	0.91, 0.14	0.60, 0.11	0.87, 0.15	0.70, 0.05
5	1.98, 0.26	1.53, 0.26	1.35, 0.11	1.15, 0.19	1.46, 0.25	1.32, 0.16
6	3.77, 0.29	2.55, 0.38	2.19, 0.28	1.80, 0.16	2.50, 0.30	2.77, 0.23
7	18.49, 1.55	12.06, 1.28	10.56, 0.90	7.76, 0.81	11.08, 0.73	12.12, 0.94
Lower-tax households						
1	1.42, 11.55	1.03, 12.34	1.32, 16.95	0.90, 12.88	0.75, 14.31	0.72, 13.74
2	0.30, 0.65	0.35, 0.69	0.39, 1.20	0.16, 0.26	0.31, 0.87	0.23, 0.88
3	0.06, 0.04	0.04, 0.02	0.08, 0.08	0.05, 0.02	0.01, 0.01	0.01, 0.02
4	0.30, 0.10	0.28, 0.12	0.19, 0.07	0.32, 0.12	0.17, 0.07	0.38, 0.18
5	0.78, 0.15	0.71, 0.16	0.63, 0.14	0.58, 0.08	0.71, 0.13	0.71, 0.10
6	1.73, 0.29	1.40, 0.19	1.50, 0.29	1.25, 0.19	1.57, 0.17	1.53, 0.15
7	12.82, 1.26	8.93, 0.91	9.85, 0.96	6.59, 0.71	8.70, 0.76	10.19, 0.80

Notes. Each cell gives two ratios, in order: outside financial assets and debt, each divided by the portfolio base. These are Out_k and $Debt_k$ in the household exposure formula of Appendix B.1. Outside financial assets exclude retirement accounts and the portfolio base of deposits, direct bonds and bond funds. Within each tax group, bins are ordered by outside assets less debt relative to the portfolio base and contain about one seventh of the group's portfolio assets. SCF waves 2001, 2004, 2007, 2010, 2013 and 2016 supply the six periods. These ratios are measured inputs. The coefficients a_H, b_A, b_D translating them into rate exposure are estimated jointly.

exposures, $h_k = \mu_R + s_R\zeta_k$, where ζ_k are fixed standardized normal quadrature nodes. The mean μ_R and dispersion s_R are estimated, while the quadrature weights are fixed. Those weights divide the Financial Accounts sector total across types. Neither the total portfolio assets nor funding is fitted, and all types have the same tax-neutral treatment.

Household balance sheets. In 2015–2017, high-tax households are about 5% of families and hold 46% of household portfolio assets. Outside financial assets exclude retirement accounts and the portfolio base. Table 7 reports the outside-asset and debt ratios used in the exposure formula, each formed at the tax-unit level before bin averaging.

Household tax treatment. Statutory rules determine after-tax shares. Household calculations use each SCF tax unit's income and filing status and include the net investment income tax and the additional tax on Social Security benefits induced by interest income.[24] Federal and state after-tax products are formed at the tax-unit level before aggregation. High-tax households face the home state's top rate, and lower-tax households face tax-unit bracket rates. The model applies a common home-state exemption in all income-tax jurisdictions and taxes out-of-state municipal interest at the resident state's rate. The baseline multiplies federal and state shares for taxable interest and includes no federal deduction of state tax on out-of-state municipal interest. Appendix D.2.1 varies this convention. The Social Security component on exempt interest is approximated by the tax response to a small ordinary-income increment less the tax on that increment. This component can make the absolute home-municipal after-tax share slightly below one.

Institutional tax treatment. Institutions have no modeled state-income-tax wedge and choose across all markets. Independent bond-specific shocks in the state municipal markets let them diversify geographically. Before 2018, institutions face a 35% corporate rate. P&C proration taxes 15% of municipal interest at that rate, giving relative after-tax share $\xi_P = 0.65/0.9475 = 0.686$. Life-insurer proration taxes one half of municipal interest, giving $\xi_L = 0.65/0.825 = 0.788$. Banks have $\xi_B = 0.65$ and also incur the tax cost of disallowed financing-interest deductions on municipal holdings. Under the calibration's non-bank-qualified treatment (Appendix D.1), this annual cost falls from 1.02% in 2000–2002 to 0.11% in 2015–2017. The residual group has $\xi = 1$.

[24] For 209 of 792 state-years, missing lower-tax filing-status cutoffs or related bracket rules are imputed from nearest-year thresholds or simple proxies. Published rates and top rates are unchanged.

Table 8: Inputs fixed from data in each calibration period

	2000–02	2003–05	2006–08	2009–11	2012–14	2015–17
Panel A. Group portfolio assets m_g (trillions of current dollars)						
High-tax households	2.02	1.97	3.47	3.82	5.72	6.97
Lower-tax households	3.99	5.94	6.50	8.44	7.97	8.09
P&C insurers	0.58	0.80	0.91	1.00	1.06	1.11
Life insurers	1.56	2.11	2.10	2.52	2.84	3.05
Banks	1.50	1.89	2.38	3.28	4.21	4.83
Tax-neutral residual	11.54	14.94	20.04	21.42	23.29	26.47
Panel B. Funding as a share of portfolio assets, $100W_g$						
Households (both tax groups)	86.9	88.0	86.2	90.6	91.1	91.4
P&C insurers	73.9	71.3	71.2	73.5	73.3	72.6
Life insurers	73.2	68.9	66.4	69.7	69.8	71.5
Banks	30.1	25.6	24.4	24.4	25.2	28.0
Tax-neutral residual	50.9	52.0	50.1	57.9	68.8	72.5
Panel C. Tax inputs and national bond supplies						
High-tax household ξ_i	0.577	0.609	0.610	0.606	0.565	0.538
Lower-tax household ξ_i	0.766	0.771	0.760	0.764	0.786	0.761
Bank financing-tax cost (%/year)	1.020	0.556	0.858	0.323	0.134	0.108
Municipal supply ($ trillion)	1.08	1.92	2.36	2.78	2.99	3.14
Taxable supply ($ trillion)	6.18	7.98	9.74	13.72	17.39	20.02

Notes. In Panel A, m_g is group g's total of type-level portfolio assets m_i, its measured debt securities and cash. Panel B is the measured funding for modeled bonds and cash divided by those portfolio assets. Dollar funding is $m_g W_g$. For example, banks' 2015–2017 funding is \$4.83 trillion times 0.280, or \$1.35 trillion. The same W_g applies to every type within a sector and to both household tax groups. Panel C's household shares are the issuer-supply-weighted averages used in the one-market estimation. The full 45-market model retains each state's tax shares. Institutional relative after-tax shares are constant before 2018: 0.686 for P&C, 0.788 for life insurers, 0.650 for banks and 1 for the residual group. Banks' financing-tax cost is additional to this relative share. Stocks are mean end-year values in each period. Taxable supply includes treasury and corporate debt. Sources and adjustments are described in Appendix B.1.

Data sources. Portfolio assets, funding and aggregate ownership are from the Financial Accounts. I assign fund holdings to their owners, separate nonprofits and adjust sector stocks for tax-exempt holdings. Ownership is measured from stocks of holdings without using tax-exempt interest flows. Mergent's dated outstanding amounts of tax-exempt bonds give municipal supply by maturity and issuer state. Financial Accounts treasury and corporate totals give taxable supply, with corporate maturity shares from the Mergent Fixed Income Securities Database (FISD) and treasury maturities from the Monthly Statement of the Public Debt. Table 8 reports the resulting quantities.

Rate process. The annual three-month treasury bill rate for 1952–2024 gives persistence 0.854 and an innovation standard deviation of 1.63 percentage points, which become $\rho = 0.729151$ and $\sigma_r = 0.042778$ over two-year periods.[25] The A-or-above corporate zero curve sets the mean rate and the observed monthly states. The mean $\bar{r} = 0.060412$ is twice the 2000–2017 mean annualized two-year corporate zero yield, and the short rate is $r(z) = \max\{\bar{r} + z, 0\}$. Each month's observed state is twice that month's continuously compounded two-year corporate zero yield, less $\bar{r}$. Long yields therefore do not determine the rate states. The same curve supplies the ten-, twenty- and thirty-year yield conditions in annual log units.

The 2018–2020 extension. For the post-2017 curve comparison, I keep all estimated coefficients fixed and update measured inputs using the 2019 SCF composition, annually inflation-adjusted incomes and mean 2018–2020 end-year stocks. Institutions face the 21% corporate rate, P&C proration of 25% and a life-insurer policyholders' share of 30%. The annual bank financing-tax cost is 0.117%, and insurer portfolio fractions use available 2018–2019 holdings. This seventh period adds no calibration targets.

[25] Annual persistence is 0.853903 and the innovation standard deviation is 0.016266. The two-year state is twice the annualized rate deviation, so ρ is the square of annual persistence and σ_r is $2\sqrt{1+\rho}$ times the annual innovation standard deviation, both evaluated at the unrounded estimates.

Algorithm 1 Equilibrium prices and portfolios at given coefficients, in the notation of equations (107)–(112)

Require: Measured taxes, portfolio assets, funding, supplies, rate dynamics, state grid and candidate coefficients

Construct investor exposures and bond-specific variances from the coefficients
Initialize portfolios and the log price functions on the grid
repeat ▷ outer loop: consistent expectations
    From the candidate prices, compute $\mathcal{M}$, Σ and c^L at every grid state
    for all grid states z_g, independently **do** ▷ inner step
        Solve (110) and (111) jointly by Newton's method for $\mu(z_g)$, holdings and cash
    end for
    Form $\widehat{p} = \mathcal{M} - \mu$ and record $\max_{j,g} |\widehat{p}_j(z_g) - p_j(z_g)|$
    If not converged, update all candidate price functions from $\widehat{p}$ by a damped, accelerated step
until the recorded maximum difference is below 10^{-10}
Recompute the moments and check portfolio conditions, clearing and funding limits
Evaluate yields at monthly states and holdings at December states

B.2 Solving equilibrium at given coefficients

An equilibrium is a set of municipal and taxable log price functions of the rate state, together with the portfolios they support at every state. Today's portfolios depend on the risk of reselling bonds next period, and that risk depends on tomorrow's prices, so equilibrium is a fixed point in the price functions. I compute it with two nested steps, summarized in Algorithm 1. The outer loop iterates on the price functions until the prices investors use to form expectations are the prices that clear markets. Within each outer iteration, the candidate prices fix every bond's resale distribution, so each rate state becomes a static problem. At each state, one Newton solve finds every investor's portfolio, including which bonds it holds, and the returns that clear all markets.

Price functions and expectations. Log prices $p_j(z) = \log P_j(z)$ are stored at eighteen rate states $\{z_g\}$: fifteen evenly spaced states and the period's three December states (Appendix B.7). Position j identifies a market and a maturity, and $p_{j,-}(Z')$ is the next-period log price of its one-period-shorter resale bond. Between grid states, log prices are linear in z. The next state is $Z' = \rho z + \sigma_r U$ with $U \sim N(0,1)$, so expectations over Z' are exact integrals of the normal density over linear segments, with no quadrature. From the candidate price functions, I compute at each grid state

$$\mathcal{M}_j(z) = \log \mathbb{E}_z \exp\{p_{j,-}(Z')\} + \tfrac{1}{2} v_j, \tag{107}$$

$$\Sigma_{jk}(z) = \mathrm{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j. \tag{108}$$

Here v_j is the bond-specific return variance. The first expression is the log expected resale payoff, including the bond-specific mean correction, so the mean adjusted holding return of Appendix A.1.1 is $\mu_j = \mathcal{M}_j - p_j = \log \mathbb{E}_z[1 + R_j]$. The covariance matrix Σ measures risk in the centered log-return innovations, and the liability credit is $c^L_{ij} = -\mathrm{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r U)$, as in (46).

Inner step: portfolios and clearing at one state. At grid state z_g, the moments $\mathcal{M}$, Σ and c^L are held fixed and the cash rate is $r(z_g) = \max\{\bar{r} + z_g, 0\}$. The unknowns are the return vector μ, every investor's bond holdings x_i and its cash x_i^0. Portfolio equations use the normalized payoff units of Section 4.2: a household's home-state municipal after-tax share and an institution's municipal share are set to one, with risk aversion and outside exposures transformed accordingly. Let $K_i = \mathrm{diag}(\xi_i^{a(j)})$ collect the after-tax shares for the market–maturity positions j, where $a(j)$ is the market of position j, and let the taxable share ξ_i^C, which equals the relative after-tax share ξ_i in these units, also apply to cash. For a household, the normalized share on out-of-state municipal interest is one minus its state rate. Fiscal calculations use absolute federal and state tax rates (Appendix D.4). Given returns μ, investor i's portfolio solves

$$\max_{x_i \geq 0,\, \mathbf{1}' x_i \leq W_i} x_i' \big(K_i \mu - \xi_i^C r \mathbf{1} - \tau_i^{\mathrm{fin}}\big) + \gamma_i x_i' K_i c_i^L - \frac{\gamma_i}{2} x_i' K_i \Sigma K_i x_i. \tag{109}$$

The three terms are the return above cash, the liability credit and the cost of portfolio risk. The financing-tax vector τ_i^{fin} is zero except on banks' municipal positions, where it is twice their measured annual cost. Cash is the uninvested funding $x_i^0 = W_i - \mathbf{1}'x_i$, and κ_i is the multiplier on the shorting constraint on cash, $x_i^0 \geq 0$. The portfolio conditions are

$$\begin{aligned} \widetilde{\nu}_i &= \gamma_i K_i[\Sigma K_i x_i - c_i^L] - K_i\mu + \xi_i^C r\mathbf{1} + \tau_i^{\text{fin}} + \kappa_i\mathbf{1}, \\ \widetilde{\nu}_{ij} &= \frac{\epsilon w_j}{x_{ij}}, \qquad \kappa_i = \frac{\epsilon}{x_i^0}, \qquad x_{ij} > 0, \quad x_i^0 > 0, \end{aligned} \tag{110}$$

and returns must clear every market,

$$\sum_i m_i x_i(z_g) = Q, \tag{111}$$

where Q is the fixed vector of bond supplies. The first line of (110) is the marginal net cost of each position. Newton's method solves (110) and (111) for all investors at once, with holdings, cash and returns as joint unknowns. Each grid state is solved independently.

How participation is determined. No participation pattern is guessed or imposed. The second line of (110) replaces exact complementarity with a small log barrier, $\epsilon = 10^{-9}$. A position with a marginal net cost $\widetilde{\nu}_{ij}$ well above zero, because its required return exceeds the offered return, is held only in the negligible amount $x_{ij} = \epsilon w_j/\widetilde{\nu}_{ij}$. A position the investor values has $\widetilde{\nu}_{ij}$ close to zero and a holding set by its portfolio condition. Which investors hold which bonds is therefore an output of each state's solve, and it can differ across rate states and change from one outer iteration to the next. Municipal barrier weights w_j equal issuer-state supply shares, and taxable weights equal one, so splitting the national municipal market leaves its total barrier weight unchanged.

Outer step: updating the price functions. The returns that clear markets imply updated current log prices

$$\widehat{p}_j(z_g) = \mathcal{M}_j(z_g) - \mu_j(z_g). \tag{112}$$

If the candidate prices were an equilibrium, $\widehat{p}$ would equal them. Otherwise I update the candidate price functions from $\widehat{p}$ by a damped, Anderson-accelerated step, recompute the moments and solve every grid state again. All grid states in an iteration use the same candidate prices, and all markets and maturities are updated together because covariances and funding link them. Two boundary conditions close the recursion. The one-period municipal claim pays one, so its continuation log price and return variance are zero. Cash has log price $-r$, which supplies the continuation price of the next-longer taxable bond. The iteration stops when the maximum difference between the candidate prices and the clearing-implied prices $\widehat{p}$ is below 10^{-10}. This checks the undamped pricing equation, so a small damped step alone cannot trigger convergence. The price functions, resale moments and portfolios then satisfy the equilibrium conditions to the stated tolerance.

Appendix B.7 gives the starting values, the update rule and the accuracy checks.

B.3 Coefficients, targets and criterion

The data do not directly measure risk aversion or how much rate risk an investor's outside assets and liabilities create. Seventeen coefficients, collected in θ, govern these objects and the small bond-specific risk. I estimate them jointly by fitting twenty-two conditions on ownership, insurer portfolios and taxable term spreads. They are common across periods and issuer states.

Coefficients. Three coefficients govern portfolio risk: common risk aversion $\gamma_i = \gamma$ in the normalized payoff units, the thirty-year bond-specific variance v_{30} and its maturity exponent p_v (Table 9). For $n = 4, 6, \ldots, 30$ calendar years, the two-year return variance is

$$v_n = v_{30}(n/30)^{p_v}, \qquad v_2 = 0.$$

The same variance applies to municipal and taxable bonds at a given maturity, in both the portfolio problem and the resale moments. Appendix B.5 discusses its normalization. The remaining fourteen coefficients set outside exposures and their distribution. Three map measured household balance sheets into exposure, eight describe the insurer types, one sets bank exposure and two set residual-group exposure (Table 9).

Targets. Twenty-two targets compare model outcomes with data. Nine concern ownership: municipal stocks held by the two household groups, P&C insurers, life insurers and banks, and taxable stocks held by households

and those three institutional sectors. Four are the mean maturities of P&C and life-insurer municipal and taxable portfolios. Six describe insurer portfolio composition: two measure the concentration of municipal shares across insurers, and four the fractions of municipal and taxable holdings beyond twenty years. Three are taxable term spreads over the two-year yield at ten, twenty and thirty years. Municipal yields, the municipal puzzle, the empirical tax-pass-through coefficients and the predictions P1–P3 are not targeted.

Which targets inform the coefficients. Section 5.2 explains these links, and the panel headings of Table 9 name the most informative targets for each group of coefficients. Each coefficient affects every target through equilibrium prices and portfolios, so the fit is determined jointly.

Constructing the conditions. Let $\mathcal{C}_{kj}(\theta)$ be the signed model-minus-data condition for target j in period k, after solving that period's equilibrium, and let $o_{kj} > 0$ be its observed normalizer. For ownership, $\mathcal{C}$ is the model stock less the observed stock, and o is the observed stock. For insurer mean maturity, $\mathcal{C}$ is $B(\overline{T} - \overline{T}^{\text{obs}})$ and $o = B^{\text{obs}}\overline{T}^{\text{obs}}$, where B is the model portfolio value, $\overline{T}$ its holdings-weighted maturity and the superscript obs marks observed values. Weighting by the model portfolio prevents a nearly empty book from counting as much as a large one. Ownership and maturity conditions include the riskless two-year municipal stock.

The six composition conditions use the risky maturities of four through thirty years. For insurer i, let M_i and C_i be municipal and taxable holdings in that window and $B_i = M_i + C_i$. The concentration statistic $\sum_i (M_i^2/B_i)/\sum_i B_i$ squares each insurer's municipal portfolio share and averages it with portfolio-size weights. At a given aggregate municipal share, a larger value means greater concentration in portfolios with high municipal shares. Its condition is the model statistic less the observed one. For either asset's long fraction, with model book B, holdings L beyond twenty years and observed long fraction f^{obs}, the condition is

$$\mathcal{C}^{\text{long}} = \frac{L - f^{\text{obs}}B}{B^{\text{obs}}_{\text{window}}}.$$

The observed window book equals the national sector stock times the measured share of its holdings in the window. The normalizers are the observed concentration statistics and long fractions.

Holdings conditions are computed at the three observed December states of each period and then averaged, with products and ratios formed before this averaging. Yield conditions use annual log yields at all 36 monthly states. Because observed two-year yields set those states, the ten-, twenty- and thirty-year yield errors are also errors in spreads over the two-year yield. Their normalizers are the observed period-mean term spreads.

The weighted distance. Each signed condition is averaged across the six periods and scaled by one tenth of its mean normalizer:

$$f_j(\theta) = \frac{6^{-1}\sum_{k=1}^{6}\mathcal{C}_{kj}(\theta)}{0.1\,(6^{-1}\sum_{k=1}^{6} o_{kj})}, \qquad \mathcal{L}(\theta) = \frac{1}{2}\sum_{j=1}^{22} f_j(\theta)^2. \tag{113}$$

A deviation equal to 10% of the mean observed normalizer gives $|f_j| = 1$. Each condition's weight in $\mathcal{L}$, the inverse square of one tenth of its mean normalizer, depends on the data alone. Estimation searches over the common coefficient vector θ to reduce $\mathcal{L}$, solving the model in all six periods with Algorithm 1 at each trial θ and holding measured inputs fixed.

B.4 Estimated coefficients and fit

Estimates. Estimation starts from the initial values in Table 9 and reaches $\mathcal{L} = 36.3575$.[26] A restart with finer derivative steps lowers the criterion to 36.2870, and the coefficients at this point are the estimates reported in Table 9.

Fit to the targets. Table 10 reports the observed and fitted quantities behind all twenty-two conditions in the one-market model used for estimation. Fourteen conditions are within 10% of their observed normalizers. The maturity and long-fraction conditions weight discrepancies by model holdings, so their contributions to the criterion differ from the differences between the averages displayed in the table.

[26] The coefficient search is a bounded trust-region method. Central differences measure how the averaged conditions respond to coefficients, with feasible one-sided differences at bounds. The search compares damped Gauss–Newton, backtracked and projected-gradient steps, using a secant estimate of the remaining curvature. Table 9 records intervals, initial values and coordinate scales. A scaled move δ_n changes coefficient n by $c_n\delta_n$. The two searches stop when the trust-region radius falls below 10^{-5}, with final improvements below 0.001.

Table 9: Measured rate parameters and estimated coefficients, with search settings

Symbol	Economic role	Value	Search interval	Start	Scale
Measured rate process, fixed before calibration					
$\bar{r}$	Central short-rate level, twice the mean annualized two-year corporate zero yield	0.06041	–	–	–
ρ	Rate-state persistence, estimated from historical short rates	0.7292	–	–	–
σ_r	Rate-state innovation standard deviation, estimated from historical short rates	0.04278	–	–	–
Risk. Main information: taxable term spreads and the allocation of long bonds					
γ	Common risk-aversion curvature, per unit of portfolio assets	9.6969	$[1, 10]$	9	1
v_{30}	Bond-specific variance of a thirty-year bond's two-year return	0.000722	$[0.000020, 0.006000]$	0.0015	0.0004
p_v	Maturity exponent of the bond-specific variance	1.2597	$[0, 3]$	1.15	0.3
Households. Main information: household municipal and taxable stocks					
a_H	Outside rate-exposure intercept	1.3045	$[-3, 3]$	1	0.3
b_A	Loading on the outside-asset ratio	2.704	$[0, 12]$	2	0.4
b_D	Loading on the debt ratio	0.8379	$[0, 20]$	0.5	0.5
P&C insurers. Main information: P&C stocks, maturities, composition and long fractions					
a_P	Low type's outside rate exposure	-1.961	$[-10, 2]$	-1.5	0.5
d_P	High-minus-low outside rate exposure	3.0311	$[0.02, 12]$	2.4	0.6
w_P	Low type's share of portfolio assets	0.0802	$[0.01, 0.99]$	0.22	0.08
Life insurers. Main information: life-insurer stocks, maturities, composition and long fractions					
a_L	Low type's outside rate exposure	-5.3927	$[-12, 2]$	-4	0.6
d_L	High-minus-low outside rate exposure	6.3007	$[0.02, 15]$	3.2	0.8
w_L	Low type's share of portfolio assets	0.2616	$[0.01, 0.99]$	0.1	0.08
t_L	Middle type's location within the exposure span	0.8625	$[0.01, 0.99]$	0.75	0.08
u_L	Middle type's share of portfolio assets excluding the low type	0.0162	$[0.01, 0.99]$	0.2	0.08
Banks and residual group. Main information: bank stocks and taxable term spreads					
h_B	Bank outside rate exposure	1.2045	$[-3, 3]$	0.6	0.3
μ_R	Residual-group mean outside rate exposure	1.0683	$[-4, 4]$	1	0.4
s_R	Residual-group outside rate-exposure dispersion	1.7168	$[0.01, 5]$	2.2	0.4

Notes. Values use the model's two-year period. The first three are measured and fixed. The remaining seventeen are fitted jointly to the twenty-two averaged conditions of Table 10, then held fixed in the full 45-market model. Each affects all conditions through equilibrium prices and clearing. Headings name the most informative conditions. Outside exposures $h_i = -\ell_i$ use bond-loading units. Positive values add rate risk and negative values hedge it, as explained in Appendix B.1. Portfolio-asset shares divide each insurer sector's measured debt securities and cash across its types. The last three columns give search bounds, initial values θ_n^0 and step scales c_n, with a scaled move δ_n changing coefficient n by $c_n\delta_n$.

Sensitivity of the estimates. Small moves from the estimates barely change the fit or the gap. The best such move, in one coefficient or several together, lowers the criterion by less than 0.001% and changes the average thirty-year gap by 0.004 basis points. Larger changes in any one group of coefficients all worsen the fit (Table 11). Each such change holds the remaining coefficients and measured inputs fixed. Lowering risk aversion by 20% has the largest effect on yields: the thirty-year gap falls by 13.6 basis points, the thirty-year taxable yield by 70 basis points and term spreads lie about 20% below their targets. Both curves fall nearly in proportion: the gap per unit of taxable yield falls only from 0.170 to 0.166, so the model's implied tax barely changes. Changing bond-specific variance by 20% moves the gap and the taxable yield by less than 0.1 basis points.

From the one-market model to the full model. Restoring the 45 municipal markets re-estimates no coefficient. Across the six periods, the full 45-market minus one-market thirty-year gaps are $(-0.85, -0.72, 2.48, -0.22, -1.15, -0.93)$ basis points, and thirty-year taxable yields differ by at most 0.52 basis

Table 10: Calibration targets and fitted quantities, 2000–2017

Quantity	Data	Model
Ownership stocks (billions of dollars)		
High-tax households, municipal	906.8	1,135.1
Lower-tax households, municipal	903.6	662.2
P&C insurers, municipal	295.0	289.1
Life insurers, municipal	21.7	22.0
Banks, municipal	251.6	266.5
Households, taxable	1,635.7	1,544.0
P&C insurers, taxable	335.8	244.7
Life insurers, taxable	1,589.7	1,624.4
Banks, taxable	537.3	522.4
Insurer mean maturities (years)		
P&C insurers, municipal	12.1	15.1
P&C insurers, taxable	7.6	6.3
Life insurers, municipal	14.6	17.9
Life insurers, taxable	12.3	11.7
Taxable term spreads (basis points)		
10-year taxable spread	209.0	214.7
20-year taxable spread	279.9	284.6
30-year taxable spread	299.3	303.6
Insurer portfolio composition		
P&C municipal-share concentration	0.4374	0.5613
Life municipal-share concentration	0.0065	0.0063
P&C municipal holdings beyond twenty years	11.2%	19.8%
P&C taxable holdings beyond twenty years	7.1%	4.1%
Life municipal holdings beyond twenty years	24.9%	28.4%
Life taxable holdings beyond twenty years	25.6%	26.4%

Notes. Model values are from the one-market model used to estimate the common coefficients. Entries average the six periods equally. Stocks and mean maturities include two-year municipal bonds. Concentration and long fractions use the risky maturities of four through thirty years. Long fractions are percentages held beyond twenty years. The concentration statistic is the portfolio-size-weighted average of squared municipal shares, defined in Appendix B.3. Term spreads are in annual log-yield units over the two-year yield. The criterion first constructs dated portfolio-weighted maturity and long-fraction discrepancies and then averages them, as in (113), so its terms are not differences of these displayed averages.

points. The full 45-market equilibria supply the municipal yield curves, cross-state pass-through and policy calculations. On the observed corporate curve and the empirical sample weights, the model explains 83% of the 117.5-basis-point thirty-year puzzle (Figure 8). These municipal outcomes are predictions of a calibration that targets ownership, insurer portfolios and taxable term spreads.

B.5 Bond-specific return risk and its normalization

The calibrated model adds the independent shock in (32) to the main-text return. The gap barely responds to its variance or to the normalization of its mean.

Role. The shock can represent an issue-specific change in trading demand that affects sale proceeds without moving aggregate rates. Its variance cost grows with position size, which makes bonds imperfect substitutes and supports joint holdings by several investors, as Figure 14 illustrates. The shock is zero for the one-period claim. In the one-market calibration, 20% changes in its variance move the thirty-year gap by less than 0.1 basis points (Table 11).

Specification. Writing the shock into realized resale proceeds, $P^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta}$, is algebraically identical to multiplying the holding return by it. Every return, conditional moment and portfolio condition is therefore the same under either form, and the shock adds no persistent price state.

Table 11: Sensitivity of the one-market calibration to estimated coefficients

Coefficient change	30-year gap (bp)	Change (bp)	Criterion $\mathcal{L}$
Estimated coefficients	102.6		36.29
Risk aversion $\gamma \times 0.8$	89.0	−13.6	55.02
Risk aversion $\gamma = 10$ (upper bound)	104.7	+2.1	36.74
Bond-specific variance $v_{30} \times 0.8$	102.7	0.0	36.61
Bond-specific variance $v_{30} \times 1.2$	102.7	0.0	36.59
Maturity exponent $p_v - 0.2$	102.6	0.0	36.55
Maturity exponent $p_v + 0.2$	102.7	0.0	36.47
Household exposure intercept $a_H - 0.25$	99.0	−3.7	78.82
Household exposure intercept $a_H + 0.25$	106.7	+4.1	160.84
Household asset loading $b_A \times 0.8$	100.9	−1.7	62.45
Household asset loading $b_A \times 1.2$	105.2	+2.6	400.24
Household debt loading $b_D \times 0.8$	106.2	+3.5	6,720
Household debt loading $b_D \times 1.2$	103.9	+1.3	82.17
P&C exposures ×0.8	102.2	−0.4	56.35
P&C exposures ×1.2	103.3	+0.6	62.51
Life-insurer exposures ×0.8	101.9	−0.7	59.68
Life-insurer exposures ×1.2	103.4	+0.8	46.24
Bank exposure ×0.8	100.5	−2.1	119.11
Bank exposure ×1.2	104.5	+1.8	241.78
Residual exposure mean ×0.8	102.6	−0.1	538.79
Residual exposure mean ×1.2	101.7	−0.9	65.73
Residual exposure s.d. ×0.8	99.3	−3.3	92.74
Residual exposure s.d. ×1.2	105.8	+3.1	1,687

Notes. Results average the six periods of 2000–2017. The $\gamma = 10$ row uses the upper bound of the search interval in Table 9. The P&C, life-insurer and bank rows scale every type's exposure, and the residual rows scale the mean or the dispersion of exposures. The gap is the thirty-year municipal yield less the household benchmark on the model's own taxable curve, before the corporate normalization. The criterion is (113).

Normalizing the mean. The shock's lognormal mean can also be normalized to one without changing holding returns. Replace the shock factor by $e^{\sigma^a_{\eta,T}\eta-(\sigma^a_{\eta,T})^2/2}$ and shift log prices to

$$\check{p}^a_T(z) = p^a_T(z) - \frac{1}{2}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2, \qquad \check{P}^a_T(z) = e^{\check{p}^a_T(z)}.$$

The adjacent-maturity shifts exactly offset the normalization:

$$\frac{\check{P}^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta-(\sigma^a_{\eta,T})^2/2}}{\check{P}^a_T(z)} = \frac{P^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta}}{P^a_T(z)}.$$

With fixed dollar supplies, portfolios, optimality and clearing are unchanged. For T periods of $\Delta = 2$ years, the annualized yield shift is

$$\check{y}^a_T - y^a_T = \frac{1}{2T\Delta}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2.$$

The fitted variance profile is common to both markets, so both thirty-year continuously compounded yields rise by 0.855 basis points and their difference is unchanged. The corporate normalization uses their ratio, so the normalized thirty-year gap rises by 0.20 basis points, from 97.39 to 97.59. The riskless boundary is unchanged.

B.6 Bond-supply sensitivity

The calibration and the puzzle accounting of Section 5.3 hold bond quantities fixed. Here supply responds to yields in both markets. This exercise uses the one-market model and reports gaps on the model's own taxable

curve, before the corporate normalization used for the full 45-market comparison. Elastic supply leaves the thirty-year gap close to its fixed-supply value.

Elastic bond supply. I allow borrowing to respond to its cost in both markets, holding the estimated coefficients fixed. For market $a \in \{M, C\}$ and maturity T, supply follows

$$Q_T^a(z) = \overline{Q}_T^a \left(\frac{y_T^a(z)}{\overline{y}_T^a} \right)^{-\chi_{a,T}},$$

where y_T^a is the annual continuously compounded yield and $\chi_{a,T}$ is the absolute yield elasticity of supply. Each schedule passes through the original quantity $\overline{Q}_T^a$ and fixed-supply yield $\overline{y}_T^a$ at the mean rate state. These anchors remain fixed when both yield curves are re-solved. Risky maturities from four to thirty years have elastic supply. The two-year riskless claims retain their baseline treatment.

St. Clair (2024) estimates that a one-percent increase in borrowing costs reduces municipal issuance by 0.28 to 0.59 percent. I use 0.45, the midpoint of the rounded range 0.3–0.6, and impose the same elasticity in both markets. The estimate concerns annual borrowing near the bank-qualified threshold. With other financing terms fixed, a proportional change in borrowing cost is locally a proportional change in yield. Applying that response to outstanding model supply defines the comparative-static exercise.

Table 12 reports gaps from the one-market model over the six calibration periods. At equal elasticities of 0.3, 0.45 and 0.6, the average thirty-year gap lies between 101.0 and 102.0 basis points, compared with 102.6 under fixed supply. At 0.45, municipal yields fall by 14.1 basis points and the household benchmark by 13.4, so the gap changes by less than one basis point.

I also consider a larger taxable response. The November 2022 estimates of Siani (2022) imply about 1.5% less corporate issuance per ten-basis-point increase in credit spreads, holding the benchmark rate fixed. I match that local slope by setting $\chi_{C,T} = 15\overline{y}_T^C$, while keeping $\chi_{M,T} = 0.45$. The thirty-year taxable elasticity is 0.86–0.97 across periods. Applying this corporate issuance response to the pooled taxable supply is an additional sensitivity assumption. The average gap is 99.2 basis points, retaining 96.7% of its fixed-supply value. Both yields adjust: municipal yields fall by 19.5 basis points and the household benchmark by 16.1. A larger taxable supply elasticity therefore need not widen the gap.

Table 12: Yield gaps with elastic supply in both markets

Municipal elasticity	Taxable elasticity	10 years	20 years	30 years
0	0	35.7	84.9	102.6
0.30	0.30	36.4	84.1	101.9
0.45	0.45	36.7	84.0	102.0
0.60	0.60	36.8	83.4	101.0
0.45	$15\overline{y}_T^C$	37.2	83.2	99.2

Notes. Gaps are in basis points, averaged over the original rate-state observations within each period and equally across the six periods of 2000–2017. All other coefficients and inputs remain fixed. Elasticities are reported as positive magnitudes: higher borrowing costs reduce supply. The last row converts a corporate spread semi-elasticity to a yield elasticity at each fixed anchor, with yields in annual decimal units.

B.7 Numerical implementation and accuracy

The baseline equilibria satisfy their conditions to tight tolerances, and the 2015–2017 thirty-year gap barely moves when the grid or the barrier changes.

B.7.1 Grid, integration and convergence

Each period's grid has fifteen equally spaced states on $[-6\sigma_z, 6\sigma_z]$, including zero, plus its three December rate states, where $\sigma_z = \sigma_r/\sqrt{1-\rho^2}$. Log prices are interpolated linearly between these eighteen nodes and extrapolated linearly along the end segments. All moments integrate this piecewise-linear function analytically. For a standard normal U and a log-price segment $a + bU$ on $l < U < u$,

$$\mathbb{E}[e^{a+bU}\mathbf{1}_{\{l<U<u\}}] = e^{a+b^2/2}\{\Phi(u-b) - \Phi(l-b)\}.$$

The moments therefore carry no quadrature error. The local rate loadings $A_T^a = -\partial p_T^a/\partial z$ satisfy equation (45) under its differentiation conditions. The baseline grid has no node exactly at the rate floor $z = -\bar{r}$, and adding

one is checked below. Yields are evaluated at the 36 monthly states and holdings at the three December nodes. The comparisons in Figures 7 and 8 read each bond's issuer-state yield at its issue-quarter-end state and use empirical bond, rating and quarter weights.

During calibration, the solution at a nearby coefficient vector supplies starting prices and portfolios, and the full 45-market solve starts from the one-market equilibrium at the same coefficients. Holdings and participation are then solved endogenously. At each node, Newton's method solves portfolios and clearing jointly until the residuals of (110) and (111) are below 10^{-11}. In the first outer iteration, each node approaches the barrier $\epsilon = 10^{-9}$ gradually from 10^{-5}. Later iterations start each node from its previous solution and solve at ϵ directly. The price update is damped and uses Anderson acceleration over the last seven iterates, with each step capped at 0.025 in log price. In the six baseline periods, the maximum price-update error falls from about 10^{-2} to 10^{-6} within forty iterations and below 10^{-10} within seventy. All moments and residuals are recomputed before an equilibrium is accepted.

B.7.2 Equilibrium accuracy

Four diagnostics check prices against resale values (R_p), market clearing (R_c), portfolio optimality (R_s) and complementarity (R_{comp}):

$$\begin{aligned} R_p &= \|p - \mathcal{M} + \mu\|_\infty, \qquad & R_c &= \|Q - \textstyle\sum_i m_i x_i\|_\infty, \\ R_s &= \|\widetilde{\nu} - \epsilon w/x\|_\infty, \qquad & R_{\text{comp}} &= \max\{\|x \odot \widetilde{\nu}\|_\infty, \|\kappa \odot x^0\|_\infty\}. \end{aligned} \tag{114}$$

Independent checks of the full 45-market equilibria at the eighteen nodes of each of the six periods give maxima of $R_p < 9.3\times10^{-11}$, $R_s < 9.9\times10^{-12}$, $R_c < 2.3\times10^{-16}$ and $R_{\text{comp}} < 1.01\times10^{-9}$, with zero violations of funding limits or nonnegativity and zero variance for riskless claims. Acceptance requires $R_p < 3\times10^{-7}$, $R_s < 10^{-7}$, $R_c < 10^{-8}$ and a funding-limit violation below 10^{-10}. Complementarity is reported separately because it equals the barrier weight at an exact barrier solution. The 2018–2020 period, solved at the same coefficients, gives $R_p = 4.52\times10^{-11}$, $R_s = 7.78\times10^{-12}$, $R_c = 2.22\times10^{-16}$ and complementarity 1.00×10^{-9}. The grid comparison in Appendix B.7.3 assesses accuracy between nodes.

Riskless municipal returns. Each issuer's riskless one-period municipal return is the lowest return any eligible investor requires on it, including that investor's multiplier on its shorting constraint on cash, as in (39). The lowest return that households require is

$$\min_{i\in\mathcal{H}} \frac{\xi_i^C r(z) + \kappa_i(z)}{\xi_i^{M_s}},$$

where $\mathcal{H}$ contains all 630 household types and s identifies the issuer. Every issuer's one-period return is within 0.01 basis points of this household frontier at every grid node. The minimum can differ from the issuing state's high-tax resident benchmark. In the six calibrated periods, the national one-period municipal yield is also within 0.01 basis points of the household benchmark at every observed month. Household taxes therefore anchor the riskless short end in the solved model, as in (12).

B.7.3 Grid and barrier sensitivity

Three independently checked 2015–2017 re-solves test whether the baseline grid and barrier are adequate. Each keeps coefficients and measured inputs fixed and makes one change: it refines the equally spaced grid from fifteen to twenty-nine nodes, lowers the barrier from 10^{-9} to 10^{-10}, or adds a node exactly at the rate floor. Table 13 reports each change relative to the baseline.

The thirty-year gap moves by less than 0.23 basis points in every case. Holdings move more: grid refinement lowers lower-tax household municipal holdings by 4.02%, and adding the floor node lowers those holdings by 1.69% and banks' taxable holdings by 2.80%. Lowering the barrier changes risky positions at December nodes by at most 0.071% of investor funding. With fixed prices and controlled stationarity error, concavity bounds the barrier portfolio's utility loss by $\epsilon(\sum_j w_j + 1)$. Cash and a riskless bond can offer the same after-tax return, so prices leave the split between them undetermined. At December nodes, lowering the barrier moves this split by up to 24.84% of an investor's funding, while complementarity falls from about 10^{-9} to 10^{-10}.

B.7.4 Policy calculations

The counterfactuals of Section 7 use the same price-and-portfolio algorithm. The home-state repeals, seven states in each of six periods, use the baseline barrier and tolerances. Independent reconstruction gives maximum

Table 13: Numerical sensitivity at fixed coefficients, 2015–2017

Comparison	Nodes	ϵ	Δ 30-year gap (bp)	Δ lower-tax household municipal holdings (%)	Δ bank taxable holdings (%)
Finer grid	32	10^{-9}	+0.199	−4.02	−2.59
Smaller barrier	18	10^{-10}	+0.0001	−0.004	+0.001
Node at the rate floor	19	10^{-9}	−0.228	−1.69	−2.80

Notes. Each row compares one change with the baseline eighteen-node, 10^{-9}-barrier solution. Nodes include the three December states, and ϵ is the barrier of (110). The gap is the national supply-weighted municipal yield minus the household benchmark, in continuously compounded yields over the period's 36 monthly states. Its baseline is 104.27 bp. Holdings are measured at the three December nodes, and changes are relative to each group's baseline holdings. The rows are numerical comparisons, distinct from the empirically weighted gap of Table 4. The rate floor affects prices in every row.

$R_p = 9.79 \times 10^{-11}$, $R_s = 9.20 \times 10^{-11}$ and $R_c = 3.33 \times 10^{-16}$, zero violations of funding limits or nonnegativity and zero variance for riskless claims. The frontier for riskless municipal returns holds within 1.83×10^{-5} basis points at every observed month. Separate checks verify the changed tax rules and fiscal identities.

The equal-treatment reform, which removes 99.9% of tax distinctions in every period, is solved first for 2015–2017, starting from the solution with 99% removal. The 2015–2017 solution initializes the other five periods, each checked independently. The equal-treatment solves use price-update tolerance 2×10^{-7}, inner tolerance 2×10^{-8}, tolerance 5×10^{-9} on each holding times its stationarity residual and linear-solve tolerance 2×10^{-9}. In a version of the model with three municipal markets, moving from 99% to 99.9% removal changes thirty-year municipal and taxable yields by 1.9 and 0.2 basis points and each group's maturity shares by at most 0.2 percentage points. Near full removal, municipal and taxable bonds become close substitutes, and within-maturity allocations between them can change by 9.5 percentage points of group assets. Table 39, Panel B, therefore reports riskless shares and the maturity composition of risky holdings.

C Adjustments and empirical tests

This appendix first constructs the five adjustments used to measure and account for the municipal puzzle. Paired issues and high-grade general-obligation bonds then check whether the puzzle depends on the taxable comparison or the security sample, and Appendix C.4 relates it to the single-tax-rate benchmark of Green (1993). The remaining subsections report the tax and interest-rate tests P1 to P3, their robustness checks and ownership evidence by tax status.

C.1 Comparability and bond-side adjustments

Before the municipal spread over the household benchmark can be compared with the model, five effects outside the model must be removed. Two concern comparability. Municipal yields are measured at issuance, and almost all long-maturity municipal bonds are callable, while the corporate curve describes seasoned, option-free bonds. Removing the value of the issuer's call right and the issuance premium puts the two curves on the same basis and defines the municipal puzzle. The other three are bond-side differences in capital-gains taxation, trading costs and default risk. Each could explain part of the puzzle independently of the model's mechanism, so I subtract them before comparing the puzzle with the model.

Table 14 reports each adjustment's contribution by maturity, along with the puzzle and the residual before the model. At thirty years, the call and issuance-premium adjustments reduce the 159.8-basis-point spread to a puzzle of 117.5 basis points. The bond-side adjustments explain 14.5 basis points, or 12% of the puzzle, and leave a residual before the model of 103.0 basis points. Taking the upper end of every adjustment's range still leaves 85.1 basis points. As a share of the puzzle, the bond-side adjustments matter more at intermediate maturities, where they explain 40% of the ten-year puzzle. Their sum stays near 12–16 basis points from twelve years onward, while the puzzle widens.

C.1.1 The common comparison

Every adjustment is a basis-point contribution to the spread

$$\text{Spread}(T) = y_T^{M,\text{raw}} - \bar{\xi}_H(T)\, y_T^C, \tag{115}$$

Table 14: What explains the municipal spread over the household benchmark: contributions in basis points to Spread(T) of equation (115). The two comparability columns remove the call option and the issuance premium from the municipal yield itself, giving the puzzle of Figure 2. The three bond-side differentials are municipal-minus-corporate.

		Comparability			Bond-side differentials					
T	Spread	Call	Issuance premium	Puzzle	Capital gains	Liquidity	Default	Bond-side adjustments	Share of puzzle	Residual before the model
2	9.9	0.1	2.9	6.9	0.0	−0.4	14.7	14.3		−7.3
6	27.9	0.7	3.3	23.9	10.1	−0.4	8.5	18.2	76%	5.8
10	52.2	4.6	4.0	43.6	9.9	−0.3	8.1	17.6	40%	26.0
14	87.0	17.9	7.2	61.9	7.1	−0.1	7.6	14.6	24%	47.3
18	110.3	23.9	9.6	76.9	4.7	−0.2	7.9	12.4	16%	64.5
22	128.8	27.0	11.4	90.5	3.6	2.8	9.0	15.4	17%	75.1
26	145.0	28.5	12.5	104.0	2.9	1.5	11.3	15.7	15%	88.3
30	159.8	29.5	12.8	117.5	2.3	0.1	12.2	14.5	12%	103.0

Notes. The puzzle is the spread less the call and issuance-premium adjustments. The bond-side column sums the three adjustments, and the share divides that sum by the puzzle (omitted where the puzzle is below 20 basis points). The residual before the model is the puzzle less this sum. Thirty-year ranges, in basis points: call $[22.8, 36.3]$, issuance premium $[8.1, 17.3]$, capital gains $[1.2, 2.5]$, liquidity $[-2.2, 2.3]$, default $[6.1, 16.4]$. Combining the range endpoints gives a thirty-year residual before the model between 85.1 and 123.8 basis points.

where $y_T^{M,\text{raw}}$ is the observed municipal zero-coupon curve, built from offering prices and contractual cash flows with each bond treated as a bullet, before either comparability adjustment and y_T^C is the corporate curve of Section 3.6. The household benchmark $\bar{\xi}_H(T)y_T^C$ averages each bond's household after-tax corporate yield at its dated federal and state tax rates, and dividing this average by the averaged corporate yield gives $\bar{\xi}_H(T)$. These rates include the net investment income tax (NIIT) from 2013 and, from 2018, the cap on state and local tax (SALT) deductions. The household wedge $1 - \bar{\xi}_H(T)$ averages 41.3%.

All curves use the same three-stage averaging: equal weights within maturity × issue-quarter × half-notch rating cells, then across ratings within quarters, then across quarters. Each adjustment is estimated on the observations that identify it, namely callable contracts, early secondary trades, resale behavior, matched trading costs or CDS quotes, and is then carried to the same maturity grid. The two comparability adjustments change the municipal yield itself. The three bond-side adjustments are municipal-minus-corporate differences in compensation, measured in the household's after-tax yield units, so compensation common to both markets cancels. The contributions are estimated separately and add up as in (24). The combined range adds the endpoints of the separate ranges and is not a joint confidence interval.

C.1.2 Adjustment I: embedded call options

What it measures. An issuer's call right lets it repay a bond before maturity at a stated call price. When rates fall, the issuer can refinance, so the holder loses part of the price gain that a noncallable bond would deliver. Investors therefore pay less for callable bonds, and callable yields exceed the yields on otherwise identical option-free bonds. More than 95% of municipal bonds beyond twenty years are callable, so dropping them would leave almost no long-maturity sample. I therefore value each bond's call right and remove it.

Calculation. I value each bond's contractual coupons, principal and call schedule on a grid of future rate states. Working backward from the final coupon and principal at maturity, the value at each earlier coupon date is the coupon plus the discounted continuation value, which is capped at the call price on call dates:

$$C_k(z) = P_\Delta(z)e^{-s^{\text{fit}}\Delta}\sum_j \Pi_{zj}V_{k+1}(z_j), \qquad V_k(z) = 100ca_k + \min\{C_k(z), K_k\}. \tag{116}$$

Here $V_k(z)$ and $C_k(z)$ are the value and the discounted continuation value at coupon date k in rate state z, c is the annual coupon rate, a_k the accrual fraction, $P_\Delta(z)$ the reference discount price over one coupon interval Δ, Π_{zj} the transition probabilities between rate states and K_k the call price, with $K_k = \infty$ at dates without a call right. The constant spread s^{fit} is set so that the callable value at the bond's starting rate state, defined below, equals its observed price. Holding this fitted spread and the cash flows fixed, I remove the cap. The

difference between the two values, the uncapped (bullet) value V_b^{bul} and the callable value V_b^{call} of bond b, is the call value,

$$CV_b \;=\; V_b^{\text{bul}} - V_b^{\text{call}} \;\geq\; 0. \tag{117}$$

Adding CV_b to the observed price gives the option-free price. I convert the observed price, treated as a bullet, and the option-free price into zero-coupon yields with the same bullet cash flows and reference curve. For each price, a flat continuously compounded spread $\hat{s}$ over the reference discount curve $D_0(t)$ at rate state $z = 0$ prices the bond's coupons and principal at that price. The zero-coupon yield is $200[\exp\{(y_0(T_b) + \hat{s})/2\} - 1]$ percent, a nominal semiannual yield, with $y_0(T_b) = -\log D_0(T_b)/T_b$ and T_b the bond's maturity. This conversion spread only changes units and is separate from the fitted spread s^{fit}, which stays fixed in the option valuation. The difference between the two resulting zero-coupon yields is the bond's call spread, which is nonnegative because the option-free price is no lower. Because the fitted spread is held fixed, the adjusted yield keeps its compensation for credit, liquidity and other features, and only the call right is removed. Noncallable bonds have a zero call spread. The call adjustment Call(T) is the three-stage average of the call spreads in the pricing sample described below, and the main puzzle subtracts it from the full observed curve.

The reference curve is the model's supply-weighted municipal discount curve for the bond's issue period, one of the model's seven periods. The rate state follows the calibrated AR(1) process. Log prices are rescaled around the rate state $z = 0$ so that pricing yield volatility equals measured municipal yield volatility times a scale m, and a pricing drift is then fitted so that the grid reproduces the curve. Measured volatility at maturity T is the standard deviation of twelve-month changes in the T-year AA-and-above municipal zero-coupon yield, so $m = 1$ prices the call at observed volatility. Each bond starts at the observed rate state in the final month of its issue quarter, $z_t = r_t - \bar{r}$, where r_t, the observed one-period cash rate, is twice the continuously compounded two-year corporate zero yield. The valuation follows each bond's contractual call calendar. The pricing sample contains 1,828,379 A-or-above fixed-rate and original-issue-discount bonds. Calendar and price screens exclude 16,295 bonds with irregular first coupons, unsupported conventions or unusable call schedules, and the eligible bonds carry 97.6%–99.8% of each maturity's weight. These screens change three-stage municipal yields by at most 1.1 basis points and the household benchmark by less than 0.1 basis points.

Why the estimate is reliable. Three features discipline the option value. First, the contract fixes the payoff: the coupon, call dates and call prices determine what the issuer gains by calling. Second, market prices of callability discipline the volatility scale m, which determines how often calling pays. I fit m so that model call spreads match observed callable-minus-noncallable yield differences among AA-and-above bonds matched by maturity, issue quarter, rating and half-point coupon at six to twenty years. I then apply it to A-or-above bonds without re-estimation. If callability is the only difference within these cells, the correct m equates the model and market differences. The estimate is $\hat{m} = 1.077$. Its conditional bootstrap 95% band, $[0.964, 1.189]$, comes from 2,000 draws that resample issue quarters independently, preserving within-quarter dependence across maturities. Each draw holds fixed the model's matched-cell differences at $m = 1$, their elasticities with respect to m and the fitting weights, which makes the band conditional. The band contains measured volatility ($m = 1$). The municipal gap over the household benchmark is not a fitting target.

Third, the adjusted curve matches bonds that never needed an adjustment. Figure 15 compares the call-adjusted curve with yields on bonds that were noncallable from issuance, in shared maturity, quarter and rating cells through twenty years. The two curves differ by less than three basis points through ten years. The largest difference on the two-year grid is eight basis points, at eighteen years, and at twenty years the adjusted curve lies 3.5 basis points below the noncallable curve.

Result and sensitivity. The call adjustment is 4.6 basis points at ten years and 29.5 at thirty (Figure 16(a)). Repricing at the endpoints of the bootstrap band for $\hat{m}$ gives a thirty-year range of $[22.8, 36.3]$ basis points, and pricing at measured volatility gives 24.9. Panel (b) of Figure 16 compares the market callable-minus-noncallable differences at the fitting maturities with model differences at measured and fitted volatility. Panel (c) shows the elasticity of the adjustment with respect to m, 2.2 at thirty years, which is why the range is wider in percentage terms than the band for $\hat{m}$. A different option model gives nearly the same adjustment. In a one-factor Hull–White model (Hull and White, 1994), the drift fits each dated discount curve exactly, annual mean reversion is fixed at 0.10, and volatility is fitted to the same callable-minus-noncallable differences. This model gives 4.7, 23.4 and 28.5 basis points at ten, twenty and thirty years, against 4.6, 25.3 and 29.5 in the baseline. It fits the market differences with a root-mean-square error of 3.7 basis points, against 4.2 for the baseline model.

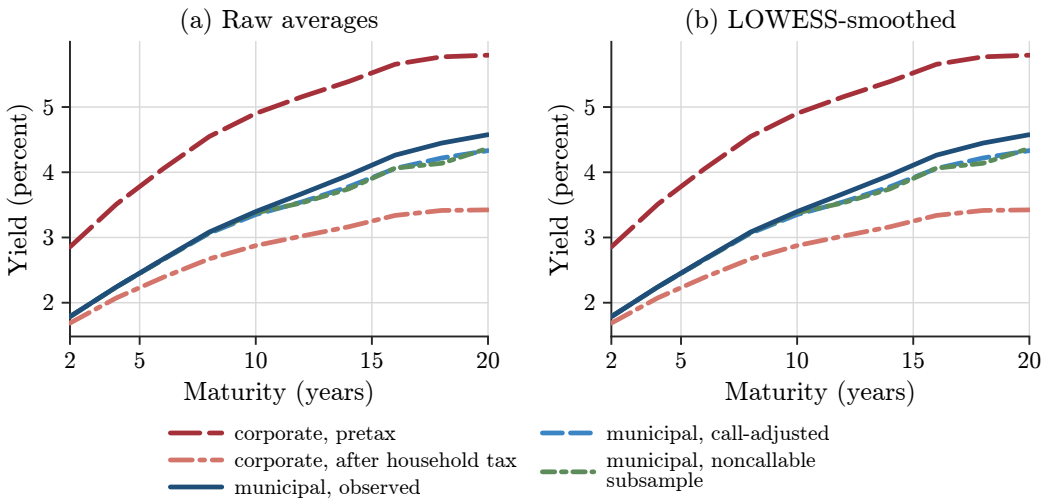


Figure 15: Noncallable validation of the call adjustment: raw means (a) and locally weighted scatterplot smoothing (LOWESS) (b). Lines show pretax and household after-tax corporate yields alongside observed, call-adjusted and noncallable municipal yields. The call-adjusted and noncallable curves share A-or-above maturity–quarter–rating cells containing noncallables, with fifty or more quarters per maturity and no extrapolation past twenty years.

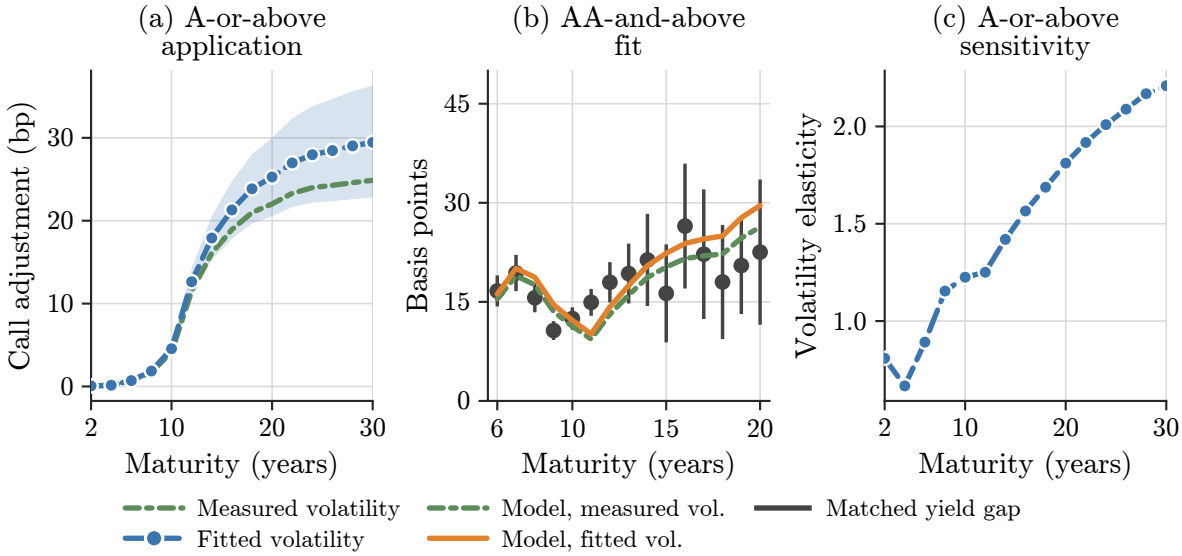


Figure 16: Call adjustment and volatility calibration. (a) A-or-above call adjustments at measured and fitted volatility, with direct repricing at the endpoints of the bootstrap band for $\hat{m}$. (b) AA-and-above callable-minus-noncallable market differences in matched cells (matched yield gap, ±1.96 standard errors) and model differences at the fitting maturities, 6–20 years. (c) Elasticity of the A-or-above call adjustment with respect to the volatility scale m.

C.1.3 Adjustment II: the issuance premium

What it measures. Municipal yields are measured at the offering, while the corporate curve describes seasoned bonds. Buyers at the offering receive a higher yield than buyers of the same bond shortly afterward, whether because of placement costs, distribution, liquidity seasoning or price discovery. The issuance premium measures this difference so that the municipal curve is on the same seasoned basis as the corporate curve.

Calculation. For bond b, let y_b^{off} be its offering yield and y_{bj}^{tr} the yield on customer purchase j within thirty days of the offering. Let $\bar{y}_k^{\text{seas}}(w)$ be the yield on seasoned bonds in the same maturity bucket k and week w,

with w_b^{off} and w_{bj} the weeks of the offering and of purchase j. The premium is the offering-to-trade yield decline less the contemporaneous decline in seasoned yields:

$$\text{Premium}_{bj} = (y_b^{\text{off}} - y_{bj}^{\text{tr}}) - [\bar{y}_k^{\text{seas}}(w_b^{\text{off}}) - \bar{y}_k^{\text{seas}}(w_{bj})]. \tag{118}$$

Comparing the bond with itself removes differences in issuer and contract terms, and subtracting the seasoned-yield change removes market-wide rate movements. The estimate uses neither the corporate yield nor the household benchmark. The early trade is a customer purchase, like the offering itself, so the comparison avoids a purchase-to-sale bid–ask difference that would overlap with the liquidity adjustment. Early and seasoned trades must have at least \$250,000 par, and when-issued trades are excluded. The seasoned benchmark is the par-weighted purchase yield of bonds older than three years, with at least 25 prints in the bucket and week.

Bonds with such an early institutional-size purchase cover 4% of the curve. For each of these bonds, I average the premia over its early purchases. I then average the bonds within each maturity bucket with the three-stage weights and apply the bucket estimates to the whole curve. Issues from 2000–2004, before the trade tape begins, receive the estimate from later issues. Bucket estimates are interpolated from bucket midpoints to the two-year grid, with end values held flat. The premium is a yield change between the offering and early trades, so I subtract it from the zero-coupon curve basis point for basis point, giving Prem(T). Offering and zero-coupon yields move closely together, with correlations of 0.998, 0.993 and 0.937 at five, ten and thirty years, and their level differences come from the zero-coupon conversion.

Result and robustness. The issuance premium rises from 2.9 basis points at two years to 12.8 at thirty (Table 15 and Figure 17(b)). A block bootstrap over issue quarters gives a thirty-year interval of [8.1, 17.3] basis points, and the intervals exclude zero at every maturity.

The rising maturity profile holds across trade windows, trade sizes and coupon types. In every trade-age window of Table 15 and Figure 17(a), premia beyond twelve years exceed those below twelve years, and from fourteen days on the two longest buckets give 9.3–12.8 basis points. In those buckets, par and discount bonds give 10.9 and 13.2 basis points and premium bonds 9.3 and 10.9, so yield-to-worst conventions for premium bonds do not drive the gradient. Including trades of all sizes raises the thirty-year estimate to 21.1 basis points. Long-end premia remain positive in every trade-size decile and are smallest for the largest trades (Figure 17(c)), consistent with small-trade markups raising the all-size estimates.

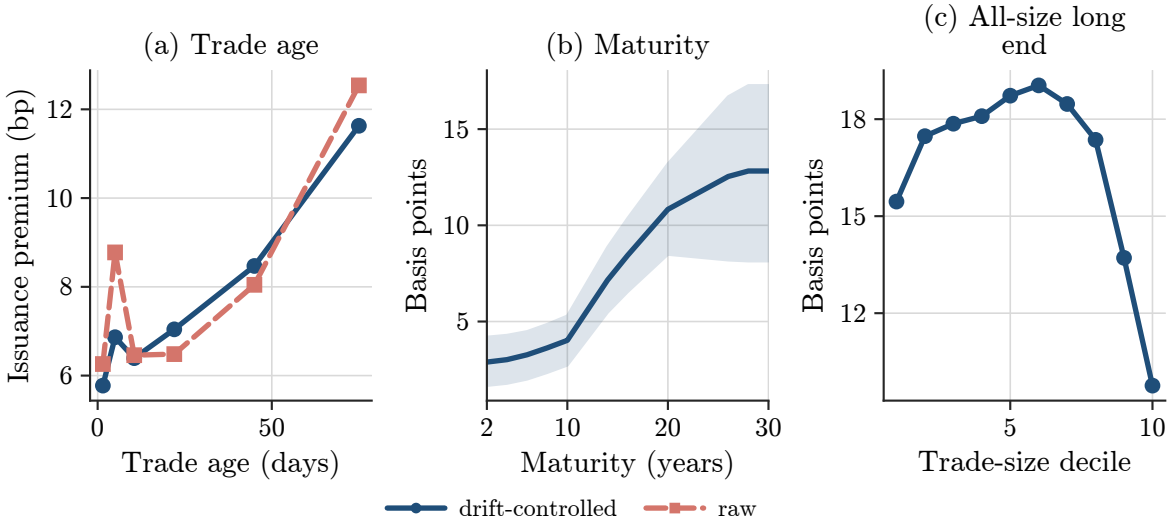


Figure 17: Issuance-premium diagnostics. (a) Institutional-size estimates by trade age, with (drift-controlled) and without (raw) the market-movement control. (b) Institutional-size maturity profile with block-bootstrap band. (c) All-size long-end estimates by trade-size decile.

C.1.4 Adjustment III: capital-gains taxation

What it measures. A bondholder earns interest and, if the bond is sold before maturity, a price gain or loss. The model applies the interest-income tax treatment to the whole holding return. In this federal comparison, it therefore treats municipal price gains as exempt and corporate price gains as ordinary income. Actual rules tax

Table 15: Issuance premium by maturity bucket (bp) under alternative trade-age windows, in days. The baseline window is thirty days. Panel A, the baseline, requires prints of at least $250,000 par in both the early print and the seasoned benchmark. Panel B uses prints of all sizes.

maturity bucket (yr)	3	7	14	30	60
Panel A: institutional-size prints (baseline)					
1–5	3.7	3.6	1.4	2.9	3.3
5–8	2.8	5.2	3.0	3.3	4.0
8–12	3.4	5.2	3.9	4.0	4.6
12–17	6.2	9.0	7.7	7.5	7.8
17–23	9.1	7.3	9.3	10.8	12.2
23–31	9.5	7.2	11.9	12.8	12.0
Panel B: prints of all sizes					
1–5	4.5	6.7	5.6	3.6	2.2
5–8	5.9	10.1	7.6	6.1	5.0
8–12	7.2	10.5	9.0	7.9	7.4
12–17	15.5	16.2	13.8	12.8	12.4
17–23	15.0	17.3	16.7	16.6	16.7
23–31	18.1	18.7	20.0	21.1	21.0

Notes. Each cell is the premium for bonds with at least one print inside the window, averaged first over each bond's prints inside the window and then over bonds with the curve's three-stage weights (Appendix C.1.1).

the two parts differently. Municipal interest is exempt, but a resale gain can be taxable, and a corporate resale gain can face the lower capital-gains rate. For example, consider a bond bought at par and sold more than a year later for a $1 gain. At the federal rates used here, the model assigns no tax to the municipal gain and 37 cents to the corporate gain, while capital-gains treatment assigns 20 cents to each. Capital-gains treatment therefore reduces the municipal bond's after-tax advantage when prices rise. Under the full loss offsets assumed below, losses have the opposite effect. The sale price also reflects the next buyer's taxes: at a given required after-tax return, a buyer who owes market-discount tax at redemption pays less for the bond.

Tax rules. Table 16 compares the model treatment B with the detailed treatment A. Coupons and original-issue discount (OID) are treated the same way in both: municipal coupons and OID are exempt, and corporate coupons and OID are ordinary income. OID raises tax basis as it accrues, so accrued interest is not taxed again as a gain. The original buyer's basis at a sale date h is the adjusted issue price AIP_h, which moves from the issue price to face at constant yield, and a secondary buyer's basis starts at the purchase price, with premium adjustments following Appendix D.3. Market discount, the shortfall of a secondary purchase price below the adjusted issue price for OID bonds or below face otherwise, is ordinary income at disposition unless it is below the de minimis threshold, in which case it is taxed as a capital gain. With n remaining years and face 100, the threshold is

$$\bar{D}_n = 0.0025 \times 100 \times n. \tag{119}$$

Both treatments use the same federal rates in all seven periods, $\tau_{oi} = 0.37$ on ordinary income and $\tau_{cg} = 0.20$ on capital gains, excluding NIIT and state tax. Under treatment A, residual gains and losses face the capital-gains rate with full offsets for losses, so the tax on a residual gain G, negative for a loss, is

$$\mathcal{T}(G) = \tau_{cg} G. \tag{120}$$

Calculation. I follow an investor who buys at issuance and first sells after h years. For each sale horizon, I find the municipal issuance yield that gives this investor the same expected after-tax holding return as the corporate bond, averaging over sale-date rate states. I do this first under the model treatment B and then under the detailed treatment A, holding corporate prices, the rate process and the reinvestment curve fixed. Terminal wealth at the sale date is the after-tax sale proceeds plus interim coupons, net of tax, reinvested to that date on the unshifted municipal curve. The holding return is the annualized log of expected terminal wealth over the offering price. Resale prices are those that give the next buyer the corporate bond's after-tax return under the same treatment, including the market-discount tax. With δ_B and δ_A the two required changes in the municipal issuance yield, the correction for a maturity-T bond sold after h years is

$$\Delta_{\mathrm{cg}}(h, T) = 10^4 \big(\delta_A - \delta_B\big) \text{ basis points.} \tag{121}$$

Table 16: Tax treatments compared in the capital-gains adjustment

Return component	Model-based treatment B	Detailed treatment A
Municipal coupons and OID	Exempt	Exempt
Corporate coupons and OID	Ordinary income	Ordinary income
Original municipal buyer's residual resale gain	Exempt	Capital gain
Original corporate buyer's residual resale gain	Ordinary income	Capital gain
Municipal secondary buyer's market discount	Exempt	Capital gain if de minimis, otherwise ordinary income
Corporate secondary buyer's market discount	Ordinary income	Capital gain if de minimis, otherwise ordinary income

Notes. Residual gains are measured relative to adjusted basis. Ordinary-income and capital-gains rates are 37% and 20%. A secondary buyer holds to maturity, with market-discount tax due at redemption.

Both yield changes are parallel shifts of the municipal log-discount curve in annual continuously compounded units. For a par bond, a higher issuance yield raises the coupon at an issue price of 100, and the municipal resale price is recalculated for the changed payments. Subtracting δ_B removes any return difference already present on the model curves, so Δ_{cg} isolates the difference in tax treatment. The additive accounting uses it as a basis-point approximation to the corresponding nominal semiannual shift.

Averaging across bonds and sale dates. Representative par and zero-coupon bonds are built from each period's supply-weighted municipal and corporate model curves at the rate state $z = 0$, and sale-date rate states follow the calibrated AR(1) distribution h years ahead, truncated to the model grid. Each corporate–municipal pair of instruments is weighted by the product of each instrument's share of issuance in its market, and at thirty years the par/par pair receives 98.7% of the weight.

The correction applies only to bonds that are sold before maturity, so I weight each horizon by the probability $\pi_h(T)$ that the original buyer first sells then:

$$\text{CG}(T) = \sum_h \pi_h(T)\, \Delta_{\text{cg}}(h, T), \tag{122}$$

with a zero correction for bonds held to maturity. Total turnover overstates these probabilities. The same dollar of bonds can trade repeatedly, and the next buyer's tax is already in the resale price, so weighting every later trade would count that tax again. I therefore let a trading segment, a share $\phi(T)$ of par, carry all resales, while the remaining par is held to maturity.

Turnover t_a is seasoned traded par over outstanding par at bond age a in the MSRB customer tape. Holding aggregate turnover fixed, the first-sale hazard within the trading segment at age a is $\lambda_a = \min(t_a/\phi, 0.99)$, and the probability of a first sale at age a is $f_\phi(a) = \phi\lambda_a \prod_{j<a}(1-\lambda_j)$. Sale horizons lie on the two-year grid, so the weight $\pi_h(T)$ sums $f_\phi(a)$ over ages $h-1$ and h. The first horizon also receives age 0, and the last also receives age $T-1$. I set ϕ so that the probability of a first sale within the observation window equals the measured ceiling on the share of par ever sold, which is cumulative seasoned volume per CUSIP capped at outstanding par. The ceiling uses issues offered from 2005 with at least three observable years, and each bond's observation window is the shorter of its original maturity and its years from offering to 2021. If a homogeneous hazard ($\phi = 1$) stays below the ceiling, the segment share is one, as at four years (Table 17). At thirty years, almost all of the trading segment sells within the window, so ϕ equals the ceiling, 0.602, and in this first-sale model the original buyer holds 40% of thirty-year par to maturity. Results average the seven periods equally.

Why the adjustment is small. The adjustment is 10.7 basis points at eight years and 2.3 at thirty (Table 17 and Figure 18). Four features keep it small at long maturities. Coupons and OID are taxed identically in both treatments, so only price gains and losses matter. Gains and losses partly offset across rate states. Bonds held to maturity, 40% of thirty-year par, have no residual gain. Finally, the resale-tax wedge is spread over the bond's remaining life, and the de minimis threshold, below which market discount is taxed at the lower capital-gains rate, rises with that life.

Sensitivity. The thirty-year range is $[1.2, 2.5]$ basis points, the envelope of eighteen variants. They cross three tape windows (the full window and its two halves) with three scalings of the measured resale ceiling (0.5,

0.75 and 1.0), and the segment share is refitted for each scaling. Each of these nine combinations is computed under two issuance weightings of the par and zero-coupon instrument pairs, matched to each maturity or pooled across maturities. Two diagnostics that count more resales also leave the adjustment small. Weighting horizons by the ages of observed trades (conditional on observed sale), which include repeat sales and exclude bonds held to maturity, gives 4.8 basis points at thirty years. A homogeneous hazard in which every bond can trade ($\phi = 1$) gives 3.9. It lies above the range because it sells more long-maturity par than the measured resale ceiling allows.

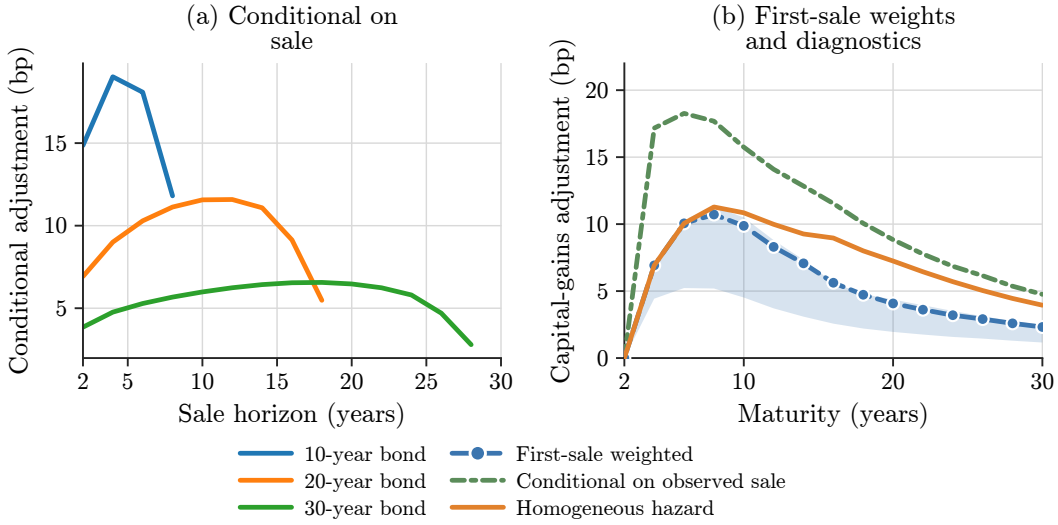


Figure 18: Capital-gains adjustment, averaged over the seven periods. (a) Issuance-weighted correction $\Delta_{cg}(h, T)$ conditional on a sale at horizon h, for ten-, twenty- and thirty-year bonds. (b) Adjustment weighted by first-sale probabilities, with the range of its eighteen variants, and the conditional-on-observed-sale and homogeneous-hazard diagnostics.

Table 17: Capital-gains adjustment at selected maturities (bp, except ϕ and P(hold)). The reported column weights sale horizons by first-sale probabilities, with the eighteen-variant envelope as its range. The cond. on sale and homogeneous columns are the diagnostics. The max column is the largest correction $\Delta_{cg}(h, T)$ across sale horizons. Columns ϕ and P(hold) give the trading-segment share and the held-to-maturity probability. At two years no sale horizon precedes maturity, so the adjustment is zero.

T	cond. on sale	max over h	ϕ	P(hold)	homogeneous	reported	range
2	0.0	0.0	–	1.00	0.0	0.0	[0.0, 0.0]
4	17.2	17.2	1.00	0.60	6.9	6.9	[4.4, 7.0]
8	17.7	19.9	0.88	0.40	11.3	10.7	[5.2, 11.3]
10	15.7	19.0	0.82	0.37	10.8	9.9	[4.5, 10.5]
20	8.8	11.6	0.57	0.43	7.2	4.1	[2.0, 4.4]
30	4.8	6.6	0.60	0.40	3.9	2.3	[1.2, 2.5]

C.1.5 Adjustment IV: liquidity

What it measures. Investors who expect to trade require compensation for trading costs. If municipal bonds cost more to trade than comparable corporate bonds, part of the municipal spread compensates for liquidity. The adjustment is small because comparable municipal and corporate bonds cost about the same to trade: municipal costs are slightly lower through twenty years, higher at twenty-one to twenty-five years and almost identical at twenty-six to thirty years.

Calculation. I compare institutional-size trades of at least $250,000 in MSRB municipal and TRACE corporate data for 2005–2020, made more than three months after issue and more than one year before maturity. The imputed round-trip cost is $(\max - \min)/\max$ price within CUSIP $\times$ day $\times$ size cells that contain both customer buys and customer sells. Costs are matched by month $\times$ rating family $\times$ maturity bin, with at least ten bonds per market in each cell. A round-trip cost is paid once per holding period, while a yield is earned

every year, so with matched municipal and corporate round-trip costs RTC^M and RTC^C in percent of price and a holding period of h years, the yield differential in basis points is

$$\Delta y^{\mathrm{liq}} = \frac{100(\mathrm{RTC}^M - \mathrm{RTC}^C)}{h}.$$

The baseline uses $h = 5$ in both markets, so each percentage point of round-trip cost is worth 20 basis points of annual yield. Trading costs reduce after-tax wealth in both markets, so the differential is not multiplied by an interest-income after-tax share.

Why the estimate is reliable. Two regressions support the conversion. A regression of A-or-above municipal offering yields on bond-average round-trip costs, within maturity-bin × rating × offer-year cells, gives 15.4 basis points per percentage point of cost, close to the 20 of the baseline conversion. Issuer-level regressions (Table 19) show that liquidity is priced within the municipal market: at twenty-six to thirty years, more liquid issuers have offering yields about 8–15 basis points lower, and the cost coefficient implies 27.3 basis points per percentage point. Applied to the twenty-six-to-thirty-year cost difference of 0.0025 percentage points, even this larger slope gives about 0.07 basis points.

Result and sensitivity. At twenty-six to thirty years, municipal and corporate costs are 0.5335% and 0.5310% of price (Table 18). The difference gives 0.050 basis points annually under the baseline, reported as 0.1, and 0.04 under the regression conversion. The thirty-year band, $[-2.2, 2.3]$ basis points, spans holding periods of two and ten years, the regression conversion and ±1.96 standard errors. The largest municipal cost excess is in the twenty-one-to-twenty-five-year bin, where the adjustment is 3.57 basis points, about 1.5 standard errors from zero. To obtain $\mathrm{Liq}(T)$, I place the six bin estimates at 3, 8, 13, 18, 23 and 28 years and interpolate linearly to the two-year grid, holding end values flat. The thirty-year adjustment is therefore the twenty-six-to-thirty-year estimate. The interpolated adjustment peaks at 2.9 basis points at twenty-four years (Figure 19). The ten-bond minimum per cell avoids thin cells. With a one-bond minimum, the twenty-six-to-thirty-year estimate becomes −6.53 basis points, which would raise the residual before the model, while a twenty-bond minimum gives 0.30.

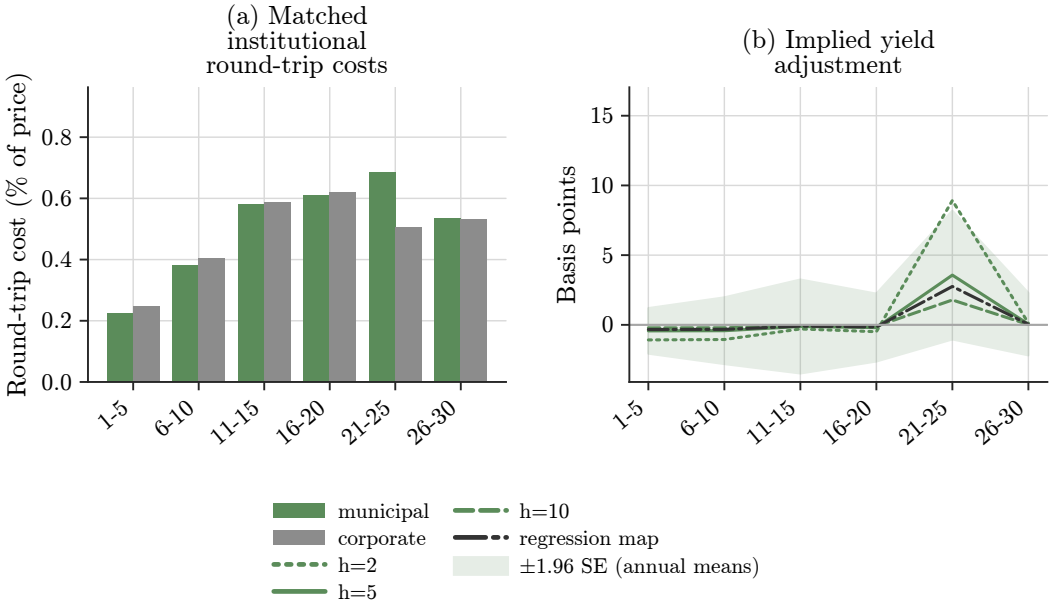


Figure 19: Liquidity adjustment from matched institutional-size customer trades of at least $250,000, 2005–2020. Panel (a): imputed round-trip costs in common month, rating and maturity cells, percent of price. Panel (b): the municipal-minus-corporate cost difference annualized under three holding periods and under the regression map (15.4 basis points per percentage point of cost). Shading is ±1.96 standard errors calculated across annual means at $h = 5$.

Issuer-level evidence. Table 19 asks whether more liquid municipal issuers have lower offering yields, controlling for maturity, rating, state, issue month and call status. Trading frequency averages, across an

Table 18: Liquidity by remaining-maturity bin (years), matched trades of 2005–2020. Round-trip costs (RTC) and their municipal-minus-corporate difference in percent of price, and adjustments in bp under three holding periods h and the regression map (regr., 15.4 bp per percentage point of cost).

Bin	Municipal RTC	Corporate RTC	Difference	$h = 2$	$h = 5$	$h = 10$	regr.
1–5	0.225	0.246	-0.022	-1.08	-0.43	-0.22	-0.33
6–10	0.382	0.403	-0.021	-1.05	-0.42	-0.21	-0.33
11–15	0.582	0.588	-0.006	-0.30	-0.12	-0.06	-0.09
16–20	0.609	0.619	-0.010	-0.49	-0.20	-0.10	-0.15
21–25	0.684	0.506	0.178	8.92	3.57	1.78	2.75
26–30	0.533	0.531	0.003	0.13	0.05	0.03	0.04

issuer's bonds, each bond's mean daily MSRB trades from its dated date to maturity. The spread measure averages each bond's mean imputed round-trip cost in the same way, in price units, so 0.01 is 1% of price. The coefficients check liquidity pricing within the municipal market and are not added as a separate adjustment.

Table 19: Issuer liquidity and offering yield, by maturity bin.

	Trading frequency		Transaction-cost spread	
	(1) Above corp median	(2) Top 25% trade	(3) Issuer-avg spread (cont.)	(4) Lowest-spread quartile
Maturity 1–5y	-0.046^{***}	-0.028^{***}	13.446^{***}	-0.057^{***}
	(0.009)	(0.004)	(1.200)	(0.003)
Maturity 6–10y	-0.007	0.021^{***}	11.624^{***}	-0.048^{***}
	(0.009)	(0.005)	(1.415)	(0.005)
Maturity 11–15y	-0.011	0.041^{***}	11.860^{***}	-0.040^{***}
	(0.009)	(0.005)	(0.830)	(0.005)
Maturity 16–20y	-0.035^{***}	0.024^{***}	13.447^{***}	-0.043^{***}
	(0.010)	(0.006)	(1.084)	(0.006)
Maturity 21–25y	-0.091^{***}	-0.041^{***}	20.427^{***}	-0.104^{***}
	(0.019)	(0.015)	(2.588)	(0.016)
Maturity 26–30y	-0.093^{***}	-0.082^{***}	27.262^{***}	-0.146^{***}
	(0.023)	(0.027)	(4.383)	(0.026)
Fixed effects	Yes	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes	Yes
Observations	785,256	785,256	779,988	785,256

Notes. AA-and-above federally exempt issues, 2000–2021. The outcome is the offering yield in percent, so -0.046 is -4.6 bp. Each row reports, for that maturity bin, the coefficient on the column's issuer-average liquidity measure: (1) an indicator that daily MSRB trades per bond exceed the TRACE median for plain-vanilla corporate debentures and medium-term notes; (2) an indicator for the top quartile of municipal trade frequency; (3) the continuous (cont.) spread, the imputed round-trip cost in price units, so 13.4 is 13.4 bp per 0.01 of spread; (4) an indicator for the lowest-spread quartile. Fixed effects interact rounded maturity, rating rounded down, state, issue year–month and call status. Standard errors are clustered by issuer (20,346 issuers; 19,806 in column (3)). $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

C.1.6 Adjustment V: default

What it measures. Both curves contain compensation for default risk. If municipal credit compensation exceeds corporate credit compensation, part of the municipal spread is a default premium. I measure each market's credit compensation with credit default swap (CDS) spreads, the annual premium for protection against default.

Calculation. Following the taxable-equivalent convention of Schwert (2017), I treat the CDS spreads as credit compensation in the comparison $y^M/\xi_H - y^C$. Multiplying this comparison by the household's after-tax share gives the puzzle, $y^M - \xi_H y^C = \xi_H(y^M/\xi_H - y^C)$, so the CDS difference receives the same multiplier. With

matched municipal and corporate spreads s_b^M and s_b^C for curve observation b, and its dated household after-tax share $\xi_{H,b}$, the adjustment is

$$\mathrm{Def}(T) = \mathrm{Avg}_T[\xi_{H,b} s_b^M] - \mathrm{Avg}_T[\xi_{H,b} s_b^C], \tag{123}$$

where Avg_T is the three-stage averaging of Appendix C.1.1 with whole-notch in place of half-notch rating cells. Equal credit spreads give a zero adjustment. A positive adjustment means that municipal credit compensation explains part of the puzzle.

State general-obligation (GO) Markit CDS supply the municipal spread and are matched to A-or-above bonds by rating, maturity and quarter. The corporate spread weights pooled AAA, AA and A Markit CDS by each year's par shares of corporate bonds with at least \$250 million outstanding (on average 8.6%, 22.8% and 68.6% over 2000–2020), so the credit comparison has the same rating mix as the corporate yield curve. The CDS sample begins in 2004 and contains 1,559,971 curve observations. On this sample, the thirty-year spread over the household benchmark is 165.7 basis points, against 159.8 in the full sample. The full-sample accounting uses the maturity profile of the adjustment estimated on this sample. Beyond quoted tenors, the cumulative spread $T\,\mathrm{CDS}(T)$ is extrapolated linearly. Exactly observed cells cover 49% of the panel at 1–10 years, 47% at 11–20 and 13% at 21–30 (Figure 20(c)).

Result and sensitivity. The default adjustment is positive at every maturity and explains 12.2 basis points of the thirty-year puzzle (Figure 20(b)). Its range, [6.1, 16.4] basis points, spans five variants: the baseline, cells with at least half of their observations quoted or interpolated, the sample from 2007 and CDS-to-cash basis factors of 0.75 and 0.50, which scale both CDS proxies (Table 20). The baseline follows the corporate curve's rating mix and lies between a pooled AA corporate leg, which gives 16.5 basis points, and a pooled A leg, which gives 6.9.

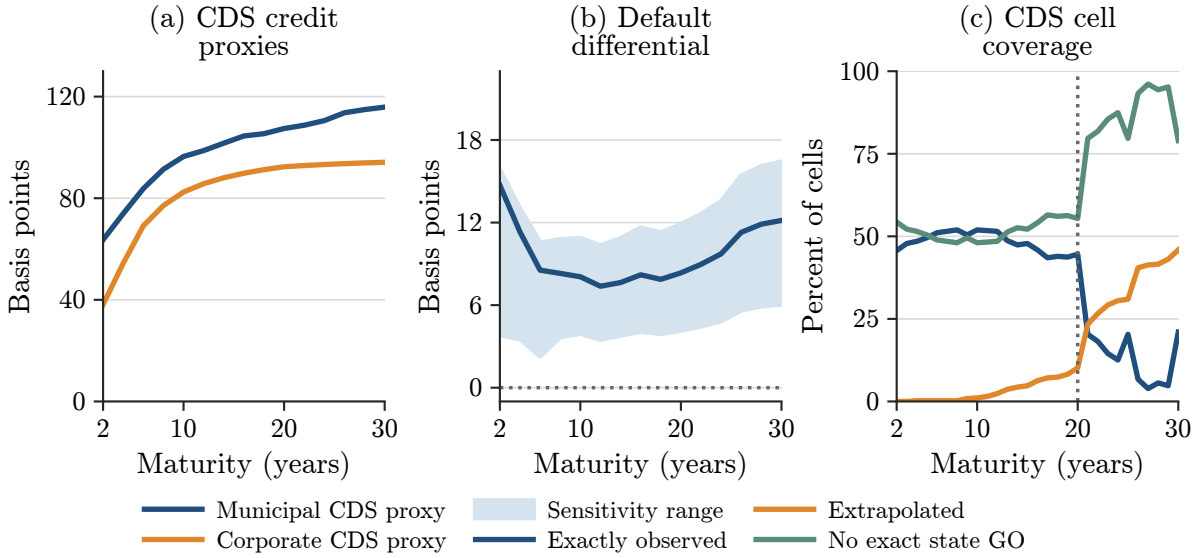


Figure 20: Default adjustment on the A-or-above CDS panel. (a) Municipal and corporate CDS proxies in taxable-equivalent units. (b) Their differential after the dated household tax mapping in (123), with the five-variant sensitivity range. (c) Shares of rating-matched CDS cells that are exactly observed, extrapolated or without a state GO observation (no exact state GO). The last two overlap, so the three shares are not a partition. The vertical rule marks twenty years.

C.1.7 Comparing the model with the adjusted data

The model comparison in Section 5.3 preserves the model's municipal-to-taxable yield ratio and applies it to the observed corporate yield. Subtracting the observed household benchmark then puts the model gap on the same basis as the empirical adjustments. With bars denoting sample averages and superscripts m for model and d for data,

$$\widetilde{y}_T^M = \frac{\bar{y}_T^{M,m}}{\bar{y}_T^{C,m}} \bar{y}_T^{C,d}, \qquad \widetilde{\mathcal{G}}_T = \widetilde{y}_T^M - B_T^d.$$

Table 20: Default sensitivity in basis points. Every column uses the dated household tax mapping of (123). The 2007+ column starts the sample in 2007, the basis columns scale both CDS proxies by the stated factor, and the quality-capped column keeps cells with at least half of their source observations quoted or interpolated.

T	2007+	basis 0.50	basis 0.75	baseline	quality-capped
2	15.79	7.35	11.03	14.71	3.85
6	10.53	4.27	6.41	8.54	2.31
10	10.85	4.03	6.05	8.06	3.95
14	10.83	3.82	5.73	7.64	4.00
18	11.25	3.94	5.91	7.87	4.53
22	12.59	4.48	6.72	8.97	6.84
26	15.37	5.65	8.47	11.29	12.64
30	16.40	6.08	9.11	12.15	13.85

Here $\bar{y}_T^{C,d}$ is the corporate curve y_T^C of (115), and $B_T^d = \bar{\xi}_H(T)\bar{y}_T^{C,d}$ is the observed household benchmark. All yields use nominal semiannual units, and dated after-tax yields are formed before the averaging of Appendix C.1.1. Figure 8 and Table 4 report the normalized comparison. For reference, with the same averaging and issue-quarter rate states, the gaps on the model's own taxable curve are −0.2, 36.2 and 107.4 basis points at two, ten and thirty years. The thirty-year value differs from the 102.4 basis points of Section 5.2, which averages the six calibration periods equally, because this averaging follows the 2000–2020 bond sample. Section 5.3 explains the distinction.

Across the fifteen maturities, the normalized root-mean-square unexplained residual is 7.53 basis points (12.94 unnormalized), and the maximum absolute residual is 10.72 at sixteen years. Fourteen of fifteen absolute residuals are below 10 basis points. The signed residuals are −7.83 at twelve years, −6.72 at twenty and +5.62 at thirty.

C.2 Paired issues: Build America Bonds and tax-exempt bonds

Build America Bonds (BABs) are taxable bonds sold by the same issuers as tax-exempt municipal bonds. Section 3.8 compares the two to check whether the widening gap over the household benchmark survives when issuer identity is held fixed and the BAB yield replaces the corporate benchmark. That comparison's sample, full estimates and robustness checks follow. The gap widens with maturity in every specification.

Design and estimating equation. The issue-level Mergent sample, one observation per CUSIP, contains non-AMT bonds issued in 2009Q2–2010Q4. Table 21 uses AA-and-above (agency score ≤ 3) and A-or-above (≤ 7) issues. Figure 3 uses the AA-and-above issues and averages yields by bond type and integer maturity $n \in \{1, \ldots, 30\}$. Multiplying BAB offering yields by 0.65, one minus the period's top federal household rate, puts them on the top-bracket household's federal after-tax scale. With y_b the tax-exempt offering yield or the scaled BAB yield, $EXMP_b$ a tax-exempt indicator and n_b maturity in years, I estimate

$$y_b \;=\; \alpha_1\, EXMP_b \;+\; \beta\,(EXMP_b \times n_b) \;+\; \varphi\,(0.65\, y^C_{n_b,t_b}) \;+\; \lambda_{j(b)\times n_b \times q_b} \;+\; \varepsilon_b. \tag{124}$$

The fixed effect compares bonds of issuer $j(b)$ with the same maturity and issue quarter q_b, so β is the change per year of maturity in the tax-exempt yield minus the scaled BAB yield. The corporate yield $y^C_{n_b,t_b}$, measured at the bond's maturity and issue month t_b, varies across issue months within matched quarters, which identifies φ. Equation (4) of Section 3.8 uses the unscaled corporate yield as its control, so only that control's coefficient, $\varphi_C = 0.65\varphi$, and its standard error change.

Identification. With issuer identity fixed, the widening gap supports the participation mechanism under the 0.65 approximation and when the two bond types have comparable security design and issuance premia. In the AA-and-above sample, optional calls appear on 49% of BABs and 39% of tax-exempt bonds, so the call-matched columns also require the same call status.

Results. Table 21 shows a positive maturity gradient in every column. All maturity interactions are significant at one percent, and the Panel B bin coefficients rise with maturity.

Robustness. Call matching lowers the longest-bin coefficient by about 15% in the AA-and-above sample and leaves the $[5, 10)$ coefficient unchanged. An exact after-tax internal rate of return would treat original-issue discount, acquisition premium, interest income and basis separately for BABs and exempt bonds, so I also check

the 0.65 scaling on bonds offered near par. Restricting the call-matched AA-and-above sample to offering prices of 99–101 leaves a maturity gradient of 2.1 basis points per year (s.e. 0.6), against 3.1 in the full call-matched sample. The 98–102 restriction gives 2.7 basis points (s.e. 0.4). Both remain significant at one percent. Finally, the scaling uses the 35% top federal rate, below the 41.1% average household wedge of the AA-and-above curve (Appendix Figure 27). Scaling BAB yields by the lower after-tax share implied by that wedge would widen the gap.

Table 21: Paired issues: the tax-exempt-versus-BAB gap widens with maturity.

	AA and above (score ≤ 3)		A or above (score ≤ 7)	
	(1) base	(2) call-matched	(3) base	(4) call-matched
Panel A. Continuous maturity				
Tax-exempt	-0.102^{***}	-0.091^{**}	-0.062^{**}	-0.043
	(0.034)	(0.036)	(0.027)	(0.030)
Tax-exempt $\times\ n_b$	0.034^{***}	0.031^{***}	0.033^{***}	0.030^{***}
	(0.003)	(0.003)	(0.002)	(0.003)
After-tax corporate yield	0.379^{***}	0.376^{***}	0.264^{*}	0.261^{*}
	(0.138)	(0.135)	(0.134)	(0.130)
Panel B. Five-year maturity bins				
Tax-exempt	-0.048	-0.050	-0.007	-0.008
	(0.033)	(0.033)	(0.024)	(0.024)
Tax-exempt $\times\ [5, 10)$ y	0.192^{***}	0.191^{***}	0.192^{***}	0.190^{***}
	(0.022)	(0.022)	(0.017)	(0.017)
Tax-exempt $\times\ [10, 15)$ y	0.396^{***}	0.364^{***}	0.368^{***}	0.342^{***}
	(0.033)	(0.033)	(0.026)	(0.030)
Tax-exempt $\times\ [15, 20)$ y	0.450^{***}	0.390^{***}	0.419^{***}	0.403^{***}
	(0.041)	(0.039)	(0.029)	(0.039)
Tax-exempt $\times\ [20, 25)$ y	0.576^{***}	0.540^{***}	0.593^{***}	0.510^{***}
	(0.061)	(0.066)	(0.071)	(0.066)
Tax-exempt $\times\ [25, 30]$ y	0.755^{***}	0.639^{***}	0.996^{***}	0.804^{***}
	(0.079)	(0.039)	(0.074)	(0.060)
After-tax corporate yield	0.380^{***}	0.378^{***}	0.264^{*}	0.262^{**}
	(0.138)	(0.135)	(0.134)	(0.130)
Issuer $\times$ maturity $\times$ quarter FE	Yes	Yes	Yes	Yes
$\times$ call status	No	Yes	No	Yes
Jurisdiction cluster	Yes	Yes	Yes	Yes
Observations	29,839	28,890	61,816	59,886

Notes. FE denotes fixed effects. Offering yields in percent, 2009Q2–2010Q4, excluding AMT bonds. BAB yields are multiplied by 0.65. The corporate-yield row reports φ, the coefficient on the after-tax control in equation (124). The omitted bin is $[1, 5)$ years. Standard errors are clustered by issuing jurisdiction (51 in columns (1)–(2), 52 in columns (3)–(4)). Call-matched columns add the optional-call flag to the absorbed issuer-by-maturity-by-quarter cell. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.3 The municipal puzzle on unlimited-tax general-obligation bonds

This check, cited in Section 3.6, asks whether the declining implied tax persists among high-grade general-obligation (GO) bonds. It does, with a decline as large as in the A-or-above full sample.

Design. Unlimited-tax GO bonds pledge the issuer's full faith and credit and unlimited taxing power, while revenue bonds pledge project revenues. Figure 21 restricts the comparison to AA-and-above unlimited-tax GO bonds[27] passing the main call-pricing and calendar screens (Appendix C.1.2) and applies only the call and

[27] Mergent security code "K".

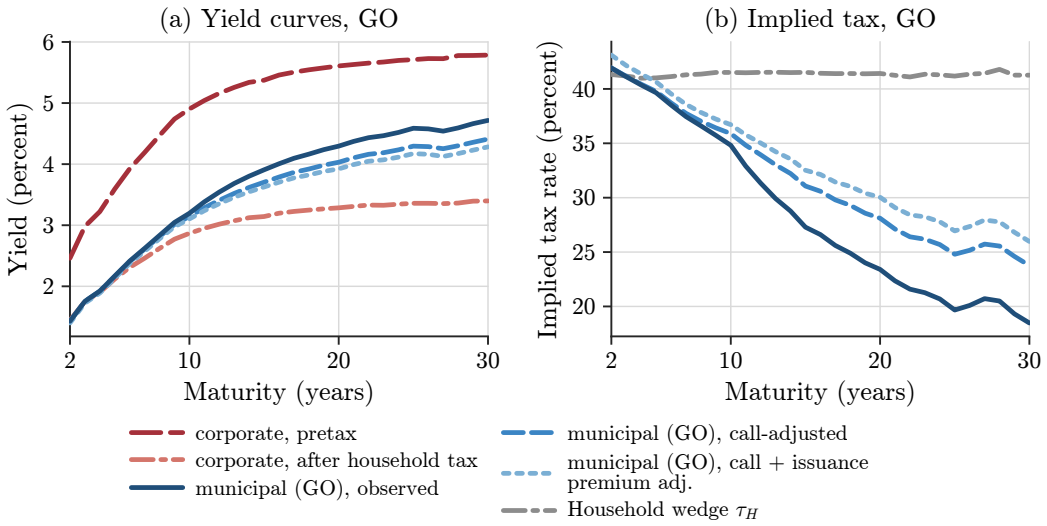


Figure 21: The municipal puzzle on AA-and-above unlimited-tax GO bonds: yields in panel (a) and the implied tax against the household wedge in panel (b). The 465,258 bonds pass the main call-pricing and calendar screens, and curves use the sample's own weights and household benchmark. Call adjustments use the calibrated model. Few GO bonds have an early institutional-size purchase, so the issuance-premium adjustment applies the all-bond estimate (Appendix C.1.3).

issuance-premium adjustments. Appendix C.1.6 separately assesses the aggregate default adjustment, which uses state-government CDS as a proxy.

Result. Adjusted GO implied tax falls from 43.1% at two years to 26.0% at thirty. This 17.1-point decline matches the A-or-above full sample and exceeds the 14.6-point decline of the AA-and-above sample. The result agrees with earlier evidence that the maturity pattern persists after default and liquidity adjustments and in high-grade or GO samples (Longstaff, 2011; Fleckenstein and Longstaff, 2025; Chalmers, 1998).

Robustness. The screens exclude 4,029 GO bonds (0.86%). Holding the issuance premium fixed, they change the decline by 0.49 points relative to the unscreened call-adjusted GO series, almost entirely at the short end.

C.4 The single-tax-rate benchmark of Green (1993)

Sections 2 and 3.7 cite Green (1993), whose single-tax-rate explanation attributes rising municipal-to-taxable yield ratios to the timing of tax on taxable bonds. His pricing relation need not bind, so I keep the household benchmark. Participation in the model also answers his critique of clientele explanations.

Green's mechanism. Green's package buys two five-percent market-discount bonds and shorts one ten-percent par bond of equal maturity, which defers tax. After purchase, the package's only net cash flow is one dollar of principal at maturity. At tax rate τ, the package costs P_T and pays $1-\tau(1-P_T)$ after tax at maturity. With d_T the price of a dollar paid at T,

$$P_T = d_T\big[1-\tau(1-P_T)\big], \qquad \text{so} \qquad P_T = \frac{(1-\tau)\,d_T}{1-\tau d_T}.$$

Asymmetric taxation of long and short positions limits the strategy, because borrowing and short-sale deductions cannot exceed the investment income they support. Green's high-tax investor prices both municipal bonds and the package. Dealers face no such limit and tie taxable par-bond prices to the package, which lowers long taxable yields below fully taxed returns. One investor can thus price municipal bonds at every maturity while the yield ratio rises. Fama (1977), by contrast, requires different price-setting investors at different maturities.

Why the relation need not bind. Deep market-discount bonds can be scarce, annual accrual of original-issue discount removes the deferral, and shorts consume deduction capacity. Households that forgo the package can accept the household benchmark $\xi_H y_T^C$, which lies below Green's relation. With sufficient capacity, these higher-benefit households set municipal prices, and investors using the strategy do not. Raising municipal yields to Green's relation would require shorting municipal bonds, which faces the frictions in Appendix D.5.

His relation therefore need not bind. Green's own 1987–1990 estimates also leave thirty-year municipal yields about twenty taxable-equivalent basis points above his relation, with a gap that increases with maturity.

Green's critique of clientele explanations. Green rejects earlier clientele explanations because long municipal yields remained high relative to after-tax taxable yields after the Tax Reform Act of 1986 reduced banks' tax advantages and municipal holdings. In the model, participation depends on exemption benefits and the duration benefit (Section 4.3), so the clientele that prices each maturity can change. Households alone hold short maturities and set their yields. Institutions with smaller exemption benefits also hold long maturities, so long yields reflect these institutions' tax treatment too. Implied taxes therefore respond to supply and risk capacity. Section 3.4 documents ownership across exemption benefits. I keep the household benchmark $\xi_H y_T^C$ and measure tax timing separately in the capital-gains adjustment (Appendix C.1.4). To distinguish participation from tax timing alone, P1 tests household state-tax pass-through and P2 tests a bank-only benefit (Sections 6.1 and 6.2).

C.5 State-tax pass-through: full regression tables

The P1 estimates of Section 6.1 use the sample, outcome and inference described below. Two tables then check whether the declining state-tax pass-through depends on the yield construction or on state credit risk. Both keep specification (26), and the decline survives both checks.

Design. The main sample is federally exempt AA-and-above issuance (agency score ≤ 3) from 2000–2017 in the 47 states retained in Section 6.1, counting the District of Columbia as a state. It excludes zero-maturity bonds and groups maturities into $[0, 5)$, $[5, 10)$, $[10, 15)$, $[15, 20)$ and $[20, 30]$ years. Every regression keeps the controls of equation (26): the six lagged state controls and the rounded average rating in $Z_{b,t}$, each interacted with all five bins, and log offering par and security-type indicators in V_b. Stock and issuance controls use all observed federally exempt bonds, regardless of rating. Outstanding par equals the offering amount until the date of the reported outstanding amount, and that amount afterward. Fixed effects are cells defined by issue year, issue month, rounded maturity and callability. Observations with missing controls are dropped without imputation, and singleton cells are then removed. The zero-yield columns have 681,838 observations, and six missing offering yields leave 681,832 in the offering-yield columns.

Call-adjusted outcome. The main outcome uses call-adjusted zero-coupon yields. As in the main adjustment (Appendix C.1.2), a constant spread over the calibrated model's supply-weighted municipal curve for the bond's issue period is fitted to each bond's observed callable price. Each bond starts at the observed rate state in the final month of its issue quarter. P1 places cash flows on a regular semiannual grid. Observed call dates map to the nearest grid date within 0.27 years, and the call is exercisable on every grid date from the grid date nearest the first call date. The main adjustment follows the contractual call calendar. Callable bonds are excluded when the first call date lies more than 0.27 years before the first semiannual grid date or the call schedule lacks a usable date and price. Table 22 shows that the decline does not depend on this adjustment.

Identification and inference. Unweighted OLS is the main estimator. For the IV comparison, the 1990 rate interacted with each maturity bin instruments the corresponding current-rate interaction. The exclusion restriction is that, conditional on the controls and fixed effects, the historical rate affects current implied taxes only through the current rate. The first stage gives a Kleibergen–Paap (KP) rk Wald F of 10.4 and Sanderson–Windmeijer (SW) conditional statistics of 76.1, 115.9, 104.4, 155.7 and 230.4 for the current-rate interactions from the shortest to the longest bin. Standard errors are clustered by state, with $t(G-1)$ critical values for G states ($t(46)$ in the main sample). Each table's decline p-value tests the endpoint equality $\beta_1 = \beta_5$.

C.5.1 Comparison across yield constructions

Table 22 compares Mergent offering yields, which for premium callable bonds can be yields to worst, with zero-coupon yields from offering prices, with and without call adjustment. Both OLS and IV reject endpoint equality at one percent for every construction, so neither the call adjustment nor the zero-coupon conversion drives the decline.

C.5.2 State credit-risk controls

Table 23 keeps pre-2018 observations that have a five-year state-government CDS quote in the preceding month and adds that quote interacted with the maturity bins. Because the CDS quotes begin in 2004, this

Table 22: P1 across yield constructions, 2000–2017.

	Offering yield		Call-adjusted zero		Unadjusted zero	
	OLS	IV	OLS	IV	OLS	IV
β_1: $[0, 5)$ years	0.985^{***}	1.084^{***}	1.069^{***}	1.088^{***}	1.080^{***}	1.102^{***}
	(0.238)	(0.249)	(0.195)	(0.225)	(0.196)	(0.227)
β_2: $[5, 10)$ years	0.635^{***}	0.648^{***}	0.677^{***}	0.694^{***}	0.683^{***}	0.693^{***}
	(0.075)	(0.093)	(0.069)	(0.086)	(0.069)	(0.087)
β_3: $[10, 15)$ years	0.414^{***}	0.410^{***}	0.484^{***}	0.447^{***}	0.515^{***}	0.479^{***}
	(0.053)	(0.075)	(0.096)	(0.118)	(0.100)	(0.117)
β_4: $[15, 20)$ years	0.268^{***}	0.257^{***}	0.390^{***}	0.326^{***}	0.428^{***}	0.360^{***}
	(0.060)	(0.067)	(0.102)	(0.116)	(0.108)	(0.121)
β_5: $[20, 30]$ years	0.193^{***}	0.152^{**}	0.305^{***}	0.229^{***}	0.335^{***}	0.255^{***}
	(0.052)	(0.057)	(0.075)	(0.081)	(0.078)	(0.085)
Decline $\beta_1 - \beta_5$	0.791^{***}	0.933^{***}	0.764^{***}	0.859^{***}	0.745^{***}	0.847^{***}
p-value of decline	0.004	< 0.001	0.002	< 0.001	0.003	0.001
Issue month × maturity × call FE	Yes	Yes	Yes	Yes	Yes	Yes
State cluster	Yes	Yes	Yes	Yes	Yes	Yes
Observations	681,832	681,832	681,838	681,838	681,838	681,838
First-stage F (Kleibergen–Paap)	—	10.4	—	10.4	—	10.4

Notes. FE denotes fixed effects. Outcomes are implied state taxes (25) from Mergent offering yields or from zero-coupon yields with and without the call adjustment. The call-adjusted zero is the main outcome. Controls, fixed effects and state-clustered inference (47 states) follow Appendix C.5, 2000–2017. IV uses bin-interacted 1990 rates. The decline row reports $\beta_1 - \beta_5$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

sample covers 2004–2017. On this common sample of 345,877 bonds from 25 states, the CDS controls raise the longest-bin OLS estimate from 0.335 to 0.416 and the IV estimate from 0.322 to 0.368. In every row, both estimators keep the longest-bin estimate far below the shortest-bin estimate and reject endpoint equality at five percent.

Table 23: P1 sample and credit controls, 2000–2017.

Sample	Obs.	States	β_1, OLS	β_5, OLS	Decline p	β_1, IV	β_5, IV	Decline p
Main sample	681,838	47	1.069^{***}	0.305^{***}	0.002	1.088^{***}	0.229^{***}	< 0.001
				(0.075)			(0.081)	
Observed CDS sample	345,877	25	1.210^{***}	0.335^{***}	0.009	1.349^{***}	0.322^{***}	0.009
				(0.081)			(0.098)	
CDS sample + CDS × bin	345,877	25	1.400^{***}	0.416^{***}	0.011	1.391^{***}	0.368^{***}	0.025
				(0.068)			(0.087)	

Notes. Controls, fixed effects and state-clustered inference follow Appendix C.5, 2000–2017. The observed-CDS sample keeps bonds with a prior-month five-year state-government CDS quote, without imputation, and covers 2004–2017. The last row adds that quote interacted with the maturity bins. IV uses bin-interacted 1990 rates. Parentheses give standard errors of β_5, and decline p-values test $\beta_1 = \beta_5$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.6 State-tax pass-through: instrument vintages, weighting and further checks

These checks ask whether P1 depends on the historical tax-rate instrument, the weight assigned to each state, fixed state tax levels or any single state, and they report the post-2018 tax regime separately. All retain the controls, fixed effects and state-clustered inference of Appendix C.5, and every check shows a declining profile.

Instrument vintages. The baseline IV instrument is the state's 1990 top income-tax rate, with the exclusion restriction of Appendix C.5. Tables 24 and 25 compare instruments from different years (vintages) in 1945–2000 on the same pre-2018 sample. Table 24 replaces the five current-rate bin interactions in (26) with

the current rate and its interaction with maturity in years, n_b. The interaction's coefficient is the pass-through gradient per year. IV instruments the current rate and its maturity interaction with the historical rate and its maturity interaction. OLS and all instrument vintages give negative gradients, unweighted and with equal total weight per state. Unweighted, every vintage's gradient is significant at five percent, with Kleibergen–Paap F statistics of 7.8 to 65.1. Table 25 compares the five-bin IV profiles with OLS, and Figure 22 plots them. Every vintage rejects endpoint equality at five percent.

Table 24: Continuous-maturity P1 checks, 2000–2017.

		Unweighted				Equal per state	
Vintage	Corr.	Gradient	s.e.	KP F	SW F	Gradient	KP F
OLS	—	-0.0392^{***}	(0.0131)	—	—	-0.0326^{**}	—
1945	0.40	-0.0535^{**}	(0.0251)	8.0	19.2	-0.0440	2.8
1950	0.31	-0.0607^{**}	(0.0273)	7.8	18.4	-0.0878	1.8
1960	0.38	-0.0620^{**}	(0.0254)	9.0	28.8	-0.0997^{*}	2.5
1970	0.51	-0.0574^{***}	(0.0201)	11.5	45.2	-0.0795^{***}	7.3
1980	0.76	-0.0424^{**}	(0.0164)	10.6	103.4	-0.0385^{**}	26.9
1990	0.82	-0.0440^{***}	(0.0127)	30.1	404.1	-0.0307^{**}	25.9
2000	0.88	-0.0399^{***}	(0.0124)	65.1	998.5	-0.0258^{*}	36.5

Notes. Controls, fixed effects and state-clustered inference follow Appendix C.5, 2000–2017. The gradient is the coefficient on the current rate times maturity n_b, the change in pass-through per year. IV instruments the current rate and its maturity interaction with the vintage's rate and its maturity interaction. KP F is the Kleibergen–Paap statistic and SW F the Sanderson–Windmeijer conditional statistic for the interaction. Corr. is the cross-state correlation between each state's mean current top rate over the estimation sample and its vintage-year top rate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 25: P1 instrument-vintage checks, 2000–2017.

Estimator/vintage	β_1	β_2	β_3	β_4	β_5	Decline p	KP F
OLS (main)	1.069^{***}	0.677^{***}	0.484^{***}	0.390^{***}	0.305^{***}	0.002	—
	(0.195)	(0.069)	(0.096)	(0.102)	(0.075)		
1945	1.257^{***}	0.868^{***}	0.551^{***}	0.363^{**}	0.245^{**}	0.027	3.1
	(0.421)	(0.158)	(0.158)	(0.143)	(0.103)		
1950	1.453^{***}	0.942^{***}	0.622^{***}	0.428^{***}	0.313^{***}	0.024	3.1
	(0.473)	(0.181)	(0.153)	(0.143)	(0.116)		
1960	1.483^{***}	0.942^{***}	0.627^{***}	0.432^{***}	0.332^{***}	0.014	3.6
	(0.439)	(0.167)	(0.147)	(0.140)	(0.113)		
1970	1.394^{***}	0.858^{***}	0.566^{***}	0.400^{***}	0.325^{***}	0.006	4.4
	(0.350)	(0.125)	(0.143)	(0.141)	(0.114)		
1980	1.084^{***}	0.756^{***}	0.528^{***}	0.374^{***}	0.280^{***}	0.010	3.8
	(0.273)	(0.083)	(0.121)	(0.120)	(0.084)		
1990	1.088^{***}	0.694^{***}	0.447^{***}	0.326^{***}	0.229^{***}	< 0.001	10.4
	(0.225)	(0.086)	(0.118)	(0.116)	(0.081)		
2000	1.072^{***}	0.678^{***}	0.480^{***}	0.378^{***}	0.280^{***}	0.001	23.5
	(0.215)	(0.079)	(0.095)	(0.096)	(0.069)		

Notes. Controls, fixed effects and state-clustered inference follow Appendix C.5, 2000–2017. IV uses the vintage-year top rate interacted with each bin. KP F is the Kleibergen–Paap statistic, and decline p-values test $\beta_1 = \beta_5$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Weighting. Unweighted OLS gives states with more bonds more influence. Every weighting in Table 26 (unweighted, equal total weight per state or per state-year, and $1/\sqrt{n_{\text{state}}}$, with n_{state} the number of observations in the bond's state) retains a declining OLS profile, with endpoint differences significant at one percent.

State fixed effects. State fixed effects absorb fixed tax levels. Adding them to the pre-2018 OLS specification, on 681,838 bonds, gives a short-bin coefficient of 0.665 (s.e. 0.201) and a long-bin coefficient of -0.112 (0.212). The decline is 0.777 (0.220, $p = 0.001$). IV is not identified with state effects. The five excluded

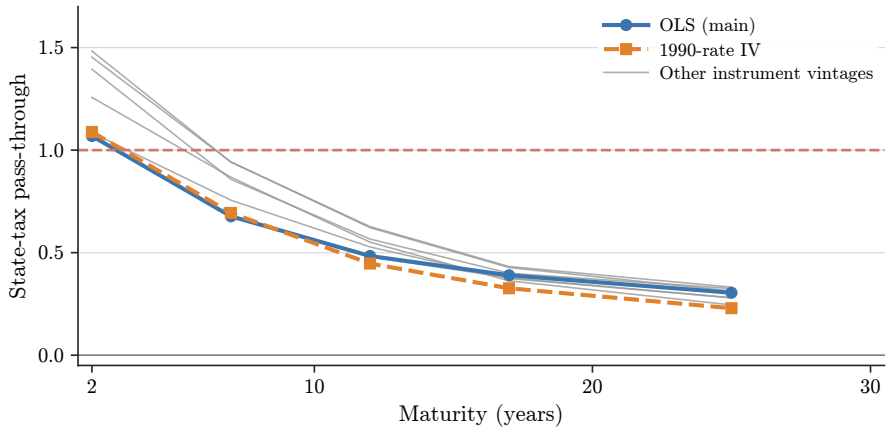


Figure 22: State-tax pass-through using historical tax-rate instruments, 2000–2017. Blue: main OLS profile. Orange: IV using the 1990 rate. Gray: IV using the 1945, 1950, 1960, 1970, 1980 and 2000 rates. Dashed red: representative-tax benchmark. Table 25 reports state-clustered uncertainty.

Table 26: P1 OLS estimates under four weighting schemes, 2000–2017.

Weighting	β_1	β_2	β_3	β_4	β_5	Decline p
Unweighted (main)	1.069^{***}	0.677^{***}	0.484^{***}	0.390^{***}	0.305^{***}	0.002
	(0.195)	(0.069)	(0.096)	(0.102)	(0.075)	
Equal per state	0.863^{***}	0.543^{***}	0.374^{***}	0.308^{***}	0.234^{**}	0.006
	(0.191)	(0.082)	(0.101)	(0.111)	(0.088)	
Equal per state-year	0.911^{***}	0.509^{***}	0.317^{***}	0.261^{***}	0.211^{***}	0.002
	(0.190)	(0.069)	(0.080)	(0.090)	(0.074)	
$1/\sqrt{n_{\text{state}}}$	0.990^{***}	0.617^{***}	0.420^{***}	0.344^{***}	0.282^{***}	0.006
	(0.202)	(0.080)	(0.109)	(0.121)	(0.091)	

Notes. OLS with the controls, fixed effects and state-clustered inference of Appendix C.5, 2000–2017. The equal-weight rows give each state or state-year the same total weight, and n_{state} is the bond's state observation count, all measured after the complete-case and singleton restrictions. Decline p-values test $\beta_1 = \beta_5$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

instruments, the 1990 rate interacted with each maturity bin, sum to the state's 1990 rate, which is collinear with the state effects.

Leave-one-state-out. All 47 single-state omissions retain a positive short-to-long decline under both OLS and IV (Figure 23). OLS long-bin estimates span 0.232–0.482, with declines 0.452–0.889. IV long-bin estimates span 0.116–0.315, with declines 0.669–1.004. The largest endpoint-test p-values are 0.062 for OLS and 0.036 for IV.

The two tax regimes. The main P1 sample ends in 2017, so its after-tax share of taxable interest, $\xi_s^{\text{st}}\xi_t^{\text{fed}}$, follows the same deduction convention as the model's six pre-2018 periods in Figure 9. From 2018, the TCJA capped state and local tax (SALT) deductions at \$10,000 per return. When the cap binds, an additional dollar of state tax is no longer federally deductible, and the combined after-tax share on taxable interest becomes $\xi_t^{\text{fed}} - \tau_s^{\text{st}}$, with $\xi_t^{\text{fed}} = 0.63$ from the 37% top rate (notation of Section 6.1). For a household whose cap binds, representative-tax indifference gives $y_{s,t}^M(T) = (\xi_t^{\text{fed}} - \tau_s^{\text{st}})\, y_t^C(T)$, and the implied state tax for this regime, which equals the statutory rate under the benchmark, is[28]

$$ImpliedStateTax_{s,t}^{\text{cap}}(T) \;\equiv\; \xi_t^{\text{fed}} - \frac{y_{s,t}^M(T)}{y_t^C(T)}. \tag{125}$$

Table 27 estimates each regime separately. Both estimators reject endpoint equality at one percent in each period.

[28] See the IRS, 2017 Schedule A instructions, and its summary of the 2018 changes.

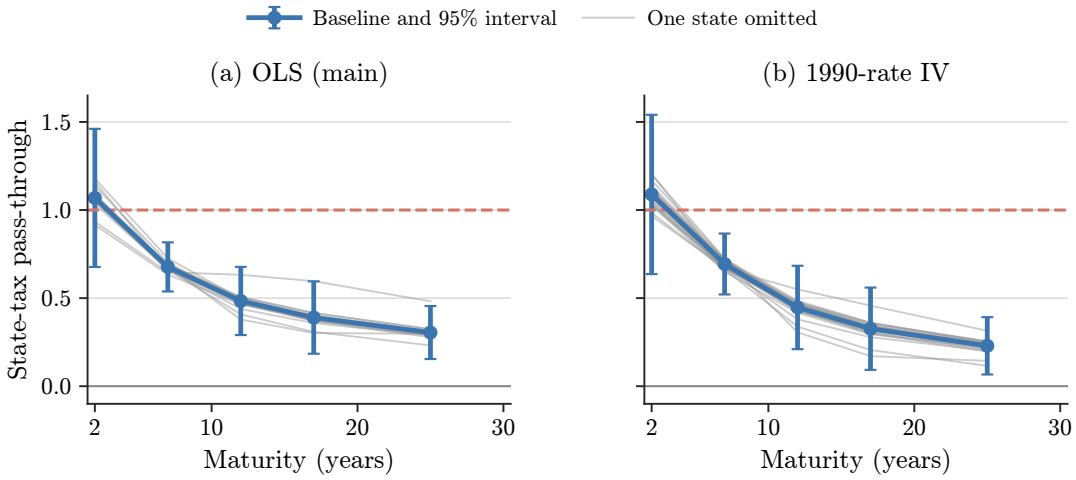


Figure 23: Leave-one-state-out P1 profiles, 2000–2017. Gray: each of the 47 state omissions. Blue: baseline estimates with state-clustered 95% intervals. Panel (a) uses OLS and panel (b) the 1990-rate instruments. Dashed: representative-tax benchmark.

Table 27: P1 under the two federal/state tax regimes.

	2000–2017 (main)		2018–2020	
	OLS	IV	OLS	IV
β_1: $[0, 5)$ years	1.069^{***}	1.088^{***}	0.948^{***}	0.909^{***}
	(0.195)	(0.225)	(0.182)	(0.246)
β_2: $[5, 10)$ years	0.677^{***}	0.694^{***}	0.834^{***}	0.879^{***}
	(0.069)	(0.086)	(0.121)	(0.143)
β_3: $[10, 15)$ years	0.484^{***}	0.447^{***}	0.446^{***}	0.466^{***}
	(0.096)	(0.118)	(0.061)	(0.074)
β_4: $[15, 20)$ years	0.390^{***}	0.326^{***}	0.312^{***}	0.323^{***}
	(0.102)	(0.116)	(0.078)	(0.105)
β_5: $[20, 30]$ years	0.305^{***}	0.229^{***}	0.088	0.082
	(0.075)	(0.081)	(0.091)	(0.109)
Decline $\beta_1 - \beta_5$	0.764^{***}	0.859^{***}	0.860^{***}	0.827^{***}
p-value of decline	0.002	< 0.001	< 0.001	0.008
Issue month × maturity × call FE	Yes	Yes	Yes	Yes
State cluster	Yes	Yes	Yes	Yes
Observations	681,838	681,838	146,174	146,174
First-stage F (Kleibergen–Paap)	—	10.4	—	7.9

Notes. FE denotes fixed effects. Controls, fixed effects and state-clustered inference (47 states) follow Appendix C.5. IV uses bin-interacted 1990 rates. The outcome is (25) before 2018 and, for a household whose SALT deduction cap binds, (125) from 2018. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.7 The bank-qualified discontinuity: construction and robustness

Bank qualification changes banks' tax benefit without changing households' exemption, so it measures what share of a benefit that only banks receive is reflected in yields at each maturity. P2 predicts that this capitalization share rises with maturity. In Table 28, the 2SLS designation premium in offering yields is 1.8 basis points at 1–2 years, indistinguishable from zero, and 18.4 at 10–20 years. With each yield divided by its own expected annual benefit before estimation, the estimated capitalization shares are 0.17 and 0.52. The gradient survives alternative windows, the construction choices of Dagostino (2025) and splits by benefit regime. A placebo cutoff at $5 million shows essentially no designation jump.

Table 28: Bank-qualified designation and yield premia by maturity.

	(1)	(2)	(3)
Dependent variable	BQ designation	Offering yield (bp)	Yield / expected benefit
Estimator	OLS	2SLS	2SLS
Regressor, interacted with each maturity bin	Below-cap indicator	BQ designation	BQ designation
Reported effect	Designation increase	Yield reduction	Scaled yield reduction
1–2 years	0.42^{***}	1.8	0.17
	(0.015)	(2.2)	(0.11)
2–5 years	0.42^{***}	8.7^{***}	0.31^{***}
	(0.015)	(2.2)	(0.08)
5–10 years	0.44^{***}	13.8^{***}	0.41^{***}
	(0.015)	(2.2)	(0.06)
10–20 years	0.45^{***}	18.4^{***}	0.52^{***}
	(0.017)	(2.3)	(0.07)
Over 20 years	0.43^{***}	18.2^{***}	0.47^{**}
	(0.027)	(6.3)	(0.19)
10–20 minus 1–2 years		16.5^{***}	0.34^{***}
		(2.7)	(0.11)
Fixed effects	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes
Observations	355,372	355,372	355,372
First-stage F (Kleibergen–Paap)		156.0	156.0

Notes. Each column is a separate pooled regression containing all five maturity interactions. Column (1) reports the designation increase associated with below-cap status, estimated by OLS. Columns (2)–(3) instrument designation by below-cap status within each bin, as in (28), and report $-\beta_g$, so positive numbers mean lower yields on designated bonds. Column (3) divides each bond's yield by its expected annual benefit $\bar{X}_b = 0.8\,\tau_{B,t}\hat{c}_b$ before estimation (Appendix C.7.2), so its entry in each maturity bin estimates the capitalization share. Contrasts use unrounded estimates. Parentheses give standard errors clustered by issuer (15,889 issuers). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fixed effects in every column. Maturity bucket × year, state × year, state × offering year–month, rating category × year, security type × year, use of proceeds, coupon bin × maturity bucket and callability × maturity bucket × year. Other controls are log tranche par, maturity-bin indicators and separate linear issuance slopes on each side of the cap within each bin.

Sample. Tax-exempt new issues in 2000–2021 within $0.5–4 million of the statutory cap ($30 million in 2009–10, $10 million otherwise). Maturity is measured at issuance. Appendix C.7.2 defines the fixed-effect buckets.

C.7.1 Capitalization of the bank benefit

The designation premium $\varpi_k = y_k^N - y_k^Q$, the non-designated minus the designated yield, measures how much of the bank benefit X is capitalized into maturity-k yields. It is zero in the representative-tax benchmark, X in the bank-only benchmark and $\theta_k X$ in general, where θ_k is the capitalization share derived below, the yield reduction as a fraction of the benefit.

The two benchmarks. Both benchmarks compare annual after-tax carry per dollar of basis on matched bonds, in the notation of Section 6.2, with dates suppressed and taxable yield $y_k^C > 0$ at maturity k.

In the representative-tax benchmark, $y_k^N = (1-\tau_H)y_k^C$, where $\tau_H = 1-\xi_H$ is the household wedge. At this yield, a designated bond earns the bank $y_k^N - 0.2\,\tau_B c$ after the tax cost of its disallowed interest deduction. A taxable bond earns $(1-\tau_B)y_k^C$, so the taxable bond has an advantage of

$$\underbrace{(1-\tau_B)y_k^C}_{\text{taxable alternative}} \quad - \quad \underbrace{\left[(1-\tau_H)y_k^C - 0.2\,\tau_B c\right]}_{\text{qualified bond at the household yield}} \quad = (\tau_H - \tau_B)y_k^C + 0.2\,\tau_B c. \tag{126}$$

The advantage is positive whenever $\tau_H > \tau_B - 0.2\,\tau_B c/y_k^C$, in particular when household rates are at least banks' and financing costs are positive. Banks then still prefer taxable bonds, and designation leaves household taxes unchanged, so $y_k^Q = y_k^N$ and $\varpi_k = 0$ at every maturity.

In the bank-only benchmark, each municipal bond gives banks the same after-tax carry as taxable bonds:

$$\begin{aligned} y_k^N - \underbrace{\tau_B c}_{\text{full disallowance}} &= (1-\tau_B)y_k^C, \\ y_k^Q - \underbrace{0.2\,\tau_B c}_{\text{qualified disallowance}} &= (1-\tau_B)y_k^C. \end{aligned} \tag{127}$$

Subtracting the two conditions cancels the taxable yield and gives $\varpi_k = 0.8\,\tau_B c = X$. With constant financing costs, the premium in basis points is flat in maturity. The expected annual benefit $\bar{X}_b$ (Appendix C.7.2) evaluates this benefit at expected annual financing costs.

Derivation of the designation premium. The derivation compares matched bonds of maturity k in supply $q_k > 0$, with outside opportunities and non-tax characteristics fixed. The designation benefit u is zero for a non-designated bond and X for a designated one. At yield y, household demand is $d_k^H(y)$ and bank demand is $d_k^B(y+u)$, both reflecting baseline financing costs, taxes and risk compensation. For each $u \in [0, X]$, assume that a clearing yield exists, that demands are continuous and nondecreasing with finitely many continuously differentiable regimes, and that total demand has a strictly positive slope at clearing. Then the clearing yield $y_k(u)$ satisfies

$$d_k^H(y_k(u)) + d_k^B(y_k(u)+u) = q_k, \qquad y_k^N = y_k(0), \quad y_k^Q = y_k(X). \tag{128}$$

Differentiating (128) on a smooth segment gives

$$-\frac{dy_k(u)}{du} = \frac{\partial d_k^B/\partial y}{\partial d_k^H/\partial y + \partial d_k^B/\partial y} \equiv w_{Bk}(u) \in [0,1], \tag{129}$$

with demand slopes at the clearing net returns, so $w_{Bk}(u)$ is the banks' share of the total demand response. A larger benefit raises bank demand, and the yield falls to offset it. Integrating across participation boundaries, where $y_k(u)$ is continuous, gives

$$\varpi_k = y_k(0) - y_k(X) = \int_0^X w_{Bk}(u)\,du = \theta_k X, \qquad \theta_k = \frac{1}{X}\int_0^X w_{Bk}(u)\,du. \tag{130}$$

If banks never participate, $\partial d_k^B/\partial y = 0$ and $\varpi_k = 0$, the representative-tax benchmark. If only banks hold the bond, $\partial d_k^H/\partial y = 0$ and $\varpi_k = X$, the bank-only benchmark. When both respond, capitalization lies between them. The capitalization share θ_k is distinct from banks' share of holdings.

The flat premium X is an annual yield-equivalent carry per dollar of tax basis. At issue, a bank's tax basis equals the price it pays, so X compares directly with offering yields.

C.7.2 Design and the benefit measures

The design compares issuers just below and just above the annual issuance cap, with below-cap status instrumenting designation within each maturity bin of (28). Banks' interest expense per dollar of assets measures the cost c in X, and the expected annual benefit $\bar{X}_b$ scales the premia.

Running variable, data and window. The running variable R_{jt} of Section 6.2 is issuance minus the year's cap, with issuance measured as all tax-exempt par within each six-digit CUSIP issuer prefix and offering calendar year, before sample restrictions. Designation and other bond characteristics come from the Mergent bond master files. The baseline window keeps issuer-years with $0.5 \le |R_{jt}| \le 4$, and the below-cap indicator is $E_{jt} = \mathbf{1}\{R_{jt} \le 0\}$.

Controls. In (28), Z_b contains the fixed effects and controls listed under Table 28. The fixed effects use seven maturity buckets, distinct from the five coefficient bins g, with upper limits of 2, 5, 10, 15, 20 and 30 years and a seventh bucket beyond 30 years.

Financing costs and the current-year benefit. Section 265(b) allocates a bank's interest expense across its assets in proportion to their average adjusted bases, without matching each bond to borrowing of the same maturity. A dollar of municipal basis held throughout the year is therefore allocated $c_{nt} = I_{nt}/\bar{A}_{nt}$ of interest expense. Here I_{nt} is bank n's annual interest expense in its December report and $\bar{A}_{nt}$ its average assets, both from Call Report data in the BankFind database of the Federal Deposit Insurance Corporation (FDIC). This expense-to-assets ratio is neither a deposit rate nor a maturity-matched financing cost. Designation reduces the nondeductible fraction from one to 0.2 and saves $X_{nt} = 0.8\,\tau_{B,t}c_{nt}$ at the federal bank tax rate $\tau_{B,t}$, 35% before 2018 and 21% thereafter. Weighting banks by their municipal-security holdings M_{nt}, held to maturity or available for sale and bank-qualified or not, gives the financing cost and the current-year benefit

$$c_t = \frac{\sum_{n\in\mathcal{B}_t} M_{nt}(I_{nt}/\bar{A}_{nt})}{\sum_{n\in\mathcal{B}_t} M_{nt}}, \qquad X_t = 0.8\,\tau_{B,t}c_t. \tag{131}$$

Here $\mathcal{B}_t$ is the set of banks with positive holdings, positive average assets, nonmissing interest expense and an expense-to-assets ratio from zero to below 15%. Mean c_t is 2.34% in 2000–04, 2.31% in 2005–08, 1.09% in 2009–10, 0.51% in 2011–13, 0.31% in 2014–17 and 0.66% in 2018–19, giving current-year benefits X_t of about 66, 65, 30, 14, 9 and 11 bp (2009–10 under the ordinary rule).

The measure uses book assets and interest expense because the statute's adjusted tax bases and otherwise-deductible interest are not observed bank by bank, and the IRS permits specified book-asset averaging conventions. X_t thus measures the ordinary-rule benefit of a C-corporation bank from observable data.[29] In 2009–10, the Section 265(b)(7) exception for holdings up to two percent of assets can reduce the disallowance on non-designated bonds and with it designation's incremental benefit, so I report those years separately.

The expected annual benefit. A bond's benefit accrues over its life. For a bond b issued in month t, the expected annual benefit is $\bar{X}_b = 0.8\,\tau_{B,t}\,\hat{c}_b$, where $\hat{c}_b$ is the expected annual financing cost over the bond's life. Regressing annual holdings-weighted financing costs on one-year expected rates over 2000–2021 ($R^2 = 0.75$) gives $\hat{c}_b = 0.38 + 0.55\,r_b$, both in percent per year. Here r_b is the issue-month Kim–Wright (Kim and Wright, 2005) expected average short rate at the bond's rounded whole-year maturity, capped at ten years. Because r_b is the fitted yield less its term premium, a term premium rising with maturity stays out of the denominator of the scaled outcome. Bin means of $\bar{X}_b$ are 33.9, 41.5, 50.5, 56.8 and 53.3 bp.

The current-year benefit X_t reflects only the issue year's financing cost. It falls from 104 bp in 2000 to an average of 9 bp in 2013–17, and premia do not track it, so ratios of premia to X_t depend on the years pooled (Appendix C.7.5). I therefore scale each bond's outcome by its own $\bar{X}_b$. The scaled premium estimates the capitalization share θ_k of (130), the share of the expected benefit reflected in yields: zero in the representative-tax benchmark and one in the bank-only benchmark.

C.7.3 Identification

The design requires that, given the controls, crossing the cap changes offering yields only through designation. Differences between measured and statutory issuance make the first stage fuzzy, the donut removes managed bunching, and the covariates that differ at the cap are controlled.

The fuzzy first stage. Statutory issuance aggregates subordinate entities and includes bank loans and leases, mostly absent from Mergent. It excludes refundings of designated bonds, which Section 265(b)(3)(D) deems designated, and private-activity bonds other than 501(c)(3) bonds. The statute also tests reasonably anticipated issuance at each sale, so a March designation stands even when an unanticipated October sale takes the year over the cap. As a result, 21% of bonds \$0.5–4 million above the cap are designated. In that window, 4% of issues designate in single-sale issuer-years, against 44% in issuer-years with multiple sales. Refundings are 46% of the bonds in that window, and measuring issuance net of refundings reduces designation in the corresponding window to 9%. Designation falls from 24% at \$0.5–1 million above the cap to 15–16% at \$4–8 million and 5% beyond \$10 million, consistent with measurement noise.

[29] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-11–13-12, for the allocation and the 20% disallowance.

Bunching and at-cap issuers. The half-million donut excludes managed bunching, with 2,081 issuer-years within \$0.25 million below the cap against 572 above. It also excludes 1,034 issuer-years with 12,076 bonds exactly at the \$10 million cap (\$30 million in 2009–10). At-cap issuers are eligible (Section 6.2) and designate 79.6% of their bonds, the below-cap rate, against 27% at \$1–50 thousand above the cap, so E_{jt} uses a weak inequality. A strict inequality would misclassify them, lowering the no-donut first stage from 0.39 to 0.30 but barely changing 2SLS, because dividing by the first stage offsets the attenuation.

Balance at the cap. Local linear covariate regressions in the baseline window, with maturity-bucket-by-year and state-by-year effects, find callability and the number of maturities per issue balanced in every bin. The remaining differences concern controlled covariates. Ratings, observed for 62% of bonds, differ only at 1–2 years (0.17 notch, $t = 2.1$). Coupons, set at pricing and covered by coupon-bin-by-maturity-bucket fixed effects, are 0.09–0.14 percentage points lower below the cap in the four bins beyond two years. Log par, always controlled, is lower below the cap at 1–2 years (-0.26) and higher at 10–20 years ($+0.12$).

C.7.4 Results

Table 29: Bank designation interacted with the expected annual benefit.

Maturity (years)	1–2	2–5	5–10	10–20	> 20
Yield reduction per benefit unit	0.058^{**}	0.096^{***}	0.154^{***}	0.175^{***}	0.233^{***}
	(0.024)	(0.025)	(0.027)	(0.031)	(0.089)
10–20 minus 1–2	0.117^{***}				
	(0.032)				
Fixed effects			Yes		
Issuer cluster			Yes		
Observations			355,372		
First-stage F (Kleibergen–Paap, joint)			122.0		

Notes. Entries are $-\beta_g$ in equation (132), the offering-yield reduction in basis points per basis point of expected annual benefit $\bar{X}_b$. Each designation–benefit interaction is instrumented by the below-cap indicator times the benefit in that bin. Controls add the benefit by bin to the fixed effects, log tranche par and running-variable slopes of Table 28, on its 2000–2021 sample within \$0.5–4 million of each year's cap. Standard errors clustered by issuer (15,889 clusters) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The premium is small and insignificant at 1–2 years and rises with maturity through 10–20 years, in basis points and in expected-benefit units.

Premia in basis points. In Table 28, below-cap status raises designation by 0.42 to 0.45 in every bin (column (1)). In column (2), the 10–20 year premium exceeds the 1–2 year premium by 16.5 basis points (s.e. 2.7). Figure 24 shows the offering-yield discontinuity separately for each maturity bin.

Premia in expected-benefit units. Column (3) of Table 28 divides each offering yield by its own $\bar{X}_b$ before any pooling and then estimates (28) by 2SLS. The estimated capitalization shares are 0.17, 0.31, 0.41, 0.52 and 0.47, with a 10–20 minus 1–2 year contrast of 0.34 (s.e. 0.11). This specification underlies the statements of Section 6.2 that about half of the benefit is priced at 10–20 years and that the rising share rejects full capitalization.

Yield-level benefit interactions. A second specification checks that the rising profile does not come from dividing yields by the expected benefit. It keeps offering yields in basis points and interacts designation with the expected annual benefit:

$$y_b = \sum_{g=1}^{5} \beta_g B_b \bar{X}_b \mathbf{1}\{n_b \in g\} + \widetilde{Z}_b'\delta + u_b, \qquad B_b\bar{X}_b\mathbf{1}\{n_b \in g\} \quad \text{instrumented by } E_{jt}\bar{X}_b\mathbf{1}\{n_b \in g\}, \qquad g = 1, \ldots, 5. \tag{132}$$

Here $\widetilde{Z}_b$ adds $\bar{X}_b$ by bin to the fixed effects and running-variable controls of (28), and u_b is the regression error. Table 29 reports the yield reductions $-\beta_g$ and the joint first-stage strength. Column (3) of Table 28 divides the outcome by $\bar{X}_b$, whereas this specification multiplies designation and its instruments by $\bar{X}_b$ and

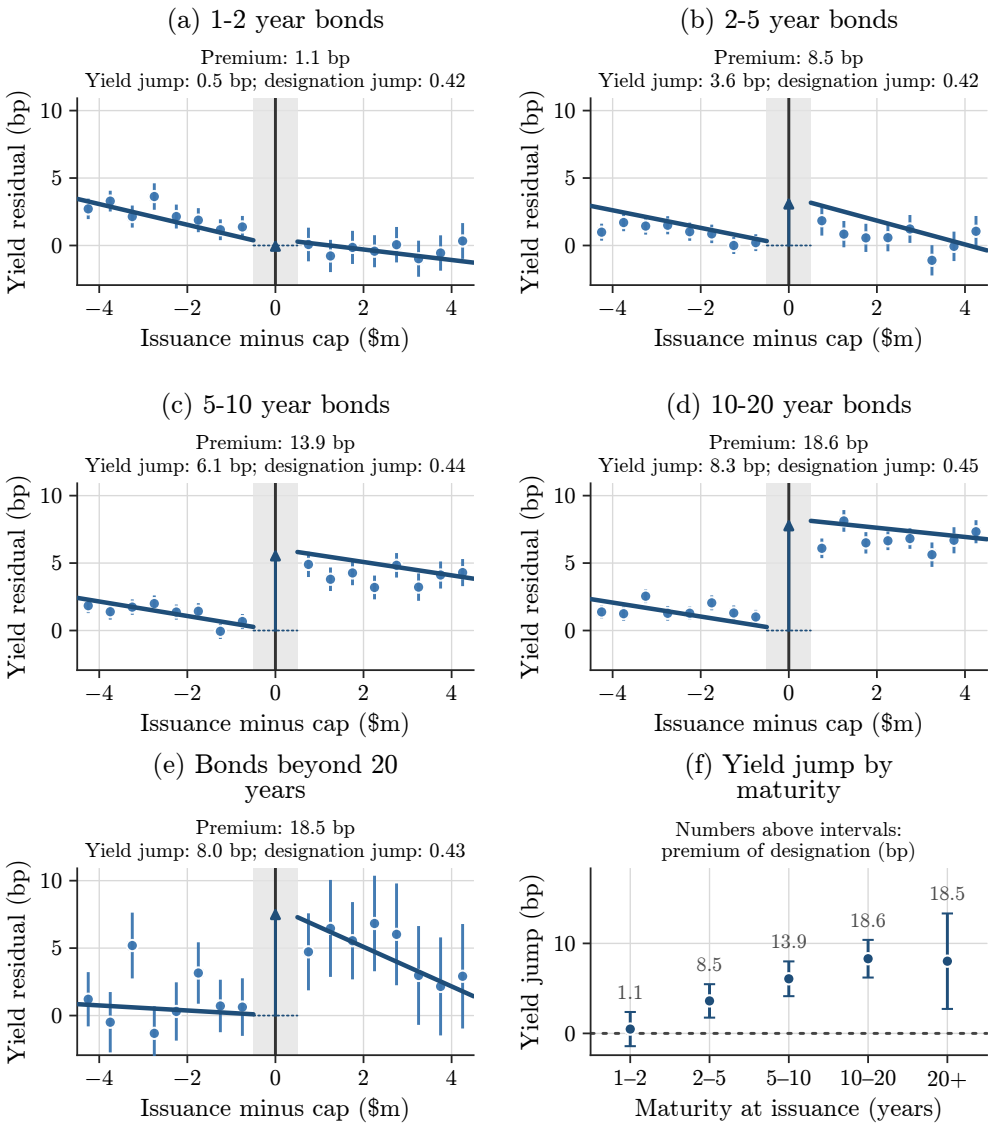


Figure 24: Offering-yield discontinuities by maturity. Panels (a)–(e) show $0.5 million issuance-bin means of offering-yield residuals after the fixed effects and log par of (28), relative to the below-cap fitted value at the cap, with local linear fits, donut shading and 95% bars. Arrows show above-cap-minus-below-cap jumps, and headers divide each yield jump by the designation jump to obtain the premium. Panel (f) collects these yield jumps and premia. Its premia are the separate-bin counterparts of the joint 2SLS estimates in column (2) of Table 28. The common scale covers the central 96 percent of binned means.

adds benefit controls. With this scaling, high-benefit years, where premia are the smallest fraction of the benefit (Appendix C.7.5), carry most of the weight, and the levels are lower. Both give rising profiles.

C.7.5 Robustness and a placebo cutoff

The maturity gradient survives changes in the window, the sample construction and the benefit measure, and the designation jump appears only at the statutory cap.

Donut, bandwidth and window. Across donut half-widths from zero to $1.5 million and bandwidths from $2 million to $8 million (Table 30), the 10–20 minus 1–2 year contrast is 13.5–18.4 bp and the 1–2 year premium is never significant. In the $5–15 million window of Dagostino (2025), without a donut, the contrast is 15.4 (1.8) bp, one of the most precise estimates. The last row of Table 30 drops the region that her bunching estimation treats as contaminated, $8.5–13.5 million at the $10 million cap. Within $|R| \leq 8$, it compares issuers

at least \$1.5 million below the cap with issuers at least \$3.5 million above it. First stages rise to 0.47–0.54, and with this longer extrapolation the contrast is 9.8 (4.2) bp.

Table 30: Bank-qualified discontinuity: sensitivity to the donut, the bandwidth and the window.

	first stage	1–2y	2–5y	5–10y	10–20y	20y+	10–20 minus 1–2
Donut half-width, bandwidth \$4 million							
none	0.39***	1.9 (1.8)	7.5*** (1.8)	13.9*** (1.8)	17.7*** (1.8)	17.7*** (4.5)	15.7*** (2.0)
\$0.25 million	0.40***	1.6 (2.0)	7.6*** (2.0)	13.4*** (2.0)	16.7*** (2.2)	16.3*** (5.5)	15.1*** (2.4)
\$0.5 million (baseline)	0.42***	1.8 (2.2)	8.7*** (2.2)	13.8*** (2.2)	18.4*** (2.3)	18.2*** (6.3)	16.5*** (2.7)
\$1 million	0.44***	1.2 (2.9)	8.2*** (2.7)	13.3*** (2.8)	19.7*** (3.1)	20.6*** (8.0)	18.4*** (3.5)
\$1.5 million	0.44***	2.0 (4.4)	7.3* (3.8)	11.8*** (4.0)	15.5*** (4.2)	19.5* (10.2)	13.5*** (5.2)
Bandwidth, donut \$0.5 million							
\$2 million	0.38***	0.8 (4.7)	7.9 (4.9)	13.3*** (4.8)	16.9*** (5.0)	21.0 (14.1)	16.1*** (4.8)
\$3 million	0.40***	2.7 (3.0)	8.9*** (3.0)	15.2*** (2.9)	19.2*** (3.1)	19.9*** (7.6)	16.5*** (3.3)
\$5 million	0.42***	−0.9 (1.9)	5.7*** (1.8)	11.4*** (1.8)	14.9*** (1.9)	15.9*** (4.9)	15.8*** (2.3)
\$6 million	0.43***	−0.5 (1.7)	5.4*** (1.5)	10.8*** (1.6)	13.2*** (1.6)	11.4*** (4.1)	13.6*** (2.1)
\$8 million	0.45***	0.0 (1.5)	6.3*** (1.2)	10.1*** (1.3)	13.5*** (1.3)	12.3*** (3.5)	13.5*** (1.8)
Other windows							
\$5–15 million, no donut	0.40***	0.1 (1.6)	5.6*** (1.5)	12.1*** (1.6)	15.5*** (1.5)	16.7*** (3.8)	15.4*** (1.8)
\$8.5–13.5 million dropped, $\lvert R\rvert \leq 8$	0.47***	0.2 (3.5)	6.0** (3.0)	7.6** (3.1)	10.0*** (3.4)	−1.8 (9.5)	9.8** (4.2)

Notes. Cells report the 2SLS premium of designation in basis points by maturity bin, pooled over 2000–2021 at each year's cap, with the baseline fixed effects and issuer-clustered standard errors in parentheses. The bandwidth and the donut half-width are the largest and smallest $|R_{jt}|$ retained, and the first-stage column refers to the 1–2 year bin. The \$5–15 million window is the one used in the spread comparison of Dagostino (2025). The last row drops the region her bunching estimation excludes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Construction checks and the relation to Dagostino (2025). Dagostino (2025) measures issuer-year issuance with Ipreo MuniIC data and documents bunching below the cap and a jump in designation at the \$30 million cap of 2009–10, as in my data. Her regressions on \$1 million issuance bins control for size, maturity, insurance, rating and use of proceeds, with year and issuer effects and standard errors clustered by year. In that study, tax-adjusted spreads over treasuries jump about 10 bp at the cap for general-obligation (GO) bonds issued before 2008. Table 31 applies her construction choices within my design and also nets refunding par from the running variable. In every row, the 10–20 year premium exceeds the 1–2 year premium at the one percent level. Issuer effects steepen the gradient, while netting refundings shifts issuer-years and widens confidence intervals.

Table 31: Bank-qualified discontinuity: construction checks.

	1–2y	2–5y	5–10y	10–20y	20y+	10–20 minus 1–2
Baseline (Table 28)	1.8 (2.2)	8.7*** (2.2)	13.8*** (2.2)	18.4*** (2.3)	18.2*** (6.3)	16.5*** (2.7)
Bond insurance and enhancement × year	0.9 (2.2)	8.0*** (2.2)	13.0*** (2.2)	17.7*** (2.3)	17.3*** (6.2)	16.8*** (2.6)
General-obligation bonds only	1.7 (2.3)	7.6*** (2.2)	13.3*** (2.3)	15.2*** (2.5)	15.8** (7.3)	13.5*** (2.9)
Issuer fixed effects	−5.0* (2.6)	1.8 (2.5)	7.5*** (2.6)	17.3*** (2.8)	16.8*** (4.1)	22.3*** (2.4)
Tax-adjusted spread over treasuries	5.3 (3.3)	12.6*** (3.2)	21.4*** (3.2)	26.9*** (3.5)	22.8*** (8.6)	21.6*** (3.8)
Insurance controls, year clusters (22)	0.9 (2.9)	8.0*** (2.6)	13.0*** (3.6)	17.7*** (3.4)	17.3** (7.6)	16.8*** (3.7)
Running variable net of refundings	0.8 (3.1)	4.9 (3.1)	8.1*** (3.1)	15.7*** (3.6)	11.0 (8.2)	14.8*** (3.7)
and refunding bonds excluded	0.9 (3.1)	6.3* (3.2)	8.9*** (3.2)	16.0*** (3.6)	10.9 (7.5)	15.1*** (3.6)

Notes. Cells report the 2SLS premium of designation in basis points, pooled over 2000–2021 at the year's cap in the window \$0.5 to \$4 million, with the baseline fixed effects and issuer-clustered standard errors unless stated. First stages are 0.40 to 0.51 in every bin of every row. Insurance and enhancement: adds bond insurance × year and credit enhancement × year (47% of window bonds are insured). The general-obligation, issuer-fixed-effect, tax-adjusted-spread and year-clustered rows also include these controls. General obligation: unlimited-tax, limited and double-barreled general-obligation bonds, 70% of the window. Issuer fixed effects: identified from issuers that cross the cap across years. Tax-adjusted spread: $100\,[y/(1-\tau_{B,t}) - y^{\text{Tsy}}]$, the outcome of Dagostino (2025), with y the offering yield and y^{Tsy} the maturity-matched treasury par yield, both in percent. Net of refundings: the running variable excludes refunding par, and the last row also drops refunding bonds from the estimation sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Applying her \$1 million issuance-bin estimator to her window, with maturity-bucket, state and rating effects by year and use-of-proceeds controls, gives flat below-cap bins and a step above the cap, matching her Figure 3. Pooled across maturities, the step is 4.2 bp in the offering yield and 5.6 bp in the tax-adjusted spread. The step

rises with maturity, from 1.1 and 1.4 bp at 1–5 years to 7.8 and 10.2 bp at 10–20 years. In her pooled-spread setting for GO bonds issued before 2008, with her controls but with state fixed effects and clustering by issuer, the reduced-form jump at the cap is 10.4 bp (s.e. 2.5), against her estimate of about 10. With her issuer fixed effects and clustering by year, which leaves eight clusters, the jump is 6.3 (3.8) bp, because within-issuer variation across the cap is thin before 2008.

The benefit measure and benefit regimes. Measuring the expected rate in $\bar{X}_b$ with the Adrian–Crump–Moench (Adrian et al., 2013) decomposition, which removes larger term premia than Kim–Wright at five and ten years, gives expected-benefit premia of 0.17, 0.35, 0.46, 0.55 and 0.48 (contrast 0.39). With years split into terciles of X_t, the 10–20 minus 1–2 year contrasts are 0.57 in low-benefit years (2013–18 and 2020–21), 0.37 in middle years (2003–04, 2009–12 and 2019) and 0.27 in high-benefit years (2000–02 and 2005–08). In the high-benefit years, expected benefits were nearly flat in maturity (62.8–71.2 bp) and yield curves often inverted, so the benefit's maturity structure does not drive the gradient. Excluding the strong bank-demand years 2020–21 gives 0.18, 0.25, 0.35, 0.39 and 0.36 (contrast 0.21). In the yield-level interaction specification of Table 29, excluding 2009–10 leaves the 10–20 minus 1–2 year contrast at 0.113 (s.e. 0.032).

Placebo cutoff. At a placebo cutoff of $5 million, first stages are 0.00–0.05 and all reduced-form yield reductions are within two bp of zero, with a 10–20 minus 1–2 year contrast of 0.4 (1.2), against 7.8 (1.2) in the reduced form at the cap.

The 2009–10 cap. The designation jump moves with the $30 million cap of the American Recovery and Reinvestment Act (ARRA). A $0.5–8 million window around it covers 16 thousand bonds issued in 2009–10. With the baseline fixed effects, the first stages are 0.19, 0.20, 0.19, 0.16 and 0.26, jointly significant at five percent. At the same cutoff, first stages are 0.01–0.04 in 2004–08 and −0.05 to 0.01 in 2011–15. At $10 million in 2009–10, they are 0.01–0.06 and not jointly significant. With designation jumps smaller than in other years, the reduced form is the informative estimate for 2009–10. Its yield reductions at the $30 million cap are 0.5, 3.2, 1.6, 10.9 and 21.5 bp, and the average reduction beyond ten years exceeds the 1–2 year reduction by 15.7 (6.7), significant at five percent.

By period. Estimates by period test whether premia track the current-year benefit. Those for 2000–04, 2005–08, 2009–10, 2011–13, 2014–17 and 2018–19 use the baseline specification, with the ARRA cap and window in 2009–10. Before 2009, when X_t is 65–66 bp, premia are at most about 0.3 of X_t at every maturity, so the benefit is weakly capitalized when the financing-tax cost of non-designated bonds approaches a percentage point. In 2011–17, when X_t is 9–14 bp, premia are 2.4–2.5 bp at 1–2 years, below X_t, and 14–28 bp at 5–20 years, at or above X_t. Ratios of the 5–20 year premium to X_t exceed two in some low-cost periods.

In 2018–19, after the cut in the bank tax rate, premia by bin are 6.9 (4.3), 6.4 (5.0), 7.7 (5.7), 14.9 (7.7) and 3.7 (24.2) bp. A difference-in-discontinuities estimate pools 2014–2019 and interacts designation with an indicator for 2018–19. It gives changes in 2018–19 of 3.1 (6.5), −9.1 (6.7), −19.2 (7.6), 3.6 (10.0) and −1.3 (28.7) bp, with unchanged first stages. The benefit measure uses the corporate rate because Subchapter S banks, whose income is taxed to their shareholders and whose tax rate the TCJA cut less, hold only 6–14% of banks' municipal holdings over 2000–2021.

Annual reduced-form yield reductions at the $10 million cap, pooled across maturities, are largest in 2016 and 2019–21 (11–18 bp), 6–8 bp in 2011–13 and within about 3 bp of zero in 2000–01, 2004–07 and 2018. First stages are 0.30–0.62 in every year except 2009–10, when the cap was $30 million. Bank holdings of municipal securities rose from $97 billion in 2000 to $370 billion in 2017, fell after the 2018 rate cut to $322 billion on average in 2018–19 and then averaged $396 billion in 2020–21. Premia do not simply track the current-year benefit, and the pattern is consistent with changing bank participation.

C.8 Investor holdings by tax status

The municipal portfolio shares of insurers and pension funds cited in Section 3.4 vary with exemption benefits (Table 2). In 2000–2017, exempt municipal bonds account for 34–48% of the average P&C insurer's bond portfolio, compared with 6–9% for life insurers (Figure 25). In the public-pension sample, mean portfolio shares never exceed 0.13% for exempt and 0.78% for taxable municipals (Figure 26). After the TCJA corporate-rate cut lowered P&C insurers' exemption benefit (Table 37), their exempt share fell from 38.0% in 2017 to 28.6% in 2019, while the share of other taxable bonds rose from 50.2% to 57.5%.

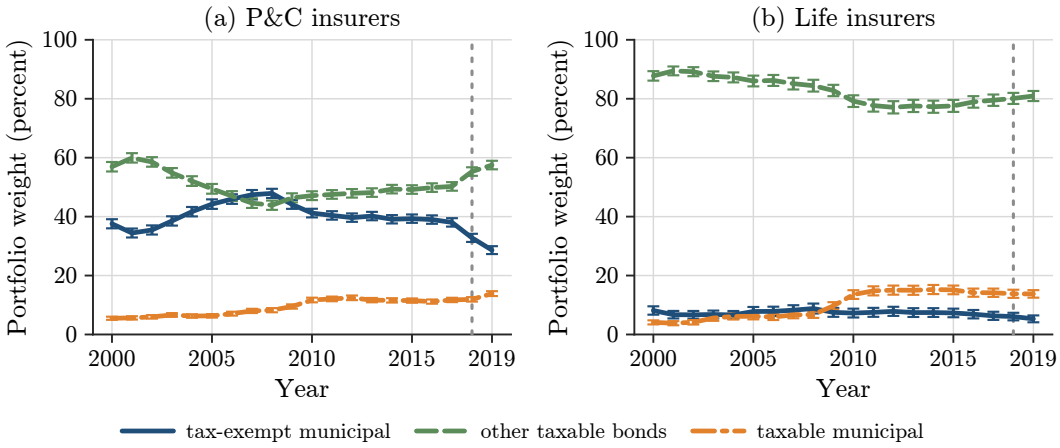


Figure 25: Tax-exempt municipals, taxable municipals and other taxable bonds as shares of insurers' total bond holdings: P&C in panel (a) and life in panel (b), 2000–2019, NAIC Schedule D. The dotted vertical line marks the TCJA corporate-rate cut. Bars are 95% intervals for yearly mean insurer portfolio shares, with standard errors clustered by insurer.

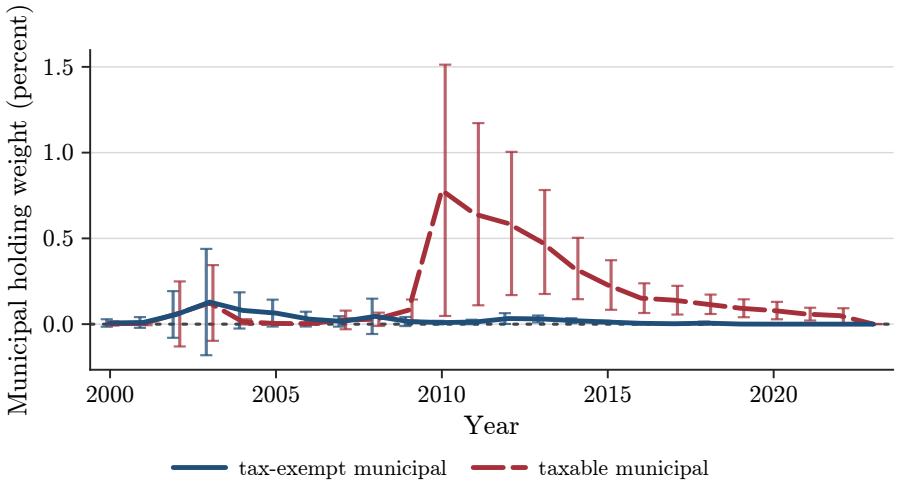


Figure 26: Municipal holdings by federal tax status as shares of total bond portfolios in the public-pension sample obtained by Freedom of Information Act (FOIA) requests (Section 3.1), 2000–2023. All thirty plans are covered from 2015. Bars are 95% intervals for yearly mean plan portfolio shares, with standard errors clustered by plan. Other taxable bonds are omitted. Taxable municipals include Build America Bonds.

C.9 P3: municipal-yield response to treasury yields

P3 (Section 6.3) tests whether municipal yields respond more to a given movement in maturity-matched treasury yields as maturity rises. The comparison concerns this relative response, so longer bonds' greater duration alone does not explain the prediction. The model motivates the maturity ordering. The empirical coefficients measure coupon-yield responses, so they differ from the model's zero-coupon loading ratios. The maturity gradient survives treasury-curve aging, alternative filters, state clustering and a taxable reference market.

Design and estimating equation. Beyond the screens in the notes to Table 32, the sample excludes trades less than 92 days after issue, when offering yields still anchor prices, and less than 365 days before maturity, where the fitted treasury curve is erratic. Consecutive trades in the same CUSIP and on the same dealer side, at most 90 days apart, form a pair. Its municipal yield change is $\Delta y^{M}_{b,t}$ and its matched treasury change is $\Delta y^{\text{Tsy}}_{n_b,t} = y^{\text{Tsy}}_{n_b(t),t} - y^{\text{Tsy}}_{n_b(t-1),t-1}$, where $t-1$ denotes the trade preceding t, $n_b(t)$ is the bond's remaining maturity in years at trade t and y^{Tsy} is the daily treasury par-yield curve of Gürkaynak et al. (2007). Column (1) of Table 32 estimates (29), which includes the maturity level n_b and no fixed effects, so the response at maturity

Table 32: P3: observed coupon-yield response to treasury par yield by remaining maturity.

	(1) Continuous $\Delta y^M_{b,t}$	(2) Bin $\Delta y^M_{b,t}$
$\Delta y^{\text{Tsy}}_{n_b,t}$	0.346^{***}	0.351^{***}
	(0.016) [0.012]	(0.013) [0.006]
$\Delta y^{\text{Tsy}}_{n_b,t} \times n_b$	0.007^{***}	
	(0.002) [0.002]	
$\Delta y^{\text{Tsy}}_{n_b,t} \times Long_b$		0.075^{***}
		(0.012) [0.009]
$Long_b$		0.016^{***}
		(0.002) [0.002]
n_b	0.002^{***}	
	(0.000) [0.000]	
Trade-date cluster	Yes	Yes
Observations	952,874	952,874
Weights	Inverse pair count	Inverse pair count

Notes. The sample is pairs of consecutive same-side MSRB customer trades within 90 days in federally tax-exempt, noncallable bonds rated AA and above (agency score ≤ 3), January 2005 to August 2020. Between the two trades of each pair, the maturity-matched treasury yield moved at least 10 basis points in either direction. The treasury yield is interpolated on the daily par-yield curve of Gürkaynak et al. (2007) at the bond's exact remaining maturity on each transaction date. Observations are weighted by the inverse count of their CUSIP's base-sample trade pairs, computed before observations with missing or trimmed outcomes are dropped. The dependent variable is trimmed at its first and ninety-ninth percentiles. $Long_b = \mathbf{1}\{n_b \geq 5\}$. Parentheses give trade-date-clustered standard errors (3,864 clusters). Brackets give state-clustered standard errors (54 clusters: the 50 states, the District of Columbia, Puerto Rico, Guam and the U.S. Virgin Islands). Stars are identical under either clustering. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

n_b is $\beta_0 + \beta_1 n_b$. Column (2) replaces maturity by $Long_b = \mathbf{1}\{n_b \geq 5\}$, so its short and long responses are β_0 and $\beta_0 + \beta_1$.

Identification. First differencing within a bond absorbs its time-invariant characteristics, and noncallability removes call-option repricing. Comparing purchases with purchases and sales with sales limits bid–ask and intraday noise.

Results. In the continuous specification, the coefficient on the treasury change is 0.346 and the maturity interaction is 0.007, both significant at one percent (Table 32, column (1)). In the binned specification, the response rises from 0.351 below five years to 0.426 from five years (Table 33, first row).

Robustness: treasury-curve aging. Part of the matched treasury change comes from the bond becoming shorter between trades. Holding the later date's curve fixed, this aging component is $y^{\text{Tsy}}_{n_b(t),t} - y^{\text{Tsy}}_{n_b(t-1),t}$. Fixing the benchmark at the first trade's maturity, $\Delta y^{\text{Tsy,fixed}}_{b,t} = y^{\text{Tsy}}_{n_b(t-1),t} - y^{\text{Tsy}}_{n_b(t-1),t-1}$, removes it. On upward-sloping curves, aging lowers the reference yield by an average of 2.6 bp below five years and 1.6 bp beyond, against absolute changes near 26 bp. Removing it lowers the short response more than the long response and so increases the gradient (Table 33). Municipal aging does not fully offset treasury aging because the two curves have different slopes. Controlling for linear municipal aging with elapsed time and its interaction with maturity raises the gradient further. The headline specification keeps the benchmark matched at each trade, the conservative choice because it gives the smallest gradients in Table 33.

Robustness: filter thresholds. Across the treasury-move cutoffs and trade windows of Table 34, every maturity interaction is positive. Every binned cell is significant at one percent and every continuous cell at five percent, twenty-seven of the thirty at one percent.

Robustness: state clustering. Brackets in Table 32 give standard errors clustered by issuing state, which allow correlation among a state's bonds across dates, while the headline trade-date clustering allows common shocks across states on the same date. Both headline interactions remain significant at one percent,

Table 33: P3: observed coupon-yield responses and treasury-curve aging.

	Bin specification		Interaction	
Treasury benchmark	< 5y	≥ 5y	Bin	Continuous
Benchmark matched at each trade's maturity (Table 32)	0.351^{***}	0.426^{***}	0.0749^{***}	0.0070^{***}
Benchmark held at the first trade's maturity	0.340^{***}	0.423^{***}	0.0825^{***}	0.0084^{***}
plus elapsed time and elapsed time × maturity	0.334^{***}	0.419^{***}	0.0850^{***}	0.0086^{***}

Notes. Each row estimates (29) on the 952,874 pairs of Table 32, with its weights and trade-date clustering. The first two columns are the bin specification's responses below and from five years, β_0 and $\beta_0 + \beta_1$, and the last two are the interactions with $Long_b$ and n_b. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 34: P3 sensitivity: observed maturity interaction across the treasury-move cutoff and trade-gap window (weighted by inverse CUSIP base-pair counts).

Panel A. Continuous interaction $\Delta y^{\text{Tsy}}_{n_b,t} \times n_b$					
Cutoff \ window	15d	30d	60d	90d	120d
≥ 0.05	0.0053^{***}	0.0057^{***}	0.0051^{**}	0.0071^{***}	0.0090^{***}
≥ 0.075	0.0061^{***}	0.0057^{***}	0.0047^{**}	0.0068^{***}	0.0087^{***}
≥ 0.10	0.0065^{***}	0.0060^{***}	0.0049^{**}	0.0070^{***}	0.0087^{***}
≥ 0.15	0.0080^{***}	0.0071^{***}	0.0059^{***}	0.0077^{***}	0.0091^{***}
≥ 0.20	0.0088^{***}	0.0082^{***}	0.0067^{***}	0.0081^{***}	0.0094^{***}
≥ 0.25	0.0104^{***}	0.0083^{***}	0.0074^{***}	0.0087^{***}	0.0097^{***}
Panel B. Binned interaction $\Delta y^{\text{Tsy}}_{n_b,t} \times Long_b$					
Cutoff \ window	15d	30d	60d	90d	120d
≥ 0.05	0.049^{***}	0.061^{***}	0.070^{***}	0.074^{***}	0.088^{***}
≥ 0.075	0.056^{***}	0.063^{***}	0.068^{***}	0.074^{***}	0.086^{***}
≥ 0.10	0.062^{***}	0.066^{***}	0.069^{***}	0.075^{***}	0.086^{***}
≥ 0.15	0.079^{***}	0.077^{***}	0.079^{***}	0.083^{***}	0.092^{***}
≥ 0.20	0.094^{***}	0.090^{***}	0.085^{***}	0.088^{***}	0.097^{***}
≥ 0.25	0.111^{***}	0.100^{***}	0.096^{***}	0.096^{***}	0.102^{***}

Notes. Each cell re-estimates column (1) (Panel A) or (2) (Panel B) of Table 32 on pairs whose maturity-matched treasury move is at least the cutoff, in percentage points, and whose trades are at most the window apart. The headline sample uses 0.10 and 90 days. Standard errors are clustered by trade date as in Table 32. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

and binned-grid significance levels are unchanged. In the continuous grid of Table 34, the 0.05, 0.075 and 0.10 cutoffs give $p < 0.05$ at thirty days and $p < 0.10$ at sixty days, and the 0.15 cutoff gives $p < 0.05$ at sixty days. All other cells keep their trade-date significance levels, so twenty-seven of the thirty continuous cells remain significant at five percent.

Robustness: taxable reference market. Section 4.6 derives P3 relative to the model's jointly priced taxable market. Against corporate zero yields, the maturity gradient is larger than against treasury par yields. Both reference curves here are monthly averages at the bond's remaining maturity in each trade month. On the 517,864 cross-month pairs whose absolute monthly treasury change is at least 10 bp, with the original weights and trade-date clustering, the maturity interaction is 0.0132 (s.e. 0.0026) against corporate changes and 0.0072 (s.e. 0.0025) against treasury changes.

C.10 Robustness to the rating cut

The yield pattern and the empirical tests use two rating cuts of the unrounded mean agency score on the twenty-two-notch scale, on which 1 = AAA, 3 = AA and 7 = A−. The curves in Sections 3 and 5 use scores of seven or better, the A-or-above sample, matching the corporate benchmark's credit band. The BAB comparison in Section 3 and the P1 and P3 tests in Section 6 use scores of three or better, the AA-and-above sample.

The curve pattern survives the narrower cut, and the BAB, P1 and P3 tests retain their gradient signs on the wider cut. P1's pre-2018 state-tax pass-through declines across all five bins under OLS and IV (Table 35),

with endpoint differences significant at five percent under OLS and one percent under IV. The wider sample gives larger P3 interactions, with similar BAB and P1 gradients.

C.10.1 Curve construction at AA and above

On the AA-and-above sample, the implied tax also falls with maturity, and the normalized model gap $\widetilde{\mathcal{G}}_T$ of (23) matches the residual before the model to within ten basis points at fourteen of fifteen maturities.

Figure 27 recomputes the AA-and-above curves with the three-stage averaging of Appendix C.1.1: equal weights within cells of maturity, issue quarter and rating, then across ratings, then across quarters. At each maturity, the implied tax is one minus the ratio of these mean municipal to taxable yields. Net of calls and issuance premia, it falls from 40.1% at two years to 25.5% at thirty, a decline of 14.6 points against 17.1 on the A-or-above sample.

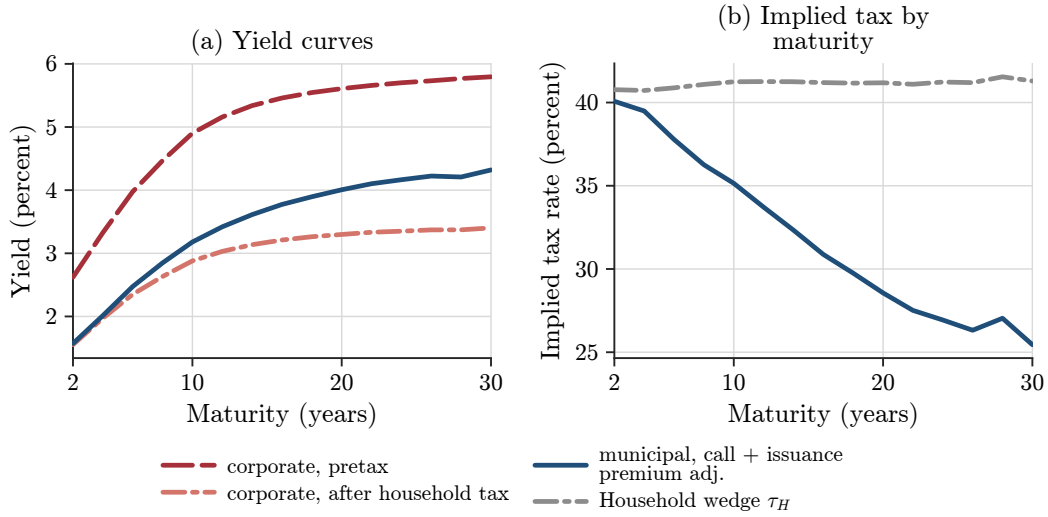


Figure 27: The AA-and-above municipal puzzle, 2000–2020, with this sample's own weights and household benchmark. Panel (a) shows pretax and household after-tax corporate zero-coupon yields and the municipal yield after removing callability and the issuance premium, and panel (b) shows the implied tax $1-y_T^M/y_T^C$ and the household wedge $1-\bar{\xi}_H(T)$, labeled τ_H in the legend. Calls use dated calibrated prices (Appendix C.1.2), and the sample's all-size issuance premium is scaled by the A-or-above sample's ratio of institutional-size to all-size premia (see text).

The model comparison repeats the accounting of Section 5.3, applying the model's ratio of municipal to taxable mean yields, averaged with this sample's weights, to observed corporate yields (the corporate normalization of (23)). Call, liquidity and default estimates are sample-specific, and capital gains use the common method. To match the institutional-size baseline of Appendix C.1.3, the issuance adjustment scales this sample's premium from all trade sizes by the A-or-above sample's ratio of institutional-size to all-size premia at each maturity, giving 11.5 bp at thirty years. The normalized model gap exceeds the residual before the model at every maturity beyond two years, 96.2 against 89.9 basis points at thirty years, so the unexplained residuals beyond two years are negative. The unexplained residual is −6.3 basis points at thirty years and 7.1 in root mean square across maturities.

C.10.2 Empirical tests on the A-or-above sample

Each test is repeated on both rating cuts. Table 21 (Appendix C.2) compares the two BAB rating cuts in columns (1) and (3), with call matching in columns (2) and (4). Table 35 repeats the P1 specification of Appendix C.5, so its AA-and-above columns reproduce the headline P1 estimates. Table 36 repeats the P3 specification of Table 32 but clusters standard errors by state. Its AA-and-above coefficients therefore equal the headline ones, and its standard errors equal the bracketed state-clustered ones of Table 32.

The wider sample also preserves the binned gradient across all thirty cutoff–window combinations in Table 34: coefficients range from 0.078 to 0.134, against 0.049 to 0.111 for AA and above. All are significant at one percent with state-clustered standard errors.

Table 35: P1 by rating cut, 2000–2017.

	OLS		IV	
	Score ≤ 3	Score ≤ 7	Score ≤ 3	Score ≤ 7
β_1: $[0, 5)$ years	1.069^{***}	0.986^{***}	1.088^{***}	1.080^{***}
	(0.195)	(0.255)	(0.225)	(0.233)
β_2: $[5, 10)$ years	0.677^{***}	0.607^{***}	0.694^{***}	0.665^{***}
	(0.069)	(0.100)	(0.086)	(0.117)
β_3: $[10, 15)$ years	0.484^{***}	0.380^{***}	0.447^{***}	0.384^{***}
	(0.096)	(0.083)	(0.118)	(0.098)
β_4: $[15, 20)$ years	0.390^{***}	0.285^{***}	0.326^{***}	0.264^{***}
	(0.102)	(0.086)	(0.116)	(0.095)
β_5: $[20, 30]$ years	0.305^{***}	0.240^{***}	0.229^{***}	0.214^{***}
	(0.075)	(0.077)	(0.081)	(0.079)
Decline $\beta_1 - \beta_5$	0.764^{***}	0.746^{**}	0.859^{***}	0.866^{***}
p-value of decline	0.002	0.011	< 0.001	< 0.001
Issue month × maturity × call FE	Yes	Yes	Yes	Yes
State cluster	Yes	Yes	Yes	Yes
Observations	681,838	1,397,768	681,838	1,397,768
First-stage F (Kleibergen–Paap)	—	—	10.4	12.6

Notes. FE denotes fixed effects. Controls, fixed effects and state-clustered inference (47 states) follow Appendix C.5, 2000–2017, with call-adjusted zero yields. Score ≤ 3 is the AA-and-above sample and score ≤ 7 the A-or-above sample. IV uses bin-interacted 1990 rates. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 36: P3: observed coupon-yield response to treasury par yields, by rating cut.

	(1) continuous		(2) bin <5y vs ≥5y	
	score ≤ 3	score ≤ 7	score ≤ 3	score ≤ 7
$\Delta y^{\text{Tsy}}_{n_b,t}$	0.346^{***}	0.281^{***}		
	(0.012)	(0.011)		
$\Delta y^{\text{Tsy}}_{n_b,t} \times n_b$	0.007^{***}	0.013^{***}		
	(0.002)	(0.002)		
$\Delta y^{\text{Tsy}}_{n_b,t} \times Long_b$			0.075^{***}	0.113^{***}
			(0.009)	(0.006)
State cluster	Yes	Yes	Yes	Yes
Observations	952,874	2,023,637	952,874	2,023,637
Weights	Inverse pair count			

Notes. The estimates use Table 32's first-differenced specification for coupon yields and par yields, weighted by inverse CUSIP pair counts and clustered by state, with $Long_b = \mathbf{1}\{n_b \geq 5\}$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D Institutional detail and data

This appendix states the tax rules behind the investor after-tax shares of Section 3 and shows how the model and the empirical adjustments implement them, separating statutory rules from model assumptions. Appendix D.4 converts these rules into tax receipts and financing costs under the two reforms, Appendix D.5 examines financing, short-sale and hedging restrictions, and Appendix D.6 lists the data sources.

D.1 Investor-by-investor federal tax treatment since 2000

The model prices bonds through each investor's relative after-tax share, so the federal rules that set this share determine which investors value the exemption most. Let $\xi_i^C = 1 - \tau_i^C$ and $\xi_i^M = 1 - \tau_i^M$ be the absolute fractions of a dollar of taxable and municipal interest that investor i keeps, before the normalization of Section 4.2. Pricing uses their ratio $\xi_i = \xi_i^C / \xi_i^M$, and the exemption benefit $1 - \xi_i = (\tau_i^C - \tau_i^M)/(1 - \tau_i^M)$ equals

τ_i^C only when municipal interest is fully exempt. Households differ in marginal rates, insurers lose part of the benefit through proration, and banks can lose the deduction for financing costs. Lost financing deductions are a separate cost, described for banks below and for other investors in Appendix D.5. Tables 2 and 37 give the pre-2018 and 2018–2020 values.

Table 37: Investor heterogeneity in the municipal market after the 2017 Act

Investor type	Federal treatment	State treatment	Net exemption benefit
High-tax household	Interest exempt	Exempt on in-state bonds only	0.35–0.41
Middle-tax household	Interest exempt	Exempt on in-state bonds only	0.22–0.32
Low-tax household	Interest exempt	Exempt on in-state bonds only	0.10–0.12
P&C insurer	Exempt, less proration	Premium taxes, no distinction	0.17
Life insurer	Exempt, less proration	Premium taxes, no distinction	0.16
Bank	Exempt, financing cost disallowed	Varies by state	0.16–0.18
Other (pensions, etc.)	Varies	Varies	< 0.05

Notes. Household values are federal marginal tax rates in three groups (35% and above, 22% to 32%, below 22%), and the top value includes the 3.8% net investment income tax. The $10,000 cap on the state and local tax deduction makes the home-state exemption more valuable to households whose cap binds. Institutions face the 21% corporate rate. P&C proration rises to 25% of exempt interest, giving $\xi = 0.79/0.9475$, and the life insurer's policyholders' share is set at 30%, giving $\xi = 0.79/0.937$. The bank value is the break-even implied tax defined under Table 2, evaluated at the 21% rate, with the 2018–2020 financing-tax cost of 0.117% a year and 2018–2020 taxable yields.

Households. Municipal interest is generally federally exempt, and taxable bond interest is taxed as ordinary income. A household's federal after-tax shares are therefore $\xi_i^C = 1 - \tau_{oi}$ and $\xi_i^M = 1$, where τ_{oi} is its ordinary-income tax rate, before the net investment income tax below and the Social Security interaction of Appendix B.1. Households in both high and lower brackets report exempt interest (Figure 1, panel (b)), so the exemption benefit varies across municipal-holding households.

The top ordinary rate in Table 38 fell under the 2001 cuts and again in 2003, when the cuts were accelerated. It rose in 2013 and fell in 2018. In P1, I set the federal after-tax share of taxable interest to one minus this dated top ordinary rate, excluding the net investment income tax. The top long-term capital-gains rate fell to 15% in 2003 and rose in 2013. Since 2013, the 3.8% net investment income tax (NIIT) has applied to taxable interest and many gains, but not to exempt interest, widening the exemption benefit of the households it covers. The calibration computes federal and state rates for each Survey of Consumer Finances (SCF) tax unit under each year's law (Appendix B.1).

Table 38: Top federal tax rates relevant to municipal pricing, 2000–2020

Period	Top ordinary rate	+ NIIT (on taxable interest)	Top corporate rate
2000	39.6%	39.6%	35%
2001	39.1%	39.1%	35%
2002	38.6%	38.6%	35%
2003–2012	35.0%	35.0%	35%
2013–2017	39.6%	43.4%	35%
2018–2020	37.0%	40.8%	21%

Banks. A bank's municipal coupons are exempt, but the interest that finances those holdings can lose its deduction. Let c be the annual financing cost, the interest expense allocated to a dollar of holdings, and τ_B the bank's tax rate, as in Section 6.2. Interest expense that finances non-bank-qualified exempt bonds is generally nondeductible. It then costs c after tax, against $c(1 - \tau_B)$ if deductible, a penalty of $\tau_B c$. This penalty is the financing-tax cost netted from the bank's benefit in Tables 2 and 37. Section 265(b)(3) excepts bank-qualified bonds, but Section 291 cuts their deduction by 20%, leaving 80% and an after-tax cost of $c(1 - 0.8\tau_B)$. Interest expense is allocated to exempt bonds pro rata, in proportion to adjusted tax basis.[30]

[30] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-9 and 13-11. Special rules apply to certain 2009–2010 issues and grandfathered obligations.

The Tax Equity and Fiscal Responsibility Act of 1982 disallowed 15% of banks' deduction for interest allocable to exempt bonds, a cut raised to 20% in 1984. The 1986 Act generally denied financial institutions' interest deductions allocable to exempt obligations acquired after August 7, 1986. The regular bank-qualified exception covers designated small-issuer obligations within the $10 million annual limit and other conditions, and temporary 2009–2010 rules expanded eligibility.

In the model, a bank's relative after-tax share is $\xi_B = 1 - \tau_B$, 0.65 before 2018 and 0.79 after, which gives the gross exemption benefit of Table 3. Each dollar of bank municipal holdings also bears the financing-tax cost $\tau_B c(1 - 0.8 q_{BQ})$, where q_{BQ} is the bank-qualified share of those holdings and c, consistent with pro rata allocation, is measured by the ratio of interest expense to assets. The aggregate calibration sets $q_{BQ} = 0$. Appendix C.7.1 derives the share of the designation benefit that is capitalized in yields.

Property and casualty insurers. P&C insurers pay the corporate rate τ_c on taxable bond interest. Proration cuts their losses-incurred deduction by a percentage of exempt interest, so that percentage of municipal interest is in effect taxed at τ_c. Before the 2017 Tax Cuts and Jobs Act, the percentage was 15%, giving an effective municipal rate of $0.15 \times 0.35 = 0.0525$. The Act replaced the percentage with 5.25% divided by the highest corporate rate, or 25% at $\tau_c = 21\%$. The municipal after-tax share thus stays at 0.9475 while the taxable share rises from 0.65 to 0.79. The relative after-tax share rises from 0.686 to 0.834, and the exemption benefit falls from 31.4% to 16.6%. Cutting the corporate rate while keeping 15% proration would not reproduce the 2018 law.

The calibration applies the top corporate rate to insurers and banks. Before 2018, marginal corporate rates were 15–34% below $100,000 and 34–39% from $100,000 to $18,333,333, including bubble rates that recapture the lower brackets' benefit, and a flat 35% above. Since 2018 the rate is 21% at every taxable-income level.

Life insurers and other institutions. Section 812 divides a life insurer's net investment income into company and policyholder shares, and exempt interest reduces its reserve deductions. With a one-half policyholders' share and a 35% corporate rate, the pre-2018 calibration has an effective municipal rate of 17.5%, after-tax shares of 0.825 on municipal and 0.65 on taxable interest, and a relative after-tax share of 0.788. The post-2017 calibration's 30% policyholders' share and 21% corporate rate give an effective municipal rate of 6.3%, after-tax shares of 0.937 and 0.79, and a relative after-tax share of 0.843. In both calibrations the life insurer's exemption benefit is below the P&C insurer's, and the Act narrows the difference. Tax-exempt institutions and tax-deferred accounts already receive taxable interest free of current tax and gain no additional benefit from the exemption.

D.2 State exemptions and their implementation in the model

State exemptions add a benefit that depends on both the issuer's state and the investor's residence. This dependence is the source of variation in P1, and the home-state repeal counterfactual removes the distinction between own-state and other-state interest.

Resident households. State income taxes generally exempt residents' interest on home-state municipal bonds and tax their interest on other states' bonds. The quantitative model applies a common home-state exemption in every income-tax market, while the P1 sample keeps only jurisdictions where the exemption applies, excluding the four states with exceptional rules (Illinois, Iowa, Wisconsin and Oklahoma).

For a household resident in state s, let τ_s^{st} be its state tax rate and ξ^{fed} its federal relative after-tax share. The baseline normalizes the home-state municipal share to one and sets the out-of-state municipal share to $1 - \tau_s^{\text{st}}$, with no federal deduction of state tax on out-of-state municipal interest, a convention varied in Appendix D.2.1. Through 2017, the share on taxable interest, formed tax unit by tax unit, is $\xi^{\text{fed}}(1 - \tau_s^{\text{st}})$, which includes the federal deduction of state tax on taxable interest. After 2017, the model treats the state-and-local-tax deduction cap as binding, so the share is $\xi^{\text{fed}} - \tau_s^{\text{st}}$, as in (125). These are pricing shares, before the conversion to tax dollars in Appendix D.4.

Funds and institutions. Municipal funds generally pass the federal exemption through to shareholders as exempt-interest dividends. Their state treatment depends on the shareholder's residence and the share of interest from that state's bonds. A single-state fund preserves the home-state exemption for resident shareholders, while a national fund passes it through only on the eligible share, which weakens the household's state-tax incentive. Among institutions, tax-deferred accounts and many pensions owe no current state tax on interest. Insurers commonly pay premium taxes, which do not distinguish municipal from taxable interest, and banks' and other corporations' state taxes depend on the tax base and business apportionment. P1 therefore exploits a household

exemption that depends jointly on issuer and holder residence (Section 6.1). The model assigns this home-state wedge to households and uses federal proration and financing deductions for insurers and banks.

D.2.1 Sensitivity to the federal deduction of state tax on out-of-state municipal interest

Granting every household the federal deduction of state tax on out-of-state municipal interest in the pre-2018 periods, which the baseline omits, lowers the level of state-tax pass-through but preserves its declining profile and the large long-maturity gap.

The rule and the test. Before 2018, itemizers could deduct state income taxes on federally exempt interest against ordinary federal income tax. Not every household received the deduction's full benefit. Taking the standard deduction, paying the alternative minimum tax or choosing a larger sales-tax deduction could eliminate the added saving.[31] I grant every tax unit the full deduction, whatever its eligibility, valued at the ordinary-income component of its baseline federal marginal rate, which excludes NIIT. With federal and state shares multiplied tax unit by tax unit, I re-solve the model in the six pre-2018 periods, holding all other inputs and estimated coefficients fixed. The post-2017 binding-cap convention is unchanged.

Why the deduction lowers pass-through. For the 2015–2017 California high-tax group, the effective after-tax shares with the deduction are 0.498 on taxable interest and 0.919 on out-of-state municipal interest. At a riskless taxable yield of 5%, a municipal yield of 2.71% equates the two after-tax returns, against 2.87% for the high-tax group in states without an income tax. The taxable share already includes the baseline deduction, so the added municipal deduction lowers the yield needed to match it. Without the added deduction, the California group's break-even out-of-state yield is also 2.87%. Residents of high-tax states can therefore accept lower out-of-state yields than residents of states without an income tax, which compresses cross-state yield differences and reduces state-tax pass-through even at the riskless maturity.

Result. With the main sample's 2000–2020 weights and corporate normalization, the thirty-year model gap falls only 0.23 basis points, from 97.39 to 97.16. The model's P1 pass-through, computed as in Figure 9 and averaged over the six periods, falls from 1.000 to 0.904 at the two-year riskless maturity and from 0.460 to 0.416 at thirty years.

D.3 Taxation over a bond's life: accrual, basis and resale

The model applies the investor's interest-income after-tax share to the whole return $\mathcal{R} = \mu + u$ in (33), without separately tracking coupons, accrual and resale gains.[32] The capital-gains adjustment of Appendix C.1.4 measures the effect of separating interest from resale gains. The adjustment relies on the basis accounting below: accrued interest changes the bond's tax basis, so the same accrual is not taxed again as a price gain.

Buying at issue and holding to maturity. For individuals, federal law separates interest accrual from gains realized at disposition (Sections 171, 1272–1278 and 1288). Consider a bond with face value 100, annual coupon payment $\bar{c}$ in dollars and annual effective issue yield y_0. Under constant-yield accrual, its adjusted issue price (AIP), which starts at the original price AIP_0, is the original buyer's tax basis after each payment and satisfies

$$\mathrm{AIP}_t = (1 + y_0)\mathrm{AIP}_{t-1} - \bar{c}, \qquad t = 1, \ldots, n, \qquad \mathrm{AIP}_n = 100. \tag{133}$$

A discount, $\mathrm{AIP}_0 < 100$, is original-issue discount (OID), and a premium, $\mathrm{AIP}_0 > 100$, is original-issue premium (OIP). Annual interest $y_0\mathrm{AIP}_{t-1} = \bar{c} + (\mathrm{AIP}_t - \mathrm{AIP}_{t-1})$, the coupon plus the noncash accrual, is taxed at the ordinary-income rate τ_{oi}:

$$\mathrm{tax}_t = \begin{cases} \tau_{oi} y_0 \mathrm{AIP}_{t-1}, & \text{taxable bond,} \\ 0, & \text{tax-exempt bond.} \end{cases} \tag{134}$$

[31] IRS Publication 550 (2017), "State income taxes"; 2017 Instructions for Form 6251, line 3; and 2017 Instructions for Schedule A, line 5. 2016 Instructions for Form 8960, line 9b, restrict NIIT deductions to taxes attributable to net investment income, which excludes exempt interest.

[32] The after-tax share multiplies both the log of the expected gross return, $\log \mathbb{E}[1 + R]$, and the centered log-return innovation. Taxing a simple return R gives the after-tax gross return $1+\xi R$, whose log is $\log(1+\xi R)$. For a deterministic two-year simple return of 5.4% and $\xi = 0.62$, this log differs from $\xi \log(1+R)$ by 3.2 basis points per two-year period (1.6 a year), and by zero for a fully exempt leg. Appendix A.1.1 reports the effect of taxing simple returns exactly in the solved model. The return convention matches the empirical implied tax and the after-tax yield mappings of Appendix C.1.4.

The basis change $\text{AIP}_t - \text{AIP}_{t-1}$ prevents taxing accrued interest again at redemption. OID raises basis, including for zero-coupon bonds, and OIP lowers it. Under this accrual convention, the original buyer's basis reaches face at maturity, leaving no residual gain for the capital-gains adjustment. Amortizing premium on a taxable bond requires a Section 171 election, which the capital-gains adjustment assumes, while amortization on an exempt bond gives no deduction. For taxable bonds, de minimis OID below a quarter point per full year can be treated as zero, with the discount recognized as capital gain at maturity. This de minimis rule generally does not apply to tax-exempt obligations.[33]

Selling before maturity. A sale at price P_h at date h realizes the gain $G_h = P_h - \text{basis}_h$, measured from adjusted basis. A holding period of more than one year makes a capital gain long-term, taxed at rate τ_{cg}.

The next buyer's basis. A secondary buyer starts with basis equal to the purchase price P. If P exceeds face, the premium amortizes at the buyer's yield, mandatorily for exempt bonds and by election for taxable bonds. On an OID bond bought at date s for $\text{AIP}_s < P \leq 100$, the acquisition premium $P - \text{AIP}_s$ reduces the remaining OID by the fraction $(P - \text{AIP}_s)/(100 - \text{AIP}_s)$, leaving basis 100 at maturity.

Market discount and the tax at sale. Market discount is the shortfall of the purchase price below AIP_s for OID bonds or below face for par and premium bonds. It accrues ratably, or at constant yield by election. Unless included in income annually, accrued market discount D_h is taxed as ordinary income at sale or redemption, up to any positive gain. This rule also covers exempt bonds bought after April 30, 1993. Below the de minimis threshold (119), the discount is treated as zero and the gain is a capital gain. The sale tax therefore separates the accrued market discount from the remaining gain:

$$\text{tax at sale} = \tau_{oi} \min\{\max(G_h, 0), D_h\} + \tau_{cg}[G_h - \min\{\max(G_h, 0), D_h\}], \qquad G_h = P_h - \text{basis}_h. \tag{135}$$

The first term taxes the positive gain up to accrued market discount at τ_{oi}. The second applies τ_{cg} to the residual gain or loss. A buyer at a market discount owes this tax, so the capital-gains adjustment builds it into the price that buyer pays (Appendix C.1.4).

Losses, institutions and the adjustment. Individual capital losses offset gains and up to $3,000 of ordinary income a year, with carryforwards (Sections 1211–1212). Corporations pay their ordinary rate on gains but offset capital losses only against gains, and banks treat gains and losses on debt as ordinary income (Section 582(c)). The model's institutional after-tax shares apply one rate to the full return. The capital-gains adjustment applies the household accrual and basis rules above with its stated federal rates and assumes full loss offsets at the capital-gains rate, as in (120).

D.4 Fiscal accounting and counterfactual evidence

This subsection supports the fiscal results of Section 7. On average, home-state repeal raises own-state financing costs by 0.68–0.88 per dollar of added state receipts across the seven states, and equal treatment raises all issuers' costs by 1.196 per dollar of added federal and state receipts.

D.4.1 Pricing shares and the annual fiscal ledger

Portfolio choice depends on relative after-tax shares, but tax receipts require absolute tax rates. For example, the pre-2018 P&C relative after-tax share of 0.6860 reflects taxes of 5.25% on municipal and 35% on taxable interest. Taxing only taxable interest, at $1 - 0.6860$ or 31.4%, would preserve the relative share but misstate both receipts. Each reform therefore changes absolute tax rates, while preferences and liability hedges stay in each investor's baseline payoff units (Section 4.2). For households, I keep the dated federal marginal rates, state-tax approximations and Social Security taxation proxy of Appendix B.1, and state receipts go to the state of residence. Before 2018, federal and state after-tax shares multiply, so the federal tax per dollar of taxable interest is the total tax less the state rate, $\tau_i^{\text{fed}}(1 - \tau_i^{\text{st}})$, where τ_i^{fed} and τ_i^{st} are investor i's federal and state tax rates.

What the ledger counts. The ledger covers interest on bonds and on taxable cash. Type i holds dollar cash $m_i(W_i - \mathbf{1}'x_i)$, its funding less its bond positions, and cash interest faces the same absolute tax rates as taxable bond interest. Aggregate funding, bond supplies and the short rate are fixed, so aggregate cash interest

[33] IRS Publication 550 (2017), "Original Issue Discount (OID)"; Publication 1212, "De minimis rule." The market-discount threshold in (119) applies separately.

is unchanged. When cash moves between investors with different tax rates, the change in its tax receipts is offset by the change in after-tax cash income, preserving identity (30). Changes in the bank financing-tax cost are recorded as federal receipts and deducted from banks' income, without counting interest expense or issuer payments twice. Other liability flows, issuer deductions, refinancing timing and fiscal feedback are outside the ledger.

Financing costs and averaging. Issuer costs are annual replacement financing at the solved yields, computed as debt outstanding at each maturity times its yield, $y_T D_T$, summed across maturities at $z = 0$. The contractual payments of existing bonds stay unchanged. For both reforms, I weight calendar years equally and divide mean dollar costs by mean dollar receipts. Yield changes average each period's monthly rate states.

D.4.2 Portfolio reallocation under the two reforms

Receipts depend on who holds each bond after a reform, so Table 39 reports the portfolio changes behind the fiscal effects.

Table 39: Portfolio allocation under the two tax reforms

Panel A. Home-state repeal: average resident allocations, 2000–2017			
	Home-state municipal	All risky	Riskless
California	20.33 → 4.35	30.02 → 31.56	69.98 → 68.44
New York	19.65 → 3.92	29.97 → 31.18	70.03 → 68.82
New Jersey	18.63 → 1.32	31.34 → 31.12	68.66 → 68.88
Massachusetts	18.96 → 1.55	29.95 → 30.99	70.05 → 69.01
Ohio	19.10 → 1.09	29.75 → 31.00	70.25 → 69.00
Pennsylvania	17.42 → 2.15	30.76 → 30.58	69.24 → 69.42
Minnesota	20.19 → 0.64	29.60 → 31.38	70.40 → 68.62
Panel B. Equal treatment: average investor allocations, 2000–2017			
	Risky maturity (years)	Share of long bonds (%)	Riskless
High-tax households	6.71 → 4.91	2.6 → 0.0	75.31 → 76.96
Lower-tax households	12.85 → 12.14	29.7 → 27.2	66.07 → 64.95
Banks	11.76 → 8.03	8.6 → 0.0	0.00 → 0.00
P&C insurers	9.93 → 6.83	5.3 → 1.7	21.70 → 0.65
Life insurers	11.81 → 12.07	14.6 → 14.0	0.38 → 0.68
Tax-neutral investors	7.73 → 8.53	39.2 → 57.1	27.90 → 29.56

Notes. Entries are baseline → reform values. Home-state, all-risky and riskless shares are percentages of each population's funding, which equals its bond holdings plus cash and excludes outside assets and liabilities. The share-of-long-bonds column reports each group's percentage of the total long-bond market. Panel A aggregates resident households within each state and averages their portfolio shares across the six periods of 2000–2017. Each row is a separate repeal. Panel B averages each investor group's allocations across the same periods under 99.9% equal treatment. Both panels weight types by their mass m_i and use rate state $z = 0$. Tax-neutral investors are the residual group of Table 3. Riskless holdings comprise cash and one-period municipal bonds (two years on the numerical grid). All longer bonds are risky. In Panel A, home-state municipal holdings overlap the other columns, and the last two columns sum to 100%. In Panel B, risky maturity is the value-weighted mean maturity in years of the group's risky municipal and taxable bonds. The long-bond market comprises all municipal and taxable bonds maturing in more than ten years, and its shares sum to 100% across groups. Under equal treatment, risky municipal and taxable bonds of the same maturity become close substitutes, so Panel B reports their combined maturity profile (Appendix B.7.4).

Home-state repeal. After repeal, home- and out-of-state municipal interest are both state-taxable, so switching issuers does not avoid state tax. Panel A combines the two resident household groups, which respond differently. High-tax residents raise their riskless share by 0.6–2.7 percentage points across the seven states. Lower-tax residents raise their taxable-bond share by 5.1–14.6 points and lower their riskless share. For both groups together, the riskless share rises in New Jersey and Pennsylvania and falls in the other five states. The high-tax group's mean bond maturity rises by 0.4–0.9 years because it replaces riskless municipal bonds with cash, which the bond-maturity average excludes, while its risky bonds become slightly shorter. In California, its mean bond maturity rises from 5.99 to 6.44 years and its mean risky-bond maturity falls from 7.05 to 6.52, while the lower-tax group's mean risky-bond maturity falls from 14.47 to 11.84 years. The added receipts thus accompany substitution across markets and tax groups, without a shift toward longer risky bonds.

Equal treatment. In Panel B, the tax-neutral group's share of the long municipal and taxable bond market rises from 39.2% to 57.1%. The high-tax household and bank shares fall to almost zero, the P&C share falls from 5.3% to 1.7%, and the lower-tax household and life-insurer shares decline slightly. High-tax households, banks and P&C insurers also shorten the average maturity of their risky bonds by 1.8 to 3.7 years, and high-tax households raise their riskless share.

D.4.3 Home-state repeal: financing costs and receipts

Panel A of Table 40 reports each state experiment by period, and Panel B decomposes the six-period averages. On average, own-state costs stay below added state receipts in every state.

Table 40: Home-state repeal: period financing ratios and the decomposition of costs and receipts

Panel A. Own-state / all-issuer financing costs per added state receipt dollar

State	2000–2002	2003–2005	2006–2008	2009–2011	2012–2014	2015–2017	Average
CA	0.80 / 1.19	0.61 / 0.84	0.63 / 0.67	0.78 / 1.13	0.94 / 1.41	0.96 / 1.52	0.81 / 1.17
NY	0.74 / 1.15	0.64 / 0.89	0.59 / 0.49	0.78 / 1.11	0.94 / 1.38	0.92 / 1.41	0.78 / 1.08
NJ	0.82 / 1.35	0.79 / 1.23	0.80 / 1.11	0.83 / 1.18	1.00 / 1.49	0.98 / 1.46	0.88 / 1.30
MA	0.66 / 1.03	0.51 / 0.75	0.56 / 0.49	0.66 / 0.95	0.80 / 1.04	0.90 / 1.20	0.68 / 0.89
OH	0.73 / 1.17	0.57 / 0.83	0.60 / 0.57	0.70 / 1.04	0.88 / 1.19	0.99 / 1.39	0.73 / 0.99
PA	0.66 / 1.11	0.56 / 0.88	0.66 / 1.02	0.63 / 0.87	0.90 / 1.12	1.02 / 1.32	0.73 / 1.03
MN	0.69 / 1.07	0.52 / 0.74	0.53 / 0.50	0.68 / 0.95	0.86 / 1.22	0.83 / 1.16	0.68 / 0.93

Panel B. Average annual changes ($million)

State	Cost at fixed y^0	Change in y^0	Total cost	Tax at y^0	Yield changes	Reallocation	Total receipts
CA	798.7	17.2	815.9	902.5	1.0	100.8	1004.3
NY	455.4	6.6	462.0	526.0	0.1	65.2	591.4
NJ	138.4	0.2	138.6	141.8	-0.2	15.2	156.9
MA	76.1	0.1	76.2	98.9	-0.1	14.0	112.8
OH	75.1	0.1	75.1	90.1	-0.1	12.7	102.6
PA	59.8	-0.3	59.5	73.0	-0.1	9.0	81.9
MN	59.5	0.1	59.6	75.8	-0.1	11.3	87.0

Notes. Each state-period cell is a separate repeal from its dated baseline. All-issuer costs add other municipal and pooled taxable issuers to the state's own issuers, and receipts are the repealing state's added receipts. The average column divides six-period mean dollar costs by mean receipts. Where the all-issuer ratio is below the own-state ratio, as in several 2006–2008 cells, residents' reallocation lowers other issuers' yields and hence their financing costs. Panel B splits the average own-state costs and receipts around each period's pooled no-income-tax-state municipal yield y_T^0, with the cost-at-fixed-y^0, change-in-y^0 and tax-at-y^0 columns defined in the text. Yield changes are the tax on residents' baseline home-state holdings at post-repeal yields less the tax at y^0, and reallocation is the net effect of residents' portfolio changes, including cash. Components sum to totals up to rounding. All dollar calculations use continuous yields and holdings at $z = 0$ and include residents' taxable cash interest.

Costs and receipts around the comparison yield. Panel B separates the fixed-yield comparison of Section 7.2 from yield changes and reallocation. Let D_T be the state's debt at maturity T, y_T^0 the pooled no-income-tax-state municipal yield and $\tau_T^{\text{impl}} = 1 - y_T^M / y_T^0$ the implied state tax, with primes marking post-repeal values. The cost increase is exactly

$$\Delta C = \sum_T D_T \, y_T^0 \left(\tau_T^{\text{impl}} - \tau_T^{\text{impl}\prime}\right) + \sum_T D_T \left(1 - \tau_T^{\text{impl}\prime}\right)\left(y_T^{0\prime} - y_T^0\right).$$

The first term holds y^0 fixed, and the second comes from the change in y^0. Averaged over the six periods, they give Panel B's cost-at-fixed-y^0 and change-in-y^0 columns. The tax at y^0 is $\sum_i \tau_i^{\text{st}} \sum_T H_{iT} y_T^0$, where H_{iT} is resident group i's baseline home-state holding and τ_i^{st} its state rate.

At the comparison yield, own-state cost per dollar of tax ranges from 0.770 in Massachusetts to 0.976 in New Jersey. Beyond the tax at the comparison yield, almost all additional receipts come from residents' reallocation, which adds 10.8% to 14.9% of that tax, so costs stay below receipts in every state. By asset, California's additional annual receipts of $1.004 billion comprise $0.227 billion from home-state bonds and net increases of

$0.300 billion from other states' bonds, $0.368 billion from taxable bonds and $0.110 billion from cash. Interest on these last three assets was already taxable, so their increases come from changes in residents' holdings and the interest they earn.

California's $100.8 million reallocation term values purchases and sales at post-repeal yields. If every bond earned the post-repeal comparison yield $y_T^{0\prime}$ at its maturity and cash kept its taxable yield, the net portfolio effect would be $4.9 million. Taxable yields above this reference add $96.8 million and municipal yield differences −$0.9 million, so most of the increase comes from substitution into higher-yielding taxable bonds. All three terms value the same solved portfolio changes.

Differences across states. At longer maturities, lower pass-through reduces the cost per resident-held dollar, but debt held by nonresidents and institutions adds cost without adding tax. New Jersey's residents hold only 49% of its 12–20 year debt, on which cost is 1.38 times the tax at the comparison yield, giving New Jersey the highest overall ratio. Minnesota's residents hold 80% of its 22–30 year debt, against 67% in New Jersey, and its lower-tax residents face a higher rate, so its cost on those bonds is 0.59 per dollar of tax, against 0.98 in New Jersey. Household composition, outside exposures and resident funding relative to local debt are common across states within each period, so these differences in resident ownership and cost come from tax schedules and market sizes.

D.4.4 Equal treatment and an issuer subsidy

Table 41 reports the equal-treatment reform by period and a subsidy that returns all added federal and state receipts to municipal issuers at the solved equal-treatment prices and portfolios. Per receipt dollar, the municipal cost after subsidy is the all-issuer cost less the taxable-issuer cost less one dollar. It averages 0.063, while taxable issuers still bear their 0.133 increase. In every period, all-issuer costs exceed the added receipts.

Table 41: Equal treatment and a revenue-financed issuer subsidy across the six pre-2018 periods

	Thirty-year yield increase (bp)		Costs per receipt dollar		
Period	Municipal	Taxable	All issuers	Taxable issuers	Municipal after subsidy
2000–2002	199.8	38.2	1.434	0.282	0.152
2003–2005	136.9	9.3	1.125	0.089	0.036
2006–2008	146.5	11.4	1.077	0.073	0.004
2009–2011	152.7	11.2	1.113	0.043	0.070
2012–2014	178.9	33.9	1.245	0.170	0.075
2015–2017	208.6	34.7	1.242	0.174	0.068
Average 2000–2017	170.6	23.1	1.196	0.133	0.063

Notes. Each row removes 99.9% of the municipal–taxable tax distinction from its own calibrated baseline. Yield changes average each period's 36 monthly rate states. Fiscal ratios use continuous yields and holdings at $z = 0$, with bond and taxable cash interest in receipts. The last column subtracts a subsidy equal to all added federal and state receipts from municipal issuer costs. The other columns describe the reform before subsidy. The subsidy leaves gross yields and portfolios unchanged because bond supplies and investors' funding are fixed. The average row divides calendar-year-weighted mean costs by mean receipts.

D.5 Financing, short sales and rate hedges

The model rules out short sales and borrowing (Sections 3.5 and 4.1). The nine cases below explain the frictions behind these constraints. Financing a municipal purchase, hedging a bond bought with the investor's own funds and shorting a municipal bond are distinct transactions, with different tax and market costs. With equal borrower and lender tax rates, the corporate-short-financed purchase in case (i) and the reverse trade in case (vi) both have nonpositive carry throughout the band (143), which contains long-end municipal coupons on the sample-average curves. The final parts connect the rate exposure left after hedging to participation and test household swap access in the quantitative model.

Household holdings. High-income households, whose top-bracket tax rates define the household benchmark, hold much of the market. In custodial data, municipal bonds are wealthy households' second-largest liquid risky asset, and their portfolio share rises with wealth (Gabaix et al., 2026). In tax years 2008, 2016

and 2019, returns with adjusted gross income (AGI) of $1 million or more, about 0.3% of all returns, reported 26–33% of tax-exempt interest.[34]

Carry notation. Define the after-tax coupon gap $g = c_M - (1-\tau)c_C$, where c_M and c_C are municipal and corporate par coupons of the same maturity and τ is the investor's marginal tax rate. This coupon measure differs from the 118-basis-point thirty-year puzzle in zero-coupon yields (Section 5.3) and from the model's one-period return gap g_T. Carry is computed for equal principal in each leg, before fees, margin and collateral costs. The legs need not match in duration or price risk.

D.5.1 Financing a municipal position

The first two cases finance municipal holdings by borrowing, directly or through a corporate short.

(i) Financing municipals with debt or a corporate short. Let c_F be the interest on a dollar borrowed to buy a dollar of municipal principal. If that interest were deductible, carry would be $c_M - (1-\tau)c_F$. Section 265(a)(2) denies deductions for interest on debt incurred or continued to purchase or carry tax-exempt obligations, so carry becomes

$$c_M - c_F = \{c_M - (1-\tau)c_F\} - \tau c_F. \tag{136}$$

At after-tax parity, $c_M = (1-\tau)c_F$, the denial turns zero carry into a loss of τc_F. Financing the position with a corporate short does not restore the tax shield. Section 265(a)(5) treats the short seller's compensatory coupon payments c_C as interest under Section 265(a)(2), which leaves carry $c_M - c_C$. This carry is negative at every maturity because the sample-average municipal zero curve, and hence the implied par yield, lies below its corporate counterpart. At thirty years with annual compounding, the municipal and corporate par yields are 4.498% and 5.443%.[35]

Municipal fund disclosures tell investors that interest on debt allocable to exempt-interest dividends is nondeductible.[36] While the deduction was available, depository institutions financed municipal holdings at scale, holding 51.8% of municipal bonds in 1972 and $235.1 billion (27.3%) in 1985. After the 1986 denial (Appendix D.1), their holdings fell to $120.4 billion in 1990 and $95.4 billion (7.5%) in 1995, while municipal debt outstanding grew from $862 billion in 1985 to $1.27 trillion in 1995.[37]

(ii) Pairing corporate positions. If the proceeds of a corporate short buy a corporate bond, the investor's own capital finances the municipal bond, and the remaining trade is the hedge in (iii). If they finance the municipal bond, the denial in (i) applies. Which case applies depends on the borrowing's purpose. Revenue Procedure 72-18 tests that purpose through tracing of the proceeds, exempt collateral or the facts and circumstances. Its presumption that business borrowing is not incurred to carry exempt bonds lapses when the borrowing exceeds business needs. If the positions form an offsetting straddle, Section 263(g) also capitalizes interest and carrying charges, net of the positions' taxable income, deferring the deductions.[38]

D.5.2 Corporate and derivative hedges

Hedging a bond's rate risk is distinct from financing its purchase: the investor buys the municipal bond with its own funds and takes an offsetting rate position elsewhere. The costs then include the hedge's tax treatment, access requirements and residual risks.

[34] IRS Statistics of Income, Individual Income Tax Returns, Table 1.4, all returns by size of adjusted gross income, tax years 2008, 2016 and 2019. Tax-exempt interest includes exempt-interest dividends from funds.

[35] 26 U.S.C. §265(a)(2), (5) and (b), 2018 U.S. Code. Section 265(a)(5)(B) excepts specified short-sale payments when the taxpayer posts cash collateral that earns no material return. That exception does not remove the opportunity cost of the collateral. See also Hines and Horton (2024) on banks' financing-tax restrictions.

[36] BlackRock Municipal Bond Fund, Inc. (2025), Part II, "Dividends and Taxes," p. II-131; iShares Trust (2026), "Taxes," p. 210; Nuveen Municipal Trust (2026), "Tax Matters," p. S-80.

[37] Board of Governors of the Federal Reserve System, Financial Accounts of the United States (Z.1), release of September 8, 2023, Table L.212, year-end levels. Depository institutions comprise U.S.-chartered depository institutions, foreign banking offices in the U.S. and banks in U.S.-affiliated areas.

[38] Rev. Proc. 72-18, 1972-1 C.B. 740, §§3.01–3.04 and 6.01, as summarized in Revenue Ruling 2004-47 and in IRS Field Service Advice 200129009, which is nonprecedential. On straddles, see 26 U.S.C. §263(g)(1)–(2) and §1092(c)(1)–(2). Positions are offsetting when holding one substantially diminishes the risk of loss from holding the other.

(iii) Hedging duration with a corporate short. An unlevered hedge of a municipal bond's duration shorts a long-maturity corporate bond and buys a short-maturity corporate bond with the proceeds. Let c_ℓ and c_s be the long and short corporate coupons, so that $g = c_M - (1-\tau)c_\ell$, and let $\delta \in [0,1]$ be the currently deductible fraction of the short payment. With equal principal in each leg, annual carry relative to holding the short corporate bond alone is

$$\Delta = \{c_M + (1-\tau)c_s - (1-\delta\tau)c_\ell\} - (1-\tau)c_s = g - (1-\delta)\tau c_\ell. \tag{137}$$

Denial under Section 265 gives $\delta = 0$ and $\Delta = c_M - c_\ell$, negative at every maturity and a loss of 95 basis points a year at thirty years on the observed sample-average par curves, larger still on the call-adjusted municipal curve. Section 263(g) capitalization and Section 163(d) carryforwards defer deductions. Current carry then approaches $c_M - c_\ell$, and later deductions bear a timing and liquidity cost.

Even where neither Section 265 nor Section 263(g) applies, Section 163(d) lets individuals deduct investment interest only if they itemize and up to net investment income, carrying the excess forward. Coupons on the short-maturity corporate bond add to this deduction capacity, while wages and exempt coupons do not. With principal q on both corporate legs, coupon difference $k = c_\ell - c_s > 0$, unused deduction capacity $\mathcal{E} \geq 0$ and no prior carryforwards, the hedge's current cash flow is

$$\Pi(q) = -(1-\tau)qk - \tau[qk - \mathcal{E}]_+, \tag{138}$$

where $[x]_+ = \max\{x, 0\}$, so $\delta < 1$ once qk exceeds $\mathcal{E}$.[39]

Even with $\delta = 1$, the hedge removes only common rate risk. Bond-specific risk (Appendix A.2.8) and credit, liquidity and call differences (Appendix C) remain. Shorting requires available bonds and incurs fees, recall risk and margin demands against an illiquid callable long position.[40] Fees for borrowing corporate bonds are 10–20 bp in the major-lender sample of Asquith et al. (2013).

(iv) Hedging with swaps. A pay-fixed, receive-floating swap hedges duration without borrowing or shorting a bond, but household access is restricted. A person that is not an eligible contract participant can trade a swap only on, or subject to the rules of, a designated contract market. An individual qualifies only with more than $10 million invested on a discretionary basis, or more than $5 million when the swap manages the risk of an asset owned or a liability incurred.[41]

Ordinary swap payments are not automatically interest on indebtedness, so Section 265(a)(2) alone does not deny their deduction. A household's Section 212 investment expenses attributable to exempt interest fall under Section 265(a)(1), which disallows them.[42] Whatever the classification, let G be a net taxable amount and $d \in [0,1]$ the denied fraction of the tax saving on its losses, so that its after-tax value is

$$A_d(G) = (1-\tau)G - d\tau G^-, \qquad G^- := \max\{-G, 0\}. \tag{139}$$

Symmetric offsets give $d = 0$. Taxing gains and denying losses gives $d = 1$, which lowers expected proceeds by $\tau\mathbb{E}[G^-]$.

Netting gives the swap a tax advantage when the corporate short payment is disallowed. Suppose both trades have matched periodic receipts $F \geq 0$ and payments $K \geq 0$. When the corporate short payment is wholly nondeductible, the bond trade's after-tax cash is $V_C = (1-\tau)F - K$. Under the rule that taxes net gains and denies losses, the swap's is $V_S = F - K - \tau[F-K]_+$. Hence

$$V_S - V_C = \tau \min\{F, K\} \geq 0. \tag{140}$$

[39] 26 U.S.C. §163(d)(1)–(3), which includes otherwise deductible short-sale expenses in investment interest; IRS Publication 550 (2016), Chapter 3, "Interest Expenses" and "Interest and carrying charges on straddles"; IRS Form 4952 and instructions (2016). Net capital gain and qualified dividends count as investment income only through an election that relinquishes their preferential rates to that extent.

[40] Financial Industry Regulatory Authority (FINRA) Rule 4210, paragraph (e)(2), sets margin requirements for bond shorts.

[41] Commodity Exchange Act §2(e) and §1a(18)(A)(xi), 7 U.S.C. §§2(e), 1a(18)(A)(xi); see also the Commodity Futures Trading Commission's 2012 final-rule summary.

[42] Treasury Regulations §1.446-3 and §1.265-1(a)(2), (c). IRS Field Service Advice 200037034 treats the ordinary swap payments before it as non-interest in the absence of a loan. The advice concerns a corporate taxpayer and is nonprecedential.

If the corporate payment is fully deductible, the two coincide for $F \geq K$. For $F < K$, the corporate trade benefits from a loss offset that the swap calculation denies. Additional deduction capacity therefore favors the corporate short.

A pay-fixed swap has the rate exposure of a short fixed-rate bond and a long floating-rate bond. It avoids the corporate short's credit and borrowing exposures but carries its own collateral and counterparty exposure. In the mean–variance objective, the swap weakly dominates any corporate hedge that delivers the same exposure with only an additional nonnegative cost and uncompensated risk orthogonal to the existing portfolio and the swap.[43] An outside dealer can also absorb swap risk without trading domestic bonds, whereas physical shorts change the gross long positions required to clear those markets.

(v) Treasury futures. Regulated futures avoid the swap eligibility rules but require cash margin and annual marking to market, with gains and losses treated as 60% long-term and 40% short-term. Individuals offset these losses fully against capital gains, deduct at most $3,000 against ordinary income and carry the excess forward. Net losses on these Section 1256 contracts can be carried back three years against net Section 1256 gains. Section 1092 defers losses to the extent of unrecognized gains on offsetting bonds. Gains can therefore be taxed before losses yield tax benefits. In present value, this timing and liquidity cost acts as a partial loss denial, $0 \leq d < 1$ in (139).[44]

D.5.3 Municipal shorts, financing structures and other entities

The remaining cases cover municipal shorts, a pass-through structure and other investors.

(vi) Shorting municipals. A short seller must pay the lender an amount equal to each coupon, called a substitute payment. Unlike the exempt coupon it replaces, this payment is taxable to the lender. A lender with tax rate τ_L who requires an after-tax payment equal to the coupon c_M must therefore receive L satisfying

$$(1-\tau_L)L = c_M, \qquad L - c_M = \frac{\tau_L}{1-\tau_L} c_M. \tag{141}$$

A short also needs a willing lender of the specific issue and must meet the contract's collateral terms.[45] The reverse trade shorts a municipal bond and buys an equal-principal corporate bond. Let τ_S be the short seller's tax rate and δ the currently deductible fraction of the substitute payment, as in (iii). Under this lender compensation, the trade's coupon carry is

$$\pi_{M\to C} = (1-\tau_S)c_C - (1-\delta\tau_S)\frac{c_M}{1-\tau_L}. \tag{142}$$

Even with full deduction, $\delta = 1$, equal borrower and lender rates τ give $\pi_{M\to C} = (1-\tau)c_C - c_M$. This carry is nonpositive when $c_M \geq (1-\tau)c_C$, and the carry $c_M - c_C$ of (i) is nonpositive when $c_M \leq c_C$, so both coupon trades have nonpositive carry throughout the band

$$(1-\tau)c_C \leq c_M \leq c_C, \tag{143}$$

and long-end municipal coupons lie in it. They exceed the household after-tax benchmark and, on the sample-average curves, fall below the corporate coupon. A municipal curve trade, which shorts long-maturity and buys short-maturity municipal bonds, pays the gross-up in (141) on its short leg. To the extent its proceeds buy the short-maturity municipal bond, that payment is nondeductible as in (i).

(vii) Tax-exempt floaters. A tender-option-bond partnership splits a long municipal bond's cash flows between a floating-rate claim and a residual inverse floater. Exempt income passes through to both classes, so the structure needs no interest deduction. The inverse holder receives the municipal curve's slope and bears its duration, so capturing g still requires one of the hedges in (iii)–(v). Demand for the floaters limits the scale of the strategy.[46]

[43] In common payoff units, let the corporate hedge be $B = \alpha S - \varsigma + \varepsilon$, where S is the swap payoff, $\alpha > 0$, $\varsigma \geq 0$, $\mathbb{E}\varepsilon = 0$, and $\mathrm{Cov}(\varepsilon, \widetilde{W}) = \mathrm{Cov}(\varepsilon, S) = 0$ for the random part $\widetilde{W}$ of the investor's existing wealth. With the certainty equivalent $\mathrm{CE}(Y) = \mathbb{E}Y - \gamma\,\mathrm{Var}(Y)/2$, one has $\mathrm{CE}(\widetilde{W} + qB) - \mathrm{CE}(\widetilde{W} + q\alpha S) = -q\varsigma - \gamma q^2\,\mathrm{Var}(\varepsilon)/2 \leq 0$ for $q \geq 0$, provided the replacement is feasible at no greater cost of financing the position.

[44] 26 U.S.C. §1256(a)(1), (a)(3), (b) and (g)(1); §1211(b); §1212(b)–(c); §1092(a)(1)(A). Section 1256(a)(4) removes Sections 1092 and 263(g) only for straddles consisting entirely of Section 1256 contracts.

[45] IRS Publication 550 (2016), "Reporting Substitute Payments"; Treasury Regulation §1.6045-2(a)(4).

[46] Revenue Procedure 2003-84, Section 2; 26 U.S.C. §702(b).

(viii) The de minimis presumption. For insubstantial holdings, Revenue Procedure 72-18 ordinarily presumes that borrowing is not incurred to carry exempt bonds, absent tracing or exempt collateral. A corporation qualifies only up to 2% of average business assets. Direct evidence overrides the presumption, and larger holdings face the facts-and-circumstances test. Sections 163(d), 263(g), 1092 and 1211(b) still constrain deductions and losses.[47]

(ix) Other entities. Financial institutions outside the bank-qualified exception face the pro rata disallowance of Appendix D.1, without tracing. Revenue Procedure 72-18 applies a similar ratio to dealers' business borrowing that cannot be traced. Insurer proration reduces the exemption regardless of financing, through P&C insurers' loss deductions (Section 832(b)(5)(B)) and life insurers' reserve deductions (Sections 807 and 812). Partnerships pass the exempt character of municipal interest through to partners under Section 702(b), but their interest expense remains subject to Section 265(a)(2) under Section 703(a). Dealers and hedge funds hold few municipal bonds. Over 2000–2020, dealers held on average 0.85% of municipal bonds, with a maximum of 1.94%. Hedge funds, reported from 2012, held at most 0.36%.[48]

D.5.4 The model constraint and risk after hedging

The frictions motivate nonnegative bond positions, while the rate risk remaining after hedging determines participation.

The model constraint. With a concave objective and linear shorting costs, a zero short position is optimal when the risk-adjusted benefit of the first dollar shorted does not exceed the tax, borrowing and financing costs. The quantitative model imposes this corner: bond positions are nonnegative and limited by measured funding, and banks bear the measured financing-tax cost. The household swap extension below keeps these bond-market constraints and lets households hedge rate risk.

Residual risk and participation after hedging. The mechanism can operate when investors also trade derivatives, because it depends on their cost of bearing additional rate risk after available hedges. To isolate this tradeoff, suppose the shorting constraint on cash is slack and bond-specific risk is zero. Let a hedge remove q_i units of rate exposure at an expected charge $f_i q_i$ and a differentiable convex additional cost $\psi_i(q_i)$, all in the investor's normalized payoff units. With residual exposure $\widehat{D}_i^{\text{hedge}} = \widehat{D}_i - q_i$, where $\widehat{D}_i$ is the net rate exposure (10) before hedging, the hedge-dependent part of the objective is $-f_i q_i - \psi_i(q_i) - \gamma_i \sigma_r^2 (\widehat{D}_i^{\text{hedge}})^2/2$. An interior hedge choice therefore satisfies

$$\lambda_i := \gamma_i \sigma_r^2 \widehat{D}_i^{\text{hedge}} = f_i + \psi_i'(q_i). \tag{144}$$

Here λ_i is the unit cost of rate risk at the post-hedge exposure, and a binding hedge limit turns the equality into a one-sided condition. At an equilibrium allocation with hedges and other holdings optimized, suppose the household holds the T-period municipal bond. The lower-tax investor's municipal slack, its required return on that bond minus the offered return, is then

$$\nu_{I,T}^M = (\xi_I - \xi_H) r - A_{T-1}^M (\lambda_H - \lambda_I), \tag{145}$$

and the lower-tax investor holds the bond as well only if this slack is zero. This is the participation comparison (59) with unit risk costs measured after hedging. Even when both investors hedge, differences in liabilities sustain unequal valuations of duration through unequal hedge access, binding limits or hedging costs that rise with the hedge.

If unlimited hedging in either direction spans rate risk, with $\psi_i = 0$ and a common normalized charge f for both investors, then $\lambda_H = \lambda_I = f$. At a positive short rate, the slack reduces to the positive tax difference $(\xi_I - \xi_H) r$, which excludes lower-tax municipal holdings. This conclusion requires a common after-tax hedging opportunity, which zero dealer fees alone do not provide. Observed insurer holdings (Figure 25) therefore require a departure from this benchmark.

[47] Rev. Proc. 72-18, §3.05, as summarized in Revenue Ruling 2004-47 and Field Service Advice 200129009.

[48] Rev. Proc. 72-18, §§5.03 and 7, as summarized in Revenue Ruling 2004-47; 26 U.S.C. §265(b)(1)–(2), §832(b)(5)(B)–(C), §807(a)–(b) and §812. Holdings: Financial Accounts of the United States (Z.1), release of September 8, 2023, Tables L.212 and B.101.f, quarterly levels.

Insurers already hedge, so their participation depends on the cost of rate risk after hedging. At year-end 2015, interest-rate swaps accounted for 64% of the notional value of U.S. insurers' interest-rate hedges. Insurers that used derivatives held 87% of life-insurer assets but 19% of P&C assets.[49]

D.5.5 Household swaps: incentives and equilibrium

Only the wealthiest households can meet the swap eligibility thresholds of case (iv). This exercise goes further and gives every household access, in the 2015–2017 period, to a pay-fixed, receive-floating swap, which gains when rates rise. Stage 1 evaluates the incentive for all 630 household types at fixed mean-state bond portfolios. Stage 2 lets households choose swaps and bonds jointly and re-solves both markets. Swap access reduces the thirty-year gap by 9.1%, so most of the gap survives household hedging. At $z = 0$, the households that hedge keep almost all their municipal holdings in riskless two-year bonds.

Swap quotes, pricing and taxes. Swap quotes come from the 95 dates on which the two-year zero rate bootstrapped from the swap curve lies within 10 bp of the model's mean rate, 3.0206% a year, all in 2001–2008, outside the 2015–2017 period. Fixed rates at ten, twenty and thirty years are 4.8217%, 5.0976% and 5.3736%, and the twenty-year rate is interpolated.[50] Floating payments are quarterly and fixed payments semiannual, and a single curve discounts payments and projects floating rates. Swaps are priced, at inception and at liquidation after two years, under a Gaussian rate state with quarter-by-quarter drifts fitted to these quotes, while physical rate dynamics preserve the calibrated two-year persistence and variance.

Annual net gains and the second-year liquidation payoff are taxed as ordinary income. Losses are denied in the baseline and offset symmetrically in a sensitivity check. Municipal funds likewise distribute swap gains as taxable ordinary income or capital gains and allocate net payments pro rata against taxable and exempt income.[51] Separately held swap cash earns taxable interest when positive and pays nondeductible interest when negative, and existing cash does not change at fixed portfolios.

Stage 1: the incentive at fixed bond portfolios. Let $\widetilde{W}_i$ be the calibrated log-return risk index (the portfolio innovation (35)), including the term $-h_i z'$ from outside rate exposure (Appendix B.1), and ξ_i^M investor i's after-tax share of home-state municipal interest (Section 4.1). A swap with notional $q \geq 0$ and after-tax cash A_i per unit notional adds qA_i/ξ_i^M to the normalized risk index. Here A_i is a cash payoff, unrelated to the bond loading A_T^q. Differentiating the mean–variance objective gives the annualized marginal benefit, or score, in basis points per unit notional,

$$b_i = \frac{10^4}{2}\left[\mathbb{E}A_i - \gamma_i \operatorname{Cov}(\widetilde{W}_i, A_i)\right]. \tag{146}$$

A positive score means that a small pay-fixed position improves the objective at the fixed bond portfolio. The score uses the model's log-return approximation, and independent bond-specific shocks have zero covariance with the rate-only swap.

Stage 1 results. At all three tenors, 270 types have positive scores and 360 negative scores. The positive-score types, those with the three largest of the seven outside rate exposures in each household tax group and home state, hold 42.9% of household portfolio assets but only 9.8% of household municipal bonds. Their larger gains make the portfolio-asset-weighted mean positive despite the negative median (Table 42). With symmetric loss offsets, 585 of the 630 scores are positive. Doubling pricing resolution and draws preserves every sign and changes scores by at most 2.57 bp, and terminal dollar payoffs and covariances also preserve the signs. A wider quote window gives between 275 and 302 positive scores, some near zero.

[49] NAIC, "Update on the Insurance Industry's Use of Derivatives and Exposure Trends," year-end 2015, Tables 1 and 8 and the accompanying discussion. Notional shares measure instrument use and are silent on how much liability risk is eliminated.

[50] Federal Reserve H.15 (Selected Interest Rates) series DSWP1–DSWP5, DSWP7, DSWP10 and DSWP30, numbered by tenor in years, are mid-market fixed rates against three-month London Interbank Offered Rate (LIBOR) and differ from household quotes.

[51] BlackRock Municipal Bond Fund, Inc. (2025), Part II, pp. II-20, II-126 and II-132; iShares Trust (2026), "Taxes," p. 213.

Table 42: Fixed-portfolio swap benefits with gains taxed and losses denied.

Household group	Years	Mean payoff	Risk benefit	Total	Median	Positive
High tax	10	−408.6	852.8	444.3	−14.4	135/315
High tax	20	−473.5	996.0	522.6	−13.1	135/315
High tax	30	−473.9	988.2	514.2	−17.2	135/315
Lower tax	10	−387.9	675.6	287.7	−28.8	135/315
Lower tax	20	−450.0	787.6	337.7	−31.5	135/315
Lower tax	30	−449.5	783.3	333.9	−33.3	135/315

Notes. Annualized scores b_i of equation (146), in basis points per unit notional. Years is the swap tenor. Mean payoff and risk benefit are the score's two terms and total is their sum, each averaged within the group with portfolio-asset weights. Positive counts types with positive scores out of all types. The median and the count of positive scores weight types equally. Entries are rounded separately.

Table 43: Household swap access and the equilibrium municipal gap.

Maturity	Baseline	Swap access	Change
2 years	0.00	0.00	0.00
10 years	23.68	22.24	−1.44
20 years	72.63	65.42	−7.20
30 years	104.27	94.79	−9.48

Notes. Entries are the supply-weighted municipal yield minus the household benchmark, in basis points, averaged over the 36 monthly rate states of 2015–2017 with continuous yields. After-tax shares, parameters and dollar supplies are fixed, and both yields adjust. Changes are computed before rounding. Refining the simulation fourfold moves either yield curve or the gap by less than 0.003 bp, and price and optimality residuals are below 10^{-10}.

From a marginal incentive to a finite hedge. At fixed bond prices and holdings, let $c_i \geq 0$ be any additional two-year certainty-equivalent cost per unit notional, in A_i's absolute units. The mean–variance objective is quadratic in q, giving

$$q_i^*(x; c_i) = \frac{\xi_i^M}{\gamma_i \operatorname{Var}(A_i)} \left[\mathbb{E}A_i - \gamma_i \operatorname{Cov}\left(\widetilde{W}_i(x), A_i\right) - c_i \right]_+ . \tag{147}$$

Here $[u]_+ = \max\{u, 0\}$, as in (138), and $\widetilde{W}_i(x)$ is the risk index of bond portfolio x. Costs reduce hedge demand at given bond holdings, and margin and position limits restrict feasible hedges.

Stage 2: equilibrium with household hedge access. With the 2015–2017 calibration fixed, every household jointly chooses bonds and a nonnegative thirty-year pay-fixed swap notional. Outside dealers quote swaps with no fee, margin or notional limit and do not trade the modeled bonds. New swaps are struck at the current state's par rate and priced through liquidation with the fitted drifts, counted from their start date. Swap gains are taxed, losses are denied ($d = 1$) and interim swap cash is financed separately. Physical bond shorts remain excluded, bonds and cash stay nonnegative and institutions' hedging is unchanged, while bond prices in both markets and all portfolios adjust across rate states.

Swap access reduces the thirty-year gap by 9.48 bp, or 9.1%, on the 2015–2017 model curves before the full-sample corporate normalization (Table 43). Thirty-year municipal yields fall 15.95 bp and taxable yields 12.04 bp. At $z = 0$, 270 types hedge, but over 99.99% of their municipal holdings remain riskless two-year bonds, because these types mainly hedge outside rate exposure. Portfolio responses at other rate states still affect continuation prices.

Scope of the equilibrium test. The test substantially expands household access. It keeps asymmetric taxes and nondeductible interest on negative swap cash but has no dealer fees, bid–ask spreads, margin, collateral opportunity costs or notional limits, and the largest mean-state notional is 6.88 times the type's funding.

D.6 Data sources and analysis coverage

Table 44 lists what each source measures and its coverage, the span assembled for the paper. Each analysis states its own estimation window. Appendix B explains the calibration inputs, and Appendix C gives the matched samples for the adjustments and empirical tests.

Table 44: Data sources, variables and coverage.

Source	Variables	Coverage
Municipal issues: Mergent Municipal Bond Database (FTSE Russell)	CUSIP, issue & maturity dates, coupon, rating, callability, tax status (federal/state), issuer state, purpose, amounts outstanding	1986–2023
Corporate issues: Mergent FISD (FTSE Russell)	CUSIP, issue & maturity dates, coupon, callability, rating, issuer industry, seniority, amounts outstanding	1986–2023
Insurers: NAIC Schedule D (annual, Q1–Q3 supplements)	CUSIP-level par, book and market values for U.S. P&C and life insurers	1991–2023
Banks: FDIC BankFind (Call Report income statement RI and securities schedule RC-B)	Annual interest expense, average assets and municipal securities holdings by bank	2000–2021
Public pensions: FOIA requests and annual financial reports	CUSIP-level municipal/taxable par and market values, 30 state/local plans (all 30 covered from 2015)	2000–2023
Sector holdings and assets: Financial Accounts (Z.1)	Quarterly municipal holdings by sector (L.212, B.101.f), sector debt securities and cash, treasury and corporate totals	1972–2022
Households: SCF, Federal Reserve	Tax-unit income and filing status, financial assets and debt	2001–2019 waves
Household taxes: IRS Statistics of Income (SOI)	Filer counts and taxable and tax-exempt interest by AGI bracket	2000–2022
Municipal trades: MSRB Real-Time Transaction Reporting System (RTRS)	Trade-level price, par volume, customer-side flag, trade-condition codes	2005–2023
Corporate trades: FINRA TRACE Enhanced (Academic)	Trade-level price, par volume, customer-side flag, condition codes	2002–2023
Treasury curve: Gürkaynak–Sack–Wright (GSW), Federal Reserve	Daily fitted par yields, 1–30y, used in P3	2000–2023
Short rates: TB3MS, FRED	Monthly three-month treasury bill rates for the rate process	1952–2024
Treasury supply: Monthly Statement of the Public Debt	Security-level amounts and maturities for the treasury maturity profile	2001–2020
Corporate zeros: A-or-above corporate zero curve (HQM, FRED)	Monthly A-or-above zero curve, 1–30y	2000–2023
Credit risk: Markit/IHS	State-government and AAA, AA and A corporate CDS at 1–30y tenors, for the matched adjustment sample	2004–2020
Expected rates: Kim–Wright, Federal Reserve; Adrian–Crump–Moench	Monthly fitted yields and term premia, expected-average-short-rate component capped at 10y	2000–2021
Swaps: Federal Reserve H.15 (FRED)	Par interest-rate swap rates, 1–30y, for household swap pricing	2000–2016
State taxes: Tax Foundation	Top marginal rate, bracket thresholds, treatment of out-of-state municipal interest	2000–2022
State income: Bureau of Economic Analysis (BEA)	Quarterly state-level personal income	2000–2022
Historical state rates: Institute on Taxation and Economic Policy (ITEP)	Top statutory rate by state, the P1 instruments	1945–2000 vintages